

Rough Polynomial Fluctuation Theory: Coefficient-Uniform Critical Jumps, Finite Radial Variation, and Endpoint Bounds

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Abstract

We prove coefficient-uniform bounds for the pointwise critical 2-jump functional of the full sharp-truncation trajectory

$$T_{\Omega, \varepsilon}^{P, b} f(x) = \int_{|x-y| > \varepsilon} e^{iP(x, y)} b(|x-y|) \frac{\Omega((x-y)/|x-y|)}{|x-y|^n} f(y) dy,$$

where P is an arbitrary real polynomial on $\mathbb{R}^n \times \mathbb{R}^n$ of bounded degree, $\Omega \in L \log L(\mathbb{S}^{n-1})$ has mean zero, and the radial factor b has finite ρ -variation for some fixed $1 \leq \rho < \infty$. For every $1 < p < \infty$ the pointwise functional

$$\sup_{\lambda > 0} \lambda N_{\lambda}(T_{\Omega, \varepsilon}^{P, b} f : \varepsilon > 0)^{1/2}$$

is bounded on $L^p(\mathbb{R}^n)$, and it maps L^1 to $L^{1, \infty}$. The constants depend only on the dimension, the degree bound, the relevant spatial exponent, and ρ ; they are independent of every coefficient of P and of where the radial variation of b is concentrated.

At $P = 0$ and $b \equiv 1$, the weak endpoint strictly strengthens, and hence resolves, the critical-jump problem posed by Jones–Seeger–Wright. For general P , the principal-value weak type $(1, 1)$ consequence answers the rough-kernel problem posed by Ding and Lai in the mean-zero $L \log L$ class. The jump theorem implies strong and weak r -variation inequalities for every $r > 2$, maximal-truncation bounds, almost-everywhere principal-value convergence, and a canonical trajectory determined by its finite-annulus increments and zero-at-infinity anchor. For radial $1 \leq \rho < 2$, a direct Young–Stieltjes argument transfers the unweighted trajectory pointwise. At and above $\rho = 2$, the proof instead uses operator-specific Fourier, correlation, and packet orthogonality. The finiteness of ρ is necessary: a bounded radial Hilbert example rules out $\rho = \infty$. An exact Mellin representation of the Poisson semigroup also shows that true output V^2 fails already for sharp Hilbert truncations.

The proof combines a finite global algebraic flag on polynomial phases modulo pure input and output phases, a phase-adapted Calderón–Zygmund decomposition with anchored phase–packet induction, and a selector-free fixed-rank Rademacher–Menshov reduction supported by concentration-stable equal- and unequal-shell correlation estimates. The finite-radial extension is obtained by proving the stability of every oscillatory, terminal-packet, weak-endpoint, and completion step under a V^p radial amplitude.

AI-use disclosure. The research direction and target theorem of this paper were formulated through iterative interaction between the author and an artificial-intelligence system. The author directed and coordinated the mathematical development, including the selection and refinement of the problems and intermediate objectives pursued during that process. The AI system generated the mathematical proof arguments and derivations establishing the results of the paper; these contributions occur throughout Sections 2–14 rather than being confined to isolated lemmas or theorems. The author examined, selected, revised, and organized the resulting arguments.

1 Introduction and main results

A maximal-truncation estimate controls only the largest excursion of an operator trajectory. An r -variation estimate with $r > 2$ also quantifies convergence, but the endpoint $r = 2$ is generally inaccessible. The critical replacement is the pointwise 2-jump functional

$$\mathbf{J}(F)(x) := \sup_{\lambda > 0} \lambda N_\lambda(F(x))^{1/2}. \quad (1)$$

The spatial norm is taken only after this pointwise supremum; in particular, the endpoint quantity is

$$\|\mathbf{J}(F)\|_{L^{1,\infty}} = \left\| \sup_{\lambda > 0} \lambda N_\lambda(F)^{1/2} \right\|_{L^{1,\infty}}.$$

This is stronger than the conventional mixed quasi-seminorm $\sup_{\lambda > 0} \|\lambda N_\lambda(F)^{1/2}\|_{L^{1,\infty}}$. The strictness of this comparison, for genuine finite jump paths, is quantified in Proposition 14.1. It measures, simultaneously at every spatial point and over all amplitudes, the number of disjoint oscillations of the prescribed size, and it implies every r -variation estimate with $r > 2$ through an abstract endpoint principle. The purpose of this paper is to prove this critical estimate, including spatial weak type $(1, 1)$, for the complete sharp-truncation trajectory of rough polynomial singular integrals.

The weak endpoint at $P = 0$ strictly strengthens, and therefore resolves, the problem posed by Jones–Seeger–Wright [19]. We prove it independently and establish, in the same framework, the coefficient-uniform extension to arbitrary real two-variable polynomial phases of bounded degree. The angular kernel is assumed only to belong to the endpoint class $L \log L$; the phase is subject to no nondegeneracy, smallness, or genericity hypothesis. Thus the theorem simultaneously treats rough angular data, genuinely nonconvolution phases, critical fluctuations, and the spatial endpoint.

Let $d\sigma$ be the usual surface measure on $\mathbb{S}^{n-1}$. Put

$$K_\Omega(z) = \frac{\Omega(z/|z|)}{|z|^n}, \quad z \neq 0.$$

For $A = \|\Omega\|_1 > 0$ define the homogeneous gauge

$$\mathfrak{L}(\Omega) = A + \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| \log^+ \left(\frac{|\Omega(\theta)|}{A} \right) d\sigma(\theta), \quad (2)$$

and set $\mathfrak{L}(0) = 0$. This is equivalent, with dimensional constants, to the homogeneous Luxemburg $L \log L$ norm. Its homogeneity and the boundedness of the odd and even projections

$$\Omega_o(\theta) = \frac{\Omega(\theta) - \Omega(-\theta)}{2}, \quad \Omega_e(\theta) = \frac{\Omega(\theta) + \Omega(-\theta)}{2}$$

give

$$\mathfrak{L}(\Omega_o) + \mathfrak{L}(\Omega_e) \leq C \mathfrak{L}(\Omega). \quad (3)$$

The elementary normalization details are recorded in Lemma 2.1.

On bounded compactly supported inputs define

$$T_{\Omega, \varepsilon}^P f(x) = \int_{|x-y| > \varepsilon} e^{iP(x,y)} K_\Omega(x-y) f(y) dy.$$

The final construction for general data is given in Section 12; it is characterized by the finite-annulus identity and by convergence to zero as $\varepsilon \rightarrow \infty$.

Let E be a linearly ordered set and let $F = (F_t)_{t \in E}$ be a scalar path. For $\lambda > 0$, its paired strict jump number is

$$N_{\lambda, E}(F) := \sup(\{0\} \cup \mathcal{W}_{\lambda, E}(F)) \in \mathbb{N}_0 \cup \{\infty\},$$

where $\mathcal{W}_{\lambda,E}(F)$ is the set of all $N \in \mathbb{N}$ for which there are $s_i, t_i \in E$, $1 \leq i \leq N$, satisfying

$$s_1 < t_1 \leq s_2 < t_2 \leq \cdots \leq s_N < t_N, \quad |F_{t_i} - F_{s_i}| > \lambda \quad (1 \leq i \leq N).$$

Thus a path with no strict λ -jump has jump number zero, while a path with arbitrarily large finite witnesses has jump number infinity. For $r > 0$, its homogeneous r -variation is

$$V_E^r(F) := \sup \left(\{0\} \cup \left\{ \left(\sum_{i=1}^N |F_{t_i} - F_{t_{i-1}}|^r \right)^{1/r} : N \in \mathbb{N}, \quad t_0 < \cdots < t_N, \quad t_i \in E \right\} \right).$$

When the index set is displayed or unambiguous, we omit the subscript E ; thus the main positive-radius paths use $N_\lambda(F)$ and $V^r(F_t : t > 0)$. For a restriction to an ordered subset $D \subset E$, put

$$\mathbf{J}_D(F) := \sup_{\lambda > 0} \lambda N_{\lambda,D}(F|_D)^{1/2},$$

and omit D when it is the full parameter set. For a measurable function H , set $\|H\|_{L^{1,\infty}} = \sup_{\tau > 0} \tau |\{x : |H(x)| > \tau\}|$. Call a path coordinatewise measurable if every F_t is measurable in the spatial variable. For such a path define $\mathbf{J}(F)$ pointwise by (1), initially as an extended-valued pointwise functional without a separate measurability assertion. We call the path jump-separable if there are a countable parameter set $D \subset E$ and one common full-measure spatial set on which, for every $\lambda > 0$, its full and D -restricted jump numbers agree. Whenever $\mathbf{J}(F)$ is measurable—in particular, for every jump-separable coordinatewise measurable path—and $1 < p < \infty$, set

$$\mathcal{J}_{2,p}(F) = \|\mathbf{J}(F)\|_p = \left\| \sup_{\lambda > 0} \lambda N_\lambda(F)^{1/2} \right\|_p, \quad \mathcal{J}_{2,1,\infty}(F) = \|\mathbf{J}(F)\|_{L^{1,\infty}}. \quad (4)$$

Because all jumps are strict, the supremum in λ may be restricted to $\mathbb{Q}_+$ without changing $\mathbf{J}(F)$. More precisely, if $D \subset E$ is countable and the coordinates F_t are measurable for $t \in D$, then, for every $N \geq 1$,

$$\{N_\lambda(F_t : t \in D) \geq N\} = \bigcup_{(s_i, t_i)_{i=1}^N \in \mathcal{O}_N(D)} \bigcap_{i=1}^N \{|F_{t_i} - F_{s_i}| > \lambda\},$$

where $\mathcal{O}_N(D)$ is the countable set of ordered paired $2N$ -tuples in D . Thus the jump count on D is measurable, and every jump-separable coordinatewise measurable path has a measurable pointwise jump functional. If the path is locally continuous in the truncation parameter, its jump counts on a dense D agree with those on all positive parameters by Lemma 2.6. These observations and the rational λ -supremum prove measurability of $\mathbf{J}(F)$.

We use the following common-representative convention throughout. Whenever one of the continuum-parameter operator trajectories constructed below is asserted pointwise on the full parameter interval, its construction is carried out on one spatial set X_0 of full measure, independent of the parameter, on which the parameter path is locally continuous (or locally absolutely continuous when that stronger property is used) and all stated finite-annulus identities hold simultaneously. All coordinates are set equal to zero on X_0^c . Thus a countable dense parameter set determines all jumps and variations by Lemma 2.6, and the preceding countable-union argument proves their measurability. This convention does not impose continuity on the abstract scalar paths used in the jump lemmas. We do not pass from coordinatewise almost-everywhere identities at uncountably many parameters to a common operator representative without an explicit finite-witness or anchored uniform-Cauchy argument. Joint measurability in the parameter is stated separately whenever Fubini or a parameter integral requires it.

Throughout, C and c may change from line to line. Every constant in a global theorem or corollary depends only on the parameters displayed there; in particular, it is independent of the support and location of the input and of every approximation used to construct the operator. Constants inside proofs may depend on fixed smooth cutoffs and normalized coordinate charts chosen once and for all. Only an

explicitly localized statement may depend on support or chart data named in that statement. No constant depends on the coefficients of P , on truncation or dyadic scales, or on auxiliary finite cutoffs. Every coefficient norm is a fixed Euclidean norm on the relevant finite-dimensional polynomial space; changing that norm changes only the permitted structural constants.

Theorem 1.1 (Coefficient-uniform pointwise critical-jump inequalities). *Let $n \in \mathbb{N}$ and $d \in \mathbb{N}_0$. Let P be a real polynomial on $\mathbb{R}^n \times \mathbb{R}^n$ of total degree at most d , and let $\Omega \in L \log L(\mathbb{S}^{n-1})$ satisfy $\int \Omega d\sigma = 0$. Then the pointwise critical-jump functional of the sharp-truncation trajectory satisfies the coefficient-uniform estimates*

$$\mathcal{J}_{2,p}(T_{\Omega,\varepsilon}^P f : \varepsilon > 0) \leq C_{n,d,p} \mathfrak{L}(\Omega) \|f\|_p, \quad 1 < p < \infty, \quad (5)$$

$$\mathcal{J}_{2,1,\infty}(T_{\Omega,\varepsilon}^P f : \varepsilon > 0) \leq C_{n,d} \mathfrak{L}(\Omega) \|f\|_1. \quad (6)$$

Estimate (5) holds for every $1 < p < \infty$ and $f \in L^p(\mathbb{R}^n)$; estimate (6) holds for every $f \in L^1(\mathbb{R}^n)$. The constants are independent of every coefficient of P .

For each fixed Ω as above and each fixed f in the data class of the asserted estimate, Proposition 4.6 in the strong case and Section 12 in the weak case construct the entire trajectory on one full-measure set, on which all of its finite-annulus identities hold and

$$T_{\Omega,\varepsilon}^P f(x) \longrightarrow 0 \quad \text{as } \varepsilon \rightarrow \infty \quad (7)$$

holds. These two properties, rather than the estimates alone, determine the trajectory uniquely: the finite-annulus identities make the difference of two such representatives independent of ε on that common full-measure set, and the zero anchor makes this difference vanish. When $1 < p < \infty$, the convergence in (7) also holds in $L^p(\mathbb{R}^n)$.

Put

$$\kappa_r = \left(\sum_{k=1}^{\infty} k^{-r/2} \right)^{1/r} \leq \left(\frac{r}{r-2} \right)^{1/r}, \quad r > 2. \quad (8)$$

Corollary 1.2 (Variation and convergence consequences). *Under the hypotheses of Theorem 1.1, for every $r > 2$,*

$$\|V^r(T_{\Omega,\varepsilon}^P f : \varepsilon > 0)\|_p \leq C_{n,d,p} \kappa_r \mathfrak{L}(\Omega) \|f\|_p, \quad 1 < p < \infty, \quad (9)$$

$$\|V^r(T_{\Omega,\varepsilon}^P f : \varepsilon > 0)\|_{L^1,\infty} \leq C_{n,d} \kappa_r \mathfrak{L}(\Omega) \|f\|_1. \quad (10)$$

Here (9) holds for every $1 < p < \infty$ and $f \in L^p(\mathbb{R}^n)$, and (10) holds for every $f \in L^1(\mathbb{R}^n)$. The factor κ_r is needed only for the V^r conclusions. The zero anchor and (20) give directly

$$\left\| \sup_{\varepsilon > 0} |T_{\Omega,\varepsilon}^P f| \right\|_p \leq C_{n,d,p} \mathfrak{L}(\Omega) \|f\|_p, \quad 1 < p < \infty,$$

$$\left\| \sup_{\varepsilon > 0} |T_{\Omega,\varepsilon}^P f| \right\|_{L^1,\infty} \leq C_{n,d} \mathfrak{L}(\Omega) \|f\|_1,$$

with no κ_r loss. The finiteness of V^r implies that $\lim_{\varepsilon \downarrow 0} T_{\Omega,\varepsilon}^P f$ exists almost everywhere in the respective ranges. Indeed, if a scalar path F_ε with finite V^r were not Cauchy as $\varepsilon \downarrow 0$, there would be $\delta > 0$ and, recursively, pairs

$$0 < s_{k+1} < t_{k+1} < s_k < t_k, \quad |F_{t_k} - F_{s_k}| > \delta.$$

For every N , inserting the endpoints of the first N pairs in increasing order into one variation partition would give $V^r(F)^r \geq N\delta^r$, a contradiction. Completeness of $\mathbb{C}$ therefore gives the asserted endpoint limit. This argument uses only finite variation; the logically separate zero anchor at infinity is what gives the maximal estimate above.

Consequently, the principal-value limit $T_{\Omega}^P f := \lim_{\varepsilon \downarrow 0} T_{\Omega, \varepsilon}^P f$ defines an operator from L^1 to $L^{1, \infty}$ with bound $C_{n, d} \mathfrak{L}(\Omega)$. For general P , this answers, in the mean-zero $L \log L$ class, the rough-kernel weak type $(1, 1)$ problem posed by Ding and Lai [15, §6.1]; the maximal-truncation and critical-jump bounds are stronger.

For $1 \leq \rho < \infty$ and a pointwise bounded radial function $b : (0, \infty) \rightarrow \mathbb{C}$, put

$$\|b\|_{\mathcal{V}^\rho} := \|b\|_\infty + V^\rho(b; (0, \infty)). \quad (11)$$

A function of finite ρ -variation is regulated. In (11), $\|b\|_\infty$ denotes the pointwise supremum of the chosen representative, consistently with the pointwise definition of V^ρ . Whenever point values enter a Stieltjes formula, we use the representative

$$b^+(t) := \lim_{s \downarrow t} b(s), \quad t > 0.$$

Indeed, simultaneous approximation of the points of any finite partition from the right gives $V^\rho(b^+; (0, \infty)) \leq V^\rho(b; (0, \infty))$, and the same right-limit argument gives $\|b^+\|_\infty \leq \|b\|_\infty$. To quantify the exceptional point values, fix $\delta > 0$ and distinct $t_1 < \dots < t_N$ such that $|b(t_i) - b^+(t_i)| > \delta$. Choose, successively and sufficiently close to the right of each t_i , points $s_i \in (t_i, t_{i+1})$ (with the evident convention for $i = N$) such that $|b(s_i) - b^+(t_i)| < \delta/2$. The ordered partition containing every t_i, s_i then gives

$$N(\delta/2)^\rho \leq V^\rho(b; (0, \infty))^\rho.$$

Thus there are only finitely many such points for each δ , and $b = b^+$ off a countable set. Polar coordinates then show that $b(|z|) = b^+(|z|)$ for almost every $z \in \mathbb{R}^n$, so replacing b by b^+ changes none of the Lebesgue integral operators below and does not increase $\|b\|_{\mathcal{V}^\rho}$. Initially for bounded compactly supported inputs define

$$T_{\Omega, \varepsilon}^{P, b} f(x) = \int_{|x-y| > \varepsilon} e^{iP(x, y)} b(|x-y|) K_\Omega(x-y) f(y) dy.$$

Theorem 1.3 (Coefficient-uniform bounds for every finite radial variation exponent). *Let the hypotheses on n, d, P , and Ω be those of Theorem 1.1. Fix $1 \leq \rho < \infty$, and let $b : (0, \infty) \rightarrow \mathbb{C}$ be pointwise bounded with $\|b\|_{\mathcal{V}^\rho} < \infty$. Then*

$$\|\mathbf{J}(T_{\Omega, \varepsilon}^{P, b} f : \varepsilon > 0)\|_p \leq C_{n, d, p, \rho} \|b\|_{\mathcal{V}^\rho} \mathfrak{L}(\Omega) \|f\|_p, \quad 1 < p < \infty, \quad (12)$$

$$\|\mathbf{J}(T_{\Omega, \varepsilon}^{P, b} f : \varepsilon > 0)\|_{L^{1, \infty}} \leq C_{n, d, \rho} \|b\|_{\mathcal{V}^\rho} \mathfrak{L}(\Omega) \|f\|_1. \quad (13)$$

Estimate (12) holds for every $f \in L^p(\mathbb{R}^n)$, and (13) holds for every $f \in L^1(\mathbb{R}^n)$. The constants are independent of every coefficient of P and of the location of the radial variation of b . The trajectory is uniquely determined by its finite-annulus increments and the zero-at-infinity anchor. For every output exponent $r > 2$, the corresponding strong and weak r -variation conclusions hold with the additional factor κ_r . The maximal-truncation estimates follow instead from the zero anchor and (20), with the same right-hand sides as (12)–(13) and no κ_r loss. Almost-everywhere convergence as $\varepsilon \downarrow 0$ follows from the finiteness of V^r .

The dependence on the fixed finite exponent ρ is not asserted to be uniform. Proposition 14.3 shows that the conclusion is false at $\rho = \infty$. The range $1 \leq \rho < 2$ also admits the direct pathwise proof of Proposition 13.1; the general proof, including the critical and supercritical radial ranges, is given in Section 13.

For a bounded function b of finite total variation, write

$$\|b\|_{\text{BV}} = \|b\|_\infty + \text{Var}_{(0, \infty)}(b).$$

The case $\rho = 1$ of Theorem 1.3 is complemented by the following direct Stieltjes transfer, whose constant is obtained without the operator-specific radial argument.

Corollary 1.4 (Radial BV transfer). *Under the hypotheses of Theorem 1.1, let $b : (0, \infty) \rightarrow \mathbb{C}$ be bounded and of finite total variation. All conclusions of Theorem 1.1 and Corollary 1.2 remain valid with $T_{\Omega, \varepsilon}^P$ replaced by $T_{\Omega, \varepsilon}^{P, b}$ and with each right-hand side multiplied by*

$$\|b\|_{\text{BV}} = \|b\|_{\infty} + \text{Var}_{(0, \infty)}(b).$$

The constants are independent of where the variation of b is concentrated.

The primary result is the weak pointwise critical-jump estimate (6). Since $\sup_{\lambda} \|\lambda N_{\lambda}^{1/2}\|_{L^{1, \infty}} \leq \|\mathbf{J}\|_{L^{1, \infty}}$, its $P = 0$ case independently resolves and strictly strengthens the Jones–Seeger–Wright problem, while the full theorem yields the coefficient-uniform nonconvolution extension for general P . The strong pointwise critical-jump estimate is proved within the same architecture. The variation, maximal, and convergence assertions follow from the abstract jump calculus and the zero anchor. Theorem 1.3 then extends both spatial bounds to every fixed finite radial variation exponent; Corollary 1.4 records the independent anchored Stieltjes proof at bounded variation.

Position in the literature

Static rough polynomial theory. For $P = 0$, the strong and weak theory of rough homogeneous singular integrals at the $L \log L$ angular scale was developed in [6, 16, 12, 29]. For arbitrary $P(x, y)$, Jiang–Lu [18] studied the static principal-value operator with $\Omega \in L \log L$ and a radial factor in BV, including an L^p boundedness criterion and weighted estimates. Ojanen [27] proved weighted strong estimates for the same polynomial-phase, angular- $L \log L$, radial-BV class under the geometric hypotheses formulated there. Al-Salman and Grafakos [3] more recently obtained static L^p bounds for arbitrary $P(x, y)$, angular $L \log L$ data, and a rough radial factor satisfying an L^{γ} -type condition. The corresponding rough-kernel weak type $(1, 1)$ endpoint for the principal-value operator was explicitly listed as an open problem by Ding and Lai [15, §6.1]. Theorem 1.1 and Corollary 1.2 answer it in the mean-zero $L \log L$ class, with the stronger critical-jump control of the complete sharp-truncation trajectory.

Smooth coefficient-uniform weak theory. Ricci and Stein [28] established coefficient-uniform strong L^p estimates for smooth kernels. Chanillo and Christ [10] proved coefficient-uniform weak type $(1, 1)$ for arbitrary polynomial phases and smooth Calderón–Zygmund convolution kernels. Their argument removes pure phases, normalizes a leading coefficient by dilation, and uses a phase-adapted double induction; their final remarks also yield L^p bounds, $1 < p < \infty$, for the maximal truncation. Rough $L \log L$ kernels fall outside that theory. The global phase flag introduced below is designed to retain coefficient uniformity while accommodating rough annular decay and phases that degenerate on the diagonal.

Maximal rough polynomial theory. Representative rough polynomial estimates include [25, 2]; polynomial maximal-truncation estimates for smooth Calderón–Zygmund kernels include [21]. Choudhary, Shrivastava, and Shuin [11] state coefficient-uniform sparse and strong maximal estimates for arbitrary P when $\Omega \in L^{\infty}$. For $P = 0$, Lai [22, Theorem 1.1] proves the weak $(1, 1)$ maximal estimate at the $L \log L$ angular endpoint. In the present paper, maximal truncation is obtained as a consequence of the stronger critical-jump theorem.

Fluctuation theory. For sharply truncated Hilbert transforms, Campbell, Jones, Reinhold, and Wierdl [7, Theorems 1.2–1.4 and Conjecture 7.2] established strong and weak r -variation estimates for $r > 2$, the associated noncritical jump bounds, and short 2-variation estimates, and isolated the critical jump exponent. Their higher-dimensional work [8] developed singular-integral variation theory at the rough $L \log L$ scale. Jones–Seeger–Wright [19, Theorem 1.6] proved the conventional strong critical-jump inequality for the unphased rough trajectory under $\Omega \in L^a$, $a > 1$, and posed the corresponding weak-type $(1, 1)$ estimate for rough homogeneous kernels [19]. Their formulation places the supremum in the jump

amplitude outside the spatial norm. The $P = 0$ case of (6) places that supremum pointwise inside $L^{1,\infty}$, and hence strictly strengthens the problem they posed.

Chen, Ding, Hong, and Liu [9, Theorems 1.1 and 1.3] proved weighted L^p bounds for the pointwise critical-jump functional of rough homogeneous singular-integral and averaging trajectories when the angular function belongs to L^q , $q > 1$. Their one-dimensional specializations provide the Hilbert-truncation and one-sided-average input in Lemma 2.4. Ding, Hong, and Liu [14, Theorem 1.2(ii)] state, with the angular size tacitly normalized, the unphased strong critical-jump theorem at the $L \log L$ endpoint. Bhojak and Shrivastava [5, Theorem 1.1] subsequently state the unphased weak critical-jump and weak r -variation endpoints. Remark 1.5 explains why neither of the latter two arguments is used here.

Recent polynomial-modulation variation theory. Becker, Jamneshan, and Thiele [4, Theorem 1.3 and Corollary 1.4] prove strong L^p r -variation estimates, $1 < p < \infty$ and $r > 2$, for sharp truncations of Hölder Calderón–Zygmund kernels satisfying exact two-sided annular cancellation. Their estimate takes the supremum over bounded-degree polynomial input modulations pointwise inside the L^p norm and holds more generally on homogeneous Lie groups. In Euclidean space, this formulation already contains arbitrary two-variable phases: for each output point x , set

$$Q_x(y) = P(x, y) - P(x, 0), \quad T_{\Omega, u}^{P, 1} f(x) = e^{iP(x, 0)} S_u(K_\Omega, Q_x, f)(x).$$

Because the output factor is independent of u , their smooth-kernel theorem is not restricted to pure-input phases. Combining their one-dimensional result with the method of rotations also gives the fixed-phase, coefficient-uniform strong $r > 2$ estimates considered here for odd rough angular kernels in L^1 (and hence for the entire mean-zero class when $n = 1$). It does not cover the general even rough component at the $L \log L$ scale, the critical 2-jump estimate, or spatial weak type $(1, 1)$. Conversely, the estimates proved here are uniform in the coefficients of each fixed phase but do not place a pointwise supremum over phases inside the spatial norm.

Contribution of the present paper. The central result is the weak critical-jump estimate (6). At $P = 0$ it independently resolves the Jones–Seeger–Wright problem at the $L \log L$ angular endpoint; for general P it gives a coefficient-uniform extension to all bounded-degree two-variable polynomial phases, with no nondegeneracy assumption. The same theorem covers convolution phases, genuinely nonconvolution phases, and phase geometries that degenerate on the diagonal. The strong critical 2-jump estimate is proved in parallel. The strong $r > 2$ consequence extends the smooth and odd-kernel subclasses covered by Becker, Jamneshan, and Thiele to the full rough $L \log L$ class, including its general even component. The pointwise critical jump, the spatial weak endpoint, the extension to every finite radial variation exponent, and the sharpness results of Section 14 are separate conclusions not contained in that result.

Remark 1.5 (Independence from prior proofs). The proof below does not use the coefficient-detection and annular-decay arguments in [2, 11, 18, 23, 24, 25] or the endpoint fluctuation arguments in [14, 5]. The observations in this remark concern the cited proofs as written, not the truth of their theorem statements. The discussion of [5] refers to version 2, accessed on August 27, 2026.

The obstruction in [11] is inherited from the one-ray reduction of Lu–Zhang [25, pp. 210–213], which [11, Lemma 3.2] explicitly adapts. For the top bidegree block

$$P_{m, \ell}(x, y) = \sum_{\substack{|\alpha|=m \\ |\beta|=\ell}} a_{\alpha, \beta} x^\alpha y^\beta,$$

that reduction detects the coefficient tensor only through its diagonal restriction

$$A_{m, \ell}(\theta) = P_{m, \ell}(\theta, \theta) = \sum_{\substack{|\alpha|=m \\ |\beta|=\ell}} a_{\alpha, \beta} \theta^{\alpha+\beta}.$$

In the genuinely mixed setting $n \geq 2$ and $m, \ell \geq 1$, the map $P_{m, \ell} \mapsto A_{m, \ell}$ is in general not injective. For example, the alternating bilinear phase $P(x, y) = x_1 y_2 - x_2 y_1$ has a nonzero top bidegree block, but

$A_{1,1} \equiv 0$. Thus the nonzero top-bidegree hypothesis in Lemma 3.2 does not supply the nonvanishing needed to apply Corollary 3.4 in the passage to (4.7). Related coefficient-detection reductions are invoked or reproduced by Jiang–Lu [18, p. 142], Lu–Wu [23, p. 54] and [24, p. 308], and Al-Salman [2, pp. 82–83]. The global phase flag used here avoids this diagonal compression and retains the full mixed coefficient data.

In [14], the squared-multiplier geometric tail preceding (2.16) carries a factor comparable to m that is not retained in the displayed estimate; after the Plancherel square root, the operator-norm estimate (2.16) is therefore missing a factor comparable to $\sqrt{m}$. Independently, Lemma 2.11 is false as stated for $0 < \theta < 1$: as $t \downarrow 0$, the ratio of its left-hand side to its right-hand side is comparable to $t^{\theta-1}$. The lemma is used in precisely this range to reduce a loss of order $m^{2-\theta}$ to the m -weighted angular summability available at the $L \log L$ endpoint. Consequently, the published proof does not justify that endpoint summation in its printed form.

In version 2 of [5], the short-jump packets $B_{i,\text{in}}^{k-s,m}$ depend on the output point x , whereas (2.2) and Lemma A.2 control sums whose coefficients are chosen before x is evaluated; the variational Rademacher–Menshov theorem does not by itself supply the missing selector estimate. In the long-jump argument, the displayed fixed-shift families $\mathcal{D}_u$ are not nested: for $n = 1$ and $u = 1/3$, the intervals $[1/3, 4/3)$ and $[2/3, 8/3)$ overlap without either containing the other. These points require additional arguments. For these reasons, the present paper proves the relevant critical-jump endpoints independently.

Endpoints and sharpness

The theorem reaches both critical output thresholds: jump exponent 2 and spatial weak type $(1, 1)$. For $n \geq 2$, the angular $L \log L$ assumption is sharp for coefficient-uniform estimates of this generality in the following inherited sense. Al-Qassem, Cheng, and Pan [1, Theorem 1.3(b)] construct mean-zero angular functions in $L(\log L)^{1-\varepsilon}$, for every $\varepsilon > 0$, whose associated scalar rough oscillatory integrals fail to be uniformly bounded over polynomial phases.

The radial and fluctuation endpoints are settled separately in Section 14. Theorem 1.3 holds for each fixed $1 \leq \rho < \infty$, but a one-dimensional Hilbert multiplier example excludes $\rho = \infty$. An abstract scalar-path example shows that the direct Young–Stieltjes transfer necessarily stops before $\rho = 2$, even though operator-specific orthogonality recovers every finite ρ . Finally, an exact Mellin representation reduces Poisson 2-variation to sharp Hilbert truncations and, together with the Jones–Wang counterexample, proves that the pointwise critical jump cannot be strengthened to true output V^2 . The precise optimal growth of the $r > 2$ constants as $r \downarrow 2$ is not addressed.

Three obstructions and their resolution

Phase degeneracy. Neither a displayed monomial coefficient nor the diagonal restriction of the phase provides a stable normalization. Simultaneous translation mixes individual coefficients, and the alternating bilinear phase $x_1 y_2 - x_2 y_1$ vanishes on the diagonal. Section 3 constructs a finite global flag on polynomial phases modulo pure input and output phases. The transverse rows are placed in reduced row-echelon form, and the final nonterminal rows consist of convolution phases and alternating bilinear forms. Dilation normalizes the first nonzero flag coordinate, while simultaneous translation preserves that coordinate and creates only later ones. This triangular structure yields coefficient-uniform annular decay and closes the local analysis by a finite backward induction over the flag.

Modulation-sensitive cancellation. The parent-dependent input modulations generated by a non-convolution phase destroy ordinary mean-zero cancellation. The phase-adapted Calderón–Zygmund decomposition replaces it by

$$\int_Q e^{iP(c_Q, y)} b_Q(y) dy = 0$$

and subtracts the same parent anchor from the operator phase. The anchored packet proposition then performs a simultaneous phase–packet induction. A large first coordinate gives scale-covariant decay

before any expansion; a small coordinate is Fourier-expanded with packet coefficients fixed before the output point is evaluated. The all-low branch terminates in the separated-packet theory. Proposition 6.2 isolates this interaction among phase reduction, modulation-sensitive cancellation, and separated-packet estimates.

Output-dependent selectors. A jump linearization chooses parameter intervals as functions of the output point. The endpoint argument first imposes finite cube, shell, and parameter cutoffs and assigns every relevant increment a fixed rank. A variational Rademacher–Menshov estimate then controls every interval sum through that ranking, without introducing an output-dependent sign choice. The phase-adapted bad functions enter directly through cancellation and the average density bound $\|h_Q\|_1 \lesssim \alpha|Q|$. On each shell, a descending homogeneous-degree packet induction combines Fourier expansion of low blocks with a concentration-stable high-block estimate. Across high shells, an exact oriented two-ray calculation retains a coefficient of size R^m after all projective cancellations and lower-degree perturbations. A one-sided polynomial incidence estimate then sums the small-cube mesh errors geometrically while the larger atom remains fixed. The resulting fixed-sign all-shell estimate is precisely the input needed by the rank argument, and it requires no L^∞ decomposition of the bad data.

The proof uses the following specialized external inputs: critical-jump normability of Mirek, Stein, and Zorin-Kranich; the pointwise-inside-supremum one-dimensional and smooth-approximate-identity jump estimates of Chen, Ding, Hong, and Liu; the long–short mechanisms of Jones, Seeger, and Wright; the Hilbert short-variation theorem of Campbell, Jones, Reinhold, and Wierdl; and Seeger’s two separated-packet estimates. The sharpness argument additionally uses the Jones–Wang Poisson counterexample and the finite-dimensional restriction principle of de Leeuw. We also use the classical M. Riesz theorem, Young integration, and the Fefferman–Stein H^1 –BMO theory. Each input is stated in the precise form required at its point of application. All polynomial, oscillatory, incidence, packet, angular-summation, radial-stability, and limiting arguments specific to the present theorems are proved here.

The proof is organized as follows. Section 2 develops the abstract jump calculus and establishes the unphased strong endpoint. Sections 3 and 4 construct the global phase flag, prove coefficient-uniform annular decay, and close the strong polynomial jump theorem by angular extrapolation and triangular induction. Sections 5 and 6 establish the terminal separated-packet estimates and the anchored phase-packet induction. Section 7 provides the concentration-stable equal- and unequal-shell estimates, while Section 8 proves the unphased weak endpoint. Sections 9–11 combine the phase-adapted Calderón–Zygmund decomposition with far-field and local homogeneous-degree induction to obtain the polynomial weak endpoint. Section 12 constructs the trajectory for general data, derives the variation and convergence consequences, and proves the bounded-variation Stieltjes transfer. Section 13 extends the complete theory to every fixed finite radial variation exponent and proves the shellwise Hilbert refinement. Section 14 contains the mixed-norm, abstract-transfer, radial-endpoint, and true V^2 obstructions.

2 Abstract jump inequalities and the unphased strong endpoint

This section assembles the jump calculus and the one-dimensional estimates used throughout the paper, then combines them with a Riesz-transform factorization to prove the unphased strong critical-jump theorem.

Lemma 2.1 (Gauge normalization). *For every $\Omega \in L \log L(\mathbb{S}^{n-1})$, the quantity $\mathcal{L}$ in (2) is equivalent, with dimensional constants, to the homogeneous Luxemburg norm associated with $\Phi_0(t) = (1+t) \log(1+t) - t$. It also satisfies (3).*

Proof. The assertion is immediate for $\Omega = 0$, so assume $\mathcal{L}(\Omega) > 0$. Put $A = \|\Omega\|_1$ and $M = \mathcal{L}(\Omega)$. The elementary bound

$$\Phi_0(t) \leq C(t + t \log^+ t), \quad t \geq 0,$$

and $M \geq A$ give, for a sufficiently large absolute C_0 ,

$$\begin{aligned} \int_{\mathbb{S}^{n-1}} \Phi_0\left(\frac{|\Omega|}{C_0 M}\right) d\sigma &\leq \frac{C}{C_0 M} \left(A + \int_{\mathbb{S}^{n-1}} |\Omega| \log^+\left(\frac{|\Omega|}{M}\right) d\sigma \right) \\ &\leq \frac{C}{C_0 M} \left(A + \int_{\mathbb{S}^{n-1}} |\Omega| \log^+\left(\frac{|\Omega|}{A}\right) d\sigma \right) \leq 1. \end{aligned}$$

This proves one direction of the Luxemburg comparison.

Conversely, suppose that $\int \Phi_0(|\Omega|/L) d\sigma \leq 1$, and put $m_n = \sigma(\mathbb{S}^{n-1})$. Jensen's inequality yields

$$\Phi_0\left(\frac{A}{m_n L}\right) \leq \frac{1}{m_n} \int \Phi_0(|\Omega|/L) d\sigma \leq \frac{1}{m_n},$$

and hence $A \leq C_n L$. Since

$$\log^+\left(\frac{|\Omega|}{A}\right) \leq \log\left(1 + \frac{|\Omega|}{L}\right) + \log^+\left(\frac{L}{A}\right),$$

while $t \log(1+t) \leq \Phi_0(t) + t$, we obtain

$$\begin{aligned} \int |\Omega| \log^+(|\Omega|/A) d\sigma &\leq L \int \frac{|\Omega|}{L} \log\left(1 + \frac{|\Omega|}{L}\right) d\sigma + A \log^+(L/A) \\ &\leq L \int \Phi_0(|\Omega|/L) d\sigma + A + L/e \leq C_n L. \end{aligned}$$

Thus the two quantities are equivalent, with dimensional constants.

If $R\Omega(\theta) = \Omega(-\theta)$, then reflection is an isometry of the Luxemburg space. Convexity gives

$$\|\Omega_o\|_{L \log L, \text{Lux}} + \|\Omega_e\|_{L \log L, \text{Lux}} \leq 2\|\Omega\|_{L \log L, \text{Lux}}.$$

Homogeneity and the equivalence just proved give (3). □

2.1 Jump calculus and one-dimensional inputs

For a scalar path indexed by positive radii, write

$$N_\lambda^{\text{dyad}}(F) := N_\lambda(F_{2^j} : j \in \mathbb{Z}), \quad \mathbf{J}^{\text{dyad}}(F) := \sup_{\lambda > 0} \lambda N_\lambda^{\text{dyad}}(F)^{1/2}. \quad (14)$$

For a finite ordered parameter set I , $\mathbf{J}_I(F)$ denotes the same pointwise functional with the path restricted to I .

Lemma 2.2 (Pointwise jump and variation calculus). *Let E be a linearly ordered set. For scalar E -paths F, G and $\lambda > 0$,*

$$N_\lambda(F + G) \leq N_{\lambda/2}(F) + N_{\lambda/2}(G), \quad (15)$$

$$\lambda N_\lambda(F)^{1/2} \leq V^2(F), \quad (16)$$

$$V^2(F + G) \leq V^2(F) + V^2(G). \quad (17)$$

Consequently,

$$\mathbf{J}(F + G) \leq 2\mathbf{J}(F) + 2\mathbf{J}(G), \quad (18)$$

$$\mathbf{J}(F) \leq V^2(F). \quad (19)$$

If $F_t \rightarrow 0$ at the upper endpoint of its ordered parameter interval (with value 0 there when that endpoint belongs to the parameter set), then

$$\sup_t |F_t| \leq \mathbf{J}(F). \quad (20)$$

For every $r > 2$,

$$V^r(F) \leq \kappa_r \mathbf{J}(F), \quad \kappa_r = \left(\sum_{k=1}^{\infty} k^{-r/2} \right)^{1/r}. \quad (21)$$

Finally, if $F = (F_t)_{t>0}$ is a positive-radius path decomposed into dyadic parameter blocks, then

$$\lambda N_\lambda(F)^{1/2} \leq C \left[\left(\sum_{j \in \mathbb{Z}} V^2(F_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2} + \lambda N_{\lambda/3}^{\text{dyad}}(F)^{1/2} \right], \quad (22)$$

and hence

$$\mathbf{J}(F) \leq C \left[\left(\sum_{j \in \mathbb{Z}} V^2(F_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2} + \mathbf{J}^{\text{dyad}}(F) \right]. \quad (23)$$

Moreover $\lambda N_\lambda^{\text{dyad}}(F)^{1/2} \leq V^2(F_{2^j} : j \in \mathbb{Z})$.

Proof. For each strict λ -jump of $F + G$, one of the two corresponding increments has modulus $> \lambda/2$. Assign the pair to such a summand; each assigned subcollection remains ordered and paired, proving (15). Inserting the endpoints of a paired collection into a variation partition gives (16); Minkowski in the finite ℓ^2 sum gives (17). Taking the supremum in λ in (15) proves (18), and (19) is (16) followed by the same supremum.

For (20), first let t lie strictly below the upper endpoint and let $s > t$ tend to that endpoint. For every $0 < \lambda < |F_t - F_s|$ there is one strict λ -jump, so $\mathbf{J}(F) \geq |F_t - F_s|$; now let s tend to the endpoint. If the upper endpoint belongs to the parameter set, its value is zero by hypothesis, so the assertion at that endpoint is immediate.

To prove (21), fix a finite variation partition and let $a_1^* \geq a_2^* \geq \dots$ be the decreasing rearrangement of the moduli of its increments. For every $0 < \lambda < a_k^*$, at least k partition increments are strict λ -jumps. Retain any k of them and restore their original temporal order. They are a subfamily of the consecutive partition intervals, so their interiors are disjoint and the right endpoint of each retained interval is no later than the left endpoint of the next. Hence they form an admissible paired family. Thus $\lambda \sqrt{k} \leq \mathbf{J}(F)$, and letting $\lambda \uparrow a_k^*$ gives $a_k^* \leq k^{-1/2} \mathbf{J}(F)$. Hence

$$\left(\sum_i a_i^r \right)^{1/r} \leq \mathbf{J}(F) \left(\sum_{k=1}^{\infty} k^{-r/2} \right)^{1/r}.$$

Taking the supremum over partitions proves the claim. The numerical bound in (8) follows from the integral test.

For (22), let $j(t)$ be the integer determined by $2^{j(t)} \leq t < 2^{j(t)+1}$. Fix an ordered paired family (s_i, t_i) of strict λ -jumps, and denote its finite cardinality by N . If $j(s_i) = j(t_i)$, assign the pair to that dyadic block. Otherwise put

$$\widehat{s}_i = 2^{j(s_i)+1}, \quad \widehat{t}_i = 2^{j(t_i)}.$$

Then $s_i \leq \widehat{s}_i \leq \widehat{t}_i \leq t_i$, and

$$F_{t_i} - F_{s_i} = (F_{t_i} - F_{\widehat{t}_i}) + (F_{\widehat{t}_i} - F_{\widehat{s}_i}) + (F_{\widehat{s}_i} - F_{s_i}).$$

One of the three terms has modulus $> \lambda/3$. Choose exactly one such term, using any fixed tie-breaking rule. The selected middle terms form an ordered paired family on the dyadic sequence. The within-block terms and the selected initial and terminal terms assigned to any fixed dyadic block have pairwise disjoint interiors, because the original pairs are ordered. Insert all their endpoints, together with the two endpoints of that dyadic block, into a single increasing partition and delete repeated points. Every selected short segment is then one increment of this partition: two such segments may share an endpoint but cannot overlap in their interiors. Therefore the sum of their squared increments is bounded by the V^2 square of

the path on that block. Consequently, if N_{sh} and N_{long} denote the numbers assigned to short and middle terms, respectively, then

$$N = N_{\text{sh}} + N_{\text{long}}, \quad \frac{\lambda^2}{9} N_{\text{sh}} \leq \sum_j V^2(F_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})^2, \quad N_{\text{long}} \leq N_{\lambda/3}^{\text{dyad}}(F).$$

It follows that

$$\lambda \sqrt{N} \leq 3 \left(\sum_j V^2(F_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2} + 3 \frac{\lambda}{3} N_{\lambda/3}^{\text{dyad}}(F)^{1/2}.$$

Taking the supremum over all finite strict λ -jump witnesses proves (22), including the case in which the jump count is infinite. Taking the supremum in λ then gives (23). The last assertion follows from (16) on the dyadic sequence. $\square$

Normability at the critical jump exponent is essential for the continuum angular factorizations below. On a linearly ordered set E , define

$$N_{\lambda,E}^{\text{ch}}(F) := \sup \left(\{0\} \cup \{J \in \mathbb{N} : \exists t_0 < \dots < t_J, \quad t_j \in E, \quad |F_{t_j} - F_{t_{j-1}}| > \lambda \ (1 \leq j \leq J)\} \right) \in \mathbb{N}_0 \cup \{\infty\}.$$

As before, the index-set subscript is omitted when it is displayed or unambiguous. Then, for every ordered E ,

$$N_{\lambda,E}^{\text{ch}}(F) \leq 2N_{\lambda,E}(F), \quad N_{\lambda,E}(F) \leq N_{\lambda/2,E}^{\text{ch}}(F). \quad (24)$$

The first inequality follows by taking alternating edges of a chain. For the second, start from $r_0 = s_1$ and $r_1 = t_1$ in an admissible family of pairs. Inductively, after r_{i-1} has been selected from the preceding pair, at least one of s_i, t_i is more than $\lambda/2$ away from $F_{r_{i-1}}$; otherwise

$$|F_{t_i} - F_{s_i}| \leq |F_{t_i} - F_{r_{i-1}}| + |F_{s_i} - F_{r_{i-1}}| \leq \lambda.$$

Choose such an endpoint as r_i . The temporal ordering of the pairs, together with the strict increment, gives $r_{i-1} < r_i$. The resulting chain has one initial λ -increment and then strict $\lambda/2$ -increments, hence has at least as many strict $\lambda/2$ -increments as the original paired family has pairs. For a linearly ordered set E , put

$$\mathbf{J}_E^{\text{ch}}(F) := \sup_{\lambda > 0} \lambda N_{\lambda,E}^{\text{ch}}(F)^{1/2}. \quad (25)$$

We omit E when it is the full parameter set. For a finite ordered set I and a spatially measurable I -path, also put

$$\mathcal{J}_{2,p}^{\text{ch}}(F_t : t \in I) := \|\mathbf{J}_I^{\text{ch}}(F)\|_p. \quad (26)$$

Thus (24) gives, pointwise and uniformly in every ordered set E ,

$$\mathbf{J}_E^{\text{ch}}(F) \leq \sqrt{2} \mathbf{J}_E(F), \quad \mathbf{J}_E(F) \leq 2 \mathbf{J}_E^{\text{ch}}(F). \quad (27)$$

Lemma 2.3 (Pathwise critical-jump normability and integration). *For every nonempty finite ordered set I , the scalar functional $\mathbf{J}_I^{\text{ch}}$ is equivalent, with absolute constants uniform in I , to a subadditive seminorm on scalar I -paths modulo constant paths. Consequently, if μ is a complex measure on a measure space Z and $F^z = (F_t^z)_{t \in I}$ is coordinatewise $|\mu|$ -integrable, then*

$$\mathbf{J}_I^{\text{ch}} \left(\int_Z F^z d\mu(z) \right) \leq C \int_Z \mathbf{J}_I^{\text{ch}}(F^z) d|\mu|(z). \quad (28)$$

If the paths also depend on x , assume that, for every $t \in I$, the map $(z, x) \mapsto F_t^z(x)$ is jointly measurable and that, for almost every x ,

$$\int_Z |F_t^z(x)| d|\mu|(z) < \infty, \quad t \in I.$$

Then (28) holds for almost every x . In particular, for $1 \leq p < \infty$,

$$\mathcal{J}_{2,p}^{\text{ch}}\left(\int_Z F^z d\mu(z)\right) \leq C \int_Z \mathcal{J}_{2,p}^{\text{ch}}(F^z) d|\mu|(z), \quad (29)$$

whenever the right-hand side is finite.

Proof. Apply [26, Corollary 2.11] with the spatial measure space X consisting of one point of mass one, with scalar Banach space $\mathbb{C}$, and with $p = q = \rho = 2$ in the notation of that corollary (the spatial, Lorentz, and jump exponents, respectively). The jump count in the cited source is the consecutive-chain count. On this one-point space the Lorentz norm is the absolute value, so its functional is exactly the scalar finite-path functional $\mathbf{J}_I^{\text{ch}}$. In particular, on a one-point space taking the λ -supremum before or after the spatial Lorentz norm produces the same scalar quantity. Thus this specialization supplies the pointwise inside-supremum functional used here, rather than merely an estimate with the amplitude supremum outside a nontrivial spatial norm. The constant in the cited corollary is uniform in the cardinality of the finite ordered set I . This gives a seminorm $\|\cdot\|_{\mathfrak{J}_I}$ on scalar paths modulo constants such that

$$C^{-1} \mathbf{J}_I^{\text{ch}}(a) \leq \|a\|_{\mathfrak{J}_I} \leq C \mathbf{J}_I^{\text{ch}}(a),$$

with C independent of I . The cited source counts increments of size $\geq \lambda$, whereas we use strict jumps; if $N_\lambda^{\geq}$ and $N_\lambda^{>}$ denote the two counts, then $N_\lambda^{>} \leq N_\lambda^{\geq} \leq N_{\lambda/2}^{>}$, so the corresponding functionals are equivalent with an absolute constant.

Write $d\mu = \omega d|\mu|$, $|\omega| = 1$, fix $t_* \in I$, and replace each F^z by $F^z - F_{t_*}^z$. Coordinatewise integrability and finiteness of I give

$$\int_Z \sum_{t \in I} |F_t^z - F_{t_*}^z| d|\mu|(z) < \infty.$$

Thus the resulting map is Bochner integrable in the finite-dimensional path quotient. Subadditivity and homogeneity give

$$\left\| \int_Z \omega(z) (F^z - F_{t_*}^z) d|\mu|(z) \right\|_{\mathfrak{J}_I} \leq \int_Z \|F^z - F_{t_*}^z\|_{\mathfrak{J}_I} d|\mu|(z).$$

The equivalence above proves (28). For finite I , the chain functional is a finite maximum of expressions obtained from coordinate differences, finite minima, and square roots; it is therefore a Borel function of the finitely many path coordinates. Under the stated joint measurability and sectionwise integrability hypotheses, the scalar inequality consequently applies for almost every x and both sides are measurable. Minkowski's integral inequality then proves (29). $\square$

We next record the pointwise critical-jump estimates for the one-dimensional model paths used in the factorization. If $u \in \mathbb{S}^{n-1}$, put

$$H_{\varepsilon,u}f(x) = \int_{|s|>\varepsilon} f(x-su) \frac{ds}{s}, \quad A_{t,u}^+f(x) = \frac{1}{t} \int_0^t f(x-su) ds.$$

Lemma 2.4 (One-dimensional pointwise critical jumps). *For every $1 < p < \infty$, every $f \in L^p(\mathbb{R}^n)$, and every $u \in \mathbb{S}^{n-1}$,*

$$\|\mathbf{J}(H_{\varepsilon,u}f : \varepsilon > 0)\|_p + \|\mathbf{J}(A_{t,u}^+f : t > 0)\|_p \leq C_p \|f\|_p, \quad (30)$$

with a constant independent of u . The same estimate holds with the chain functional $\mathbf{J}^{\text{ch}}$ in place of $\mathbf{J}$.

Proof. Chen, Ding, Hong, and Liu [9, Theorems 1.1 and 1.3] prove the stronger weighted estimates with the supremum in λ taken pointwise before the $L^p(w)$ norm for rough homogeneous singular integrals and rough averaging families whose angular function belongs to L^q , $q > 1$. Specialize to $n = 1$ and $w \equiv 1$. On $\mathbb{S}^0 = \{-1, 1\}$, the choice

$$\Omega(1) = 1, \quad \Omega(-1) = -1$$

gives the sharp Hilbert truncation, while

$$\Omega(1) = 1, \quad \Omega(-1) = 0$$

gives A_t^+ . Both angular functions belong to every $L^q(\mathbb{S}^0)$. For a given $p > 1$, choose $q > \max\{p, p'\}$. Then $w \equiv 1$ satisfies condition (ii) in each cited theorem, and the $L^q(\mathbb{S}^0)$ norms of the two fixed angular functions are absolute constants. With the usual counting measure on $\mathbb{S}^0$, the specializations are exactly

$$T_\varepsilon^{(1,-1)} = H_\varepsilon, \quad M_t^{(1,0)} = A_t^+.$$

Thus the cited estimates have a constant depending only on p , and they prove the one-dimensional estimates. We spell out the common continuum-parameter representatives used in this specialization. Fix a Borel representative of f and, for general u , write $x = x_\perp + ru$ with $x_\perp \in u^\perp$. For almost every $x_\perp$, the function $g(r) = f(x_\perp + ru)$ belongs to $L^p(\mathbb{R})$. On each such line, for every r and every $\varepsilon > 0$, Hölder's inequality gives

$$\int_{|s|>1} \frac{|g(r-s)|}{|s|} ds \leq \|g\|_p \left\| |s|^{-1} \mathbf{1}_{\{|s|>1\}} \right\|_{p'} < \infty,$$

while the portion $\varepsilon < |s| \leq 1$ is finite by local integrability. Hence the sharp Hilbert truncation is absolutely defined. Moreover $\varepsilon \mapsto H_\varepsilon g(r)$ is locally absolutely continuous on $(0, \infty)$, since on compact ε -intervals its derivative is represented almost everywhere by

$$\frac{g(r+\varepsilon) - g(r-\varepsilon)}{\varepsilon}.$$

Likewise $t \mapsto A_t^+ g(r)$ is continuous because it is a locally absolutely continuous primitive divided by $t > 0$. Set both paths equal to zero on the exceptional lines. These choices give one common full-measure set on which the parameter paths are continuous, so their pointwise jump functionals are measurable by the finite-witness principle. Applying the one-dimensional bounds on each good line and then Fubini gives (30). The chain form follows from (27). $\square$

Lemma 2.5 (Radial profiles). *Let $1 < p < \infty$, let $f \in L^p(\mathbb{R}^n)$, let $u \in \mathbb{S}^{n-1}$, and let φ be represented by a right-continuous function that is locally of bounded variation on $(0, \infty)$, $\varphi(t) \rightarrow 0$ as $t \rightarrow \infty$, and*

$$B(\varphi) = \int_{(0, \infty)} t d|D\varphi|(t) < \infty.$$

For $P_{\varepsilon, u}^\varphi f(x) = \int_0^\infty \varphi(s) f(x - \varepsilon su) ds$, the pointwise path is represented by the compatible Stieltjes representative constructed in the proof, and

$$\|\mathbf{J}(P_{\varepsilon, u}^\varphi f : \varepsilon > 0)\|_p \leq C_p B(\varphi) \|f\|_p. \quad (31)$$

Suppose, more generally, that $(\varphi_u)_{u \in \mathbb{S}^{n-1}}$ is a family satisfying the preceding hypotheses, with right-continuous representatives chosen so that $(u, s) \mapsto \varphi_u(s)$ is jointly measurable on $\mathbb{S}^{n-1} \times (0, \infty)$, and suppose that $\int_{\mathbb{S}^{n-1}} B(\varphi_u) d\sigma(u) < \infty$. Then, for every $\varepsilon > 0$, the map $u \mapsto P_{\varepsilon, u}^{\varphi_u} f$ is strongly measurable in L^p , its Bochner integral is well defined, and

$$\left\| \mathbf{J} \left(\int_{\mathbb{S}^{n-1}} P_{\varepsilon, u}^{\varphi_u} f d\sigma(u) : \varepsilon > 0 \right) \right\|_p \leq C_p \left(\int_{\mathbb{S}^{n-1}} B(\varphi_u) d\sigma(u) \right) \|f\|_p. \quad (32)$$

Both conclusions remain valid with the chain functional.

Proof. Fix a Borel representative of f . Stieltjes integration and the zero limit at infinity give, for every $s > 0$, the following identity with no endpoint ambiguity. For $R > s$ our right-continuous convention gives

$$D\varphi((s, R]) = \varphi(R) - \varphi(s), \quad |D\varphi|((s, \infty)) \leq s^{-1} B(\varphi) < \infty.$$

Letting $R \rightarrow \infty$ and using continuity from below of the finite measure $|D\varphi|$ gives

$$\varphi(s) = -D\varphi((s, \infty)).$$

Moreover, Tonelli's theorem gives

$$\int_0^\infty |\varphi(s)| ds \leq \int_0^\infty \int_{(s, \infty)} d|D\varphi|(t) ds = \int_{(0, \infty)} \int_0^t ds d|D\varphi|(t) = B(\varphi).$$

Thus the defining integral for $P_{\varepsilon, u}^\varphi f$ is a Bochner integral in L^p . The L^p -valued Bochner–Fubini theorem then yields

$$P_{\varepsilon, u}^\varphi f = - \int_{(0, \infty)} t A_{\varepsilon t, u}^+ f d\varphi(t) \quad \text{in } L^p. \quad (33)$$

Indeed, the total variation of the right-hand side is bounded in L^p by $B(\varphi)\|f\|_p$. Below we show that the pointwise Stieltjes integral on the right is simultaneously defined for every $\varepsilon > 0$ on one full-measure set; that integral is the compatible representative used here.

First fix a finite ordered parameter set $I \subset (0, \infty)$. Applying (28) to (33), and then applying Minkowski's integral inequality in L^p , gives

$$\begin{aligned} \|\mathbf{J}_I^{\text{ch}}(P_{\varepsilon, u}^\varphi f : \varepsilon \in I)\|_p &\leq C \int_{(0, \infty)} t \|\mathbf{J}_I^{\text{ch}}(A_{\varepsilon t, u}^+ f : \varepsilon \in I)\|_p d|D\varphi|(t) \\ &\leq C_p B(\varphi) \|f\|_p. \end{aligned} \quad (34)$$

For the second inequality, multiplication by a fixed $t > 0$ preserves the order of the parameters, and we used Lemma 2.4. In particular, the bound is uniform in I .

It remains to justify passage from a dense parameter set to the entire pointwise path. Define the measurable directional one-sided maximal function

$$\mathcal{M}_u^+ f(x) := \sup_{r \in \mathbb{Q}_+} |A_{r, u}^+ f(x)|, \quad \|\mathcal{M}_u^+ f\|_p \leq C_p \|f\|_p. \quad (35)$$

The estimate is the one-dimensional one-sided Hardy–Littlewood maximal theorem, applied on almost every line parallel to u . On those same lines the restriction of f is locally integrable and $r \mapsto A_{r, u}^+ f(x)$ is continuous on $(0, \infty)$. Consequently, the rational supremum in (35) also dominates $|A_{r, u}^+ f(x)|$ for every real $r > 0$.

Put $d\nu(t) = t d\varphi(t)$. This is a finite complex measure with $|\nu|((0, \infty)) = B(\varphi)$. If $\varepsilon_k \rightarrow \varepsilon > 0$, then for almost every x the integrand in (33) converges pointwise in t , and

$$|A_{\varepsilon_k t, u}^+ f(x) - A_{\varepsilon t, u}^+ f(x)| \leq 2\mathcal{M}_u^+ f(x).$$

Dominated convergence with respect to $|\nu|$ proves that $\varepsilon \mapsto P_{\varepsilon, u}^\varphi f(x)$ is continuous for almost every x . Let $I_m \uparrow \mathbb{Q}_+$ be an increasing exhaustion by finite ordered sets. Every strict chain witness is finite, so

$$\mathbf{J}_{I_m}^{\text{ch}}(P^\varphi)(x) \uparrow \mathbf{J}_{\mathbb{Q}_+}^{\text{ch}}(P^\varphi)(x).$$

Monotone convergence applied to the p th powers in (34) yields the same estimate on $\mathbb{Q}_+$. The continuity just proved and the chain form of the finite-witness principle in Lemma 2.6 then give the chain form of (31) on the full parameter set.

We now treat the parameterized family. For rational $0 < a < b < \infty$, the identity

$$D\varphi_u((a, b]) = \varphi_u(b) - \varphi_u(a)$$

shows that $u \mapsto D\varphi_u((a, b])$ is measurable. On each $K_m = [m^{-1}, m]$, these rational half-open intervals form a countable generating algebra and $|D\varphi_u|(K_m) \leq mB(\varphi_u)$. The monotone class theorem therefore

shows that $u \mapsto D\varphi_u(E)$ is measurable for every Borel $E \subset K_m$. Approximating $t\mathbf{1}_E(t)$ by bounded simple functions on K_m , and then letting $m \rightarrow \infty$, shows that the measures

$$\nu_u(dt) := t D\varphi_u(dt)$$

form a measurable finite complex-measure kernel. For completeness, total variation on a Borel set is the supremum of $\sum_k |\nu_u(E_k)|$ over finite partitions by members of the preceding countable algebra; regular approximation then gives the same supremum as the full Borel definition. Hence $(u, E) \mapsto |\nu_u|(E)$ is a measurable positive-measure kernel, and

$$|\nu_u|((0, \infty)) = B(\varphi_u).$$

The map

$$(u, t) \mapsto A_{\varepsilon t, u}^+ f = \int_0^1 f(\cdot - \varepsilon t r u) dr$$

is strongly measurable with values in the separable space L^p : in fact strong continuity of translations and approximation of the r -integral by simple functions show that it is continuous in (u, t) . Moreover,

$$\begin{aligned} & \int_{\mathbb{S}^{n-1}} \int_{(0, \infty)} \|A_{\varepsilon t, u}^+ f\|_p d|\nu_u|(t) d\sigma(u) \\ & \leq \|f\|_p \int_{\mathbb{S}^{n-1}} B(\varphi_u) d\sigma(u) < \infty. \end{aligned}$$

Integration against the preceding kernel now shows that

$$u \mapsto P_{\varepsilon, u}^{\varphi_u} f := - \int_{(0, \infty)} A_{\varepsilon t, u}^+ f d\nu_u(t)$$

is strongly measurable. It agrees in L^p with the original profile integral by the same Bochner–Fubini argument as above, and

$$\|P_{\varepsilon, u}^{\varphi_u} f\|_p \leq B(\varphi_u) \|f\|_p.$$

Hence its integral in u is a well-defined Bochner integral.

For every finite ordered I , apply (28) first to the measure ν_u and then to $d\sigma(u)$, and finally use Minkowski's integral inequality and Lemma 2.4. This gives

$$\begin{aligned} & \left\| \mathbf{J}_I^{\text{ch}} \left(\int_{\mathbb{S}^{n-1}} P_{\varepsilon, u}^{\varphi_u} f d\sigma(u) : \varepsilon \in I \right) \right\|_p \\ & \leq C \int_{\mathbb{S}^{n-1}} \int_{(0, \infty)} \left\| \mathbf{J}_I^{\text{ch}} (A_{\varepsilon t, u}^+ f : \varepsilon \in I) \right\|_p d|\nu_u|(t) d\sigma(u) \\ & \leq C_p \left(\int_{\mathbb{S}^{n-1}} B(\varphi_u) d\sigma(u) \right) \|f\|_p. \end{aligned} \tag{36}$$

For completeness, we verify that the integrated path also has one pointwise representative continuous in ε . For the fixed Borel representative of f , the formula

$$A_{r, u}^+ f(x) = \int_0^1 f(x - r s u) ds$$

is jointly measurable in (u, r, x) . Consequently the rational supremum $(u, x) \mapsto \mathcal{M}_u^+ f(x)$ is measurable. The set on which $r \mapsto A_{r, u}^+ f(x)$ is continuous can be expressed through Cauchy conditions on rational sequences and is therefore measurable. Moreover,

$$G(x) := \int_{\mathbb{S}^{n-1}} B(\varphi_u) \mathcal{M}_u^+ f(x) d\sigma(u) \quad \text{satisfies} \quad \|G\|_p \leq C_p \left(\int_{\mathbb{S}^{n-1}} B(\varphi_u) d\sigma(u) \right) \|f\|_p.$$

Fubini's theorem supplies a common full-measure set of x on which $G(x) < \infty$ and, for almost every u , the map $r \mapsto A_{r,u}^+ f(x)$ is continuous. On this set define

$$\mathcal{P}_\varepsilon f(x) := - \int_{\mathbb{S}^{n-1}} \int_{(0,\infty)} A_{\varepsilon t,u}^+ f(x) d\nu_u(t) d\sigma(u).$$

The integral is absolutely convergent because it is bounded by $G(x)$, and it represents the preceding L^p -valued Bochner integral. If $\varepsilon_k \rightarrow \varepsilon$, its integrand converges for almost every (u, t) and its difference is dominated by $2\mathcal{M}_u^+ f(x)$ with respect to $d\sigma(u) d|\nu_u|(t)$. Dominated convergence therefore proves that $\varepsilon \mapsto \mathcal{P}_\varepsilon f(x)$ is continuous on the same full-measure set. For an increasing finite exhaustion $I_m \uparrow \mathbb{Q}_+$, the finite-parameter chain functionals increase pointwise to the chain functional on $\mathbb{Q}_+$. Monotone convergence in (36), followed by the continuity just proved and the chain form of Lemma 2.6, now proves (32) for the full parameter set. Finally, (27) converts both chain estimates into the stated paired estimates (and gives the asserted chain versions as well). $\square$

We repeatedly use the following elementary compactness principle.

Lemma 2.6 (Finite witnesses and dense parameter sets). *Let I be a finite ordered set, and suppose that $F_t^{(m)} \rightarrow F_t$ for every $t \in I$. Then, for every $\lambda > 0$ and $r \geq 1$,*

$$N_\lambda(F_t : t \in I) \leq \liminf_{m \rightarrow \infty} N_\lambda(F_t^{(m)} : t \in I), \quad N_\lambda^{\text{ch}}(F_t : t \in I) \leq \liminf_{m \rightarrow \infty} N_\lambda^{\text{ch}}(F_t^{(m)} : t \in I),$$

$$V^r(F_t : t \in I) \leq \liminf_{m \rightarrow \infty} V^r(F_t^{(m)} : t \in I),$$

and

$$\mathbf{J}_I(F) \leq \liminf_{m \rightarrow \infty} \mathbf{J}_I(F^{(m)}). \quad (37)$$

Moreover,

$$\mathbf{J}_I^{\text{ch}}(F) \leq \liminf_{m \rightarrow \infty} \mathbf{J}_I^{\text{ch}}(F^{(m)}). \quad (38)$$

If a path F is continuous on compact subintervals of an interval J and $D \subset J$ is dense, then its paired and chain strict jump counts, its paired and chain pointwise critical-jump functionals, and its r -variation on D agree with those on J .

Proof. Any finite family of strict λ -jumps of F has a positive minimum excess over λ ; all of its witnessing increments therefore persist for sufficiently large m . This applies verbatim to both paired and chain witnesses and proves the two jump-count inequalities. Applying the same argument to a fixed finite partition and then taking the supremum proves the variation inequality. If $a < \mathbf{J}_I(F)$, choose a jump amplitude λ and a finite strict witness with $\lambda N_\lambda(F)^{1/2} > a$; that witness persists for all sufficiently large m , proving (37). The identical persistence argument for a finite chain witness proves (38). Finally, continuity permits the finitely many points in any paired witness, chain witness, or variation partition to be perturbed into D while preserving their strict order and changing every increment by an arbitrarily small amount. This proves all dense-set assertions. $\square$

Lemma 2.7 (Homogeneous tail counting). *Let $\Omega \in L \log L(\mathbb{S}^{n-1})$ and put $A = \|\Omega\|_1$. For every $\delta > 0$,*

$$\sum_{s \geq 0} \|\Omega \mathbf{1}_{\{|\Omega| > 2^{\delta s} A\}}\|_1 \leq C_\delta \mathfrak{L}(\Omega). \quad (39)$$

If $\int \Omega = 0$, put $m_n = \sigma(\mathbb{S}^{n-1})$ and $E_s = \{|\Omega| > 2^{\delta s} A\}$, $c_s = m_n^{-1} \int_{E_s} \Omega d\sigma$. Then

$$\Omega_s^\sharp = \Omega \mathbf{1}_{E_s} - c_s, \quad \omega_s = \Omega \mathbf{1}_{E_s^c} + c_s \quad (40)$$

have mean zero and satisfy

$$\|\omega_s\|_\infty \leq C_n 2^{\delta s} A, \quad \|\omega_s\|_1 \leq 2A, \quad \|\Omega_s^\sharp\|_1 \leq 2\|\Omega \mathbf{1}_{E_s}\|_1. \quad (41)$$

Proof. If $A = 0$, all assertions are immediate. Hence assume $A > 0$. Tonelli gives

$$\sum_s \|\Omega \mathbf{1}_{E_s}\|_1 = \int |\Omega(\theta)| \#\{s : 2^{\delta s} A < |\Omega(\theta)|\} d\sigma(\theta),$$

and the counting function is bounded by $C_\delta(1 + \log^+(|\Omega|/A))$. This proves (39). The identities and bounds in (40)–(41) follow from $\int_{E_s^c} \Omega = -m_n c_s$ and $m_n |c_s| \leq \|\Omega \mathbf{1}_{E_s}\|_1 \leq A$. $\square$

Lemma 2.8 (Compact $L \log L$ data in H^1). *Suppose that F is supported in a fixed ball and $\int F = 0$. Define $\mathcal{E}(0) = 0$, while for $F \neq 0$ put*

$$\mathcal{E}(F) = \|F\|_1 + \int |F(x)| \log \left(e + \frac{|F(x)|}{\|F\|_1} \right) dx.$$

If $\mathcal{E}(F) < \infty$, then $\|F\|_{H^1(\mathbb{R}^n)} \leq C\mathcal{E}(F)$, with a constant depending only on the dimension and the supporting ball.

Proof. If $F = 0$ there is nothing to prove. Put $A = \|F\|_1$. By translation invariance we may suppose that $\text{supp } F \subset B(0, R)$, where R is fixed. Let $\phi \in C_c^\infty(B(0, 1))$ have integral one and let $\mathcal{M}_\phi F = \sup_{t>0} |F * \phi_t|$. On a fixed enlargement of the support, $\mathcal{M}_\phi F \leq CMF$. The refined weak estimate

$$|\{MF > 2s\}| \leq \frac{C}{s} \int_{\{|F|>s\}} |F|$$

follows by decomposing $F = F \mathbf{1}_{\{|F| \leq s\}} + F \mathbf{1}_{\{|F| > s\}}$: the maximal function of the first summand is at most s , and the weak $(1, 1)$ maximal theorem applies to the second. If B_R is a fixed enlargement of $B(0, R)$, layer cake, with threshold $s_0 = A$, gives

$$\begin{aligned} \int_{B_R} MF &= \int_0^\infty |B_R \cap \{MF > s\}| ds \\ &\leq CA + C \int_A^\infty \frac{1}{s} \int_{\{|F|>s/2\}} |F(x)| dx ds \\ &\leq CA + C \int |F(x)| \log^+ \left(\frac{2|F(x)|}{A} \right) dx \leq C\mathcal{E}(F), \end{aligned}$$

where Tonelli justifies the interchange. Outside that ball, cancellation gives

$$|F * \phi_t(x)| \leq \int |F(y)| |\phi_t(x-y) - \phi_t(x)| dy \leq C_{\phi,R} |x|^{-n-1} \|F\|_1.$$

Indeed, whenever the integrand is nonzero, $t \geq c_R |x|$, and the mean-value theorem gives

$$|\phi_t(x-y) - \phi_t(x)| \leq |y| \|\nabla \phi_t\|_\infty \leq C_\phi R t^{-n-1} \leq C_{\phi,R} |x|^{-n-1}.$$

The last majorant is integrable off B_R . Thus $\|\mathcal{M}_\phi F\|_1 \leq C\mathcal{E}(F)$. The radial maximal characterization of real H^1 is [17, Theorem 11 and the following Remark 2]; this proves the assertion. The Riesz-transform characterization $R_j : H^1 \rightarrow L^1$ is recorded in [17, Section II.2, p. 144]. $\square$

Proposition 2.9 (Unphased strong critical-jump inequality). *If $\int \Omega = 0$ and $\Omega \in L \log L(\mathbb{S}^{n-1})$, then, for every $1 < p < \infty$ and every $f \in L^p(\mathbb{R}^n)$,*

$$\mathcal{J}_{2,p}(T_{\Omega,\varepsilon}^0 f : \varepsilon > 0) \leq C_{n,p} \mathfrak{L}(\Omega) \|f\|_p. \quad (42)$$

Proof. Fix $1 < p < \infty$. We first take $f \in \mathcal{S}(\mathbb{R}^n)$, restrict the truncation parameter to a finite ordered set $I \subset (0, \infty)$, and use the chain convention. All constants below are independent of I .

We record the measurability convention for the angular integrations. For $M > \varepsilon$, put

$$H_{\varepsilon,u}^M f = \int_{\varepsilon < |s| < M} f(\cdot - su) \frac{ds}{s}.$$

Strong continuity of translations makes $u \mapsto H_{\varepsilon,u}^M f$ continuous as an L^p -valued map. The one-dimensional fixed-truncation theorem, followed by slicing along lines parallel to u , gives

$$\sup_{u \in \mathbb{S}^{n-1}} \|H_{\varepsilon,u} f\|_p \leq C_p \|f\|_p,$$

and $H_{\varepsilon,u}^M f \rightarrow H_{\varepsilon,u} f$ in L^p for every u . Hence $u \mapsto H_{\varepsilon,u} f$ is strongly measurable and Bochner integrable against every L^1 angular density. Likewise, for every $a \in L^1(0, \infty)$, the map

$$u \mapsto \int_0^\infty a(s) f(\cdot - \varepsilon su) ds$$

is strongly continuous in L^p , by dominated convergence. Thus every application of (29) below satisfies its strong measurability and Bochner-integrability hypotheses.

By density, both conclusions extend to every $h \in L^p(\mathbb{R}^n)$. Indeed, the fixed-truncation estimate for $H_{\varepsilon,u}$ is uniform in u , while

$$\left\| \int_0^\infty a(s) h(\cdot - \varepsilon su) ds \right\|_p \leq \|a\|_1 \|h\|_p$$

uniformly in u . Approximating h in L^p by Schwartz functions therefore makes each corresponding L^p -valued map a uniform L^p limit, in u , of strongly measurable maps. In particular these conclusions apply to the functions $g_j = -R_j f$ introduced below.

Decompose $\Omega = \Omega_o + \Omega_e$ into its odd and even parts. The gauge is stable under this decomposition: this is (3).

The odd component.

For odd Ω , polar coordinates give the exact rotation identity

$$T_{\Omega,\varepsilon}^0 f = \frac{1}{2} \int_{\mathbb{S}^{n-1}} \Omega(u) H_{\varepsilon,u} f d\sigma(u). \quad (43)$$

Lemma 2.3 and Lemma 2.4 yield

$$\mathcal{J}_{2,p}^{\text{ch}}(T_{\Omega,\varepsilon}^0 f : \varepsilon \in I) \leq C_p \|\Omega\|_1 \|f\|_p. \quad (44)$$

The even component and the Riesz-transform factorization.

Assume now that Ω is even; when $n = 1$ its mean-zero condition makes it identically zero. Let K be the homogeneous principal-value distribution

$$K(x) = \Omega(x/|x|) |x|^{-n}.$$

Its action on a Schwartz function is obtained by integrating over $\{\delta < |x| < R\}$ and passing to $\delta \downarrow 0$, $R \uparrow \infty$. More explicitly, angular cancellation gives the absolutely convergent formula

$$\langle K, \phi \rangle = \int_{\mathbb{S}^{n-1}} \Omega(u) \int_0^\infty [\phi(ru) - \phi(0) \mathbf{1}_{(0,1)}(r)] \frac{dr}{r} d\sigma(u), \quad \phi \in \mathcal{S}(\mathbb{R}^n).$$

Near zero the bracket is $O(r)$, while at infinity Schwartz decay applies; thus this formula also proves that the principal-value limit exists and is tempered.

Choose a smooth radial function Φ with $\Phi = 0$ on $[0, 1/4]$ and $\Phi = 1$ on $[3/4, \infty)$, and put $\psi = 1 - \Phi$. We now fix the distributional normalization of every Riesz transform used below. Write

$$\mathfrak{r}_j(\xi) = -i \frac{\xi_j}{|\xi|}, \quad \xi \neq 0,$$

so that $\sum_j \tau_j(\xi)^2 = -1$.

The following rough-multiplier estimate fixes all low-frequency choices:

$$\|\widehat{K}\|_{L^\infty(\mathbb{R}^n \setminus \{0\})} + \|\widehat{K\psi}\|_{L^\infty(\mathbb{R}^n)} \leq C_n \mathfrak{L}(\Omega). \quad (45)$$

For completeness, if $e = \xi/|\xi|$, the polar-coordinate formula for the even kernel gives, up to a harmless Fourier-normalization constant,

$$\widehat{K}(\xi) = c_n \int_{\mathbb{S}^{n-1}} \Omega(u) \log |e \cdot u| d\sigma(u).$$

The term containing $\log |\xi|$ is annihilated by $\int \Omega = 0$, and the sine term vanishes because Ω is even. Also

$$\widehat{K\psi}(\xi) = \int_{\mathbb{S}^{n-1}} \Omega(u) a_\psi(\xi \cdot u) d\sigma(u), \quad a_\psi(t) = \int_0^\infty \psi(r) (\cos(rt) - 1) \frac{dr}{r}.$$

One has $|a_\psi(t) + \log^+ |t|| \leq C_\psi$ and, uniformly for $R > 0$ and $|\tau| \leq 1$,

$$|a_\psi(R\tau) + \log^+ R| \leq C_\psi (1 + |\log |\tau||).$$

The functions $u \mapsto 1 + |\log |e \cdot u||$ have uniformly bounded exponential-Orlicz norm. The $L \log L$ -exponential-Orlicz Hölder inequality first gives

$$\left| \int \Omega(u) \log |e \cdot u| d\sigma(u) \right| \leq C_n \mathfrak{L}(\Omega).$$

Writing $\xi = Re$ and using $\int \Omega = 0$ also gives the exact cancellation

$$\widehat{K\psi}(Re) = \int_{\mathbb{S}^{n-1}} \Omega(u) [a_\psi(Re \cdot u) + \log^+ R] d\sigma(u),$$

so the preceding uniform estimate with $\tau = e \cdot u$ and the same Orlicz Hölder inequality prove (45).

To justify the formulas without a boundedness assumption on Ω , put $m_n = \sigma(\mathbb{S}^{n-1})$ and set

$$\Omega_M = \Omega \mathbf{1}_{\{|\Omega| \leq M\}} - m_n^{-1} \int_{\mathbb{S}^{n-1}} \Omega \mathbf{1}_{\{|\Omega| \leq M\}} d\sigma.$$

Then Ω_M is bounded, even, and mean zero, and $\mathfrak{L}(\Omega_M - \Omega) \rightarrow 0$ by absolute continuity of the $L \log L$ Luxemburg norm. The displayed distributional formulas hold for Ω_M ; their uniform Orlicz bounds and convergence in $\mathcal{S}'$ allow $M \rightarrow \infty$, proving them for the original Ω .

We define, as tempered distributions,

$$\widehat{N_j} = \tau_j \widehat{K}, \quad \widehat{R_j(K\psi)} = \tau_j \widehat{K\psi},$$

and

$$\widehat{R_j(K\Phi)} = \tau_j (\widehat{K} - \widehat{K\psi}).$$

Values assigned at $\xi = 0$ are irrelevant, since all three multipliers are locally integrable functions. In particular,

$$R_j(K\Phi) = N_j - R_j(K\psi)$$

holds exactly in $\mathcal{S}'$, with no term supported at the frequency origin. Because τ_j is homogeneous of degree zero, these operators commute with L^1 -normalized dilations and with $\mathcal{A}_0 = x \cdot \nabla + n$. On separated supports the same normalization agrees with the absolutely convergent Riesz-kernel pairing, and on H^1 pieces it agrees with the usual $H^1 \rightarrow L^1$ Riesz transform. Both assertions follow by smoothing τ_j near $\xi = 0$ and passing to the limit.

We repeatedly use the following explicit fixed-annulus consequence of Lemma 2.8. If $\eta \in C_c^\infty((0, \infty))$ is supported in $c \leq r \leq C$, then

$$F_\eta(x) = \eta(|x|)K(x), \quad \int F_\eta = 0, \quad \|F_\eta\|_{H^1} \leq C_{\eta,n} \mathfrak{L}(\Omega). \quad (46)$$

The cancellation follows in polar coordinates. Put $c_\eta = \int_0^\infty |\eta(r)|dr/r$. If $c_\eta \|\Omega\|_1 = 0$, then $F_\eta = 0$ and the claim is immediate. Otherwise

$$\|F_\eta\|_1 = c_\eta \|\Omega\|_1.$$

Since r stays in a fixed compact subinterval of $(0, \infty)$, polar coordinates also give

$$\begin{aligned} & \int |F_\eta| \log \left(e + \frac{|F_\eta|}{\|F_\eta\|_1} \right) \\ & \leq C_{\eta,n} \int_{\mathbb{S}^{n-1}} |\Omega(u)| \log \left(e + C_{\eta,n} \frac{|\Omega(u)|}{\|\Omega\|_1} \right) d\sigma(u) \leq C_{\eta,n} \mathfrak{L}(\Omega). \end{aligned}$$

The asserted H^1 bound now follows from Lemma 2.8.

We first establish

$$N_j(x) = \frac{v_j(x/|x|)}{|x|^n}, \quad v_j \text{ odd}, \quad \|v_j\|_1 \leq C_n \mathfrak{L}(\Omega). \quad (47)$$

It suffices to estimate N_j on $\{3/4 \leq |x| \leq 3/2\}$. Choose a smooth radial partition $\chi_{<} + \chi_0 + \chi_{>} = 1$ with

$$\text{supp } \chi_{<} \subset [0, 2/3), \quad \text{supp } \chi_0 \subset (1/2, 3), \quad \text{supp } \chi_{>} \subset (2, \infty),$$

so that the supports of the first and last pieces are separated from the compact x -annulus. If $\rho_j(z) = c_n z_j |z|^{-n-1}$ is the Riesz kernel, angular cancellation gives for the first piece

$$\int_0^{2/3} \chi_{<}(r) \int \Omega(u) [\rho_j(x - ru) - \rho_j(x)] d\sigma(u) \frac{dr}{r},$$

whose absolute value is at most $C\|\Omega\|_1$. The last piece is absolutely bounded by the same quantity, because $|x - y| \simeq |y|$ and the remaining radial integral is $\int_2^\infty r^{-n} dr/r$. Indeed, on the first piece the mean-value theorem gives $|\rho_j(x - ru) - \rho_j(x)| \leq Cr$, so its radial integral is $C \int_0^{2/3} dr$. The middle piece is controlled by (46); hence its Riesz transform is in L^1 with the same bound. Integrating over the compact x -annulus, and then using homogeneity and parity of the Riesz multiplier, gives

$$\int_{3/4 \leq |x| \leq 3/2} |N_j(x)| dx = \log 2 \int_{\mathbb{S}^{n-1}} |v_j(u)| d\sigma(u) \leq C_n \mathfrak{L}(\Omega),$$

which proves (47). The annular calculation gives an actual locally integrable representative away from the origin. Two homogeneous extensions that agree there can differ only by a finite linear combination of derivatives of δ_0 . Homogeneity of degree $-n$ permits only a multiple of δ_0 , which is even, whereas the Fourier-defined N_j is odd. Thus there is no origin-supported ambiguity.

Define the commutator correction

$$E_j = R_j(K\psi) - \psi N_j. \quad (48)$$

By the Fourier normalization above and the identity $K\Phi = K - K\psi$ in $\mathcal{S}'$, one has the exact distributional identity

$$R_j(K\Phi) = \Phi N_j - E_j. \quad (49)$$

Let $\mathcal{A}_0 = x \cdot \nabla + n$, the generator of L^1 -normalized dilations. We next prove

$$\|E_j\|_1 + \|(\mathcal{A}_0 - 1)E_j\|_1 \leq C_n \mathfrak{L}(\Omega). \quad (50)$$

For $|x| < 1/8$, formula (49) reads $E_j = -R_j(K\Phi)$ and the support separation gives an absolutely convergent bound. More precisely, splitting at $|y| = 1$ gives

$$|R_j(K\Phi)(x)| \leq C\|\Omega\|_1 \left(\int_{1/4}^1 \frac{dr}{r} + \int_1^\infty r^{-n} \frac{dr}{r} \right).$$

For $|x| > 2$, use $E_j = R_j(K\psi)$ and subtract $\rho_j(x)$ inside the angular integral. Here ψ is supported in $[0, 3/4]$ and $\int K(y)\psi(|y|)dy = 0$ in the principal-value sense, so the mean-value theorem applied to

$$\int \Omega(u) \int_0^{3/4} \psi(r) [\rho_j(x - ru) - \rho_j(x)] \frac{dr}{r} d\sigma(u)$$

gives $C\|\Omega\|_1|x|^{-n-1}$. On the intervening annulus, $E_j = \Phi N_j - R_j(K\Phi)$. The first term is integrable by (47). Choose $\zeta \in C_c^\infty((1/8, 8))$ equal to one on $[1/4, 4]$. The term $\zeta K\Phi$ is controlled by (46) and $R_j : H^1 \rightarrow L^1$. The remainder $(1 - \zeta)K\Phi$ is separated from the intervening x -annulus and is absolutely integrable after applying the Riesz kernel. This proves $\|E_j\|_1 \leq C\mathfrak{L}(\Omega)$.

Since $\mathcal{A}_0$ commutes with every R_j and annihilates the homogeneous distributions K, N_j ,

$$\mathcal{A}_0 E_j = R_j(K(r\psi'(r))) - r\psi'(r)N_j.$$

The factor $r\psi'(r)$ is supported in the fixed annulus $1/4 \leq r \leq 3/4$. The first term is controlled by (46) and $R_j : H^1 \rightarrow L^1$, and the second by (47). Together with the bound for E_j , this proves (50).

We next choose the ray representatives in a jointly measurable way. In logarithmic polar coordinates put

$$G_j(s, u) = e^{(n-1)s} E_j(e^s u), \quad D_j(s, u) = e^{(n-1)s} (\mathcal{A}_0 - 1) E_j(e^s u).$$

Both functions are locally integrable in (s, u) , and $\partial_s G_j = D_j$ in the distributional sense, because, with $t = e^s$,

$$\partial_s [t^{n-1} E_j(tu)] = t^{n-1} (\mathcal{A}_0 - 1) E_j(tu).$$

Applying Fubini first on each compact s -interval gives this one-dimensional distributional identity for almost every u . Choose $\vartheta \in C_c^\infty(\mathbb{R})$ with $\int \vartheta = 1$, and, for every u for which the following integrals are defined, set

$$c_j(u) = \int_{\mathbb{R}} \vartheta(s) \left(G_j(s, u) - \int_0^s D_j(v, u) dv \right) ds,$$

$$\tilde{G}_j(s, u) = c_j(u) + \int_0^s D_j(v, u) dv.$$

Let $\mathcal{N}_j^{(0)}$ be the exceptional null set of directions on which either these integrals are not defined or the one-dimensional distributional identity above fails, and set $\tilde{G}_j = 0$ on $\mathcal{N}_j^{(0)}$. Fubini and the one-dimensional $W_{\text{loc}}^{1,1}$ theorem show that $\tilde{G}_j = G_j$ almost everywhere. This construction makes $(s, u) \mapsto \tilde{G}_j(s, u)$ jointly measurable and makes $s \mapsto \tilde{G}_j(s, u)$ locally absolutely continuous for almost every u .

Define

$$\varphi_{j,u}(t) = \tilde{G}_j(\log t, u), \quad t > 0.$$

The map $(u, t) \mapsto \varphi_{j,u}(t)$ is jointly measurable. For every u , the map $t \mapsto \varphi_{j,u}(t)$ is continuous (locally absolutely continuous for every nonexceptional direction and identically zero on the exceptional set), and hence is a right-continuous representative. Moreover,

$$\varphi'_{j,u}(t) = t^{n-2} (\mathcal{A}_0 - 1) E_j(tu)$$

for almost every (t, u) . Polar coordinates give

$$\int_{\mathbb{S}^{n-1}} B(\varphi_{j,u}) d\sigma(u) = \|(\mathcal{A}_0 - 1) E_j\|_1 \leq C\mathfrak{L}(\Omega),$$

and also

$$\int_{\mathbb{S}^{n-1}} \int_0^\infty |\varphi_{j,u}(t)| dt d\sigma(u) = \|E_j\|_1.$$

For almost every u , the derivative of $\varphi_{j,u}$ is integrable on $(1, \infty)$, since

$$\int_1^\infty |\varphi'_{j,u}(t)| dt \leq \int_1^\infty t^{n-1} |(\mathcal{A}_0 - 1)E_j(tu)| dt.$$

Integrating this inequality in u gives

$$\int_{\mathbb{S}^{n-1}} \int_1^\infty |\varphi'_{j,u}(t)| dt d\sigma(u) \leq \|(\mathcal{A}_0 - 1)E_j\|_1 < \infty.$$

To make the quantifier in the parameterized profile lemma literal, enlarge the exceptional set to

$$\begin{aligned} \mathcal{N}_j := \mathcal{N}_j^{(0)} \cup \{u : B(\varphi_{j,u}) = \infty\} \cup \left\{ u : \int_0^\infty |\varphi_{j,u}(t)| dt = \infty \right\} \\ \cup \left\{ u : \int_1^\infty |\varphi'_{j,u}(t)| dt = \infty \right\}. \end{aligned}$$

For $u \notin \mathcal{N}_j^{(0)}$, local absolute continuity and the displayed derivative identity give

$$B(\varphi_{j,u}) = \int_0^\infty t^{n-1} |(\mathcal{A}_0 - 1)E_j(tu)| dt,$$

On $\mathcal{N}_j^{(0)}$ the profile is zero, so its B -value is zero; extend the displayed integral by zero there as well. Thus $u \mapsto B(\varphi_{j,u})$ is measurable by joint measurability of the integrand; the other two defining integrals are measurable for the same reason. Tonelli and the three displayed integral bounds now show that every set in the union is null. Redefine both $\tilde{G}_j(s, u)$ and $\varphi_{j,u}(t)$ to be zero for $u \in \mathcal{N}_j$. This preserves joint measurability and changes the functions only on a product-null set. For $u \notin \mathcal{N}_j$, the integrable tail derivative gives a limit at infinity, and $L^1(0, \infty)$ integrability forces that limit to be zero; for $u \in \mathcal{N}_j$ the same conclusion is immediate from the zero definition. Thus every direction, not merely almost every direction, satisfies the profile hypotheses. The product-null modification does not change the polar-coordinate convolution identity, so all hypotheses of the parameterized form of Lemma 2.5 are satisfied.

For a distribution F write $F^{(\varepsilon)}(x) = \varepsilon^{-n} F(x/\varepsilon)$. Define

$$T_\varepsilon^0 f(x) = \int_{|y| < \varepsilon} K(y) \Phi(|y|/\varepsilon) f(x-y) dy.$$

Since $(K\Phi)^{(\varepsilon)}(x) = K(x)\Phi(|x|/\varepsilon)$,

$$T_{\Omega, \varepsilon}^0 f = (K\Phi)^{(\varepsilon)} * f - T_\varepsilon^0 f. \tag{51}$$

For the current Schwartz function f , put $g_j = -R_j f$, so that $f = \sum_j R_j g_j$. The multiplier normalization (45) makes the following calculation an L^2 identity. Since $\widehat{g}_j = -\mathfrak{r}_j \widehat{f}$ and $\mathfrak{r}_j(\varepsilon \xi) = \mathfrak{r}_j(\xi)$,

$$\begin{aligned} \sum_{j=1}^n (R_j(\widehat{K\Phi}))^{(\varepsilon)} * g_j(\xi) &= -\widehat{K\Phi}(\varepsilon \xi) \sum_{j=1}^n \mathfrak{r}_j(\xi)^2 \widehat{f}(\xi) \\ &= \widehat{K\Phi}(\varepsilon \xi) \widehat{f}(\xi). \end{aligned}$$

Thus

$$(K\Phi)^{(\varepsilon)} * f = \sum_{j=1}^n (R_j(K\Phi))^{(\varepsilon)} * g_j$$

in L^2 and in $\mathcal{S}'$. Substituting (49) into this exact identity and then using (51) gives

$$T_{\Omega,\varepsilon}^0 f = \sum_{j=1}^n (\Phi N_j)^{(\varepsilon)} * g_j - \sum_{j=1}^n E_j^{(\varepsilon)} * g_j - T_\varepsilon^0 f. \quad (52)$$

Each of the three families is now controlled without a dyadic summation. First, let $q(t) = \Phi(t)t^{-1}\mathbf{1}_{(0,1)}(t)$, with the right-continuous value $q(1) = 0$. Since Φ vanishes near zero,

$$B(q) = \int_0^1 t \left| \frac{d}{dt} \frac{\Phi(t)}{t} \right| dt + |\Phi(1)| < \infty.$$

Moreover,

$$((\Phi N_j)^{(\varepsilon)} * g)(x) = \frac{1}{2} \int_{\mathbb{S}^{n-1}} v_j(u) \int_{\mathbb{R}} \Phi(|s|/\varepsilon) g(x - su) \frac{ds}{s} d\sigma(u),$$

while

$$\int_{\mathbb{R}} \Phi(|s|/\varepsilon) g(x - su) \frac{ds}{s} = H_{\varepsilon,u} g + P_{\varepsilon,u}^q g - P_{\varepsilon,-u}^q g.$$

Thus (47), Lemma 2.3, Lemma 2.4, and Lemma 2.5 control the first sum in (52) by $C\mathfrak{L}(\Omega) \sum_j \|g_j\|_p$, uniformly for $\varepsilon \in I$. For the second sum, polar coordinates give the explicit Bochner representation

$$(E_j^{(\varepsilon)} * g)(x) = \int_{\mathbb{S}^{n-1}} \int_0^\infty \varphi_{j,u}(t) g(x - \varepsilon t u) dt d\sigma(u).$$

The profiles $\varphi_{j,u}$ and Lemma 2.5 therefore control it by the same quantity, uniformly on I . Finally,

$$T_\varepsilon^0 f(x) = \int_{\mathbb{S}^{n-1}} \Omega(u) \int_0^1 q(t) f(x - \varepsilon t u) dt d\sigma(u),$$

which is the parameterized profile family $\varphi_u(t) = \Omega(u)q(t)$ and satisfies

$$\int_{\mathbb{S}^{n-1}} B(\varphi_u) d\sigma(u) = B(q) \|\Omega\|_1.$$

Thus Lemmas 2.3 and 2.5 give $C\|\Omega\|_1 \|f\|_p$. The scalar L^p bounds for the Riesz transforms [30] prove the required estimate for the even component. Combining it with (44) by the subadditive jump seminorm in Lemma 2.3, and then using (3), proves (42) for Schwartz functions, uniformly over every finite ordered parameter set I .

Extension in the truncation parameter and in the input.

All estimates above are uniform over finite parameter sets, and the same factorization with ordinary L^p norms bounds every fixed coordinate. Fix a countable dense set $D \subset (0, \infty)$, and choose $f_m \in \mathcal{S}(\mathbb{R}^n)$ with $f_m \rightarrow f$ in L^p so rapidly that

$$\sum_m \|f_{m+1} - f_m\|_p < \infty.$$

For every $\varepsilon \in D$, the right-hand side of (52) is Cauchy in L^p ; denote its limit by $T_{\Omega,\varepsilon}^0 f$.

Let $I \subset D$ be finite. After passing to a subsequence, the coordinates for f_m converge almost everywhere simultaneously on I . Lemma 2.6, in the form (38), and Fatou's lemma give

$$\mathcal{J}_{2,p}^{\text{ch}}(T_{\Omega,\varepsilon}^0 f : \varepsilon \in I) \leq C_{n,p} \mathfrak{L}(\Omega) \|f\|_p.$$

The subsequence may depend on I , which is harmless because the bound is uniform in I . If $I_1 \subset I_2 \subset \dots$ are finite and exhaust D , then $\mathbf{J}_{I_k}^{\text{ch}}$ increases pointwise to the chain functional on D : every strict jump witness is finite and is therefore contained in some I_k . Monotone convergence in the spatial variable gives the same estimate on all of D .

It remains to record compatibility of the chosen coordinates. If $\varepsilon < \eta$ lie in D , then for every m

$$T_{\Omega,\varepsilon}^0 f_m - T_{\Omega,\eta}^0 f_m = \int_{\varepsilon < |z| \leq \eta} K_\Omega(z) f_m(\cdot - z) dz. \quad (53)$$

The finite-annulus kernel is in L^1 , with norm $\|\Omega\|_1 \log(\eta/\varepsilon)$. Young's inequality passes this identity in L^p to the limiting coordinates. Since D is countable, one null set may be discarded so that the identity holds simultaneously for every pair in D .

Fix $\varepsilon_0 \in D$. For arbitrary $\varepsilon > 0$ define

$$T_{\Omega,\varepsilon}^0 f = \begin{cases} T_{\Omega,\varepsilon_0}^0 f + \int_{\varepsilon < |z| \leq \varepsilon_0} K_{\Omega}(z) f(\cdot - z) dz, & \varepsilon < \varepsilon_0, \\ T_{\Omega,\varepsilon_0}^0 f - \int_{\varepsilon_0 < |z| \leq \varepsilon} K_{\Omega}(z) f(\cdot - z) dz, & \varepsilon > \varepsilon_0, \\ T_{\Omega,\varepsilon_0}^0 f, & \varepsilon = \varepsilon_0. \end{cases}$$

The preceding identity leaves all coordinates of D unchanged. On each $0 < a < b < \infty$ and every bounded measurable $U \subset \mathbb{R}^n$, Fubini and Hölder's inequality give

$$\begin{aligned} \int_U \int_a^b \int_{\mathbb{S}^{n-1}} |\Omega(u)| |f(x - ru)| d\sigma(u) \frac{dr}{r} dx \\ \leq |U|^{1/p'} \|f\|_p \|\Omega\|_1 \log(b/a) < \infty. \end{aligned}$$

Thus, for almost every x ,

$$\int_a^b \int_{\mathbb{S}^{n-1}} |\Omega(u)| |f(x - ru)| d\sigma(u) \frac{dr}{r} < \infty.$$

Taking a countable exhaustion by rational a, b and bounded balls U produces one full-measure set on which this holds on every compact subinterval of $(0, \infty)$. The resulting path is locally absolutely continuous, with almost-everywhere derivative

$$-\varepsilon^{-1} \int_{\mathbb{S}^{n-1}} \Omega(u) f(x - \varepsilon u) d\sigma(u).$$

Its strict jump counts on D equal those on all positive parameters by Lemma 2.6. Finally use (24).

The zero-at-infinity anchor.

For Schwartz data, the defining tail integral tends to zero pointwise as $\varepsilon \rightarrow \infty$ by absolute convergence and Schwartz decay. Equation (20) therefore gives

$$\sup_{\varepsilon > 0} |T_{\Omega,\varepsilon}^0 f| \leq \mathbf{J}(T_{\Omega,\varepsilon}^0 f : \varepsilon > 0).$$

The right-hand side belongs to L^p by the estimate already proved, so dominated convergence also gives the L^p anchor on the Schwartz core.

For arbitrary $f \in L^p$, use the rapidly convergent Schwartz sequence chosen above and put

$$M_m(x) = \sup_{\varepsilon > 0} |T_{\Omega,\varepsilon}^0 (f_{m+1} - f_m)(x)|.$$

Each difference is a Schwartz function and its trajectory is anchored at infinity; hence (20) and the pointwise jump estimate give

$$\sum_m \|M_m\|_p \leq C_{n,p} \mathfrak{L}(\Omega) \sum_m \|f_{m+1} - f_m\|_p < \infty.$$

Thus $\sum_m M_m < \infty$ almost everywhere, and the core trajectories converge uniformly in $\varepsilon > 0$ outside one null set. Their limit agrees at every parameter in D with the coordinates constructed above, preserves the finite-annulus identities and local continuity, and is anchored at infinity: first approximate uniformly by one core trajectory and then send its truncation radius to infinity. The same estimate shows convergence in L^p uniformly over ε , so the anchor also holds in L^p . Finally use (27) to return to the paired convention. This completes the proof. $\square$

3 The polynomial phase flag and scale-covariant annular decay

This section develops the coefficient-uniform oscillatory input. Pure output and input phases do not affect operator norms or pointwise jump counts, because

$$P(x, y) = p(x) + q(y) + \tilde{P}(x, y) \implies T_{\Omega, \varepsilon}^P f = e^{ip} T_{\Omega, \varepsilon}^{\tilde{P}} (e^{iq} f). \quad (54)$$

3.1 Algebraic flag

Let $\mathcal{P}_d$ be the space of real polynomials of degree at most d . Set

$$\Delta P(y, u, h) = P(y + u, y + h) - P(y + u, y), \quad \mathfrak{D}P = \text{Mix} \Delta P,$$

where Mix retains precisely the monomials having positive degree in both y and u .

Lemma 3.1 (Kernel of the transverse map). *For every $P \in \mathcal{P}_d$, one has $\mathfrak{D}P = 0$ if and only if*

$$P(x, y) = p(x) + q(y) + Q(x - y) + x^T B y, \quad B^T = -B. \quad (55)$$

Equivalently, for every $a \in \mathbb{R}^n$ there are polynomials p_a, q_a such that, for all $X, Y \in \mathbb{R}^n$,

$$P(a + X, a + Y) - P(X, Y) = p_a(X) + q_a(Y).$$

Proof. The phases in (55) are annihilated by $\mathfrak{D}$. Conversely, $\mathfrak{D}P = 0$ means that ΔP has no monomial of positive degree in both y and u . Equivalently,

$$\partial_{y_j} \partial_{u_k} \Delta P(y, u, h) = 0 \quad (1 \leq j, k \leq n).$$

Differentiate this identity in h_i at $h = 0$ and put $x = y + u$. Since

$$\partial_{h_i} \Delta P(y, u, h)|_{h=0} = \partial_{y_i} P(y + u, y),$$

the chain rule shows that the mixed Hessian $M_{ki} = \partial_{x_k} \partial_{y_i} P$ satisfies $(\partial_{x_j} + \partial_{y_j}) M_{ki} = 0$, so $M(x, y) = F(x - y)$. Commutation of x - and y -derivatives gives

$$\partial_{z_\ell} F_{ki} = \partial_{z_k} F_{\ell i}, \quad \partial_{z_\ell} F_{ki} = \partial_{z_i} F_{k\ell}.$$

Put $B = (F - F^T)/2$. The two compatibility identities imply, for every i, k, ℓ ,

$$\partial_{z_\ell} F_{ik} = \partial_{z_i} F_{\ell k} = \partial_{z_k} F_{\ell i} = \partial_{z_i} F_{k\ell}.$$

Thus every entry of B is constant; in particular, B is a constant skew-symmetric matrix. Put $S = F - B = (F + F^T)/2$. The symmetric potential is produced by the following polynomial Poincaré construction. For each k the polynomial one-form $\sum_i S_{ki}(z) dz_i$ is closed, and therefore

$$v_k(z) = \int_0^1 \sum_i S_{ki}(tz) z_i dt$$

satisfies $\partial_i v_k = S_{ki}$. Symmetry of S makes $\sum_k v_k(z) dz_k$ closed, so

$$Q(z) = - \int_0^1 \sum_k v_k(tz) z_k dt$$

is a polynomial with $-\partial_k \partial_i Q = S_{ki}$. Subtracting $Q(x - y) + x^T B y$ leaves zero mixed Hessian, because

$$\partial_{x_k} \partial_{y_i} (Q(x - y) + x^T B y) = -\partial_k \partial_i Q(x - y) + B_{ki} = S_{ki}(x - y) + B_{ki} = F_{ki}(x - y).$$

A polynomial with zero mixed Hessian is a sum of a pure x -polynomial and a pure y -polynomial, which proves the converse. The resulting decomposition plainly has the stated translation property for every a . Conversely, that property implies, after taking the mixed Hessian, that $M(x + a, y + a) = M(x, y)$ for every a ; hence $M(x, y) = F(x - y)$, and the compatibility and polynomial Poincaré argument just given recovers (55), and therefore $\mathfrak{D}P = 0$. $\square$

Order the mixed monomials in (y, u, h) first by decreasing y -degree, then by decreasing u -degree, decreasing h -degree, decreasing total degree, and finally by a fixed lexicographic order. Columns are listed in this order; throughout, the pivot of a nonzero row means its first nonzero column, normalized to have coefficient one, and rows are listed in the order of their pivots. The substitutions defining Δ are linear and contain no constant term, and Mix only deletes monomials. Consequently $\mathfrak{D}$ preserves total homogeneous degree and $\text{im } \mathfrak{D}$ is the direct sum of its homogeneous components. In each such component take reduced row-echelon form with the preceding column and pivot convention. Merge the rows from the separate homogeneous components in the global order of their pivot monomials, and call them $F_1, \dots, F_M$. Here homogeneous means homogeneous in the total degree of (y, u, h) , and the global monomial order restricts to each homogeneous component. To check the echelon property after merging, let $\pi_\nu \prec \pi_\mu$ be two pivot monomials. Every monomial in F_μ has the same total degree as π_μ . If one of them, say τ , also satisfied $\tau \prec \pi_\nu$, then $\tau \prec \pi_\mu$ and the reduced row-echelon construction in the component containing F_μ would force its coefficient to vanish. If the two rows have the same total degree, reduced row-echelon form also clears the π_ν column from F_μ ; if their total degrees differ, F_μ cannot contain π_ν at all. Thus no row later than F_ν contains a monomial preceding or equal to the pivot of F_ν . We record this global echelon property for later use: if π_ν is the pivot of F_ν , then

$$\mu > \nu \implies \text{supp } F_\mu \cap \{\tau : \tau \preceq \pi_\nu\} = \emptyset. \quad (56)$$

This includes rows of a different total degree, since a row and all of its monomials lie in the same homogeneous component. Homogeneous preimages may be chosen without an extra assumption: if F_μ has degree r and $\mathfrak{D}H = F_\mu$, write $H = \sum_k H^{[k]}$ as a sum of homogeneous parts. The graded property gives $\mathfrak{D}H^{[r]} = F_\mu$ and $\mathfrak{D}H^{[k]} = 0$ for $k \neq r$. We therefore take $H_\mu = H^{[r]}$, so that $\mathfrak{D}H_\mu = F_\mu$ and H_μ is homogeneous. Replacing $H_\mu(x, y)$ by $H_\mu(x, y) - H_\mu(y, y)$, which changes only a pure input phase, arrange

$$H_\mu(y, y) = 0. \quad (57)$$

Append to this list, in decreasing homogeneous degree and fixed monomial order, the convolution monomials $(x - y)^\alpha$ with $2 \leq |\alpha| \leq d$, and, when $d \geq 2$, a fixed basis $x_i y_j - x_j y_i$ of skew bilinear forms. Denote the full finite list again by $H_1, \dots, H_{N_d}$, and write $m_\mu = \deg H_\mu$. Modulo pure phases every $P \in \mathcal{P}_d$ has a unique first nonzero flag coordinate c_ν .

Theorem 3.2 (Global polynomial phase flag). *For each $n \in \mathbb{N}$ and $d \in \mathbb{N}_0$ there is a finite ordered family of homogeneous real polynomials $H_1, \dots, H_{N_d}$ with the following properties.*

- (i) *Every $P \in \mathcal{P}_d$ has a unique representation modulo pure input and output phases,*

$$P(x, y) = p(x) + q(y) + \sum_{\mu=1}^{N_d} c_\mu(P) H_\mu(x, y). \quad (58)$$

Thus every nonpure phase has a well-defined first nonzero flag coordinate.

- (ii) *The coordinates lying in $\ker \mathfrak{D}$ occur after the transverse coordinates and, modulo pure phases, span exactly*

$$Q(x - y) + x^T B y, \quad B^T = -B.$$

In particular, the same flag incorporates convolution and alternating-bilinear phases as its final nonterminal strata.

- (iii) *If ν is the first nonzero coordinate of P and $m_\nu = \deg H_\nu$, then, for every dilation factor $\rho > 0$, the simultaneous dilation $(x, y) = (\rho X, \rho Y)$ changes that coordinate to $c_\nu \rho^{m_\nu}$. For every $a \in \mathbb{R}^n$,*

$$P(a + X, a + Y) = p_a(X) + q_a(Y) + c_\nu H_\nu(X, Y) + \sum_{\mu > \nu} c_{\mu, a} H_\mu(X, Y). \quad (59)$$

Thus translation preserves the first coordinate and creates only later coordinates.

- (iv) Suppose that H_v is transverse and that $y^\alpha u^\beta h^\gamma$ is the pivot monomial of $\mathfrak{D}H_v$ in the reduced-row-echelon construction above. If P has first coordinate $c_v = \pm 1$, then the coefficient of $y^\alpha u^\beta$ in $\mathfrak{D}P$ is a polynomial $a_P(h)$ of degree $|\gamma|$. Its h^γ coefficient is exactly $c_v = \pm 1$, and its top homogeneous coefficient tensor has norm at least $c_{n,d} > 0$. This conclusion is uniform in every later flag coefficient.

Proof. Lemma 3.1 gives spanning; we record explicitly why the resulting coordinates are unique modulo pure phases. The reduced row-echelon rows $F_1, \dots, F_M$ are linearly independent. Hence, if a linear combination of their preimages together with the appended kernel elements is pure, applying $\mathfrak{D}$ first forces every transverse coefficient to vanish. It remains to consider an identity

$$Q(x - y) + x^T B y = p(x) + q(y), \quad B^T = -B,$$

where Q is a linear combination of the appended convolution monomials and therefore has no terms of degree at most one. Taking the mixed Hessian in (x, y) gives

$$-\nabla^2 Q(x - y) + B = 0.$$

The first matrix is symmetric and the second is skew-symmetric, so both vanish. Thus $B = 0$ and Q is affine; by the chosen degree range for Q , this means $Q = 0$. This proves linear independence of the appended kernel basis in the quotient and, together with Lemma 3.1, proves the representation and the description of the kernel strata. Homogeneity gives the dilation statement. For a transverse row,

$$\mathfrak{D}(P(a + \cdot, a + \cdot)) = \text{Mix}[(\mathfrak{D}P)(y + a, u, h)].$$

Translation preserves the highest y -degree term and creates only lower y -degree terms, which are later in the chosen order. Reduced echelon form, specifically (56), prevents a later row from altering the pivot coefficient. A convolution monomial is unchanged by simultaneous translation, and a skew bilinear form changes only by pure input and output phases. This proves (iii).

For (iv), project $\mathfrak{D}P$ onto the scalar tensor component indexed by $y^\alpha u^\beta$. The pivot row contributes $\pm h^\gamma$. No later row contains the pivot monomial, by (56), and no later row contains a monomial with the same y - and u -component and higher h -order, because such a monomial would precede the pivot in the defining order. Thus the h^γ coefficient remains ± 1 , and later coefficients can change only terms of lower h -degree or other multiindices of the same h -degree. In particular, even in the latter case the fixed h^γ coordinate of the top homogeneous coefficient tensor remains ± 1 . The coordinate functional selecting h^γ has norm at most $C_{n,d}$ on each of the finitely many fixed coefficient spaces that occur. Consequently a_P has degree $|\gamma|$, and the norm of its top homogeneous coefficient tensor is at least $c_{n,d} > 0$, uniformly in all later coefficients. $\square$

For packet arguments it is convenient to use anchored representatives

$$\widehat{H}_\mu(X, Y) = H_\mu(X, Y) - H_\mu(0, Y). \quad (60)$$

They have the same flag coordinates modulo pure input phases and satisfy $\widehat{H}_\mu(0, Y) = 0$.

3.2 Polynomial oscillation estimates

We give the elementary estimates needed for rough annular decay.

Lemma 3.3 (Sublevel and negative moments). *Let $k \geq 1$, let $p(t) = a_k t^k + \dots + a_0$, $a_k \neq 0$, on an interval I of length $L > 0$, and let $\eta > 0$. Then*

$$|\{t \in I : |p(t)| \leq \eta\}| \leq C_k(\eta/|a_k|)^{1/k}.$$

Consequently, if $0 < \sigma < 1/k$, then, uniformly in c and the lower coefficients,

$$\int_I |p(t) - c|^{-\sigma} dt \leq C_{k,\sigma} |a_k|^{-\sigma} L^{1-k\sigma}. \quad (61)$$

If a is a polynomial of degree k on $\mathbb{R}^n$ whose top homogeneous coefficient norm is at least one, then, for every $R > 0$ and every fixed $C_0 < \infty$,

$$\int_{|h| \leq C_0 R} |a(h)|^{-\sigma} dh \leq CR^{n-k\sigma}, \quad 0 < \sigma < 1/k. \quad (62)$$

Here the last constant may depend on C_0 . If A_k is homogeneous of degree k and $\|A_k\|_{\text{coeff}} = 1$, then

$$\int_{\mathbb{S}^{n-1}} |A_k(\theta)|^{-\beta} d\sigma(\theta) \leq C, \quad 0 < \beta < 1/k. \quad (63)$$

Proof. Let $E = \{t \in I : |p(t)| \leq \eta\}$ and suppose $|E| > 0$. Set $\delta = |E|/[2(k+1)]$. A greedy selection gives $k+1$ distinct points of E at mutual distance at least δ : after $j \leq k$ points have been selected, the union of their open δ -neighborhoods has measure at most $2j\delta < |E|$ and therefore cannot cover E . After relabeling, we have $t_0 < \dots < t_k$ and

$$|t_i - t_j| \geq \delta = c_k |E| \quad (0 \leq i, j \leq k, i \neq j).$$

Lagrange interpolation at these points gives

$$|a_k| = \left| \sum_{j=0}^k \frac{p(t_j)}{\prod_{\ell \neq j} (t_j - t_\ell)} \right| \leq C_k \eta |E|^{-k}.$$

This proves the sublevel estimate.

Apply it to $p - c$ and combine it with the trivial bound by L :

$$|\{t \in I : |p(t) - c| \leq \eta\}| \leq \min \left\{ L, C_k (\eta / |a_k|)^{1/k} \right\}.$$

The layer-cake identity gives

$$\int_I |p(t) - c|^{-\sigma} dt = \sigma \int_0^\infty \eta^{-\sigma-1} |\{t \in I : |p(t) - c| < \eta\}| d\eta.$$

Split the last integral where the two bounds in the minimum agree, at a value $\eta_0 \asymp_k |a_k| L^k$. The part below η_0 is bounded by

$$C_k |a_k|^{-1/k} \int_0^{\eta_0} \eta^{1/k-\sigma-1} d\eta,$$

and the part above it by $L \int_{\eta_0}^\infty \eta^{-\sigma-1} d\eta$. Both are at most $C_{k,\sigma} |a_k|^{-\sigma} L^{1-k\sigma}$ precisely when $0 < \sigma < 1/k$, proving (61).

For (62), write $a = A_k + \text{lower-order terms}$. Finite-dimensional norm equivalence supplies $v \in \mathbb{S}^{n-1}$ such that

$$|A_k(v)| \geq c_{n,k} \|A_k\|_{\text{coeff}} \geq c_{n,k}.$$

Write $h = w + tv$ with $w \in v^\perp$. For fixed w , the polynomial $t \mapsto a(w + tv)$ has degree k and leading coefficient $A_k(v)$, independent of w . Its intersection with $|h| \leq C_0 R$ is an interval of length at most $2C_0 R$. Hence (61), followed by integration over the w -projection of that ball, gives

$$\int_{|h| \leq C_0 R} |a(h)|^{-\sigma} dh \leq CR^{1-k\sigma} R^{n-1} = CR^{n-k\sigma}.$$

Finally apply this estimate with $a = A_k$ and $R = 2$ to the annulus $1 \leq |x| \leq 2$. Homogeneity and polar coordinates give

$$\int_{1 \leq |x| \leq 2} |A_k(x)|^{-\beta} dx = \left(\int_1^2 r^{n-1-k\beta} dr \right) \int_{\mathbb{S}^{n-1}} |A_k(\theta)|^{-\beta} d\sigma(\theta).$$

The radial factor is positive and finite, so (63) follows. $\square$

Lemma 3.4 (Leading-coefficient van der Corput). *Let $k \geq 1$, let I be an interval of length at most $L_0 < \infty$, and let*

$$\phi(t) = a_k t^k + \cdots + a_0, \quad a_k \neq 0.$$

For every bounded-variation amplitude ψ supported in I ,

$$\left| \int e^{i\phi(t)} \psi(t) dt \right| \leq C_{k,L_0} \min(1, |a_k|^{-1/k}) (\|\psi\|_\infty + \text{Var}(\psi)).$$

The constant is independent of all lower coefficients.

Proof. Extend ψ by zero off I . Its distributional derivative is a finite measure and

$$|D\psi|(\mathbb{R}) \leq \text{Var}_I(\psi) + 2\|\psi\|_\infty.$$

For $k = 1$, the trivial estimate applies when $|a_1| < 1$, while Stieltjes integration by parts applies when $|a_1| \geq 1$ and gives the asserted bound. Suppose $k \geq 2$. The case $|a_k| < 1$ follows from the trivial estimate $L_0\|\psi\|_\infty$, so assume $|a_k| \geq 1$ and let $\mu > 0$. Put $E = \{t \in I : |\phi'(t)| \leq \mu\}$. Lemma 3.3, applied to ϕ' , gives

$$|E| \leq C_k (\mu/|a_k|)^{1/(k-1)}.$$

The boundary of E is contained in the roots of $\phi' = \mu$ and $\phi' = -\mu$ together with the endpoints of I . Consequently, after also splitting at the at most $k-2$ roots of ϕ'' , $I \setminus E$ is the union of $O_k(1)$ intervals J on each of which $|\phi'| \geq \mu$ and ϕ' is monotone. On every such interval, with one-sided endpoint values (equivalently, on half-open intervals so that jumps are counted only once), Stieltjes integration by parts yields

$$\int_J e^{i\phi(t)} \psi(t) dt = \left[\frac{e^{i\phi(t)} \psi(t)}{i\phi'(t)} \right]_{\partial J} - \int_J \frac{e^{i\phi(t)}}{i\phi'(t)} d\psi(t) + \int_J \frac{e^{i\phi(t)} \psi(t) \phi''(t)}{i\phi'(t)^2} dt.$$

Since ϕ' is monotone on J ,

$$\int_J \frac{|\phi''(t)|}{|\phi'(t)|^2} dt \leq \frac{2}{\mu}.$$

Summing the endpoint, Stieltjes, and absolutely continuous terms over the $O_k(1)$ intervals, and estimating the integral over E trivially, gives

$$\left| \int e^{i\phi(t)} \psi(t) dt \right| \leq C_{k,L_0} \left\{ (\mu/|a_k|)^{1/(k-1)} \|\psi\|_\infty + \mu^{-1} (\|\psi\|_\infty + \text{Var}_I(\psi)) \right\}.$$

Taking $\mu = |a_k|^{1/k}$ proves the result, uniformly in all lower coefficients. $\square$

Lemma 3.5 (Compact polynomial operator). *Let $R > 0$, let Q, U be cubes of diameter at most $C_0 R$, and let $\Phi(y, u)$ be a real polynomial. Suppose the largest mixed y -degree is $m \geq 1$, and among those terms the largest u -degree is $\ell \geq 1$. If the Euclidean norm M of the coefficient tensor of bidegree (m, ℓ) is nonzero, then*

$$\left\| 1_Q(y) \int_U e^{i\Phi(y,u)} g(u) du \right\|_2 \leq C R^n \min\{1, (R^{m+\ell} M)^{-\varepsilon_{m,\ell}}\} \|g\|_2 \quad (64)$$

for every $g \in L^2(U)$, where $\varepsilon_{m,\ell} > 0$ and the constant may depend on C_0 but is independent of all pure terms and all lower mixed coefficients.

Proof. Remove pure terms by unitary multiplication. Let $\Phi_{m,\ell}(y, u)$ be the component homogeneous of degree m in y and degree ℓ in u . On this fixed finite-dimensional space, equivalence of the coefficient norm with the supremum norm on $\mathbb{S}^{n-1} \times \mathbb{S}^{n-1}$ gives unit vectors v, w such that

$$|\Phi_{m,\ell}(v, w)| \geq c_{n,m,\ell} M.$$

Slice $y = y_\perp + sv$, $u = u_\perp + tw$. Direct expansion shows that

$$[s^m t^\ell] \Phi(y_\perp + sv, u_\perp + tw) = \Phi_{m,\ell}(v, w). \quad (65)$$

In particular, this coefficient is independent of the transverse variables and has modulus comparable to M . We give the fiber calculation, including the T^*T square root that determines the exponent.

Fix $y_\perp, u_\perp$ and denote the resulting operator from the t -line to the s -line by $T_{y_\perp, u_\perp}$. Cross-sections of Q and U are intervals of length at most CR ; their characteristic functions have variation at most two. Write $p_m(t)$ for the coefficient of s^m in the sliced phase. It is a polynomial of degree ℓ whose leading coefficient has modulus at least cM . The kernel of $T_{y_\perp, u_\perp}^* T_{y_\perp, u_\perp}$ is

$$\mathbf{1}_U(t) \mathbf{1}_U(t') \int_{I_Q} e^{i[\Phi(s,t) - \Phi(s,t')]} ds.$$

After $s = R\tau$, Lemma 3.4, including its bounded-variation amplitude form, gives

$$|K(t, t')| \leq CR \min \left\{ 1, (R^m |p_m(t) - p_m(t')|)^{-1/m} \right\}. \quad (66)$$

Choose

$$0 < \delta < \min\{1/m, 1/\ell\}.$$

Since $\min(1, X^{-1/m}) \leq X^{-\delta}$, the negative-moment estimate (61), applied to the polynomial in t' for fixed t , yields the required uniform bound: indeed, $t' \mapsto p_m(t) - p_m(t')$ has degree ℓ and leading coefficient $-\Phi_{m,\ell}(v, w)$, independent of $t, y_\perp, u_\perp$. Therefore

$$\begin{aligned} \sup_t \int |K(t, t')| dt' &\leq CR^{1-m\delta} M^{-\delta} R^{1-\ell\delta} \\ &= CR^2 (R^{m+\ell} M)^{-\delta}. \end{aligned}$$

The same estimate with t, t' interchanged follows by symmetry. The trivial kernel estimate gives CR^2 instead, so Schur's test and then the T^*T identity imply

$$\|T_{y_\perp, u_\perp}\|_{2 \rightarrow 2} \leq CR \min\{1, (R^{m+\ell} M)^{-\delta/2}\}. \quad (67)$$

Reassemble the slices, without invoking a direct-integral identification. Minkowski's inequality in $u_\perp$ and (67) give, for fixed $y_\perp$,

$$\|Tg(y_\perp, \cdot)\|_{L^2_s} \leq B_R \int_{\pi_{w^\perp} U} \|g(u_\perp, \cdot)\|_{L^2_t} du_\perp, \quad B_R = CR \min\{1, (R^{m+\ell} M)^{-\delta/2}\}.$$

Cauchy-Schwarz in $u_\perp$ makes the right side at most

$$CB_R R^{(n-1)/2} \left(\int_{\pi_{w^\perp} U} \|g(u_\perp, \cdot)\|_{L^2_t}^2 du_\perp \right)^{1/2} = CB_R R^{(n-1)/2} \|g\|_2.$$

The $y_\perp$ -projection of Q has measure at most CR^{n-1} . Squaring the preceding estimate, integrating in $y_\perp$, and taking a square root therefore give

$$\|Tg\|_2 \leq CR^{n-1} B_R \|g\|_2,$$

which is (64) with $\varepsilon_{m,\ell} = \delta/2$. Every bound used only the displayed leading coefficient, so all lower mixed coefficients remain unrestricted. $\square$

For $R > 0$ and $1/2 \leq u \leq 1$, put

$$A_{R,u}^{P,\Theta} f(x) = \int_{uR < |x-y| < R} e^{iP(x,y)} K_\Theta(x-y) f(y) dy. \quad (68)$$

Proposition 3.6 (Normalized rough annular decay). *Let $P \in \mathcal{P}_d$. For every flag position μ there are $a_\mu > 0$ and $C_{n,d}$ such that, if the first nonzero flag coordinate of P is ± 1 at position μ , then, for $R \geq 1$ and $\Theta \in L^\infty(\mathbb{S}^{n-1})$,*

$$\sup_{1/2 \leq u \leq 1} \|A_{R,u}^{P,\Theta}\|_{2 \rightarrow 2} \leq C_{n,d} R^{-a_\mu} \|\Theta\|_\infty. \quad (69)$$

Every later coefficient is unrestricted.

Proof. We treat the three parts of the flag.

For a transverse pivot, partition the input into cubes of side comparable to R ; the corresponding output enlargements have bounded overlap. On one input cube Q , write $A_Q = A_{R,u} \mathbf{1}_Q$. Its output is supported in $X_Q = Q + B(0, R)$, and the family $\{X_Q\}$ has bounded overlap. The kernel of $A_Q^* A_Q$ is

$$L_Q(y, y+h) = \mathbf{1}_Q(y) \mathbf{1}_Q(y+h) \int e^{i[P(x,y+h)-P(x,y)]} \overline{k_{R,u}(x-y)} k_{R,u}(x-y-h) dx,$$

where $k_{R,u} = K_\Theta \mathbf{1}_{\{uR < |\cdot| < R\}}$. Put $x = y + v$ and remove the pure y - and v -terms in the phase difference. Let v be the current transverse flag position. Since its normalized coefficient is ± 1 and all earlier coordinates vanish,

$$\mathfrak{D}P(y, v, h) = \pm F_v(y, v, h) + \sum_{v < \mu \leq M} c_\mu F_\mu(y, v, h). \quad (70)$$

Let $y^\alpha v^\beta h^\gamma$ be the pivot monomial of F_v , and put

$$m = |\alpha|, \quad \ell = |\beta|, \quad k = |\gamma|.$$

Because Mix retains only terms of positive y - and v -degree, while $\Delta P(y, v, 0) = 0$, one has $m, \ell, k \geq 1$. Thus every use below of the compact-operator and negative-moment lemmas lies in their stated positive-degree range. Write $\mathsf{T}_{m,\ell}(h)$ for the coefficient tensor of bidegree (m, ℓ) in (y, v) in the mixed part of $P(y+v, y+h) - P(y+v, y)$, and let $a(h)$ be its (α, β) -component. By Theorem 3.2(iv), a has degree k , the coefficient of h^γ is exactly ± 1 , and no later flag coefficient can change that coefficient or introduce a term of higher h -degree in the same component. Consequently

$$\|\mathsf{T}_{m,\ell}(h)\|_{\text{coeff}} \geq c_{n,d} |a(h)|, \quad (71)$$

and no mixed bidegree preceding (m, ℓ) occurs. Here is the precise link between the flag order and the hypothesis of Lemma 3.5. A mixed monomial with y -degree greater than m , or with y -degree m and v -degree greater than ℓ , precedes the pivot regardless of its h -degree. Such a monomial is absent from F_v because its pivot is the first nonzero column, absent from every later transverse row by (56), and absent from the kernel strata because $\mathfrak{D}$ vanishes there. All earlier flag coordinates are zero by hypothesis. Thus (m, ℓ) is exactly the leading bidegree to which Lemma 3.5 applies, uniformly in all unrestricted later coefficients. Notice that this argument detects a fixed component of the full (m, ℓ) tensor; it never replaces a nonzero tensor by its diagonal contraction. Such a replacement would be false in general (the alternating-bilinear contraction vanishes identically), which is why the convolution and skew kernel strata are treated separately below. For fixed h , define

$$(E_h G)(y) = \mathbf{1}_Q(y) \mathbf{1}_Q(y+h) \int_{|v| \leq R} e^{i[P(y+v,y+h)-P(y+v,y)]} G(v) dv.$$

To apply Lemma 3.5, extend $G \mathbf{1}_{\{|v| \leq R\}}$ by zero to a cube of diameter $O(R)$; this leaves the operator unchanged. Put

$$g_h(v) = \overline{k_{R,u}(v)} k_{R,u}(v-h).$$

Let $E_h^{(0)}$ denote the operator in the preceding display with the factor $\mathbf{1}_Q(y+h)$ omitted. Since

$$E_h = M_{\mathbf{1}_{Q-h}} E_h^{(0)}, \quad \|E_h\|_{2 \rightarrow 2} \leq \|E_h^{(0)}\|_{2 \rightarrow 2},$$

the additional output cutoff is harmless. Moreover $L_Q(y, y+h) = E_h g_h(y)$. If $a(h) = 0$, Cauchy–Schwarz in the input and output cubes gives the trivial estimate $\|E_h\|_{2 \rightarrow 2} \leq CR^n$, which is the desired bound below with the minimum interpreted as 1. If $a(h) \neq 0$, then (71) makes the leading tensor nonzero, so Lemma 3.5 applies to $E_h^{(0)}$. Absorbing the fixed factor $c_{n,d}$ into the structural constant, the two cases give

$$\|E_h\|_{2 \rightarrow 2} \leq CR^n \min\{1, (R^{m+\ell}|a(h)|)^{-\varepsilon}\}.$$

The actual input g_h satisfies, uniformly in h and in $u \in [1/2, 1]$,

$$\|g_h\|_2 \leq \|k_{R,u}\|_\infty \|k_{R,u}\|_2 \leq CR^{-3n/2} \|\Theta\|_\infty^2. \quad (72)$$

Consequently

$$\int |L_Q(y, y+h)|^2 dy \leq CR^{-n} \|\Theta\|_\infty^4 \min\{1, (R^{m+\ell}|a(h)|)^{-2\varepsilon}\}.$$

Choose $0 < \sigma < \min(2\varepsilon, 1/k)$ and integrate in $|h| \lesssim R$. The top homogeneous coefficient tensor of a has norm at least $c_{n,d}$ by Theorem 3.2(iv); rescaling a by this fixed constant puts it in the normalization of (62). Thus, using $\min(1, X^{-2\varepsilon}) \leq X^{-\sigma}$ and absorbing the resulting factor $c_{n,d}^{-\sigma}$, (62) gives

$$\begin{aligned} R^{-n} R^{-\sigma(m+\ell)} \int_{|h| \lesssim R} |a(h)|^{-\sigma} dh \\ \leq CR^{-n} R^{-\sigma(m+\ell)} R^{n-k\sigma} = CR^{-\sigma(m+\ell+k)}. \end{aligned}$$

Consequently,

$$\iint |L_Q(y, y+h)|^2 dy dh \leq C \|\Theta\|_\infty^4 R^{-\sigma(m+\ell+k)}. \quad (73)$$

The change of variables $(y, h) \mapsto (y, y+h)$ has unit Jacobian, so the left side is exactly $\|A_Q^* A_Q\|_{\text{HS}}^2$. Taking its square root first gives

$$\|A_Q^* A_Q\|_{2 \rightarrow 2} \leq \|A_Q^* A_Q\|_{\text{HS}} \leq C \|\Theta\|_\infty^2 R^{-\sigma(m+\ell+k)/2},$$

and the $A_Q^* A_Q$ identity supplies the second square root:

$$\|A_Q\|_{2 \rightarrow 2} \leq C \|\Theta\|_\infty R^{-\sigma(m+\ell+k)/4}.$$

Finally, bounded overlap of the X_Q and disjointness of the input cubes give, with $a = \sigma(m+\ell+k)/4$,

$$\begin{aligned} \left\| \sum_Q A_Q(f \mathbf{1}_Q) \right\|_2^2 &\leq C \sum_Q \|A_Q(f \mathbf{1}_Q)\|_2^2 \\ &\leq CR^{-2a} \|\Theta\|_\infty^2 \sum_Q \|f \mathbf{1}_Q\|_2^2 = CR^{-2a} \|\Theta\|_\infty^2 \|f\|_2^2. \end{aligned}$$

This reassembles the cubes and proves (69) with a positive exponent independent of the later coefficients.

For a convolution pivot, after pure phases are removed write $P(x, x-z) = Q(z) - x^T Bz$. Let Q_s be the first nonzero homogeneous convolution component. Its pivot monomial has coefficient ± 1 , and later convolution basis vectors do not change that coordinate; boundedness of the corresponding coordinate functional therefore gives $\|Q_s\|_{\text{coeff}} \geq c_{n,d} > 0$. Here Q_s includes every later convolution coordinate of the same homogeneous degree s ; those coordinates may change the other entries of its coefficient tensor but cannot remove the normalized pivot entry. Only components of degree strictly below s become lower coefficients in the one-dimensional van der Corput argument. Polar coordinates give

$$A_{R,u} f = \int_{\mathbb{S}^{n-1}} \Theta(\theta) \int_{uR}^R e^{i[Q(r\theta) - rx^T B\theta]} f(x - r\theta) \frac{dr}{r} d\sigma(\theta).$$

On lines parallel to θ this is a one-dimensional convolution multiplier; the skew modulation depends only on the transverse line since $\theta^T B \theta = 0$. More explicitly, write $x = x_\perp + t\theta$, with $x_\perp \in \theta^\perp$, and use the Fourier convention $\widehat{g}(\xi) = \int e^{-it\xi} g(t) dt$ on that line. The fiber operator has multiplier

$$m_{\theta, x_\perp, u}(\xi) = \int_{uR}^R e^{i[Q(r\theta) - rx_\perp^T B \theta - r\xi]} \frac{dr}{r}. \quad (74)$$

Indeed, $(x_\perp + t\theta)^T B \theta = x_\perp^T B \theta$ because $\theta^T B \theta = 0$. After the change $r = R\rho$, both the Fourier frequency and the transverse skew modulation therefore enter only the coefficient of the linear term in ρ ; neither can alter the leading coefficient. The amplitude

$$\psi_u(\rho) = \rho^{-1} \mathbf{1}_{[u, 1]}(\rho)$$

satisfies $\|\psi_u\|_\infty + \text{Var}(\psi_u) \leq C$ uniformly for $1/2 \leq u \leq 1$. The leading coefficient of the resulting polynomial phase is $R^s Q_s(\theta)$. Lemma 3.4 and Plancherel therefore give the fiber norm

$$C \min\{1, (R^s |Q_s(\theta)|)^{-1/s}\}.$$

Choose $0 < \beta < 1/s$. The last display is bounded by $CR^{-s\beta} |Q_s(\theta)|^{-\beta}$. Minkowski in θ and (63) therefore give the required angular integral explicitly: writing $Q_s = M_s A_s$ with $\|A_s\|_{\text{coeff}} = 1$ and $M_s \geq c_{n,d}$,

$$\int_{\mathbb{S}^{n-1}} |Q_s(\theta)|^{-\beta} d\sigma(\theta) = M_s^{-\beta} \int_{\mathbb{S}^{n-1}} |A_s(\theta)|^{-\beta} d\sigma(\theta) \leq C_{n,s,\beta}.$$

Thus Minkowski gives $CR^{-s\beta} \|\Theta\|_\infty$. All lower homogeneous components of Q occur only as lower coefficients in van der Corput and are unrestricted.

For a skew pivot, all transverse and convolution coordinates vanish, because they precede the skew coordinates in the flag. Thus, modulo pure phases, the remaining phase is $x^T B y$ with $B^T = -B$. The normalized pivot coefficient gives $\|B\| \geq c_{n,d} > 0$. Choose an orthogonal matrix O that puts B in real skew-normal form. The change of variables $x = O\tilde{x}$, $y = O\tilde{y}$ replaces Θ by $\Theta_O(\theta) = \Theta(O\theta)$; rotation invariance of surface measure and of the annulus gives $\|\Theta_O\|_2 = \|\Theta\|_2$ and $\|\Theta_O\|_\infty = \|\Theta\|_\infty$. We may therefore assume that one two-dimensional block is βJ with $|\beta| \gtrsim_{n,d} 1$. For the associated twisted convolution

$$Sg(u, v) = \int k(a, b) e^{i\beta(ub - va)} g(u - a, v - b) da db,$$

take the Fourier transform $\widehat{g}(u, \eta) = \int e^{-iv\eta} g(u, v) dv$ in v . A direct calculation gives

$$\widehat{Sg}(u, \eta) = \int_{\mathbb{R}} \widehat{k}^b(a, \eta + \beta a - \beta u) \widehat{g}(u - a, \eta + \beta a) da,$$

where $\widehat{k}^b(a, \xi) = \int e^{-ib\xi} k(a, b) db$. The quantity $q = \eta + \beta u$ is invariant under the passage from (u, η) to the input pair $(u - a, \eta + \beta a)$. Setting $s = u - a$, the operator on the fixed- q fiber has kernel

$$K_q(u, s) = \widehat{k}^b(u - s, q - \beta(u + s)).$$

The change of variables

$$(u, s) \mapsto (a = u - s, \rho = q - \beta(u + s))$$

has Jacobian of absolute value $2|\beta|$. Therefore Plancherel in the b variable gives, uniformly in q ,

$$\begin{aligned} \|K_q\|_{L^2_{u,s}}^2 &= \frac{1}{2|\beta|} \iint |\widehat{k}^b(a, \rho)|^2 da d\rho \\ &\leq C|\beta|^{-1} \|k\|_2^2. \end{aligned}$$

The operator norm of each fiber is bounded by its Hilbert–Schmidt norm. Since $(u, \eta) \mapsto (u, q)$ has unit Jacobian, direct integration in q and Plancherel yield

$$\|S\|_{2 \rightarrow 2} \leq C|\beta|^{-1/2} \|k\|_2.$$

For the remaining coordinates, write $B = \beta J \oplus B'$ and $z = (z_0, z') \in \mathbb{R}^2 \times \mathbb{R}^{n-2}$. For each fixed z' , the factor coming from B' is unimodular and the translation $x' \mapsto x' - z'$ is an L^2 isometry. Minkowski therefore reduces the estimate to the preceding two-dimensional bound for the sliced kernel $k_{z'}(z_0) = k_{R,u}(z_0, z')$. Cauchy–Schwarz gives

$$\int_{|z'| \leq R} \|k_{z'}\|_2 dz' \leq |\{|z'| \leq R\}|^{1/2} \left(\int \|k_{z'}\|_2^2 dz' \right)^{1/2} \leq CR^{(n-2)/2} \|k_{R,u}\|_2 \leq CR^{-1} \|\Theta\|_2.$$

Therefore $\|A_{R,u}\|_{2 \rightarrow 2} \leq CR^{-1} \|\Theta\|_2$, hence $CR^{-1} \|\Theta\|_\infty$. This proves the proposition in every flag stratum. There are only finitely many strata. $\square$

Lemma 3.7 (Short variation from partial-annulus decay). *Under the hypotheses of Proposition 3.6, after reducing the exponent if necessary, every $f \in L^2(\mathbb{R}^n)$ satisfies*

$$\|V^2(A_{R,u}^{P,\Theta} f : 1/2 \leq u \leq 1)\|_2 \leq CR^{-b_\mu} \|\Theta\|_\infty \|f\|_2 \quad (75)$$

for some $b_\mu > 0$.

Proof. Let $F(u) = A_{R,u} f$. Notice that $F(1) = 0$. For test functions,

$$F'(u, x) = -u^{-1} \int_{\mathbb{S}^{n-1}} \Theta(\theta) e^{iP(x, x - uR\theta)} f(x - uR\theta) d\sigma(\theta),$$

so $\sup_u \|F'(u)\|_2 \leq C \|\Theta\|_1 \|f\|_2$. If $a = \sup_u \|F(u)\|_2$ and $b = \sup_u \|F'(u)\|_2$, the fundamental theorem of calculus and the anchor at 1 give pointwise

$$\sup_{1/2 \leq u \leq 1} |F(u, x)|^2 \leq 2 \int_{1/2}^1 |F(v, x)| |F'(v, x)| dv.$$

Integration in x and Cauchy–Schwarz therefore give $\|\sup_u |F(u)|\|_2 \leq Ca^{1/2} b^{1/2}$. For every partition,

$$\sum |F(u_i) - F(u_{i-1})|^2 \leq 2 \sup_u |F(u)| \int_{1/2}^1 |F'(u)| du.$$

To record the exponents, put

$$M(x) = \sup_{1/2 \leq u \leq 1} |F(u, x)|, \quad A(x) = \int_{1/2}^1 |F'(u, x)| du.$$

The preceding partition estimate says $V^2(F)^2 \leq 2MA$. Minkowski gives $\|A\|_2 \leq b/2$, while the anchored fundamental theorem of calculus gives $\|M\|_2 \leq Ca^{1/2} b^{1/2}$. Consequently

$$\|V^2(F)\|_2^2 \leq 2\|M\|_2 \|A\|_2 \leq Ca^{1/2} b^{3/2},$$

and hence $\|V^2(F)\|_2 \leq Ca^{1/4} b^{3/4}$. Insert (69); since $\|\Theta\|_1 \leq C_n \|\Theta\|_\infty$, one may take $b_\mu = a_\mu/4$.

For completeness, the argument extends from test functions to all $f \in L^2$. Choose $f_j \in C_c^\infty(\mathbb{R}^n)$ converging to f in L^2 so rapidly that $\sum_j \|f_{j+1} - f_j\|_2 < \infty$, and let F_j be the corresponding paths. For the path associated with $f_j - f_\ell$,

$$V^2 \leq V^1 \leq \int_{1/2}^1 |F'_{j,\ell}(u)| du,$$

so Minkowski and the derivative bound give

$$\|V^2(F_j - F_\ell)\|_2 \leq C \|\Theta\|_1 \|f_j - f_\ell\|_2.$$

In particular,

$$W(x) := \sum_j V^2(F_{j+1} - F_j)(x) \in L^2(\mathbb{R}^n).$$

All difference paths vanish at $u = 1$, and hence, off the single null set $\{W = \infty\}$,

$$\sup_{1/2 \leq u \leq 1} |F_{j+1}(u, x) - F_j(u, x)| \leq V^2(F_{j+1} - F_j)(x).$$

Thus $F_j(u, x)$ converges uniformly in the entire parameter interval on one common full-measure set. Denote its continuous limit by $\tilde{F}(u, x)$ and set every coordinate equal to zero on the exceptional set.

This limit represents the intended operator at every parameter. Indeed, the absolute-value kernel on the fixed annulus has both row and column integrals bounded by $C\|\Theta\|_1$, so Schur's test gives, uniformly in u ,

$$\|F_j(u) - A_{R,u}^{P,\Theta} f\|_2 \leq C\|\Theta\|_1 \|f_j - f\|_2.$$

On the other hand, the preceding summable variation bound gives

$$\sup_u \|F_j(u) - \tilde{F}(u)\|_2 \leq \sum_{k \geq j} \|V^2(F_{k+1} - F_k)\|_2 \longrightarrow 0.$$

Consequently $\tilde{F}(u)$ is an L^2 representative of $A_{R,u}^{P,\Theta} f$ for every u , with the representative chosen on one common spatial set.

For clarity, this common representative may also be taken locally absolutely continuous. With a fixed Borel representative of f , define in L_x^2

$$D_f(u, x) = -u^{-1} \int_{\mathbb{S}^{n-1}} \Theta(\theta) e^{iP(x, x - uR\theta)} f(x - uR\theta) d\sigma(\theta).$$

The map $u \mapsto D_f(u)$ is strongly measurable and $\sup_u \|D_f(u)\|_2 \leq C\|\Theta\|_1 \|f\|_2$. Approximation by f_j in the fundamental theorem of calculus gives, in L_x^2 ,

$$A_{R,v}^{P,\Theta} f - A_{R,u}^{P,\Theta} f = \int_u^v D_f(w) dw.$$

Fubini therefore makes $-\int_u^1 D_f(w, x) dw$ an absolutely continuous pointwise representative for almost every x . It agrees with $\tilde{F}$ at all rational u on one full-measure set and hence, by continuity, at every u .

Finally, uniform pointwise convergence implies for each finite partition π that its ℓ^2 increment sum for $\tilde{F}$ is the limit of the corresponding sums for F_j . Taking the supremum over π , then using Fatou's lemma, yields

$$\|V^2(\tilde{F})\|_2 \leq \liminf_{j \rightarrow \infty} \|V^2(F_j)\|_2 \leq CR^{-b_\mu} \|\Theta\|_\infty \|f\|_2,$$

which proves (75) for general f without exchanging an uncountable supremum with an unsupported coordinatewise limit. $\square$

Corollary 3.8 (Scale-covariant form). *Let P be a real polynomial of degree at most d whose first nonzero flag coordinate is $c_\mu \neq 0$ of degree m_μ . Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$. For a shell radius $R > 0$ set*

$$L_\mu = |c_\mu| R^{m_\mu}.$$

If $L_\mu \geq 1$, then

$$\sup_{1/2 \leq u \leq 1} \|A_{R,u}^{P,\Theta}\|_{2 \rightarrow 2} + \|V_u^2(A_{R,u}^{P,\Theta})\|_{2 \rightarrow 2} \leq C\|\Theta\|_\infty L_\mu^{-\beta_\mu} \quad (76)$$

for some $\beta_\mu > 0$. Here the variation-operator norm means

$$\|V_u^2(A_{R,u}^{P,\Theta})\|_{2 \rightarrow 2} := \sup_{f \neq 0} \frac{\|V^2(A_{R,u}^{P,\Theta} f : 1/2 \leq u \leq 1)\|_2}{\|f\|_2}. \quad (77)$$

This is the norm of a homogeneous sublinear map; linearity of the variation map is not asserted. For the later packet rescaling only (the following variables are not hypotheses of (76)), fix a packet side

$q > 0$ and a truncation parameter $\varepsilon > 0$, and set $t = \log_2(R/q)$. Under the normalizing dilation $x = \rho X$, $y = \rho Y$, $\rho = |c_\mu|^{-1/m_\mu}$,

$$R \mapsto \tilde{R} = R/\rho = L_\mu^{1/m_\mu}, \quad q \mapsto \tilde{q} = q/\rho, \quad \varepsilon \mapsto \tilde{\varepsilon} = \varepsilon/\rho, \quad \log_2(\tilde{R}/\tilde{q}) = t. \quad (78)$$

If $h_\rho(Y) = h(\rho Y)$, then

$$\|h_\rho\|_1 = \rho^{-n} \|h\|_1, \quad \|h_\rho\|_2 = \rho^{-n/2} \|h\|_2, \quad \|h_\rho\|_\infty = \|h\|_\infty. \quad (79)$$

Thus a normalized bound $\tilde{R}^{-a_\mu}$ becomes $L_\mu^{-a_\mu/m_\mu}$. For any packet family h_Q supported on q -cubes, put $\tilde{Q} = \rho^{-1}Q$ and $\tilde{h}_Q(Y) = h_Q(\rho Y)$. Then

$$\|\tilde{h}_Q\|_1 \leq \Lambda|\tilde{Q}| \iff \|h_Q\|_1 \leq \Lambda|Q|,$$

so the density condition is invariant under the same rescaling.

Proof. Homogeneity of K_Θ gives the exact conjugacy

$$A_{R,u}^{P,\Theta} h(\rho X) = A_{R/\rho,u}^{P,\Theta} h_\rho(X).$$

Here and below $P_\rho(X, Y) = P(\rho X, \rho Y)$ and $h_\rho(Y) = h(\rho Y)$. The displayed transformations and norm identities follow by change of variables. Apply Proposition 3.6 and Lemma 3.7 at radius $\tilde{R}$ and take $\beta_\mu = \min(a_\mu, b_\mu)/m_\mu$. $\square$

4 Far-field strong estimates and triangular induction

We convert the scale-covariant annular decay into an angular $L \log L$ estimate and close the finite induction over the phase flag. Proposition 2.9 provides the terminal unphased estimate, and Section 3 provides oscillatory decay on every nonterminal stratum.

Call a phase *normalized nonterminal* if its first nonzero flag coordinate is ± 1 . Since the flag is finite, there is a number

$$a_0 = a_0(n, d) > 0 \quad (80)$$

not exceeding the exponent in Proposition 3.6 at any nonterminal flag position. If there is no such position, this section is vacuous and we set $a_0 = 1$.

4.1 Angular interpolation and short variation

Lemma 4.1 (q' -tracked partial-annulus estimates). *Let P be a normalized nonterminal phase, let $R \geq 1$, and let $1 < q \leq 2$. Suppose that $\Theta \in L^q(\mathbb{S}^{n-1})$. Uniformly in every later coefficient of P ,*

$$\sup_{1/2 \leq u \leq 1} \|A_{R,u}^{P,\Theta}\|_{2 \rightarrow 2} \leq C_{n,d} R^{-a_0/q'} \|\Theta\|_q, \quad q' = \frac{q}{q-1}. \quad (81)$$

Moreover, for every $1 < p < \infty$ put

$$\gamma_p = \frac{a_0}{2 \max(p, p')} > 0 \quad (82)$$

and let $f \in L^p(\mathbb{R}^n)$. Then

$$\sup_{1/2 \leq u \leq 1} \|A_{R,u}^{P,\Theta} f\|_p \leq C_{n,d,p} R^{-\gamma_p/q'} \|\Theta\|_q \|f\|_p, \quad (83)$$

$$\|V^2(A_{R,u}^{P,\Theta} f : 1/2 \leq u \leq 1)\|_p \leq C_{n,d,p} R^{-\gamma_p/q'} \|\Theta\|_q \|f\|_p. \quad (84)$$

All prefactors are uniform for $1 < q \leq 2$.

Proof. The trivial annular estimate is

$$\sup_u \|A_{R,u}^{P,\Theta}\|_{2 \rightarrow 2} \leq \|K_\Theta \mathbf{1}_{\{R/2 < |\cdot| < R\}}\|_1 \leq C \|\Theta\|_1. \quad (85)$$

For fixed P, R, u , the map $\Theta \mapsto A_{R,u}^{P,\Theta}$ is linear from the angular space into $\mathcal{B}(L^2)$. Complex interpolation of (85) with Proposition 3.6 gives (81). Indeed, if ϑ denotes the weight of the $L^\infty(\mathbb{S}^{n-1})$ endpoint, then

$$\frac{1}{q} = \frac{1 - \vartheta}{1} + \frac{\vartheta}{\infty}, \quad \text{so} \quad \vartheta = \frac{1}{q'}.$$

Thus R^{-a_0} becomes $R^{-a_0/q'}$, with no q -dependent interpolation loss. This interpolation is first performed for each fixed u in the Banach target $\mathcal{B}(L^2)$; both endpoint bounds are uniform in u , so the supremum in u may then be taken. The complex interpolation constants are uniform in q .

For the short variation put $F(u) = A_{R,u}^{P,\Theta} f$. Initially take f and Θ bounded. The path is absolutely continuous, $F(1) = 0$, and polar coordinates give, for almost every $u \in [1/2, 1]$,

$$F'(u, x) = -\frac{1}{u} \int_{\mathbb{S}^{n-1}} \Theta(\theta) e^{iP(x, x - uR\theta)} f(x - uR\theta) d\sigma(\theta). \quad (86)$$

Consequently, for every $1 \leq s \leq \infty$,

$$\sup_{1/2 \leq u \leq 1} \|F'(u)\|_s \leq C \|\Theta\|_1 \|f\|_s. \quad (87)$$

Set

$$a = \sup_u \|F(u)\|_2, \quad b = \sup_u \|F'(u)\|_2.$$

Since $F(1) = 0$, the fundamental theorem of calculus yields pointwise

$$\sup_u |F(u, x)|^2 \leq 2 \int_{1/2}^1 |F(v, x) F'(v, x)| dv.$$

After integration in x ,

$$\|\sup_u |F(u)|\|_2 \leq Ca^{1/2} b^{1/2}. \quad (88)$$

For every partition $1/2 \leq u_0 < \dots < u_N \leq 1$,

$$\sum_{i=1}^N |F(u_i) - F(u_{i-1})|^2 \leq 2 \sup_u |F(u)| \int_{1/2}^1 |F'(v)| dv.$$

Taking the supremum over partitions, integrating, and using (88) gives

$$\|V^2(F(u) : 1/2 \leq u \leq 1)\|_2 \leq Ca^{1/4} b^{3/4}. \quad (89)$$

Equations (81), (87), and $\|\Theta\|_1 \leq C_n \|\Theta\|_q$ therefore imply

$$\|V^2(F)\|_2 \leq CR^{-a_0/(4q')} \|\Theta\|_q \|f\|_2. \quad (90)$$

The map

$$S_R f := V^2(A_{R,u}^{P,\Theta} f : 1/2 \leq u \leq 1)$$

is nonnegative, absolutely homogeneous, and subadditive. Moreover, by (87),

$$S_R f \leq V^1(A_{R,u}^{P,\Theta} f : 1/2 \leq u \leq 1) \leq \int_{1/2}^1 |F'(u)| du,$$

and hence S_R has nondecaying strong $(1, 1)$ and (∞, ∞) bounds with norm $C \|\Theta\|_1$. We may therefore apply the Marcinkiewicz interpolation theorem for homogeneous sublinear operators. For $1 < p < 2$,

interpolation between the strong $(1, 1)$ bound and (90) puts weight $2/p'$ on the L^2 endpoint and gives decay $a_0/(2p'q')$. For $2 < p < \infty$, interpolation between (90) and the strong (∞, ∞) bound puts weight $2/p$ on the L^2 endpoint and gives decay $a_0/(2pq')$. These exponents are exactly γ_p/q' , since

$$\frac{a_0}{4q'} \frac{2}{p'} = \frac{a_0}{2p'q'} \quad (1 < p < 2), \quad \frac{a_0}{4q'} \frac{2}{p} = \frac{a_0}{2pq'} \quad (2 < p < \infty).$$

This proves (84). The same spatial interpolation, now starting from (81), proves the stronger corresponding bound for the annular operator; weakening its exponent to γ_p/q' gives (83).

To remove the boundedness assumptions, choose bounded $f_m \rightarrow f$ in $L^p(\mathbb{R}^n)$ and bounded $\Theta_m \rightarrow \Theta$ in $L^q(\mathbb{S}^{n-1})$, and put $F_m(u) = A_{R,u}^{P,\Theta_m} f_m$. By (87), linearity, and $V^2 \leq V^1$,

$$\|V^2(F_m - F_\ell)\|_p \leq C\|\Theta_m\|_1\|f_m - f_\ell\|_p + C\|\Theta_m - \Theta_\ell\|_1\|f_\ell\|_p.$$

The right-hand side tends to zero, since the sphere has finite measure. Thus the paths are Cauchy in the metric $\|V^2(F_m - F_\ell)\|_p$. We spell out the completion because the parameter set is uncountable. Every difference path is anchored at $u = 1$. Choose a subsequence, still denoted by F_m , such that

$$\sum_m \|V^2(F_{m+1} - F_m : 1/2 \leq u \leq 1)\|_p < \infty.$$

After deleting one null set, the corresponding pointwise series of variations is finite. The anchor then gives

$$\sup_{1/2 \leq u \leq 1} |F_{m+1}(u, x) - F_m(u, x)| \leq V^2(F_{m+1} - F_m)(x),$$

so $F_m(u, x)$ converges uniformly in u on that common full-measure set. For a fixed rational u , Young's inequality on the fixed annulus identifies the limit with $A_{R,u}^{P,\Theta} f$ in L^p . Passing to a further diagonal subsequence and deleting one countable union of null sets makes these identities simultaneous for all rational $u \in [1/2, 1]$. Polar coordinates and Fubini give the latter annular integrals a jointly measurable, locally absolutely continuous representative in u : its radial derivative is the right side of (86), whose $L_u^1(L_x^p)$ norm is bounded by $C\|\Theta\|_1\|f\|_p$. Continuity therefore identifies the uniform limit at every real u . On each finite partition the increment sum is continuous under this uniform convergence. Taking first the limit, then the supremum over the nested finite rational partitions, and using Fatou proves (84). The same common representative and coordinatewise convergence prove (83). $\square$

4.2 The far path and the q' loss

For a normalized nonterminal phase define

$$\mathcal{F}_\varepsilon^{P,\Theta} f(x) = \int_{|x-y| > \max(\varepsilon, 1)} e^{iP(x,y)} K_\Theta(x-y) f(y) dy \quad (91)$$

on bounded compactly supported data. Equivalently, put

$$B_j^{P,\Theta} = A_{2^{j+1}, 1/2}^{P,\Theta}, \quad j \geq 0. \quad (92)$$

If $2^j \leq \varepsilon \leq 2^{j+1}$, then

$$\mathcal{F}_\varepsilon^{P,\Theta} f = A_{2^{j+1}, \varepsilon/2^{j+1}}^{P,\Theta} f + \sum_{k=j+1}^{\infty} B_k^{P,\Theta} f, \quad (93)$$

whereas for $0 < \varepsilon \leq 1$ the path is the constant $\sum_{k \geq 0} B_k^{P,\Theta} f$.

We use the elementary long-short inequality

$$V^2(F_\varepsilon : \varepsilon > 0) \leq CV^2(F_{2^j} : j \in \mathbb{Z}) + C \left(\sum_{j \in \mathbb{Z}} V^2(F_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2}. \quad (94)$$

Indeed, insert the adjacent dyadic endpoints into an arbitrary variation partition. Each increment is the sum of two within-block increments and one increment between dyadic endpoints. The between-block increments form an ordered subpartition, and every dyadic block contributes at most two within-block increments. The inequality follows from $|z_1 + z_2 + z_3|^2 \leq 3 \sum |z_i|^2$.

Proposition 4.2 (q' -tracked far V^2). *Let P be normalized and nonterminal, let $1 < p < \infty$, and let $1 < q \leq 2$. Suppose that $\Theta \in L^q(\mathbb{S}^{n-1})$; no cancellation of Θ is required. Then, for every $f \in L^p(\mathbb{R}^n)$,*

$$\|\mathcal{F}_1^{P,\Theta} f\|_p + \|V^2(\mathcal{F}_\varepsilon^{P,\Theta} f : \varepsilon > 0)\|_p \leq C_{n,d,p} q' \|\Theta\|_q \|f\|_p. \quad (95)$$

The constant is uniform in q and in every unused coefficient of P .

Proof. Lemma 4.1 gives

$$\|B_j^{P,\Theta} f\|_p \leq C_{n,d,p} 2^{-\gamma_p j/q'} \|\Theta\|_q \|f\|_p. \quad (96)$$

Thus the series in (93) converges in L^p and

$$\sum_{j \geq 0} \|B_j^{P,\Theta} f\|_p \leq C_{n,d,p} q' \|\Theta\|_q \|f\|_p, \quad (97)$$

because, for $q' \geq 2$,

$$\sum_{j \geq 0} 2^{-\gamma_p j/q'} = \frac{1}{1 - 2^{-\gamma_p/q'}} \leq C_{\gamma_p} q'.$$

This proves the asserted bound for $\mathcal{F}_1$. Choose measurable representatives of the countable family $B_j f$. By Minkowski and monotone convergence,

$$G(x) := \sum_{j \geq 0} |B_j^{P,\Theta} f(x)| \in L^p(\mathbb{R}^n), \quad \|G\|_p \leq \sum_{j \geq 0} \|B_j^{P,\Theta} f\|_p.$$

Consequently the series is absolutely convergent outside one null set, and all of its (countably many) dyadic tail identities hold there simultaneously. On every block we use the jointly measurable, locally absolutely continuous partial-annulus representative constructed in Lemma 4.1. Formula (93) then defines one common representative of the full far trajectory, with adjacent block definitions agreeing at their shared dyadic endpoint. All pointwise variation estimates below refer to this representative.

At the dyadic parameters,

$$\mathcal{F}_{2^j}^{P,\Theta} f = \sum_{k=j}^{\infty} B_k^{P,\Theta} f, \quad j \geq 0,$$

and the sequence is constant for $j < 0$. Hence, pointwise,

$$V^2(\mathcal{F}_{2^j}^{P,\Theta} f : j \in \mathbb{Z}) \leq \sum_{j \geq 0} |B_j^{P,\Theta} f|. \quad (98)$$

Its L^p norm is bounded by (97).

On the block $[2^j, 2^{j+1}]$, the tail in (93) is constant and the nonconstant part is $A_{2^{j+1},u}^{P,\Theta}$, $1/2 \leq u \leq 1$. Therefore (84) gives

$$\|V^2(\mathcal{F}_\varepsilon : 2^j \leq \varepsilon \leq 2^{j+1})\|_p \leq C 2^{-\gamma_p j/q'} \|\Theta\|_q \|f\|_p.$$

Pointwise, $(\sum_j V_j^2)^{1/2} \leq \sum_j V_j$; hence the L^p norm of the short term in (94) is again bounded by the geometric sum in (97). Combining the long and short parts proves (95). $\square$

4.3 Finite-trajectory extrapolation to $L \log L$

Let

$$\Phi_0(t) = (1+t) \log(1+t) - t, \quad \Psi_0(s) = e^s - s - 1.$$

These are complementary Young functions. On the finite measure space $\mathbb{S}^{n-1}$, the Luxemburg norm for Φ_0 is equivalent, with dimensional constants, to the homogeneous gauge $\mathfrak{L}$ in (2); the Luxemburg norm for Ψ_0 will be denoted by $\|\cdot\|_{\exp L}$. The Orlicz–Hölder inequality therefore gives

$$\int_{\mathbb{S}^{n-1}} |uv| d\sigma \leq 2 \|u\|_{L \log L} \|v\|_{\exp L}. \quad (99)$$

For the quantitative direction used below the assertion is trivial when $\Theta = 0$; otherwise let $A = \|\Theta\|_1$ and $B = \mathfrak{L}(\Theta)$. Since $B \geq A$ and $\Phi_0(t) \leq Ct \log(e+t)$,

$$\int_{\mathbb{S}^{n-1}} \Phi_0\left(\frac{|\Theta|}{B}\right) d\sigma \leq \frac{C}{B} \left[A + \int_{\mathbb{S}^{n-1}} |\Theta| \log^+\left(\frac{|\Theta|}{A}\right) d\sigma \right] \leq C.$$

Convexity and $\Phi_0(0) = 0$ imply $\Phi_0(t/C_1) \leq C_1^{-1} \Phi_0(t)$ for $C_1 \geq 1$. Choosing C_1 larger than the preceding absolute constant gives

$$\|\Theta\|_{L \log L} \leq C_1 \mathfrak{L}(\Theta). \quad (100)$$

Proposition 4.3 (Angular endpoint for the far path). *Let P be normalized and nonterminal, let $1 < p < \infty$, and let $\Theta \in L \log L(\mathbb{S}^{n-1})$. No cancellation of Θ is required. Then, for every $f \in L^p(\mathbb{R}^n)$,*

$$\|\mathcal{F}_1^{P,\Theta} f\|_p + \|V^2(\mathcal{F}_\varepsilon^{P,\Theta} f : \varepsilon > 0)\|_p \leq C_{n,d,p} \mathfrak{L}(\Theta) \|f\|_p. \quad (101)$$

Consequently,

$$\mathcal{J}_{2,p}(\mathcal{F}_\varepsilon^{P,\Theta} f : \varepsilon > 0) \leq C_{n,d,p} \mathfrak{L}(\Theta) \|f\|_p. \quad (102)$$

Proof. Fix f and a finite ordered set $D \subset \mathbb{Q}_+$ containing 1. On $\mathbb{C}^D$ define

$$\|z\|_{E_D} = |z_1| + V_D^2(z_\varepsilon : \varepsilon \in D). \quad (103)$$

This is a norm: vanishing variation makes the trajectory constant, and the value at 1 then detects that constant. Since D is finite, E_D is a finite-dimensional Banach space, hence $L^p(\mathbb{R}^n; E_D)$ is Banach and its norm is recovered by Hahn–Banach duality. For bounded angular data define, after complexification if necessary, the linear map

$$\mathcal{A}_{f,D} \Theta = (\mathcal{F}_\varepsilon^{P,\Theta} f)_{\varepsilon \in D} \quad \text{in } L^p(\mathbb{R}^n; E_D).$$

Proposition 4.2, uniformly in D , says

$$\|\mathcal{A}_{f,D}\|_{L^q(\mathbb{S}^{n-1}) \rightarrow L^p(E_D)} \leq C_{n,d,p} q' \|f\|_p, \quad 1 < q \leq 2. \quad (104)$$

Let $\Lambda \in [L^p(\mathbb{R}^n; E_D)]^*$ have norm one. For $s \geq 2$ take $q = s/(s-1)$ in (104). The scalar functional $\Theta \mapsto \Lambda(\mathcal{A}_{f,D} \Theta)$ is represented by a function $h_s \in L^s(\mathbb{S}^{n-1})$ satisfying

$$\|h_s\|_s \leq Cs \|f\|_p. \quad (105)$$

The representatives h_s agree almost everywhere. Indeed, two of them represent the same functional on the common subspace of bounded angular functions: all of these extensions agree there with the original finite-trajectory formula. Since the sphere has finite measure, $h_s - h_t \in L^1$ for any $s, t \geq 2$. Its bounded complex sign belongs to every relevant L^q space, and testing with that sign gives $\int |h_s - h_t| d\sigma = 0$. Thus the representatives agree almost everywhere; denote their common representative by h .

Put $A = \|f\|_p$. If $A = 0$ there is nothing to prove. For every integer $k \geq 2$, $\|h\|_k \leq CkA$. Hence, by Tonelli and the exponential series,

$$\int_{\mathbb{S}^{n-1}} \Psi_0\left(\frac{|h|}{LA}\right) d\sigma = \sum_{k=2}^{\infty} \frac{\|h\|_k^k}{(LA)^k k!} \leq \sum_{k=2}^{\infty} \frac{(Ck/L)^k}{k!}. \quad (106)$$

Since $k^k/k! \leq C_0 e^k \sqrt{k}$, the last series is at most one once $L = L(n, d, p)$ is sufficiently large. Thus

$$\|h\|_{\exp L} \leq C_{n,d,p} \|f\|_p. \quad (107)$$

Equations (99) and (107) give

$$|\Lambda(\mathcal{A}_{f,D}\Theta)| \leq C_{n,d,p} \|\Theta\|_{L \log L} \|f\|_p.$$

Taking the supremum over Λ proves

$$\|\mathcal{A}_{f,D}\Theta\|_{L^p(E_D)} \leq C_{n,d,p} \mathfrak{L}(\Theta) \|f\|_p, \quad (108)$$

uniformly in the finite set D .

Bounded angular functions are dense in the Luxemburg $L \log L$ norm. Indeed, for $\Theta_N = \Theta \mathbf{1}_{\{|\Theta| \leq N\}}$, the Δ_2 condition for Φ_0 and dominated convergence for the corresponding modular give

$$\|\Theta - \Theta_N\|_{L \log L} \longrightarrow 0.$$

Thus the finite-trajectory maps extend to all $\Theta \in L \log L$. To check consistency, if $D \subset D'$ and both contain 1, coordinate restriction satisfies

$$|z_1| + V_D^2(z) \leq |z_1| + V_{D'}^2(z),$$

so it is a contraction from $E_{D'}$ to E_D . The two extensions agree on bounded angular functions and hence, by density, on all of $L \log L$. Thus the rational trajectory is independent of the exhausting sequence. Fix once and for all an enumeration of $\mathbb{Q}_+$ beginning with 1, and let D_N be its first N elements. Choose representatives of the countable family of rational coordinates simultaneously. For every N , (108) applies to these same coordinates, and

$$|\mathcal{F}_1 f| + V_{D_N}^2(\mathcal{F}_\varepsilon f) \uparrow |\mathcal{F}_1 f| + V_{\mathbb{Q}_+}^2(\mathcal{F}_\varepsilon f) \quad \text{pointwise.}$$

Monotone convergence (equivalently, Fatou followed by monotonicity) now passes the uniform finite- D_N estimate to all positive rational parameters. The finite-dimensional extension is completed first, and only then is $N \rightarrow \infty$ taken. For $0 < a < b < \infty$, finite-shell differences satisfy

$$\|(\mathcal{F}_a - \mathcal{F}_b)f\|_p \leq \|\Theta\|_1 \log(b/a) \|f\|_p, \quad (109)$$

including when $[a, b]$ meets 1: the difference is zero if $b \leq 1$, and if $a < 1 < b$ its kernel has L^1 norm $\|\Theta\|_1 \log b \leq \|\Theta\|_1 \log(b/a)$.

The difference identity and estimate above are first obtained on rational parameters; approximation by bounded angular data and Young's inequality preserve that identity. Put

$$\mathcal{A}_x(r) = \int_{\mathbb{S}^{n-1}} e^{iP(x, x-r\theta)} \Theta(\theta) f(x-r\theta) d\sigma(\theta).$$

For every $0 < a < b < \infty$, polar coordinates give

$$\int_a^b |\mathcal{A}_x(r)| \frac{dr}{r} \leq \int_{a < |z| < b} |K_\Theta(z)| |f(x-z)| dz.$$

The right-hand side belongs to L^p . Applying this observation to a countable family of annuli covering $(0, \infty)$, discard one null set on whose complement all these radial densities are locally integrable. Choose one representative of $\mathcal{F}_1 f$, and further remove the countable union of the null sets on which a rational finite-shell identity fails. On the resulting common full-measure set define

$$\mathcal{F}_\varepsilon f = \begin{cases} \mathcal{F}_1 f, & 0 < \varepsilon \leq 1, \\ \mathcal{F}_1 f - \int_{1 < |z| \leq \varepsilon} e^{iP(x, x-z)} K_\Theta(z) f(x-z) dz, & \varepsilon > 1. \end{cases}$$

The rational finite-shell identities show that this leaves every rational coordinate unchanged. The resulting path is locally absolutely continuous and satisfies, for almost every ε ,

$$\frac{d}{d\varepsilon} \mathcal{F}_\varepsilon f(x) = -\mathbf{1}_{(1,\infty)}(\varepsilon) \frac{\mathcal{A}_x(\varepsilon)}{\varepsilon}.$$

Lemma 2.6 therefore identifies the rational variation with the full-parameter variation. This proves (101). Finally (16) gives (102). $\square$

Remark 4.4 (The far anchor). The construction above is anchored at infinity. For bounded Θ , take $q = 2$ in Lemma 4.1 and interpolate spatially as in its proof. More explicitly, for $\varepsilon \geq 1$ choose $j \geq 0$ with $2^j \leq \varepsilon \leq 2^{j+1}$. Then the current partial annulus is bounded in L^p by

$$C_{n,d,p} 2^{-\gamma_p j/2} \|\Theta\|_2 \|f\|_p,$$

and the remaining annular tail is bounded by

$$C_{n,d,p} \sum_{k \geq j+1} 2^{-\gamma_p k/2} \|\Theta\|_2 \|f\|_p.$$

Both quantities tend to zero as $j \rightarrow \infty$, and hence as $\varepsilon \rightarrow \infty$. For general $\Theta \in L \log L$, choose bounded $\Theta_m \rightarrow \Theta$ in that norm. Proposition 4.3 and

$$\sup_{\varepsilon > 0} |F_\varepsilon| \leq |F_1| + V^2(F_\varepsilon : \varepsilon > 0)$$

make the approximation uniform in L^p over ε . Hence $\mathcal{F}_\varepsilon^{P,\Theta} f \rightarrow 0$ in L^p . Finite V^2 supplies an almost-everywhere limit at infinity, and an L^p -convergent subsequence identifies that limit as zero.

4.4 Local triangular reduction

For a normalized phase put

$$\mathcal{L}_\varepsilon^{P,\Omega} f(x) = \begin{cases} \int_{\varepsilon < |x-y| < 1} e^{iP(x,y)} K_\Omega(x-y) f(y) dy, & 0 < \varepsilon < 1, \\ 0, & \varepsilon \geq 1. \end{cases} \quad (110)$$

We record two elementary endpoint facts. First, if a path E is zero at one endpoint and $V^1(E) \leq W$, then

$$\lambda N_\lambda(E) \leq W, \quad N_\lambda(E) > 0 \implies \lambda \leq W, \quad \lambda N_\lambda(E)^{1/2} \leq W. \quad (111)$$

Indeed, the sum of the increments in any ordered family of jump pairs is at most the total 1-variation. Second, if

$$L_\varepsilon = \begin{cases} F_\varepsilon - F_1, & 0 < \varepsilon < 1, \\ 0, & \varepsilon \geq 1, \end{cases} \quad (112)$$

and $F_\varepsilon \rightarrow F_1$ as $\varepsilon \uparrow 1$, then

$$N_\lambda(L) \leq N_\lambda(F). \quad (113)$$

To see this, retain every jump pair lying below 1; if a pair crosses 1, it is necessarily the last nonzero pair and may be replaced by the pair with upper endpoint 1.

Proposition 4.5 (Triangular local reduction in the strong range). *Let P be a real polynomial of degree at most d whose first nonzero flag coordinate is $c_V = \pm 1$. Let $\Omega \in L^1(\mathbb{S}^{n-1})$ and let f be bounded and compactly supported. On each unit output cube, the local path (110) is, modulo pure input and output phases, the sum of*

- (i) a localized path of the form (112) whose residual phase either has all flag coordinates equal to zero, or has first nonzero flag coordinate strictly later than v ;
- (ii) an error path E satisfying

$$V^1(E)(x) \leq C_{n,d} \int_{|x-y|<1} \frac{|\Omega((x-y)/|x-y|)|}{|x-y|^{n-1}} |f(y)| dy. \quad (114)$$

The majorant in (114) has L^p norm at most $C_{n,d} \|\Omega\|_1 \|f\|_p$ for every $1 \leq p \leq \infty$.

Proof. Let Q_a be the output unit cube centered at a , and let Q_a^* be the fixed input enlargement relevant to Q_a . Write $x = a + X$, $y = a + Y$. Theorem 3.2(iii) gives

$$P(a+X, a+Y) = p_a(X) + q_a(Y) + c_v H_v(X, Y) + P_{a,>v}(X, Y),$$

where every flag coordinate of $P_{a,>v}$ is later than v . Set

$$f_a(Y) = e^{iq_a(Y)} f(a+Y) \mathbf{1}_{Q_a^*-a}(Y).$$

The coordinate translation identifies the original path at $a+X$ with the translated path at X and preserves Lebesgue measure. Multiplication of the output path by $e^{ip_a(X)}$ leaves every pointwise jump unchanged, while the pure input phase is absorbed into f_a . In particular,

$$|f_a(Y)| = |f(a+Y)| \mathbf{1}_{Q_a^*-a}(Y), \quad \|f_a\|_p = \|f \mathbf{1}_{Q_a^*}\|_p.$$

Thus the local path is the path with phase $c_v H_v + P_{a,>v}$ applied to f_a . If X lies in the reference output cube and $|X - Y| < 1$, both variables belong to a fixed bounded set. Since $H_v(Y, Y) = 0$,

$$|H_v(X, Y)| \leq C_{n,d} |X - Y|. \quad (115)$$

Thus

$$|e^{ic_v H_v(X, Y)} - 1| \leq C_{n,d} |X - Y|.$$

The total 1-variation of the difference of the two nested truncation paths is therefore bounded by (114). Its kernel has L^1 norm

$$C_{n,d} \int_0^1 \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| d\sigma(\theta) dr = C_{n,d} \|\Omega\|_1,$$

which proves the asserted L^p bound. Removing this difference leaves the local truncation of $P_{a,>v}$ applied to f_a . To identify it precisely with (112), let

$$F_\varepsilon = T_{\Omega, \varepsilon}^{P_{a,>v}} f_a$$

at this bounded compactly supported stage. The finite-shell identity gives

$$\mathcal{L}_\varepsilon^{P_{a,>v}, \Omega} f_a = F_\varepsilon - F_1 \quad (0 < \varepsilon < 1), \quad \mathcal{L}_\varepsilon^{P_{a,>v}, \Omega} f_a = 0 \quad (\varepsilon \geq 1).$$

Moreover $F_\varepsilon \rightarrow F_1$ as $\varepsilon \uparrow 1$ by the L^1 norm of the intervening shell kernel. Thus the remaining path is exactly the localized full path in (112), and (113) applies to it. $\square$

4.5 Coefficient-uniform strong jumps for every phase

Proposition 4.6 (Strong polynomial critical jumps). *Let $1 < p < \infty$, let P be any real polynomial of degree at most d , and let $\Omega \in L \log L(\mathbb{S}^{n-1})$ have mean zero. Then for every $f \in L^p(\mathbb{R}^n)$,*

$$\mathcal{J}_{2,p}(T_{\Omega, \varepsilon}^P f : \varepsilon > 0) \leq C_{n,d,p} \mathfrak{L}(\Omega) \|f\|_p, \quad (116)$$

where the constant is independent of every coefficient of P . Consequently, for every $r > 2$,

$$\|V^r(T_{\Omega, \varepsilon}^P f : \varepsilon > 0)\|_p \leq C_{n,d,p,r} \mathfrak{L}(\Omega) \|f\|_p. \quad (117)$$

Proof. We first take bounded compactly supported data, for which every finite-shell identity is classical. At this stage the induction concerns only these classical finite-shell trajectories and uses no extension at the flag position currently being proved. We use backward induction on the finite phase flag. If every flag coordinate vanishes, then P is a sum of a pure input phase and a pure output phase. Equation (54) and Proposition 2.9 give (116).

Assume the estimate for every phase whose first coordinate is later than v , and suppose that the first coordinate of P is $c_v \neq 0$, of homogeneous degree m_v . Under $x = \rho X$, $y = \rho Y$, where

$$\rho = |c_v|^{-1/m_v}, \quad (118)$$

the first coordinate becomes ± 1 . Work first with this normalized phase. Proposition 4.3 controls its far path.

For the local path partition the output into disjoint unit cubes Q_a . Only the restriction of f to a fixed enlargement Q_a^* contributes on Q_a . Proposition 4.5 gives a residual phase $P_{a,>v}$, the modulated translated input f_a used in its proof, and an error E_a . If all flag coordinates of $P_{a,>v}$ vanish, Proposition 2.9 together with the pure input/output factorization (54) supplies the base case. Otherwise its first nonzero coordinate is strictly later than v , so the induction hypothesis applies. In either case, (113) and the supremum over the jump amplitude give

$$\|\mathbf{J}(\mathcal{L}_\varepsilon^{P_{a,>v},\Omega} f_a : \varepsilon > 0)\|_p \leq C\mathfrak{L}(\Omega)\|f_a\|_p = C\mathfrak{L}(\Omega)\|f\mathbf{1}_{Q_a^*}\|_p \quad (119)$$

uniformly in a and in all translated residual coefficients. Equations (111), (19), and (114) give the same estimate for $\mathbf{J}(E_a)$, with $\|\Omega\|_1 \leq C\mathfrak{L}(\Omega)$.

Apply the pointwise quasi-triangle inequality (18) on every output cube, sum the p -th powers over the disjoint Q_a , and use bounded overlap of the Q_a^* . More explicitly, the overlap multiplicity is bounded by a dimensional constant and hence

$$\sum_a \|f\mathbf{1}_{Q_a^*}\|_p^p = \int |f(x)|^p \sum_a \mathbf{1}_{Q_a^*}(x) dx \leq C_n \|f\|_p^p.$$

Combining this identity with (119) and the corresponding error estimate gives

$$\mathcal{J}_{2,p}(\mathcal{L}_\varepsilon^{P,\Omega} f : \varepsilon > 0) \leq C_{n,d,p} \mathfrak{L}(\Omega) \|f\|_p. \quad (120)$$

Adding the local and far paths and using (18) proves the normalized case.

It remains to undo the dilation. If $f_\rho(Y) = f(\rho Y)$, homogeneity of the kernel gives the exact conjugacy

$$T_{\Omega,\varepsilon}^P f(\rho X) = T_{\Omega,\varepsilon/\rho}^{P_\rho} f_\rho(X). \quad (121)$$

The map $\varepsilon \mapsto \varepsilon/\rho$ is an order-preserving bijection of $(0, \infty)$, so it leaves every jump number unchanged. Changing variables in the input and output norms gives the explicit identities

$$\|f_\rho\|_{L_X^p} = \rho^{-n/p} \|f\|_{L_X^p}, \quad \|\mathbf{J}(T_{\Omega,\varepsilon/\rho}^{P_\rho} f_\rho)\|_{L_X^p} = \rho^{-n/p} \|\mathbf{J}(T_{\Omega,\varepsilon}^P f)\|_{L_X^p}.$$

Thus the input and output acquire the same factor $\rho^{-n/p}$. Thus the constant does not depend on c_v . The finite backward induction proves (116) for every phase.

The variation estimate is now a pointwise consequence rather than a mixed-norm extrapolation: by (21), (8), and (116),

$$\|V^r(T_{\Omega,\varepsilon}^P f : \varepsilon > 0)\|_p \leq \kappa_r \mathcal{J}_{2,p}(T_{\Omega,\varepsilon}^P f : \varepsilon > 0) \leq C_{n,d,p} \kappa_r \mathfrak{L}(\Omega) \|f\|_p.$$

This proves (117), with the stated r -dependence absorbed in its constant.

We next construct the anchored trajectory and pass to general data. For a normalized nonterminal phase, Proposition 4.3 defines the far path on L^p , including its value at 1; add the finite-shell local path

(110). Their sum has the classical finite-shell increments, agrees with the far path for $\varepsilon \geq 1$, and is anchored at infinity by Remark 4.4. For a nonnormalized nonterminal phase use the conjugacy (121). If every flag coordinate vanishes, use the anchored factorization of Proposition 2.9 and the pure modulations (54).

Choose bounded compactly supported $f_m \rightarrow f$ in L^p . Write

$$\tilde{\Omega}_m = \Omega \mathbf{1}_{\{|\Omega| \leq m\}}, \quad c_m = \frac{1}{m_n} \int_{\mathbb{S}^{n-1}} \tilde{\Omega}_m d\sigma, \quad m_n = \sigma(\mathbb{S}^{n-1}),$$

and put $\Omega_m = \tilde{\Omega}_m - c_m$. Since the defining Orlicz function satisfies the Δ_2 condition, $\tilde{\Omega}_m \rightarrow \Omega$ in the Luxemburg $L \log L$ norm. Moreover, using $\int \Omega d\sigma = 0$ and $L \log L \hookrightarrow L^1$,

$$c_m = -\frac{1}{m_n} \int_{\mathbb{S}^{n-1}} \Omega \mathbf{1}_{\{|\Omega| > m\}} d\sigma \rightarrow 0.$$

On the finite-measure sphere the Luxemburg norm of the constant function c_m is $C_n |c_m|$. Hence the bounded mean-zero functions Ω_m converge to Ω in $L \log L$. By Lemma 2.1, this convergence is equivalently

$$\mathfrak{L}(\Omega_m - \Omega) \rightarrow 0. \quad (122)$$

The lattice property of the Luxemburg norm, the embedding $L \log L \hookrightarrow L^1$ on the sphere, and the triangle inequality also give

$$\|\tilde{\Omega}_m\|_{L \log L} \leq \|\Omega\|_{L \log L}, \quad \|c_m \mathbf{1}\|_{L \log L} \leq C_n |c_m| \leq C_n \|\Omega\|_1 \leq C_n \|\Omega\|_{L \log L}.$$

Lemma 2.1 therefore yields the uniform bound

$$\sup_{m \geq 1} \mathfrak{L}(\Omega_m) \leq C_n \mathfrak{L}(\Omega). \quad (123)$$

For a normalized nonterminal phase, bilinearity gives

$$\mathcal{F}_\varepsilon^{P, \Omega_m} f_m - \mathcal{F}_\varepsilon^{P, \Omega} f = \mathcal{F}_\varepsilon^{P, \Omega_m} (f_m - f) + \mathcal{F}_\varepsilon^{P, \Omega_m - \Omega} f.$$

Proposition 4.3, (123), and (122) show that both terms tend to zero in the finite-trajectory $L^p(E_D)$ norm, uniformly over every fixed finite parameter set D . Thus the far coordinates for (Ω_m, f_m) converge in L^p to those for (Ω, f) . The same bilinear decomposition and Young's inequality give convergence of every local finite-shell coordinate. The same conclusion follows after dilation, and in the pure case from the factorization estimates in Proposition 2.9. The finite-shell bound used here is, for $0 < a < b < \infty$,

$$\left\| \int_{a < |z| < b} e^{iP(x, x-z)} K_\Theta(z) g(x-z) dz \right\|_p \leq \|\Theta\|_1 \log(b/a) \|g\|_p.$$

Enumerate $\mathbb{Q}_+ = \{r_1, r_2, \dots\}$ and put $D_N = \{r_1, \dots, r_N\}$. A single diagonal subsequence of the coordinatewise L^p convergence is almost everywhere simultaneous at every rational parameter. For each fixed N , apply (37) on the finite coordinate space indexed by D_N and then Fatou's lemma. This transfers (116) to the limit with a bound independent of N . Only after this finite-dimensional passage do we let $N \rightarrow \infty$. The pointwise functionals increase monotonically, and every strict rational jump witness is contained in one D_N ; monotone convergence therefore yields the bound on $\mathbb{Q}_+$. The finite-shell integrability used in (109) gives local absolute continuity in the truncation parameter almost everywhere, so Lemma 2.6 recovers all positive parameters. The variation estimate then follows pointwise from (21). The construction above supplies the L^p and almost-everywhere anchor at infinity. $\square$

5 Separated-packet estimates for the terminal phase

All packet families in this and the following packet sections are finite; we suppress this qualifier in individual statements. Rankings and randomizations are always performed at this finite level. In the endpoint arguments, cube, shell, and parameter cutoffs are imposed explicitly and removed only after the estimates have been established, by coordinatewise convergence on finite annuli and lower semicontinuity of strict jump counts. Thus no infinite randomization or infinite-rank argument is used.

For $n \geq 2$, we use the following two microlocal estimates of Seeger [29, Lemmas 2.1 and 2.2]. Let H_j be supported where $2^{j-2} \leq |x| \leq 2^{j+2}$, and assume that $r \mapsto H_j(r\theta)$ is C^N on $(0, \infty)$ for every θ , with the radial derivatives satisfying

$$M_N = \sup_{j, \theta} \sup_{r, 0 \leq \ell \leq N} r^{n+\ell} |\partial_r^\ell H_j(r\theta)| < \infty. \quad (124)$$

For an integer $s > 3$ and a fixed $0 < \kappa < 1$, Seeger's construction [29, equation (2.2)] is $H_j = \Gamma_j^s + (H_j - \Gamma_j^s)$. Put $L(Q) = m$ when $2^{m-1} < \ell(Q) \leq 2^m$. If the cubes have disjoint interiors, $F_m = \sum_{L(Q)=m} f_Q$, and $\text{supp } f_Q \subset Q$, $\|f_Q\|_1 \leq \Lambda|Q|$, then, for $0 < \kappa < 1$,

$$\left\| \sum_j \Gamma_j^s * F_{j-s} \right\|_2^2 \leq C M_0^2 2^{-s(1-\kappa)} \Lambda \sum_Q \|f_Q\|_1. \quad (125)$$

If additionally $\text{supp } b_Q \subset Q$, $\int b_Q = 0$, $\ell(Q) = 2^{j-s}$, $N \geq n+1$, and $0 \leq \varepsilon \leq 1$, then

$$\|(H_j - \Gamma_j^s) * b_Q\|_1 \leq C_N \left[M_0 2^{-s\varepsilon} + M_N 2^{s(n+(\varepsilon-\kappa)N)} \right] \|b_Q\|_1. \quad (126)$$

The constants may depend on the fixed value of κ and on Seeger's fixed auxiliary cutoffs, but not on j, s , the cubes, or the functions. These are precisely the support hypotheses in [29, Lemmas 2.1–2.2]: the first lemma requires no cancellation, whereas the second uses the displayed mean-zero condition. In dimension one, the corresponding one-shell estimates below are proved directly from cancellation, radial derivative bounds, and the same L^1 –BMO interpolation mechanism. This direct branch is what closes the one-dimensional weak endpoint; no one-dimensional weak pointwise-jump theorem is imported.

We also need a geometric observation.

Lemma 5.1 (Thin radial layers). *Fix $0 < c_- < c_+ < \infty$. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, let $R > 0$, let $t \in \mathbb{N}_0$, put $q = R2^{-t}$, and let $\Lambda \geq 0$. Suppose $d_U \in L^1$ are supported on pairwise disjoint q -cubes and satisfy $\|d_U\|_1 \leq \Lambda|U|$. If $E \subset [c_-, c_+R]$ is a union of M_E intervals of total length $|E|$, then*

$$\left\| K_\Theta \mathbf{1}_{\{|z| \in E\}} * \sum_U d_U \right\|_\infty \leq C_{n, c_-, c_+} \Lambda \|\Theta\|_\infty \left(\frac{|E|}{R} + M_E 2^{-t} \right), \quad (127)$$

$$\left\| K_\Theta \mathbf{1}_{\{|z| \in E\}} * \sum_U d_U \right\|_1 \leq C_{n, c_-, c_+} \|\Theta\|_1 \frac{|E|}{R} \sum_U \|d_U\|_1. \quad (128)$$

Here and in the proof the implicit constants depend only on n, c_-, c_+ .

Proof. For fixed x , every contributing cube lies in a Cq -neighborhood of the translated radial set. That neighborhood has volume at most $CR^{n-1}(|E| + M_E q)$. Disjointness and the mass-density bound prove (127). Equation (128) is Young's inequality and $\int_E dr/r \leq C|E|/R$. $\square$

Lemma 5.2 (Variational Rademacher–Menshov). *For a finite sequence $g_1, \dots, g_M$ in L^2 ,*

$$\left\| V^2 \left(\sum_{\ell \leq m} g_\ell : 0 \leq m \leq M \right) \right\|_2 \leq C \log(2+M) \sup_{\varepsilon_\ell \in \{-1, 0, 1\}} \left\| \sum_\ell \varepsilon_\ell g_\ell \right\|_2. \quad (129)$$

Proof. Pad the sequence with zeros to length 2^L and let $\mathcal{D}_r$ be the fixed partition of $\{1, \dots, 2^L\}$ into dyadic index intervals of length 2^r , $0 \leq r \leq L$. Put $G_I = \sum_{\ell \in I} g_\ell$. The canonical dyadic decomposition of an arbitrary index interval uses at most two members of each $\mathcal{D}_r$. If a collection of index intervals is pairwise disjoint, then all dyadic pieces appearing in their canonical decompositions are pairwise disjoint. Consequently, for every fixed variation partition, Minkowski in the outer finite ℓ^2 sum gives the pointwise estimate

$$\left(\sum_j \left| \sum_{\ell \in J_j} g_\ell \right|^2 \right)^{1/2} \leq \sqrt{2} \sum_{r=0}^L \left(\sum_{I \in \mathcal{D}_r} |G_I|^2 \right)^{1/2}.$$

The right side is independent of the variation partition, so taking the supremum proves the deterministic domination

$$V^2 \left(\sum_{\ell \leq m} g_\ell : 0 \leq m \leq 2^L \right) \leq \sqrt{2} \sum_{r=0}^L \left(\sum_{I \in \mathcal{D}_r} |G_I|^2 \right)^{1/2}. \quad (130)$$

For a fixed level, independent Rademacher signs ε_I and Fubini give

$$\left\| \left(\sum_{I \in \mathcal{D}_r} |G_I|^2 \right)^{1/2} \right\|_2^2 = \mathbb{E} \left\| \sum_{I \in \mathcal{D}_r} \varepsilon_I G_I \right\|_2^2.$$

Each randomized block sum is $\sum \varepsilon_\ell g_\ell$ with $\varepsilon_\ell \in \{-1, 1\}$ fixed on its dyadic block; hence its norm is bounded by the supremum in (129). Take L^2 norms in (130) and sum the $L+1 \leq C \log(2+M)$ levels. $\square$

We shall use the following elementary completion fact whenever a finite Fourier expansion is removed. Its anchored formulation avoids treating V^r as a norm modulo constants and also avoids any appeal to an uncountable-parameter Bochner space.

Lemma 5.3 (Anchored variation completion). *Let $I = [a, b]$, let $1 \leq p, r < \infty$, and let $H_N = (H_{N,t})_{t \in I}$ be paths of measurable functions satisfying $H_{N,b} = 0$. Assume that the variation functions of the paths and their differences are measurable; this holds, in particular, when their variation is detected by one fixed countable dense subset of I , as in all applications below. Suppose*

$$\|V^r(H_N - H_M : t \in I)\|_p \rightarrow 0 \quad (N, M \rightarrow \infty). \quad (131)$$

Then there is an anchored path $H = (H_t)_{t \in I}$ such that

$$\|V^r(H_N - H : t \in I)\|_p \rightarrow 0. \quad (132)$$

After passage to a subsequence, $H_{N,t}(x) \rightarrow H_t(x)$ uniformly in $t \in I$ for almost every x . Moreover,

$$\|V^r(H : t \in I)\|_p \leq \liminf_{N \rightarrow \infty} \|V^r(H_N : t \in I)\|_p. \quad (133)$$

Consequently, if $P_N = G_N + E_N$, both (G_N) and (E_N) satisfy the corresponding anchored Cauchy hypotheses (possibly with different p, r), and $P_{N,t} \rightarrow P_t$ for every t almost everywhere, then the limits satisfy $P_t = G_t + E_t$ for every t on one common full-measure set whenever the convergence of P_N is uniform in t there.

Proof. Choose a subsequence N_k such that

$$\sum_k \|V^r(H_{N_{k+1}} - H_{N_k} : t \in I)\|_p < \infty.$$

Minkowski's inequality shows that, outside one null set,

$$\sum_k V^r(H_{N_{k+1}} - H_{N_k} : t \in I) < \infty.$$

Since every difference path vanishes at b ,

$$\sup_{t \in I} |H_{N_{k+1},t} - H_{N_k,t}| \leq V^r(H_{N_{k+1}} - H_{N_k} : t \in I).$$

Thus the subsequence converges uniformly in t almost everywhere to an anchored path H . We record explicitly the measurability needed below. On the same full-measure set put

$$S_k = \sum_{\ell \geq k} V^r(H_{N_{\ell+1}} - H_{N_\ell} : t \in I).$$

The functions S_k are measurable, finite almost everywhere, and decrease to zero. Applying the variation triangle inequality first to each finite tail and then passing to its uniform limit gives

$$V^r(H_{N_k} - H : t \in I) \leq S_k.$$

Consequently, for every fixed N ,

$$|V^r(H_N - H : t \in I) - V^r(H_N - H_{N_k} : t \in I)| \leq S_k \longrightarrow 0.$$

Hence $V^r(H_N - H : t \in I)$ is the pointwise limit of measurable functions and is measurable. For every finite parameter partition, its increment sum passes to the pointwise limit. Taking the supremum over finite partitions and applying Fatou now gives

$$\|V^r(H_N - H : t \in I)\|_p \leq \liminf_{k \rightarrow \infty} \|V^r(H_N - H_{N_k} : t \in I)\|_p.$$

Indeed, for every $\delta > 0$ the Cauchy hypothesis supplies N_0 such that

$$\|V^r(H_N - H_M : t \in I)\|_p < \delta \quad \text{whenever } N, M \geq N_0.$$

For $N \geq N_0$, the preceding inequality with $M = N_k$ and then $k \rightarrow \infty$ yields $\|V^r(H_N - H : t \in I)\|_p \leq \delta$. This proves (132) for the full sequence. Pointwise, the variation triangle inequality gives

$$|V^r(H_N : t \in I) - V^r(H : t \in I)| \leq V^r(H_N - H : t \in I).$$

The reverse triangle inequality in L^p and (132) therefore give convergence of the two variation norms, which is stronger than (133). For the last assertion, choose one diagonal subsequence along which the required summability holds for both G_N and E_N ; passing the finite identities to their common uniform pointwise limits gives the claimed identity. $\square$

Because Seeger's microlocal kernels Γ_j^s have noncompact spatial support, localization and ranking are carried out with the original compactly supported kernels H_j .

Lemma 5.4 (Fixed-sign full-kernel estimate). *Let $n \geq 1$, $N = 4n + 1$, and let H_j be supported where $2^{j-2} \leq |x| \leq 2^{j+2}$, radially C^N as above, and satisfy (124). Let $\mathcal{Q}$ be a finite family of cubes with disjoint interiors, and suppose*

$$\text{supp } b_Q \subset Q, \quad \int b_Q = 0, \quad \|b_Q\|_1 \leq \Lambda |Q|.$$

For $s \in \mathbb{N}$, $s \geq 5$, pair H_j with the cubes of side 2^{j-s} . Then, for every fixed choice $|c_Q| \leq 1$,

$$\left\| \sum_j H_j * \sum_{\ell(Q)=2^{j-s}} c_Q b_Q \right\|_2^2 \leq C_n M_N^2 2^{-s/24} \Lambda \sum_Q \|b_Q\|_1. \quad (134)$$

Proof. By homogeneity assume $M_N = 1$, and put $\Sigma = \sum_Q \|b_Q\|_1$. If $\Sigma = 0$ there is nothing to prove. Write

$$F = \sum_j H_j * \sum_{\ell(Q)=2^{j-s}} c_Q b_Q.$$

We first establish an L^3 bound valid uniformly at every scale and in every dimension. Choose a fixed smooth radial cutoff equal to one on $\{2^{-2} \leq |x| \leq 2^2\}$ and dilate a sufficiently large multiple of 2^{-jn} times that cutoff. The resulting nonnegative radial majorants v_j satisfy

$$|H_j| \leq v_j, \quad \|v_j\|_1 \leq C_n, \quad \|\nabla v_j\|_\infty \leq C_n 2^{-j(n+1)}.$$

Set

$$\mathcal{P} = \sum_j v_j * \sum_{\ell(Q)=2^{j-s}} |b_Q|.$$

Then $|F| \leq \mathcal{P}$ and $\|\mathcal{P}\|_1 \leq C_n \Sigma$. We claim that

$$\|\mathcal{P}\|_{\text{BMO}} \leq C_n \Lambda. \quad (135)$$

Fix a cube R_0 with center x_0 . For $2^j < \ell(R_0)$, compact support of v_j shows that every Q contributing on R_0 lies in a fixed enlargement of R_0 . Since the cubes in $\mathcal{Q}$ are disjoint across all scales and $\|b_Q\|_1 \leq \Lambda|Q|$, integration over R_0 bounds the total contribution of these scales by $C_n \Lambda |R_0|$.

For $2^j \geq \ell(R_0)$ subtract the value at x_0 . Every cube contributing to either convolution at $x \in R_0$ or at x_0 lies in a ball of radius $C_n 2^j$ about x_0 . Disjointness and the density bound therefore give total relevant mass at most $C_n \Lambda 2^{jn}$. The mean-value theorem and the gradient estimate yield

$$\sup_{x \in R_0} \left| v_j * \sum_{\ell(Q)=2^{j-s}} |b_Q|(x) - v_j * \sum_{\ell(Q)=2^{j-s}} |b_Q|(x_0) \right| \leq C_n \Lambda \ell(R_0) 2^{-j}.$$

Summing in j proves (135). We next justify the endpoint L^1 –BMO estimate used below directly. The Fefferman–Stein interpolation identity [17, Section III.5, equation (5.1)] explicitly includes the endpoint choice $p = \infty$ and yields the same estimate; the following distributional argument makes that endpoint instance and its constants transparent. Let $h \in L^1(\mathbb{R}^n) \cap \text{BMO}(\mathbb{R}^n)$, and put $A = \|h\|_1$ and $B = \|h\|_{\text{BMO}}$. If $B = 0$, then h is almost everywhere constant, and its integrability forces $h = 0$. Suppose $B > 0$ and apply the dyadic Calderón–Zygmund decomposition to $|h|$ at height B . Its maximal cubes Q satisfy

$$\sum_Q |Q| \leq A/B, \quad B < \frac{1}{|Q|} \int_Q |h| \leq 2^n B,$$

while $|h| \leq B$ almost everywhere off their union. In particular writing $h_Q := |Q|^{-1} \int_Q h$, we have $|h_Q| \leq 2^n B$. The John–Nirenberg inequality in [17, Section II.1], applied on each Q , therefore gives constants $c_n, C_n > 0$ such that, whenever $\lambda \geq 2^{n+1} B$,

$$|\{x : |h(x)| > \lambda\}| \leq C_n \frac{A}{B} \exp(-c_n \lambda/B).$$

For $0 < \lambda < 2^{n+1} B$, Chebyshev’s inequality gives the complementary bound $|\{|h| > \lambda\}| \leq A/\lambda$. The layer-cake formula, split at $2^{n+1} B$, now yields

$$\int_{\mathbb{R}^n} |h|^3 = 3 \int_0^\infty \lambda^2 |\{|h| > \lambda\}| d\lambda \leq C_n A B^2.$$

Thus we have proved the form used below:

$$\|h\|_3^3 \leq C_n \|h\|_1 \|h\|_{\text{BMO}}^2, \quad h \in L^1 \cap \text{BMO}. \quad (136)$$

Consequently,

$$\|F\|_3^3 \leq \|\mathcal{P}\|_3^3 \leq C_n \Lambda^2 \Sigma. \quad (137)$$

Suppose first that $n = 1$. Let x_Q be the center of Q . Cancellation and the $W^{1,1}$ translation estimate give

$$\begin{aligned} \|H_j * (c_Q b_Q)\|_1 &= \left\| \int_Q c_Q b_Q(y) [H_j(\cdot - y) - H_j(\cdot - x_Q)] dy \right\|_1 \\ &\leq C \ell(Q) \|H'_j\|_1 \|b_Q\|_1. \end{aligned}$$

Since H_j is supported where $|x| \simeq 2^j$, the radial derivative bound in (124) gives $\|H'_j\|_1 \leq C 2^{-j}$. For $\ell(Q) = 2^{j-s}$ it follows that

$$\|F\|_1 \leq C 2^{-s} \Sigma. \quad (138)$$

Interpolation between (138) and (137) now gives

$$\|F\|_2^2 \leq \|F\|_1^{1/2} \|F\|_3^{3/2} \leq C 2^{-s/2} \Lambda \Sigma \leq C 2^{-s/24} \Lambda \Sigma.$$

This proves (134) in dimension one.

Assume henceforth that $n \geq 2$. Apply Seeger's decomposition with $\kappa = 1/2$ and write $F = A + R$. Equation (125) gives

$$\|A\|_2^2 \leq C_n 2^{-s/2} \Lambda \Sigma. \quad (139)$$

Summing (126) with $\varepsilon = 1/4$ gives

$$\|R\|_1 \leq C_n 2^{-s/4} \Sigma, \quad (140)$$

because $n - (4n + 1)/4 = -1/4$. The coefficients c_Q preserve cancellation and the mass-density estimate.

Put $u = \Lambda 2^{-s/8}$ and $E = \{|R| > u\}$. By (140),

$$|E| \leq C_n \Lambda^{-1} 2^{-s/8} \Sigma.$$

On E^c , equations (139) and (140) give

$$\int_{E^c} |F|^2 \leq 2 \|A\|_2^2 + 2u \|R\|_1 \leq C_n 2^{-3s/8} \Lambda \Sigma.$$

On E , Hölder's inequality and (137) give

$$\int_E |F|^2 \leq \|F\|_3^2 |E|^{1/3} \leq C_n 2^{-s/24} \Lambda \Sigma.$$

This proves (134) for $M_N = 1$ in every dimension. Rescaling the kernels restores the factor M_N^2 . $\square$

Proposition 5.5 (Selector-free terminal packet trajectory). *There are $t_0 \in \mathbb{N}$ and $\delta_T > 0$, depending only on n , such that the following holds. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, let $R > 0$, $t \in \mathbb{N}$, $q = R 2^{-t}$, $t \geq t_0$, and let $\{U\}$ be pairwise disjoint q -cubes. Let $\Lambda \geq 0$ and suppose*

$$\text{supp } d_U \subset U, \quad \int d_U = 0, \quad \|d_U\|_1 \leq \Lambda |U|, \quad M = \sum_U \|d_U\|_1.$$

For fixed complex coefficients $|a_U| \leq 1$, define

$$F_\varepsilon(x) = \int_{\varepsilon < |x-y| \leq 2R} K_\Theta(x-y) \sum_U a_U d_U(y) dy, \quad R \leq \varepsilon \leq 2R.$$

Then

$$\|V^2(F_\varepsilon : R \leq \varepsilon \leq 2R)\|_2^2 \leq C \|\Theta\|_\infty^2 2^{-\delta_T t} \Lambda M. \quad (141)$$

The same estimate holds after multiplication of the entire output path by $e^{i\xi \cdot x}$, for any fixed $\xi \in \mathbb{R}^n$. It also holds after replacing each $d_U(y)$ by $e^{i\eta_U \cdot y} d_U(y)$, with arbitrary $\eta_U \in \mathbb{R}^n$, provided $\int e^{i\eta_U \cdot y} d_U(y) dy = 0$ for every U . In the phase-adapted application below, the input modulation is frozen packet by packet and the resulting error is estimated separately.

Proof. We first justify the reduction to $R = 1$, including the normalization of the density and mass. Put $x = RX$, $y = RY$, $\tilde{U} = R^{-1}U$, and $\tilde{d}_U(Y) = d_U(RY)$. Homogeneity gives $F_\varepsilon(RX) = \tilde{F}_{\varepsilon/R}(X)$, while

$$\|\tilde{d}_U\|_1 = R^{-n}\|d_U\|_1, \quad \|\tilde{d}_U\|_1 \leq \Lambda|\tilde{U}|, \quad \tilde{M} := \sum_U \|\tilde{d}_U\|_1 = R^{-n}M.$$

Thus the unit-scale squared L_X^2 estimate has right side $C\|\Theta\|_\infty^2 2^{-\delta r^t} \Lambda R^{-n}M$. Returning to $x = RX$ multiplies the squared output norm by R^n , exactly cancelling the input mass factor R^{-n} . It is therefore enough to prove the estimate at $R = 1$. Throughout the unit-scale proof write $K_{|\Theta|}(z) = |\Theta(z/|z|)|/|z|^n$. First consider $n = 1$. For a radially smoothed one-shell kernel, we use the following fixed convention. Choose a nonnegative function $\phi \in C_c^\infty((-1/10, 1/10))$ with $\int_{\mathbb{R}} \phi = 1$, put $\phi_h(r) = h^{-1}\phi(r/h)$, and extend every radial step profile below by zero to $\mathbb{R}$ before setting $a_h = a * \phi_h$. Then

$$\|a_h\|_\infty \leq \|a\|_\infty, \quad \|\partial_r^\ell a_h\|_\infty \leq C_\ell h^{-\ell} \|a\|_\infty.$$

If a is constant on M_t intervals, then $a - a_h$ is supported in the $O(h)$ -neighborhood of its jump set, hence in $O(M_t)$ intervals of total length $O(M_t h)$. Every mollification in this proof is understood in this sense.

For a radially smoothed one-shell kernel H at relative scale h , cancellation gives

$$H * d_U(x) = \int_U [H(x - y) - H(x - c_U)] d_U(y) dy.$$

Since $\|\tau_z H - H\|_1 \leq |z| \|H'\|_1$ and $\|H'\|_1 \leq C\|\Theta\|_\infty h^{-1}$, summing over U yields

$$\left\| H * \sum_U a_U d_U \right\|_1 \leq C\|\Theta\|_\infty 2^{-t} h^{-1} M.$$

Lemma 5.1 gives the simultaneous L^∞ bound $C\Lambda\|\Theta\|_\infty$; hence the squared L^2 norm is at most $C\|\Theta\|_\infty^2 \Lambda M 2^{-t} h^{-1}$. Take $\eta = 1/8$ and $\gamma = 1/16$. Partition $[1, 2]$ into $M_t \simeq 2^\eta$ fixed cells I_ℓ of length $\delta = 2^{-\eta}$, and take $h = 2^{-\eta t}$. Put

$$G_\ell = K_\Theta \mathbf{1}_{\{|z| \in I_\ell\}} * \sum_U a_U d_U.$$

Inserting all cell endpoints into an arbitrary variation partition gives the pointwise deterministic inequality

$$V^2(F_\varepsilon : 1 \leq \varepsilon \leq 2) \leq C V^2 \left(\sum_{\ell \leq m} G_\ell : 0 \leq m \leq M_t \right) + C \left(\sum_\ell V^2(F_\varepsilon : \varepsilon \in I_\ell)^2 \right)^{1/2}. \quad (142)$$

Reversing the order of the partial sums, if required by the convention $\varepsilon < |z|$, does not change their variation. Thus every x -dependent choice of variation intervals is confined to the second term; the first term is one fixed finite sequence. Within a cell $V^2 \leq V^1$. If W_ℓ is the positive convolution over that cell, Lemma 5.1 gives

$$\sup_\ell \|W_\ell\|_\infty \leq C\Lambda\|\Theta\|_\infty(\delta + 2^{-t}), \quad \sum_\ell \|W_\ell\|_1 \leq C\|\Theta\|_\infty M.$$

Consequently,

$$\int_{\mathbb{R}} \sum_\ell V^2(F_\varepsilon : \varepsilon \in I_\ell)^2 dx \leq C\Lambda\|\Theta\|_\infty^2 M 2^{-\eta}.$$

For a fixed choice of signs on the cells, write the resulting radial step function as

$$a(r) = \sum_{\ell=1}^{M_t} \varepsilon_\ell \mathbf{1}_{I_\ell}(r), \quad \varepsilon_\ell \in \{-1, 0, 1\},$$

with a extended by zero outside $[1, 2]$, and smooth it at scale h . Let $\mathcal{D}_a$ consist of the endpoints 1, 2 together with all interior jump points of a , and set

$$E_{\text{bdry}} = \{r \in \mathbb{R} : \text{dist}(r, \mathcal{D}_a) \leq Ch\},$$

where C is fixed large enough in terms of $\text{supp } \phi$. Then

$$\#\text{components}(E_{\text{bdry}}) \leq CM_t, \quad |E_{\text{bdry}}| \leq CM_th, \quad \text{supp}(a - a_h) \subset E_{\text{bdry}}, \quad |a - a_h| \leq 2\mathbf{1}_{E_{\text{bdry}}}. \quad (143)$$

After increasing t_0 if necessary, $E_{\text{bdry}} \subset [1/2, 5/2]$, so Lemma 5.1 applies with fixed radial constants. Set $H(z) = K_\Theta(z)a_h(|z|)$. The cancellation estimate above and the L^∞ part of Lemma 5.1 give

$$\|H * \sum_U a_U d_U\|_2^2 \leq C\Lambda \|\Theta\|_\infty^2 M 2^{-(1-\eta)t}.$$

Let G_{bdry} be the output obtained by replacing the radial profile by $a - a_h$. The last inequality in (143) gives the pointwise positive-kernel domination

$$|G_{\text{bdry}}(x)| \leq 2 \left(K_{|\Theta|} \mathbf{1}_{\{|z| \in E_{\text{bdry}}\}} * \sum_U |d_U| \right) (x). \quad (144)$$

The functions $|d_U|$ have the same supports and L^1 bounds as d_U , so both estimates of Lemma 5.1 apply to the right side of (144). Together with $\|G\|_2^2 \leq \|G\|_\infty \|G\|_1$, they give

$$\|G_{\text{bdry}}\|_2^2 \leq C\Lambda \|\Theta\|_\infty^2 M (M_th + M_t 2^{-t}) M_th.$$

With the stated values of η and γ , the last factor is bounded by

$$C(2^{-2(\eta-\gamma)t} + 2^{-(1+\eta-2\gamma)t}) = C(2^{-t/8} + 2^{-t}).$$

The smoothed signed sum decays with exponent $1 - \eta = 7/8$, the boundary term with exponent at least $1/8$, and the within-cell term with exponent $\gamma = 1/16$. The signs ε_ℓ above are fixed before evaluation in x and are independent of the already fixed packet coefficients a_U . Hence Lemma 5.2 applies to the first term of (142). Since

$$\log^2(2 + M_t) \lesssim (1 + \gamma t)^2 \lesssim t^2, \quad t^2 2^{-ct} \leq C_{c,c'} 2^{-c't} \quad (0 < c' < c), \quad (145)$$

its squared cost is absorbed by reducing the positive decay exponent. In particular, $t^2 2^{-t/8} \lesssim 2^{-t/16}$; together with the within-cell estimate this proves (141) in dimension one with the explicit safe choice $\delta_T = 1/32$.

Now assume $n \geq 2$ and choose $N = 4n + 1$, $\eta = (16N)^{-1}$ and $\gamma = \eta/2$. Partition $[1, 2]$ into $M_t \simeq 2^\eta$ fixed cells I_ℓ . Inserting the cell endpoints into a variation partition gives

$$V^2(F_\varepsilon : 1 \leq \varepsilon \leq 2) \leq CV^2 \left(\sum_{\ell \leq m} G_\ell : 0 \leq m \leq M_t \right) + C \left(\sum_\ell V^2(F_\varepsilon : \varepsilon \in I_\ell)^2 \right)^{1/2}, \quad (146)$$

where $G_\ell = K_\Theta \mathbf{1}_{\{|z| \in I_\ell\}} * \sum_U a_U d_U$.

Inside one cell, $V^2 \leq V^1$ and Lemma 5.1 gives

$$\int \sum_\ell V^2(F : I_\ell)^2 \leq C \|\Theta\|_\infty^2 \Lambda M 2^{-\eta}.$$

Indeed, the supremum of one absolute cell convolution is $C\Lambda \|\Theta\|_\infty (2^{-\eta} + 2^{-t})$, while the sum of its L^1 norms is $C \|\Theta\|_1 M$.

It remains to estimate the grid partial sums. Fix signs $\varepsilon_\ell \in \{-1, 0, 1\}$ and let $a(r) = \sum \varepsilon_\ell \mathbf{1}_{I_\ell}(r)$. Mollify a at radial scale $h = 2^{-\eta t}$, using the zero-extension convention above, and set

$$a_h = a * \phi_h, \quad H(z) = K_\Theta(z)a_h(|z|).$$

The kernel H satisfies $M_N(H) \leq C \|\Theta\|_\infty h^{-N}$. Moreover, $|a| \leq 1$ and $\text{supp } a \subset [1, 2]$ imply

$$|a_h(r)| \leq \mathbf{1}_{[1-Ch, 2+Ch]}(r).$$

Thus the possibly signed smooth profile is pointwise dominated by one fixed enlarged annulus. In particular, Lemma 5.1, applied to that annulus, gives the uniform bound used below. Apply (125)–(126) with $\kappa = 1/2$, $\varepsilon = 1/4$, embedding the one-shell kernel into Seeger’s family by taking $H_0 = H$ and $H_j = 0$ for $j \neq 0$. The microlocal part A obeys

$$\|A\|_2^2 \leq C\|\Theta\|_\infty^2 2^{-t/2} \Lambda M,$$

while the remainder obeys

$$\|R\|_1 \leq C\|\Theta\|_\infty 2^{-at} M, \quad a := \frac{1}{4} - \eta N = \frac{3}{16}.$$

The full smoothed convolution $S = A + R$ therefore has $\|S\|_\infty \leq C\Lambda\|\Theta\|_\infty$. $\Lambda\|\Theta\|_\infty = 0$ only if either $K_\Theta = 0$ almost everywhere or $\Lambda = 0$, in which case $\|d_U\|_1 \leq \Lambda|U|$ forces every $d_U = 0$ almost everywhere. Thus all the trajectories in the proposition vanish in this case. In what follows assume $\Lambda\|\Theta\|_\infty > 0$, put

$$b := \frac{1}{8} - \frac{\eta N}{2} = \frac{3}{32}, \quad u = \Lambda\|\Theta\|_\infty 2^{-bt} > 0,$$

and let $E = \{|R| > u\}$. On E^c , use $|S|^2 \leq 2|A|^2 + 2u|R|$; on E , use $\int_E |S|^2 \leq \|S\|_\infty^2 |E|$ and $|E| \leq u^{-1} \|R\|_1$. This yields

$$\|S\|_2^2 \leq C\|\Theta\|_\infty^2 2^{-3t/32} \Lambda M. \quad (147)$$

Indeed, the three contributions have respective decay exponents $1/2$, $a + b = 9/32$, and $a - b = 3/32$. This records both square-root losses and fixes the exponent in place of an unspecified c_1 .

Let $\mathcal{D}_a$ consist of 1, 2 and all interior jump points of a , and put

$$E_{\text{bdry}} = \{r \in \mathbb{R} : \text{dist}(r, \mathcal{D}_a) \leq Ch\}.$$

After increasing the fixed C according to $\text{supp } \phi$, one has

$$\#\text{components}(E_{\text{bdry}}) \leq CM_t, \quad |E_{\text{bdry}}| \leq CM_t h, \quad \text{supp}(a - a_h) \subset E_{\text{bdry}}, \quad |a - a_h| \leq 2\mathbf{1}_{E_{\text{bdry}}}. \quad (148)$$

After increasing t_0 if necessary, $E_{\text{bdry}} \subset [1/2, 5/2]$, so the radial constants in Lemma 5.1 remain fixed. Let G_{bdry} denote the output generated by $a - a_h$. Positivity and (148) give, pointwise,

$$|G_{\text{bdry}}(x)| \leq 2 \left(K_{|\Theta|} \mathbf{1}_{\{|z| \in E_{\text{bdry}}\}} * \sum_U |d_U| \right) (x). \quad (149)$$

Both bounds in Lemma 5.1, applied to the right side of (149), and $\|G_{\text{bdry}}\|_2^2 \leq \|G_{\text{bdry}}\|_\infty \|G_{\text{bdry}}\|_1$ give

$$\begin{aligned} \|G_{\text{bdry}}\|_2^2 &\leq C\|\Theta\|_\infty^2 \Lambda M (M_t h + M_t 2^{-t}) M_t h \\ &\leq C\|\Theta\|_\infty^2 \Lambda M \left(2^{-2(\eta-\gamma)t} + 2^{-(1+\eta-2\gamma)t} \right) \\ &= C\|\Theta\|_\infty^2 \Lambda M (2^{-\eta t} + 2^{-t}), \end{aligned}$$

where the last identity uses $\gamma = \eta/2$. Therefore the right side of (147) holds for the unsmoothed signed sum with exponent $\min(3/32, \eta) = \eta$, uniformly in the signs. Lemma 5.2 contributes a factor $O(t^2)$ after squaring, by (145); absorb it by reducing the exponent. Here too the cell signs are fixed globally before evaluation in x and are independent of the fixed packet coefficients a_U . Together with the within-cell exponent $\gamma = \eta/2$ and (146), this proves (141). A single explicit choice valid in this branch is

$$\delta_T = \frac{\eta}{4} = \frac{1}{64(4n+1)} \quad (n \geq 2),$$

since $t^2 2^{-\eta t} \lesssim 2^{-\eta t/2}$ and $\delta_T < \gamma$. Together with the choice $\delta_T = 1/32$ above, this also makes the dependence of δ_T only on n manifest. $\square$

6 An anchored phase–packet induction

After removal of the parent-dependent input modulation, the phase-adapted decomposition recovers a mean-zero packet structure. We formulate the corresponding estimate in this anchored normalization.

Fix a real polynomial $P \in \mathcal{P}_d$ and $\Theta \in L^\infty(\mathbb{S}^{n-1})$. Let $t \in \mathbb{N}$, and let $R > 0$ and $q = R2^{-t}$ be dyadic. Let $\mathcal{Q}$ be a family of pairwise disjoint dyadic cubes, each contained in a dyadic R -cube, and for $Q \in \mathcal{Q}$ let c_Q denote its center. Suppose d_U are mean-zero descendants, supported on pairwise disjoint q -cubes $U \subset Q$, with

$$\|d_U\|_\infty \leq \Lambda, \quad M = \sum_U \|d_U\|_1. \quad (150)$$

Define the phase-adapted shell trajectory

$$\mathcal{P}_\varepsilon(x) = \sum_{Q \in \mathcal{Q}} \sum_{U \subset Q} \int_{\varepsilon < |x-y| \leq 2R} e^{i[P(x,y) - P(c_Q,y)]} K_\Theta(x-y) d_U(y) dy, \quad R \leq \varepsilon \leq 2R. \quad (151)$$

We fix the Fourier localization once, rather than separately for each block. After the normalization by an interacting R -block used below, Y and every parent center ξ_Q lie in one fixed cube, while X lies in one of only $O_n(1)$ unit cubes whose difference from that cube meets $\{1 < |Z| \leq 2\}$. Choose structural boxes $\mathfrak{X}, \mathfrak{Y}$ containing all these cubes with a fixed margin, and smooth cutoffs equal to one on the boxes. The anchored flag contains only finitely many polynomials (depending on n, d); throughout the proof we use these same boxes and cutoffs and take the maximum of the resulting Fourier constants over that finite list.

Lemma 6.1 (Fourier expansion of a low pivot). *Let H be one of the fixed anchored flag polynomials, and let X, Y, ξ range in the structural boxes just fixed. Let $\chi, \tilde{\chi}$ be the corresponding fixed smooth cutoffs. Then, for every $L \in \mathbb{R}$,*

$$\chi(X) \tilde{\chi}(Y) e^{iL[H(X,Y) - H(\xi,Y)]} = \sum_{u,v \in \mathbb{Z}^n} a_{u,v}(L, \xi) e^{iu \cdot X} e^{iv \cdot Y}.$$

There exist numbers $A_{u,v}(L)$ independent of ξ such that $|a_{u,v}(L, \xi)| \leq A_{u,v}(L)$ and

$$\sum_{u,v} A_{u,v}(L) (1 + |v|) \leq C(1 + |L|)^{C_F}. \quad (152)$$

The constants are uniform in L and ξ apart from the displayed power and, with the preceding convention, depend only on n, d .

Proof. Extend the cutoff function periodically on a fixed torus. Repeated integration by parts in X and Y gives, uniformly in ξ , $|a_{u,v}| \leq C_N(1 + |L|)^N(1 + |u| + |v|)^{-N}$. Take the right side as $A_{u,v}(L)$, choose $N > 2n + 2$, and sum (increasing C_F if needed). $\square$

Proposition 6.2 (Anchored phase–packet principle). *There are $c_P, c_E > 0$ and t_0 , depending only on n, d , such that every trajectory (151) under the finite-packet setup above, with $P \in \mathcal{P}_d$, $\Theta \in L^\infty(\mathbb{S}^{n-1})$, and $t \geq t_0$, admits a decomposition $\mathcal{P} = G + E$ satisfying*

$$\|V^2(G_\varepsilon : R \leq \varepsilon \leq 2R)\|_2 \leq C\|\Theta\|_\infty 2^{-c_P t} (\Lambda M)^{1/2}, \quad (153)$$

$$\|V^1(E_\varepsilon : R \leq \varepsilon \leq 2R)\|_1 \leq C\|\Theta\|_1 2^{-c_E t} M. \quad (154)$$

The constants are independent of R, q , every coefficient of P , the locations of the cubes, and the parent labels Q .

Proposition 6.2 is included as a self-contained bounded-density model; it is not used in the weak-endpoint induction. Its purpose is to isolate the phase–packet mechanism from the additional concentration difficulty. The actual Calderón–Zygmund bad functions satisfy only cancellation and an average mass bound. They are handled instead by the concentration-stable estimates of Section 7.

Proof. Let $\mathcal{D}_R$ be the dyadic partition into cubes of side R , and let $\mathcal{J}_R \subset \mathcal{D}_R$ be the finite family of cubes containing at least one parent $Q \in \mathcal{Q}$. Define the finite set of interacting block pairs by

$$\mathcal{A}_R = \{(I, J) \in \mathcal{D}_R \times \mathcal{J}_R : \{x - y : x \in I, y \in J\} \cap \{z : R < |z| \leq 2R\} \neq \emptyset\}. \quad (155)$$

For $(I, J) \in \mathcal{A}_R$, set

$$\mathcal{P}_\varepsilon^{I,J}(x) = \mathbf{1}_I(x) \sum_{\substack{Q \in \mathcal{Q}, Q \subset J \\ U \subset Q}} \int_{\varepsilon < |x-y| \leq 2R} e^{i[P(x,y) - P(c_Q, y)]} K_\Theta(x-y) d_U(y) dy. \quad (156)$$

The annular support gives, almost everywhere, the exact block decomposition $\mathcal{P}_\varepsilon = \sum_{(I,J) \in \mathcal{A}_R} \mathcal{P}_\varepsilon^{I,J}$. All block sums in this proof are henceforth restricted to $\mathcal{A}_R$.

For a fixed $(I, J) \in \mathcal{A}_R$, let a be the center of J and scale $x = a + RX$, $y = a + RY$. Put

$$M_{I,J} = \sum_{\substack{Q \in \mathcal{Q}, Q \subset J \\ U \subset Q}} \|d_U\|_1.$$

Because I and J have side R and their difference meets a fixed annulus of radii R and $2R$, elementary lattice geometry gives the two-sided adjacency estimate

$$\sup_{I \in \mathcal{D}_R} \#\{J \in \mathcal{J}_R : (I, J) \in \mathcal{A}_R\} + \sup_{J \in \mathcal{J}_R} \#\{I \in \mathcal{D}_R : (I, J) \in \mathcal{A}_R\} \leq C_n. \quad (157)$$

In particular, each packet mass is counted for only $O_n(1)$ output blocks, and hence

$$\sum_{(I,J) \in \mathcal{A}_R} M_{I,J} \leq C_n M. \quad (158)$$

More explicitly, if every path $H_{I,J}$ is supported in I , the variation triangle inequality, Cauchy–Schwarz in the at most C_n adjacent cubes J , and disjointness of the output cubes give

$$\begin{aligned} \left\| V^2 \left(\sum_{(I,J) \in \mathcal{A}_R} H_{I,J} \right) \right\|_2^2 &= \sum_I \left\| V^2 \left(\sum_{J: (I,J) \in \mathcal{A}_R} H_{I,J} \right) \right\|_{L^2(I)}^2 \\ &\leq C_n \sum_{(I,J) \in \mathcal{A}_R} \|V^2(H_{I,J})\|_2^2. \end{aligned} \quad (159)$$

Similarly,

$$\left\| V^1 \left(\sum_{(I,J) \in \mathcal{A}_R} H_{I,J} \right) \right\|_1 \leq \sum_{(I,J) \in \mathcal{A}_R} \|V^1(H_{I,J})\|_1. \quad (160)$$

For $Q \subset J$ and $U \subset Q$, set

$$\tilde{U} = R^{-1}(U - a), \quad \tilde{d}_U(Y) = d_U(a + RY), \quad \tilde{I} = R^{-1}(I - a).$$

Then $\tilde{U}$ has side 2^{-t} and

$$\|\tilde{d}_U\|_\infty = \|d_U\|_\infty \leq \Lambda, \quad \|\tilde{d}_U\|_1 = R^{-n} \|d_U\|_1, \quad \|\tilde{d}_U\|_1 \leq \Lambda |\tilde{U}|, \quad \sum_{\substack{Q \in \mathcal{Q}, Q \subset J \\ U \subset Q}} \|\tilde{d}_U\|_1 = R^{-n} M_{I,J}. \quad (161)$$

Homogeneity, $K_\Theta(RZ)R^n = K_\Theta(Z)$, cancels the Jacobian, so the scaled block trajectory has the unit-annulus kernel. Since the output cutoff is independent of ε , for every $r \geq 1$ one has $V^r(\mathbf{1}_I H_\varepsilon) = \mathbf{1}_I V^r(H_\varepsilon)$ pointwise. Thus restricting a unit-scale output to $\tilde{I}$ cannot increase its variation norm. If $\tilde{G}$ and $\tilde{E}$ are paths in the scaled variables, then

$$\|\mathbf{1}_I V^2(G)\|_{L_X^2} = R^{n/2} \|\mathbf{1}_{\tilde{I}} V^2(\tilde{G})\|_{L_X^2}, \quad \|\mathbf{1}_I V^1(E)\|_{L_X^1} = R^n \|\mathbf{1}_{\tilde{I}} V^1(\tilde{E})\|_{L_X^1}. \quad (162)$$

Thus unit-scale factors $(\Lambda R^{-n} M_{I,J})^{1/2}$ and $R^{-n} M_{I,J}$ become $(\Lambda M_{I,J})^{1/2}$ and $M_{I,J}$ in physical variables. Put

$$\Phi(X, Y) = P(a + RX, a + RY) - P(a, a + RY).$$

Then $\Phi(0, Y) = 0$. If $Q \subset J$ and $\xi_Q = (c_Q - a)/R$, the phase in (151) is exactly

$$\Phi(X, Y) - \Phi(\xi_Q, Y). \quad (163)$$

Write the anchored flag decomposition, retaining its pure output part,

$$\Phi(X, Y) = p(X) + \sum_{\mu=1}^{N_d} C_\mu \hat{H}_\mu(X, Y). \quad (164)$$

All C_μ are dimensionless; they may depend arbitrarily on a and R . The list and its length depend only on n, d . The exact phase difference is

$$\Phi(X, Y) - \Phi(\xi_Q, Y) = p(X) - p(\xi_Q) + \sum_{\mu=1}^{N_d} C_\mu [\hat{H}_\mu(X, Y) - \hat{H}_\mu(\xi_Q, Y)]. \quad (165)$$

Hence $e^{ip(X)}$ is one common output modulation, whereas $s_Q := e^{-ip(\xi_Q)}$ is a parent-dependent scalar of modulus one. We work with the demodulated block, retain s_Q in every packet coefficient below, and at the end multiply both constructed component paths by $e^{ip(X)}$. This last operation preserves pointwise V^2 , pointwise V^1 , and the identity $\mathcal{P} = G + E$.

If $N_d = 0$, the common output modulation is harmless and the scalars s_Q are admissible fixed packet coefficients in Proposition 5.5; that proposition gives the conclusion directly. We therefore assume $N_d \geq 1$ below.

Let $\beta_* = \min_\mu \beta_\mu > 0$, where the β_μ are from Corollary 3.8. Let C_F be from Lemma 6.1. Choose $B > 1 + 8C_F/\beta_*$, then choose $\varepsilon_* > 0$ so small that

$$C_F N_d \varepsilon_* < \min(\delta_T/8, 1/8), \quad \kappa_\mu = \varepsilon_* B^{\mu - N_d}. \quad (166)$$

Thus the κ_μ increase with μ and

$$C_F \sum_{r < \mu} \kappa_r \leq \frac{1}{4} \beta_* \kappa_\mu. \quad (167)$$

Starting at the first flag coordinate, call a coordinate low if $|C_\mu| \leq 2^{\kappa_\mu t}$. Expand every low factor

$$e^{iC_\mu [\hat{H}_\mu(X, Y) - \hat{H}_\mu(\xi_Q, Y)]}$$

by Lemma 6.1. To record the product estimate explicitly, let S be any set of low positions already expanded and index a term by tuples $\mathbf{u} = (u_r)_{r \in S}$ and $\mathbf{v} = (v_r)_{r \in S}$. Its parent-dependent coefficient and combined frequencies are

$$a_Q(\mathbf{u}, \mathbf{v}) = s_Q \prod_{r \in S} a_{u_r, v_r}^{(r)}(C_r, \xi_Q), \quad U(\mathbf{u}) = \sum_{r \in S} u_r, \quad V(\mathbf{v}) = \sum_{r \in S} v_r.$$

With $A(\mathbf{u}, \mathbf{v}) = \prod_{r \in S} A_{u_r, v_r}^{(r)}(C_r)$, if $A(\mathbf{u}, \mathbf{v}) = 0$, then $|a_Q(\mathbf{u}, \mathbf{v})| \leq A(\mathbf{u}, \mathbf{v})$ forces $a_Q(\mathbf{u}, \mathbf{v}) = 0$ for every Q , and the corresponding term is omitted. Thus every division by $A(\mathbf{u}, \mathbf{v})$ below is made only when $A(\mathbf{u}, \mathbf{v}) > 0$. Moreover, the inequality $1 + |\sum_{r \in S} v_r| \leq \prod_{r \in S} (1 + |v_r|)$ and Lemma 6.1 give

$$\begin{aligned} \sum_{\mathbf{u}, \mathbf{v}} A(\mathbf{u}, \mathbf{v}) (1 + |V(\mathbf{v})|) &\leq \prod_{r \in S} \sum_{u_r, v_r} A_{u_r, v_r}^{(r)}(C_r) (1 + |v_r|) \\ &\leq C \prod_{r \in S} (1 + |C_r|)^{C_F} \leq C 2^{C_F t \sum_{r \in S} \kappa_r}. \end{aligned} \quad (168)$$

The same estimate without the weight controls the ordinary Fourier loss. Keeping the tuple indexing and applying the triangle inequality yields the same bound while preserving uniformity in every ξ_Q . In particular, the product of the expansions preceding position μ has coefficient ℓ^1 norm at most

$$C2^{C_F t \sum_{r < \mu} \kappa_r}. \quad (169)$$

We now make the limiting convention needed for the nonlinear variation norms explicit. No infinite Fourier series is inserted directly into a V^2 or V^1 norm. For $N \in \mathbb{N}$, whenever the factor at a low position r is expanded, first retain only

$$\mathfrak{F}_N = \{(u_r, v_r) \in \mathbb{Z}^n \times \mathbb{Z}^n : |u_r| + |v_r| \leq N\}.$$

For a set S of expanded positions, put

$$\omega_N(S) = \sum_{\mathbf{u}, \mathbf{v}: \max_{r \in S} (|u_r| + |v_r|) > N} A(\mathbf{u}, \mathbf{v})(1 + |V(\mathbf{v})|), \quad \omega_N(\emptyset) = 0. \quad (170)$$

The weighted ℓ^1 estimate (168) implies $\omega_N(S) \rightarrow 0$. All component estimates below are first applied to these finite mode sums; their passage to the full Fourier expansions is carried out at the end of the proof.

Suppose first that μ is the first high coordinate. After fixing a Fourier mode from the preceding low factors, the remaining common operator phase is $\sum_{r \geq \mu} C_r \hat{H}_r(X, Y)$; its first flag coordinate is C_μ . The factor obtained by evaluating the remaining phase at $X = \xi_Q$ is a unimodular input multiplier on the descendants of Q . After division by $A(\mathbf{u}, \mathbf{v})$, the parent-dependent coefficients have modulus at most one. Absorb them and the preceding unimodular factor into input multipliers $\vartheta_{Q, \mathbf{u}, \mathbf{v}}$. Disjointness of the descendants and (161) give, in scaled variables,

$$\left\| \sum_{\substack{Q \in \mathcal{Q}, Q \subset J \\ U \subset Q}} \vartheta_{Q, \mathbf{u}, \mathbf{v}}(Y) \tilde{d}_U(Y) \right\|_2^2 \leq \sum_{\substack{Q \in \mathcal{Q}, Q \subset J \\ U \subset Q}} \|\tilde{d}_U\|_\infty \|\tilde{d}_U\|_1 \leq \Lambda R^{-n} M_{I, J}. \quad (171)$$

Thus Corollary 3.8 applies to one common operator phase. The fixed output modulation $e^{iU(\mathbf{u}) \cdot X}$ does not change variation. Summing the majorants produces exactly the loss in (169), and (162) restores physical scale. The scaled truncation interval is $1 \leq \varepsilon/R \leq 2$, so it is the partial annulus of radius 2 in Corollary 3.8. Its exact scale parameter is therefore

$$L_\mu = 2^{m_\mu} |C_\mu| \asymp_{n, d} |C_\mu|,$$

not merely $|C_\mu|$. Consequently we obtain

$$C \|\Theta\|_\infty (2^{m_\mu} |C_\mu|)^{-\beta_\mu} (\Lambda M_{I, J})^{1/2} \leq C \|\Theta\|_\infty 2^{-\beta_\mu \kappa_\mu t} (\Lambda M_{I, J})^{1/2}$$

for the entire short trajectory. Multiplication by the loss (169) and (167) leave decay at least $2^{-\beta_\mu \kappa_\mu t/2}$. Equations (159) and (158) prove (153) in every high branch, with $E = 0$.

If every coordinate is low, expand all factors. For a combined tuple and a descendant U , let ζ_U be the center of $\tilde{U}$. Write

$$e^{iV(\mathbf{v}) \cdot Y} \tilde{d}_U(Y) = e^{iV(\mathbf{v}) \cdot \zeta_U} \tilde{d}_U(Y) + \tilde{r}_{U, \mathbf{v}}(Y), \quad \|\tilde{r}_{U, \mathbf{v}}\|_1 \leq C |V(\mathbf{v})| 2^{-t} \|\tilde{d}_U\|_1. \quad (172)$$

The first term is still mean zero. For each fixed mode, its coefficients are fixed before evaluation in x . More precisely, divide the parent-dependent coefficient by $A(\mathbf{u}, \mathbf{v})$; the resulting packet coefficients have modulus at most one, so Proposition 5.5 applies to the scaled unit-annulus packets. Equation (168) makes the summation uniform in every parent label ξ_Q . The total Fourier loss is at most $2^{C_F N_d \varepsilon_\star t}$. Proposition 5.5 gives $2^{-\delta_{rt}/2}$ at the (unsquared) L^2 level, so (166) leaves at least $2^{-3\delta_{rt}/8}$ after the Fourier summation.

The remainders in (172) are controlled by total radial variation and Young's inequality. Using (168) and (161), their total L^1 variation in scaled variables is

$$C\|\Theta\|_1 2^{-t} 2^{C_F N_d \varepsilon_* t} R^{-n} M_{I,J}.$$

After (162) and (166), this is at most $C\|\Theta\|_1 2^{-3t/4} M_{I,J}$. The good term similarly returns with the factor $(\Lambda M_{I,J})^{1/2}$. Apply (159) to the squared L^2 bounds and (160) to the L^1 errors, followed by (158). This proves (153)–(154).

The exponents can now be fixed uniformly rather than left implicit. Take

$$c_P = \min \left\{ \min_{1 \leq \mu \leq N_d} \frac{\beta_* \kappa_\mu}{2}, \frac{3\delta_T}{8} \right\}, \quad c_E = \frac{3}{4}. \quad (173)$$

The first term in c_P covers every high branch, the second covers the all-low branch, and c_E is permitted because $1 - C_F N_d \varepsilon_* > 7/8$. All quantities in (173), as well as the single system of Fourier cutoffs fixed above, depend only on n, d . In the already settled $N_d = 0$ terminal case one takes $c_P = 3\delta_T/8$ and $c_E = 3/4$; its actual good decay $\delta_T/2$ is stronger.

It remains to remove the finite-mode truncation. Let $\mathcal{P}_N$ be the trajectory in which every low Fourier expansion is restricted to $\mathfrak{F}_N$, and let $\mathcal{P}_N = G_N + E_N$ be the decomposition just constructed, summed over $(I, J) \in \mathcal{A}_R$. These are genuine finite sums. If a block has first high position μ and $S = \{r : r < \mu\}$, then for $N' > N$ the same component estimate used above gives

$$\|G_{N'}^{I,J} - G_N^{I,J}\|_{L^2(V^2)} \leq C\|\Theta\|_\infty (2^{m_\mu} |C_\mu|)^{-\beta_\mu} (\Lambda M_{I,J})^{1/2} \omega_N(S). \quad (174)$$

Here and below the superscript I, J includes the fixed output restriction $\mathbf{1}_I$. If every position is low, the terminal-packet estimate and the remainder estimate in (172) give instead

$$\|G_{N'}^{I,J} - G_N^{I,J}\|_{L^2(V^2)} \leq C\|\Theta\|_\infty 2^{-\delta_T t/2} (\Lambda M_{I,J})^{1/2} \omega_N(S), \quad (175)$$

$$\|E_{N'}^{I,J} - E_N^{I,J}\|_{L^1(V^1)} \leq C\|\Theta\|_1 2^{-t} M_{I,J} \omega_N(S). \quad (176)$$

The harmless Fourier losses already displayed in (169) and (168) may be retained on the right sides; for the fixed block they do not depend on N , while $\omega_N(S) \rightarrow 0$. There are only finitely many interacting blocks. Therefore (159), (160), and (174)–(176) show that (G_N) is Cauchy in $L^2(V^2)$ and (E_N) is Cauchy in $L^1(V^1)$. Every component path is anchored at the upper shell endpoint: $G_N(2R) = E_N(2R) = 0$. Hence

$$\sup_{R \leq \varepsilon \leq 2R} |H_\varepsilon| \leq V^a(H_\varepsilon : R \leq \varepsilon \leq 2R), \quad a \in \{1, 2\},$$

for every component or difference path H in the corresponding variation space. Lemma 5.3 therefore constructs anchored paths G, E and gives convergence of the full sequences in the displayed variation norms, with no uncountable-parameter Bochner-space assumption.

On the other hand, the Weierstrass majorant in (168) shows that the Fourier products defining $\mathcal{P}_N^{I,J}$ converge absolutely and locally uniformly in (X, Y, ξ_Q) to the original phase factor. Since the packets are finite, bounded, and supported in the fixed block, a constant multiple of the integrable majorant

$$\mathbf{1}_I(x) \left(K_{|\Theta|} \mathbf{1}_{\{R < |z| \leq 2R\}} * \sum_{\substack{Q \in \mathcal{Q}, Q \subset J \\ U \subset Q}} |d_U| \right) (x)$$

is independent of N and of ε , where $K_{|\Theta|}(z) = |\Theta(z/|z|)|/|z|^n$. Indeed, disjointness and (150) give $\sum_U |d_U| \leq \Lambda$ almost everywhere, so the displayed function is bounded by $C\Lambda\|\Theta\|_1 \mathbf{1}_I$ and belongs to both L^1 and L^2 . The same majorant gives convergence uniformly in ε for almost every x (and in $L^1 + L^2$ on the fixed output block). The last assertion of Lemma 5.3 therefore identifies the anchored trajectory limit with the original one: $\mathcal{P} = G + E$. Its Fatou conclusion gives

$$\|V^2(G)\|_2 \leq \liminf_{N \rightarrow \infty} \|V^2(G_N)\|_2, \quad \|V^1(E)\|_1 \leq \liminf_{N \rightarrow \infty} \|V^1(E_N)\|_1.$$

Thus the uniform finite-mode bounds pass to G, E with no change in any decay exponent, completing the proof. $\square$

The proof is a simultaneous phase–packet induction. At every later phase, regardless of the sizes of its coefficients, one either reaches a high coordinate and obtains scale-covariant oscillatory decay, or expands a low coordinate and proceeds. The finite hierarchy (167) ensures that Fourier losses accumulated before a later high coordinate are strictly smaller than its oscillatory gain. Translation-generated coefficients enter as unrestricted later coordinates in (164), and the flag hierarchy supplies their control.

7 Concentration-stable degree-block packet estimates

The original phase-adapted Calderón–Zygmund blocks do not satisfy the pointwise packet-density hypothesis of Proposition 6.2. This section replaces that hypothesis by estimates using only cancellation and an average L^1 mass bound. Throughout, coefficient norms are fixed Euclidean norms on the relevant bounded-degree polynomial spaces.

7.1 Polynomial incidence and rational finite type

Lemma 7.1 (Multivariable sublevels and discrete smearing). *Let $I \subset \mathbb{R}^N$ be a fixed compact box with nonempty interior, and let $\mathbf{p}$ be a nonzero vector of polynomials of degree at most D on I . Put $A = \|\mathbf{p}\|_{\text{coeff}}$. There are $c, C > 0$, depending only on I, N, D and the vector length, such that*

$$|\{x \in I : |\mathbf{p}(x)| \leq \tau A\}| \leq C\tau^c, \quad (177)$$

$$\int_I \min \left\{ 1, \left(\frac{L|\mathbf{p}(x)|}{A} \right)^{-\sigma} \right\} dx \leq CL^{-\delta_\sigma}, \quad \delta_\sigma := \frac{c\sigma}{c+\sigma} > 0, \quad (178)$$

for $0 < \tau \leq 1$, $L \geq 1$, and $\sigma > 0$. At a zero of $\mathbf{p}$, the integrand in (178) is defined to be 1.

Let $0 < q \leq 1$, let $\mathcal{Q}$ be a family of pairwise disjoint q -cubes in I , choose $y_Q \in Q$ for every $Q \in \mathcal{Q}$, and attach weights $0 \leq w_Q \leq \Lambda q^N$. If $M = \sum_{Q \in \mathcal{Q}} w_Q$, then every polynomial p on $\mathbb{R}^N \times \mathbb{R}^N$ of degree at most D and coefficient norm one satisfies, for $0 < \tau \leq 1$,

$$\sum_{Q, Q' \in \mathcal{Q}} w_Q w_{Q'} \mathbf{1}_{\{|p(y_Q, y_{Q'})| \leq \tau\}} \leq CAM(\tau + q)^c. \quad (179)$$

There is also the following chart version. Let $U_v \subseteq U_v^* \subset \mathbb{R}^N$ be fixed boxes and let $\chi_v : U_v^* \rightarrow \mathcal{M}_v^*$ be a fixed finite family of rational charts. Assume, uniformly in v , that

$$c|z - z'| \leq |\chi_v(z) - \chi_v(z')| \leq C|z - z'|, \quad c \leq J_v(z) \leq C, \quad (180)$$

where J_v is the density of the induced measure under χ_v . Thus, for a coordinate q -cube $Q^\flat \subset U_v$ and its image $Q = \chi_v(Q^\flat)$,

$$cq^N \leq |Q|_{\mathcal{M}_v} = \int_{Q^\flat} J_v(z) dz \leq Cq^N. \quad (181)$$

The chart pair version permits selected points in such chart cubes and weights $w_Q \leq \Lambda|Q|_{\mathcal{M}_v}$; by (181) this is, up to a fixed constant, the same cap as $w_Q \leq \Lambda q^N$. It also permits physical q -cubes after a bounded proximity coloring: by (180), their inverse images have diameter $O(q)$ and volume comparable to q^N , while the selected coordinate centers in each color admit disjoint smearing balls of radius cq in the fixed enlarged box.

In a given application let $\mathcal{V}_v$ be a specified fixed finite-dimensional space of polynomial vectors on $\mathcal{M}_v$, and let $\mathcal{V}_{v,v'}^{(2)}$ be the specified fixed finite-dimensional pair-polynomial space on $\mathcal{M}_v \times \mathcal{M}_{v'}$. Choose a fixed common denominator $\mathcal{D}_v$ which clears every pullback of an element of $\mathcal{V}_v$ and is bounded above and away from zero on U_v^* , and assume explicitly that

$$T_v : \mathcal{V}_v \longrightarrow \mathcal{P}_{\leq D_v}(U_v), \quad T_v \mathbf{p} = \mathcal{D}_v \mathbf{p} \circ \chi_v, \quad \text{is injective.} \quad (182)$$

Then (177)–(178) hold after pullback to these charts. The pair estimate (179) also holds provided the analogous cleared pullback on $\mathcal{V}_{v,v'}^{(2)}$ is injective. The constants may depend on the finitely many least singular values in (182), in addition to the fixed chart data. No injectivity assertion is made for all ambient polynomials restricted to a lower-dimensional algebraic chart.

Proof. We first prove (177). A scalar polynomial of degree at most D on a fixed interval satisfies a uniform sublevel estimate. Indeed, when a sublevel set E has positive measure, one may choose $D + 1$ of its points with consecutive separations bounded below by a constant multiple of $|E|$; Lagrange interpolation then bounds the coefficient norm by $C\tau|E|^{-D}$. The case $D = 0$ is immediate, so assume henceforth that $D \geq 1$. For several variables, write $p(x', t) = \sum_{j=0}^D a_j(x')t^j$ and choose a_j with coefficient norm bounded below. If the $(N - 1)$ -variable exponent already obtained is c_{N-1} , split the base according to $|a_j(x')| \leq s$. The induction hypothesis bounds the first part by $Cs^{c_{N-1}}$, while on the complement the one-dimensional estimate bounds every fibre by $C(\tau/s)^{1/D}$. Taking $s = \tau^{1/(1+Dc_{N-1})}$ gives a positive exponent depending only on N and D , and proves (177). For a polynomial vector, one component has coefficient norm at least a fixed multiple of A , and normalization of that component completes the proof. Choose c in the statement no larger than this exponent and no larger than one half of the corresponding scalar sublevel exponent in dimension $2N$; for the fixed finite chart family, take the minimum of the finitely many resulting exponents.

For (178), set $\theta = \sigma/(c + \sigma)$ and

$$E_L = \{x \in I : |\mathbf{p}(x)| \leq AL^{-\theta}\}.$$

Equation (177) gives $|E_L| \leq CL^{-c\theta} = CL^{-\delta_\sigma}$. On $I \setminus E_L$ the integrand is at most $L^{-\sigma(1-\theta)} = L^{-\delta_\sigma}$. This proves (178), with a constant uniform in σ .

For (179), join two cubes when their distance is less than $10q$. Since the cubes have pairwise disjoint interiors and equal volume, a packing argument in a fixed enlargement of any one cube bounds the degree of this proximity graph by a constant depending only on N . It therefore has a bounded proper coloring. Within one color the selected points are cq -separated. Put disjoint cq -balls around them and spread each weight uniformly on its ball. The resulting density on a fixed enlargement of I is bounded by $C\Lambda$ and has integral at most M . A normalized bounded-degree polynomial has bounded gradient on a fixed enlargement of I^2 , so every bad selected pair smears into the $C(\tau + q)$ -sublevel. For any ordered pair of colors, if F, G are the two smeared densities and E is this sublevel, then

$$\iint_E F(x)G(y) dx dy \leq \min\{M^2, C\Lambda^2|E|\} \leq C\Lambda M|E|^{1/2}.$$

If $\tau + q$ is bounded below by a fixed constant, the desired estimate follows directly from $M \leq C_I\Lambda$. Otherwise apply the scalar sublevel estimate just proved, now in $2N$ variables, to E and sum the finite color pairs. The choice of c made above proves (179).

For the chart assertion, fix one of the specified rational charts and the application-specific space $\mathcal{V}_v$. The denominator and its reciprocal are uniformly bounded on the enlarged box. The required injectivity is precisely hypothesis (182); it is not inferred merely from vanishing on a possibly lower-dimensional chart image. The least singular value of this fixed map and finite-dimensional norm equivalence give

$$c_v \|\mathbf{p}\|_{\text{coeff}} \leq \|T_v \mathbf{p}\|_{\text{coeff}} \leq C_v \|\mathbf{p}\|_{\text{coeff}}, \quad |T_v \mathbf{p}(\zeta)| \simeq |\mathbf{p}(\chi_v(\zeta))|.$$

The change-of-variables formula and (180) give, for every measurable $E \subset \mathcal{M}_v$ contained in the chart,

$$c|\chi_v^{-1}(E)| \leq |E|_{\mathcal{M}_v} = \int_{\chi_v^{-1}(E)} J_v(z) dz \leq C|\chi_v^{-1}(E)|.$$

Consequently (177) and (178) transfer to the chart. For (179), use the product chart and the explicitly assumed injectivity of its cleared pullback on $\mathcal{V}_{v,v'}^{(2)}$. In coordinate variables a chart cube has mass cap $C\Lambda q^N$ by (181). Moreover, (180) converts a physical separation cq into a coordinate separation comparable to q , and a coordinate ball of radius cq has both coordinate and induced volume comparable

to q^N . Thus the proximity coloring gives disjoint coordinate smearing balls; spreading w_Q on such a ball produces density at most

$$C \frac{\Lambda q^N}{q^N} = C\Lambda.$$

The coefficient-one cleared polynomial has bounded coordinate gradient on the fixed enlarged product box, so a bad selected pair and the product of its two smearing balls lie in the $C(\tau + q)$ -sublevel. The preceding coordinate proof of (179) now applies, with only the displayed fixed metric and Jacobian constants. Taking the minimum over the finite chart family proves all asserted chart versions. $\square$

We first fix the rational quotient and its norm. This is needed because all later rational phases have parameters which may approach degenerate values, and a qualitative assertion that a phase is not separated does not by itself give a uniform oscillatory constant.

Fix compact intervals I, J and slightly larger intervals I^*, J^* . For fixed $D_0 \in \mathbb{N}$ and $0 < \eta < 1 < M$, call

$$\Psi(u, v) = \frac{N(u, v)}{Q(u, v)} \quad (183)$$

an admissible rational presentation if N, Q are real polynomials of degree at most D_0 ,

$$\|Q\|_{\text{coeff}} \leq M, \quad \eta \leq |Q(u, v)| \leq M \quad ((u, v) \in I^* \times J^*). \quad (184)$$

The denominator in (183) is part of the presentation. In every application below it is the displayed common denominator produced by the coordinate change.

For an admissible presentation define the polynomial

$$\mathfrak{C}_Q \Psi(u, u', v) = \{Q(u, v)Q(u', v)\}^2 \partial_v \{\Psi(u, v) - \Psi(u', v)\} \quad (185)$$

and set

$$\|[\Psi]\|_{\text{rat}, Q} := \|\mathfrak{C}_Q \Psi\|_{\text{coeff}, (u, u', v)}. \quad (186)$$

This definition is invariant under addition of $F(u) + G(v)$. Conversely, if (185) vanishes, then $\partial_v \Psi(u, v)$ is independent of u , and integration on the rectangle gives $\Psi(u, v) = F(u) + G(v)$. Thus (186) is a genuine norm on the quotient by separated phases. If the same rational function is written using two admissible common denominators of bounded degree, the resulting norms are uniformly equivalent. Indeed, if the denominators are Q_1, Q_2 , then pointwise on the fixed enlarged box

$$\mathfrak{C}_{Q_2} \Psi(u, u', v) = \left\{ \frac{Q_2(u, v)Q_2(u', v)}{Q_1(u, v)Q_1(u', v)} \right\}^2 \mathfrak{C}_{Q_1} \Psi(u, u', v). \quad (187)$$

The factor in braces and its reciprocal are bounded by constants depending only on η, M . All difference numerators belong to one fixed bounded-degree polynomial space, on which coefficient and supremum norms are uniformly equivalent. Consequently

$$\|\mathfrak{C}_{Q_2} \Psi\|_{\text{coeff}} \gtrsim \|\mathfrak{C}_{Q_2} \Psi\|_{\infty} \gtrsim \|\mathfrak{C}_{Q_1} \Psi\|_{\infty} \gtrsim \|\mathfrak{C}_{Q_1} \Psi\|_{\text{coeff}}.$$

Interchanging Q_1, Q_2 gives the reverse inequality and proves the claimed uniform equivalence.

We will also use the mixed numerator

$$\mathfrak{M}_Q \Psi(u, v) = Q(u, v)^3 \partial_u \partial_v \Psi(u, v). \quad (188)$$

It is a polynomial of uniformly bounded degree and satisfies

$$\|[\Psi]\|_{\text{rat}, Q} \geq c \|\mathfrak{M}_Q \Psi\|_{\text{coeff}}. \quad (189)$$

To verify this, differentiate (185) in u' and then put $u' = u$. Since the phase difference and its v derivative vanish on the diagonal, derivatives falling on the prefactor make no contribution, and one obtains exactly

$$\partial_{u'} \mathfrak{C}_Q \Psi(u, u', v) \big|_{u'=u} = -Q(u, v)^4 \partial_u \partial_v \Psi(u, v) = -Q(u, v) \mathfrak{M}_Q \Psi(u, v).$$

Differentiation and diagonal restriction are bounded maps on coefficient spaces. Moreover $\|QH\|_{\text{coeff}} \geq c\|H\|_{\text{coeff}}$ for every polynomial H of the relevant bounded degree: this follows from $\|QH\|_{L^\infty(I^* \times J^*)} \geq \eta\|H\|_{L^\infty(I^* \times J^*)}$ and finite-dimensional norm equivalence. This proves (189).

Lemma 7.2 (Rational van der Corput with a BV amplitude). *Let $\psi = N/q$ on a fixed compact interval, where N, q have degree at most D_0 , $\|q\|_{\text{coeff}} \leq M$, and $\eta \leq |q| \leq M$ on a fixed enlargement. Put*

$$G = \|q^2 \psi'\|_{\text{coeff}} = \|N'q - Nq'\|_{\text{coeff}}.$$

If a is a zero-extended bounded-variation function supported in the interval, then

$$\left| \int e^{i\psi(v)} a(v) dv \right| \leq C(\|a\|_\infty + \text{Var}(a)) \min\{1, G^{-c}\}. \quad (190)$$

When $G = 0$, the minimum on the right is understood to be 1. The same assertion holds after a bounded partition into intervals, with the sum of the sectional $L^\infty + \text{BV}$ norms on the right. The constants and the positive exponent depend only on D_0, η, M and the fixed interval and its enlargement.

Proof. Only $G > 1$ requires proof; the case $G \leq 1$ is the trivial L^1 bound. Write $p = q^2 \psi'$. By Lemma 7.1, applied to p/G , the set

$$E = \{v : |p(v)| \leq G^{1/2}\}$$

has measure at most CG^{-c_0} . Its contribution is bounded by $C\|a\|_\infty G^{-c_0}$. On the complement, $|\psi'| \geq cG^{1/2}$. The numerator of ψ'' has bounded degree. Apart from the two endpoints of the ambient interval, the boundary of E is contained in the union of the zero sets of the two bounded-degree polynomials $p - G^{1/2}$ and $p + G^{1/2}$; hence E has only $O_{D_0}(1)$ components. Partition its complement at those boundary points and at the zeros of the numerator of ψ'' ; the number of intervals is therefore bounded in terms of D_0 . On each remaining interval ψ' is monotone and bounded away from zero. The first derivative van der Corput estimate, proved for a BV amplitude by one Stieltjes integration by parts, gives $C(\|a\|_\infty + \text{Var}(a))G^{-1/2}$. Summing the boundedly many pieces and decreasing the exponent proves (190). If the numerator of ψ'' vanishes identically, ψ' is already monotone and the same argument applies without this partition. $\square$

Lemma 7.3 (Coefficient-norm polynomial van der Corput). *Let I be a fixed compact interval, let $D \geq 1$, let p be a real polynomial of degree at most D , and put*

$$\Lambda_p = \|p - p(0)\|_{\text{coeff}}.$$

If a is supported in I and is of bounded variation, then

$$\left| \int_I e^{ip(t)} a(t) dt \right| \leq C_D \{\|a\|_\infty + \text{Var}(a)\} \min\{1, \Lambda_p^{-c_D}\}. \quad (191)$$

The value $p(0)$ may be replaced by any constant; equivalently, Λ_p may be any fixed coefficient norm on the space of real polynomials of degree at most D , modulo constants. When $\Lambda_p = 0$, the minimum in (191) is understood to be 1.

Proof. Apply Lemma 7.2 with denominator identically one. Differentiation is injective from the finite-dimensional quotient of real polynomials of degree at most D by the constants into the space of polynomials of degree at most $D - 1$, and therefore $\|p'\|_{\text{coeff}} \geq c_D \Lambda_p$ by finite-dimensional norm equivalence. Decreasing the exponent in (190) proves (191). $\square$

Lemma 7.4 (Uniform rational oscillatory operator). *Let I, J be fixed compact intervals, let $A : I \times J \rightarrow \mathbb{C}$ be measurable, and define $S : L^2(J) \rightarrow L^2(I)$ initially on bounded functions by*

$$Sf(u) = \int_J e^{i\Psi(u,v)} A(u,v) f(v) dv.$$

Assume that (183)–(184) hold with fixed data. Suppose $B \geq 1$ and

$$\sup_{u \in I} \int_J |A(u, v)| dv \leq B, \quad \sup_{v \in J} \int_I |A(u, v)| du \leq B.$$

Assume also that, for every $u \in I$, there is a partition of J into at most M_0 intervals on each of which

$$\|A(u, \cdot)\|_\infty + \text{Var}(A(u, \cdot)) \leq B.$$

The endpoints may depend on u . Put $L = \|\Psi\|_{\text{rat}, Q}$. Then

$$\|S\|_{2 \rightarrow 2} \leq CB^C \min\{1, L^{-c}\}. \quad (192)$$

When $L = 0$, the minimum is again understood to be 1. The constants depend only on the fixed intervals, D_0, η , the admissibility bound M , and M_0 .

Proof. The trivial part follows from Schur's test, so assume $L \geq 1$. The kernel of SS^* is

$$K(u, u') = \int_J e^{i[\Psi(u, v) - \Psi(u', v)]} A(u, v) \overline{A(u', v)} dv.$$

For fixed (u, u') , an admissible common denominator of the phase difference is $q_{u, u'}(v) = Q(u, v)Q(u', v)$; its degree, coefficient norm, and upper and lower bounds are controlled by the fixed admissibility data. Its derivative numerator is precisely (185). If

$$G(u, u') = \|\mathfrak{C}_Q \Psi(u, u', \cdot)\|_{\text{coeff}, v},$$

then refine the two partitions of J belonging to u and u' . There are at most $2M_0$ resulting intervals, and

$$\|A(u, \cdot) \overline{A(u', \cdot)}\|_{L^\infty + \text{BV}} \leq CB^2$$

on each of them. Lemma 7.2, summed over this partition, gives

$$|K(u, u')| \leq CB^2 \min\{1, G(u, u')^{-c_0}\}. \quad (193)$$

The entries of the coefficient vector defining G are polynomials in (u, u') of uniformly bounded degree, and their joint coefficient norm is exactly L . Consequently (178), applied on $I \times I$, yields

$$\iint_{I \times I} |K(u, u')|^2 du du' \leq CB^C L^{-c_1}.$$

There is no deletion of the diagonal or of any algebraically degenerate branch: all such points are included in this negative-moment integral. Finally,

$$\|S\|_{2 \rightarrow 2}^2 = \|SS^*\|_{2 \rightarrow 2} \leq \|K\|_{L^2(I \times I)}.$$

Taking the two square roots and decreasing c_1 proves (192). $\square$

7.2 Rough two-ray correlation and a flat high block

Lemma 7.5 (Universal rough polynomial correlation). *Let $n \geq 1$, let $D \geq 1$, let $K = [r_-, r_+] \Subset (0, \infty)$, and let a, b be radial BV profiles supported in K with $L^\infty + \text{BV}$ norm at most $B \geq 1$. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$ and put*

$$k_a(z) = |z|^{-n} \Theta(z/|z|) a(|z|), \quad k_b(z) = |z|^{-n} \Theta(z/|z|) b(|z|).$$

Let S be a real polynomial of degree at most D , and let Λ be the coefficient norm of its nonconstant part. Then, for every $h \in \mathbb{R}^n$, if $n \geq 2$,

$$\left| \int e^{iS(z)} \overline{k_a(z)} k_b(z-h) dz \right| \leq CB^C \|\Theta\|_\infty^2 \min\{1, (\Lambda|h|^D)^{-c}\}. \quad (194)$$

For $n \geq 2$ the trivial bound is understood when $h = 0$ or $\Lambda = 0$. In dimension one, for every h , the precise unit-scale analogue is

$$\left| \int e^{iS(z)} \overline{k_a(z)} k_b(z-h) dz \right| \leq CB^C \|\Theta\|_\infty^2 \min\{1, \Lambda^{-c}\}.$$

Lemma 7.6 (Quantitative equal-shell pullback). *Let $D \geq 1$ and let $\phi(t, r) = \sum_{m=0}^D \phi_m(t, r)$, where ϕ_m is homogeneous of degree m , and put*

$$\Gamma = \|\phi - \phi_0\|_{\text{coeff}}.$$

Let $0 < \alpha \leq 1$, $0 < H \leq C_0$, $|U_0| \leq C_0$, let $\Delta \in \mathbb{R}$, and let $q(X, Y) = \Delta + X - Y$ satisfy $c_0 \leq |q| \leq C_0$ on a fixed enlargement of a fixed rectangle. Define

$$\Psi(X, Y) = \phi\left(\frac{U_0 + HX}{q(X, Y)}, \frac{\alpha}{q(X, Y)}\right) = \sum_{m=0}^D \frac{\phi_m(U_0 + HX, \alpha)}{q(X, Y)^m}. \quad (195)$$

Use the canonical common denominator $Q = q^D$. Then

$$\|[\Psi]\|_{\text{rat}, Q} \geq c(\alpha H)^D \Gamma. \quad (196)$$

The constant depends only on D, c_0, C_0 and the fixed rectangles.

Proof of Lemma 7.6. Put $A(X) = U_0 + HX$ and use (X, q) rather than (X, Y) as polynomial coordinates. Since X, Y range in fixed rectangles and $c_0 \leq |q| \leq C_0$ on their enlargement, Δ is uniformly bounded. Thus $(X, Y) \mapsto (X, q)$ and its inverse have uniformly bounded coefficient matrices on every bounded-degree polynomial space. For $m \geq 1$, direct differentiation gives

$$q^{D+2} \partial_X \partial_Y \{ \phi_m(A(X), \alpha) q^{-m} \} = q^{D-m} \{ m H q \partial_t \phi_m(A(X), \alpha) - m(m+1) \phi_m(A(X), \alpha) \}. \quad (197)$$

This identity also identifies all possible cancellation between adjacent homogeneous degrees. After summing (197), the coefficient of q^{D-m} is

$$-m(m+1) \phi_m(A, \alpha) + (m+1) H \partial_t \phi_{m+1}(A, \alpha), \quad \phi_{D+1} = 0. \quad (198)$$

Starting with $m = D$ and descending, the triangular system (198) recovers every $\phi_m(A, \alpha)$, $1 \leq m \leq D$. More explicitly, on the finite direct sum $\bigoplus_{m=1}^D \mathcal{P}_{\leq m}(X)$ the diagonal maps are the nonzero scalars $-m(m+1)$ and the only off-diagonal maps are $(m+1)H \partial_t$ from level $m+1$ to level m . Descending substitution and $0 < H \leq C_0$ therefore give

$$\sum_{m=1}^D \|\phi_m(A, \alpha)\|_{\text{coeff}, X} \leq C_{D, C_0} \|q^{D+2} \partial_X \partial_Y \Psi\|_{\text{coeff}, (X, q)}.$$

The uniformly invertible affine coefficient change above makes the last norm equivalent to its (X, Y) coefficient norm. Hence

$$\|q^{D+2} \partial_X \partial_Y \Psi\|_{\text{coeff}, (X, Y)} \geq c \left(\sum_{m=1}^D \|\phi_m(U_0 + HX, \alpha)\|_{\text{coeff}, X}^2 \right)^{1/2}. \quad (199)$$

In particular, no choice of a largest homogeneous term is being made: all homogeneous degrees are recovered simultaneously.

The affine map $t = U_0 + HX$ and evaluation at $r = \alpha$ are quantitatively injective on homogeneous two-variable polynomials. Indeed, recover $\phi_m(t, \alpha)$ from its composition with $U_0 + HX$ and then recover the coefficient of $t^j r^{m-j}$ by division by α^{m-j} . Since $|U_0| \leq C_0$, $0 < \alpha \leq 1$, and $0 < H \leq C_0$, finite-dimensional norm equivalence gives

$$\|\phi_m(U_0 + HX, \alpha)\|_{\text{coeff}, X} \geq c(\alpha H)^D \|\phi_m\|_{\text{coeff}}. \quad (200)$$

Indeed, the inverse of the affine substitution $t = U_0 + HX$ has norm $O_{m, C_0}(H^{-m})$, while evaluation at $r = \alpha$ multiplies the coefficient of $t^j r^{m-j}$ by α^{m-j} . The left side is thus at least $c H^m \alpha^m \|\phi_m\|_{\text{coeff}}$. If $\alpha H \leq 1$, this is at least the right side of (200); if $\alpha H > 1$, the bounded upper ranges of α and H allow the remaining factor to be absorbed into c . Combining (199) and (200) gives the desired lower bound for the reduced mixed numerator $q^{D+2} \Psi_{XY}$.

For the presentation with denominator $Q = q^D$, the mixed numerator $Q^3 \Psi_{XY}$ differs from the reduced numerator by multiplication by q^{2D-2} . This factor is bounded above and away from zero on the enlarged rectangle, so their coefficient norms are uniformly equivalent. Equation (189) now proves (196). $\square$

Proof of Lemma 7.5. Assume $n \geq 2$. Since the radial profiles are supported in a fixed compact subset of $(0, \infty)$, kernel overlap implies $0 < |h| \leq C$. The cases $h = 0$ and $\Lambda = 0$ are covered by the stated trivial bound. Put $H = |h|$ and choose $\mathcal{O} \in O(n)$ such that $h = H\mathcal{O}e_1$. Under the change of variables $z = \mathcal{O}\tilde{z}$ replace

$$S(z) \quad \text{by} \quad \tilde{S}(\tilde{z}) = S(\mathcal{O}\tilde{z}), \quad \Theta(\xi) \quad \text{by} \quad \tilde{\Theta}(\xi) = \Theta(\mathcal{O}\xi).$$

The radial profiles are unchanged, Lebesgue measure is preserved, and $\|\tilde{\Theta}\|_\infty = \|\Theta\|_\infty$. On the finite-dimensional space of nonconstant polynomials of degree at most D , composition with $\mathcal{O}$ is invertible with inverse composition by $\mathcal{O}^T$. Compactness of $O(n)$ therefore gives, uniformly in $\mathcal{O}$,

$$c_{n,D}\Lambda \leq \|\tilde{S} - \tilde{S}(0)\|_{\text{coeff}} \leq C_{n,D}\Lambda.$$

Thus the desired estimate is invariant, up to its allowed structural constant, under this rotation. We may rename the transformed quantities, assume $h = He_1$, and write

$$z = (t, \rho\omega), \quad u = t/\rho, \quad v = (t - H)/\rho, \quad \omega \in \mathbb{S}^{n-2}.$$

Then exactly

$$\rho = \frac{H}{u - v}, \quad z = \frac{H(u, \omega)}{u - v}, \quad z - h = \frac{H(v, \omega)}{u - v}. \quad (201)$$

The two rough factors are respectively functions of (u, ω) and (v, ω) , and the Jacobian times the two homogeneous kernels is

$$H^{-n}|u - v|^{n-1}(1 + u^2)^{-n/2}(1 + v^2)^{-n/2} du dv d\omega. \quad (202)$$

Thus the rough angular factors are input and output L^2 functions and are never differentiated.

Choose once and for all a nonnegative $\beta \in C_c^\infty((c, C))$ such that, on the relevant positive range,

$$\sum_{k \in \mathbb{Z}} \beta(2^k \rho) = 1.$$

After combining the boundedly many largest pieces, write $\alpha = 2^{-k} \leq 1$ and $\beta_\alpha(\rho) = \beta(\rho/\alpha)$. This is a genuine partition of unity on the overlap region, its supports have uniformly bounded overlap, and

$$\sup_\alpha \{\|\beta_\alpha\|_\infty + \text{Var}_\rho(\beta_\alpha)\} \leq C. \quad (203)$$

Thus the transverse radius is decomposed into bands $\rho \simeq \alpha$, $0 < \alpha \leq 1$. Put $U = \alpha u, V = \alpha v$. In these variables (202) becomes

$$H^{-n}\alpha^{n-1}|U - V|^{n-1}(\alpha^2 + U^2)^{-n/2}(\alpha^2 + V^2)^{-n/2} dU dV d\omega. \quad (204)$$

Annular support gives $|U - V| \simeq H$ in a fixed bounded set and also $\alpha^2 + U^2 \simeq \alpha^2 + V^2 \simeq 1$. Thus, after the factor α^{n-1} is removed, the kernel in (U, V) has size $O(H^{-1})$ and is supported in a strip of width $O(H)$.

Choose a fixed $\varepsilon_0 > 0$, smaller than one tenth of the lower constant in $|U - V| \simeq H$, and partition the U - and V -axes into half-open intervals of length $\varepsilon_0 H$. Each input interval interacts with only boundedly many output intervals and conversely. On an interacting pair write

$$U = U_0 + HX, \quad V = V_0 + HY, \quad \Delta = \frac{U_0 - V_0}{H};$$

then X, Y range over fixed intervals of length ε_0 and

$$q(X, Y) = \Delta + X - Y$$

is bounded above and away from zero on a fixed enlargement of their full product. This last assertion concerns the whole product rectangle, not only the support of the amplitude, and is the reason for the small fixed choice of ε_0 .

The phase on this block is exactly

$$\Psi_{\omega, U_0, V_0, H, \alpha}(X, Y) = S \left(\frac{(U_0 + HX)e_1 + \alpha\omega}{q(X, Y)} \right). \quad (205)$$

The factor $dV = H dY$ cancels the H^{-1} in the normalized kernel. The two angular factors are, up to the fixed sign of q ,

$$\overline{\Theta \left(\frac{Ue_1 + \alpha\omega}{\sqrt{U^2 + \alpha^2}} \right)}, \quad \Theta \left(\frac{Ve_1 + \alpha\omega}{\sqrt{V^2 + \alpha^2}} \right);$$

they are respectively functions of the output and input variables and are never differentiated. The radial factors are

$$a \left(\frac{\sqrt{U^2 + \alpha^2}}{|q|} \right), \quad b \left(\frac{\sqrt{V^2 + \alpha^2}}{|q|} \right).$$

Here is the uniformity in this assertion. The sign of q is fixed on the enlarged block. Put $s_U = (U^2 + \alpha^2)^{1/2}$ and $s_V = (V^2 + \alpha^2)^{1/2}$. Up to nonzero factors of fixed sign, the four sectional derivatives of $s_U/|q|$ and $s_V/|q|$ are therefore controlled by those of s_U/q and s_V/q . Directly,

$$\begin{aligned} \partial_X \frac{s_U}{q} &= \frac{HUq - s_U^2}{s_U q^2}, & \partial_Y \frac{s_U}{q} &= \frac{s_U}{q^2}, \\ \partial_X \frac{s_V}{q} &= -\frac{s_V}{q^2}, & \partial_Y \frac{s_V}{q} &= \frac{HVq + s_V^2}{s_V q^2}. \end{aligned}$$

Thus a derivative can vanish only through

$$HUq - s_U^2 = 0 \quad \text{or} \quad HVq + s_V^2 = 0;$$

the other two derivatives never vanish. Since $U = U_0 + HX$, $V = V_0 + HY$, and $q = \Delta + X - Y$, the two displayed critical-point equations are polynomial equations of degree at most two. Their zero sets therefore split every section into $O(1)$ monotonicity intervals (with an identically vanishing derivative requiring no split). On an interval I on which a profile argument γ is monotone, the one-dimensional BV pullback inequality gives

$$\text{Var}_I(a \circ \gamma) \leq \text{Var}_{\gamma(I)}(a) \leq \text{Var}(a),$$

and the same conclusion is immediate if γ is constant. Summing over the uniformly bounded number of intervals gives

$$\text{Var}(a \circ \gamma) + \text{Var}(b \circ \tilde{\gamma}) \leq C_D \{ \text{Var}(a) + \text{Var}(b) \}.$$

The annular support gives $\sqrt{U^2 + \alpha^2} \simeq \sqrt{V^2 + \alpha^2} \simeq 1$ and $|q| \simeq 1$, so all remaining algebraic factors in (204) and their sectional derivatives are uniformly bounded. Moreover, on the block $\rho/\alpha = H/(U - V) = 1/q(X, Y)$, so a smooth dyadic cutoff in ρ/α is the fixed function $\beta(1/q)$ of q . Since q is bounded above and away from zero on the enlarged rectangle, this cutoff and all of its sectional derivatives are uniformly bounded, consistently with (203). Thus the amplitude in each normalized block satisfies Lemma 7.4 with bound CB^C .

For completeness, the block synthesis has no hidden dependence on H . If T_{ij} is the operator between an interacting V -interval and U -interval and

$$(\mathcal{U}_i g)(X) = H^{1/2} g(U_i + HX), \quad (\mathcal{V}_j f)(Y) = H^{1/2} f(V_j + HY),$$

then $\mathcal{U}_i T_{ij} \mathcal{V}_j^{-1}$ is precisely the normalized operator above: the factor $dV = H dY$ cancels the kernel factor H^{-1} . If d_{in} and d_{out} are the two degrees of the block interaction graph, Cauchy–Schwarz gives

$$\|T\|_{2 \rightarrow 2} \leq (d_{\text{in}} d_{\text{out}})^{1/2} \sup_{i,j} \|T_{ij}\|_{2 \rightarrow 2}.$$

Both graph degrees are absolute constants. The same calculation with Schur's test proves the trivial band bound $CB^C \alpha^{n-1}$.

For fixed ω , put

$$\phi_\omega(t, r) = S(te_1 + r\omega), \quad \Gamma(\omega) = \|\phi_\omega - \phi_\omega(0, 0)\|_{\text{coeff}}.$$

Formula (205) is (195). Lemma 7.6 gives, for its canonical denominator q^D ,

$$\|[\Psi_{\omega, U_0, V_0, H, \alpha}]\|_{\text{rat}, q^D} \geq c\Gamma(\omega)(\alpha H)^D.$$

Lemma 7.4 gives the fixed- ω block gain, and the preceding almost-orthogonal synthesis preserves it. We now spell out the angular synthesis. Let $I_0 \subset \mathbb{R}$ be a fixed compact interval containing every possible U - and V -coordinate on the band, and let β denote the fixed zero-extended cutoff defining this band in $\rho/\alpha = H/(U - V)$. After the factor α^{n-1} in (204) is removed, put

$$\begin{aligned} A_{\alpha, H}(U, V) &= \beta \left(\frac{H}{U - V} \right) H^{-n} (U - V)^{n-1} (\alpha^2 + U^2)^{-n/2} (\alpha^2 + V^2)^{-n/2} \\ &\quad \times a \left(\frac{H\sqrt{U^2 + \alpha^2}}{U - V} \right) b \left(\frac{H\sqrt{V^2 + \alpha^2}}{U - V} \right), \end{aligned}$$

extended by zero off the band support. Here $U - V > 0$ by (201). Define

$$\begin{aligned} (T_{\alpha, \omega} f)(U) &= \int_{I_0} e^{iS(H(Ue_1 + \alpha\omega)/(U - V))} A_{\alpha, H}(U, V) f(V) dV, \\ f_{\alpha, \omega}(V) &= \Theta \left(\frac{Ve_1 + \alpha\omega}{\sqrt{V^2 + \alpha^2}} \right) \mathbf{1}_{I_0}(V), \\ g_{\alpha, \omega}(U) &= \Theta \left(\frac{Ue_1 + \alpha\omega}{\sqrt{U^2 + \alpha^2}} \right) \mathbf{1}_{I_0}(U). \end{aligned}$$

With the convention $\langle F, G \rangle = \int F \overline{G}$, the contribution $\mathcal{J}_\alpha$ of this transverse band is exactly

$$\mathcal{J}_\alpha = \alpha^{n-1} \int_{\mathbb{S}^{n-2}} \langle T_{\alpha, \omega} f_{\alpha, \omega}, g_{\alpha, \omega} \rangle_{L^2(I_0)} d\omega. \quad (206)$$

The functions in this formula are jointly measurable, and $\omega \mapsto f_{\alpha, \omega}, g_{\alpha, \omega}$ are strongly measurable $L^2(I_0)$ -valued maps. Moreover

$$\|f_{\alpha, \omega}\|_2 + \|g_{\alpha, \omega}\|_2 \leq C\|\Theta\|_\infty.$$

The block argument above, Lemma 7.6, and Lemma 7.4 give, uniformly in ω, α, H ,

$$\|T_{\alpha, \omega}\|_{2 \rightarrow 2} \leq CB^C \min\{1, [\Gamma(\omega)(\alpha H)^D]^{-c_0}\}. \quad (207)$$

The trivial Schur bound supplies an integrable majorant in (206); hence ordinary Fubini and Cauchy–Schwarz yield

$$|\mathcal{J}_\alpha| \leq CB^C \|\Theta\|_\infty^2 \alpha^{n-1} \int_{\mathbb{S}^{n-2}} \min\{1, [\Gamma(\omega)(\alpha H)^D]^{-c_0}\} d\omega. \quad (208)$$

No measurable-field or abstract direct-integral assertion is being used.

It remains to estimate the scalar angular integral. When $n = 2$, $\mathbb{S}^0 = \{-1, 1\}$, and the two restrictions are related by $r \mapsto -r$; either restriction has coefficient norm comparable to Λ , so the required bound follows by summing the two points. Assume henceforth in this paragraph that $n \geq 3$. Split $\mathbb{S}^{n-2}$ by a fixed measurable partition subordinate to a finite rational atlas $\omega = \omega_v(\zeta)$; the chart Jacobians are bounded above and away from zero on fixed enlargements. In the v -th chart let $\mathbf{g}_v(\zeta)$ be the vector of all nonconstant (t, r) coefficients of $S(te_1 + r\omega_v(\zeta))$, after multiplication by one fixed common chart denominator. The denominator and its reciprocal are bounded on the enlargement, so pointwise

$$|\mathbf{g}_v(\zeta)| \simeq \Gamma(\omega_v(\zeta)). \quad (209)$$

For each nonempty chart, the restriction map

$$S - S(0) \mapsto \mathbf{g}_v$$

is injective: if its nonconstant (t, r) coefficients vanish throughout the chart, then $S - S(0)$ vanishes on the corresponding open cone and therefore identically. Thus, for the application-specific polynomial space consisting of these restrictions, this verifies exactly the injectivity hypothesis (182); no injectivity of all ambient polynomials restricted to the sphere is being used. With $A_v = \|\mathbf{g}_v\|_{\text{coeff}, \zeta}$, finite-dimensional norm equivalence and the finite atlas give

$$A_v \geq c\Lambda \quad \text{for every } v. \quad (210)$$

If $(\alpha H)^D A_v \geq 1$, apply (178) to $\mathbf{g}_v/A_v$, with $L = (\alpha H)^D A_v$ and $\sigma = c_0$; if $(\alpha H)^D A_v < 1$, use the trivial bound. In view of (209)–(210), summing the finitely many charts gives

$$\int_{\mathbb{S}^{n-2}} \min\{1, [\Gamma(\omega)(\alpha H)^D]^{-c_0}\} d\omega \leq C \min\{1, [\Lambda(\alpha H)^D]^{-c_1}\}. \quad (211)$$

All constants here are uniform as $\alpha, H \downarrow 0$: the normalized rectangles, denominator bounds, block-graph degrees, and numbers of sectional monotonicity intervals are fixed. Combining (208) and (211) proves the band bound

$$CB^C \|\Theta\|_\infty^2 \alpha^{n-1} \min\{1, [\Lambda(\alpha H)^D]^{-c}\}.$$

The original correlation is the sum of these band contributions because the preceding cutoffs form a partition of unity. Taking absolute values only after this identity and using their bounded overlap, write $A = \Lambda H^D$. The dyadic band sum is bounded by

$$\sum_{\alpha \in 2^{-\mathbb{N}_0}} \alpha^{n-1} \min\{1, (A\alpha^D)^{-c}\}.$$

If $A \leq 1$, this is $O(1)$. If $A > 1$, choose $\alpha_0 \simeq A^{-1/D}$. The bands below the threshold satisfy

$$\sum_{\alpha \leq \alpha_0} \alpha^{n-1} \lesssim A^{-(n-1)/D}.$$

After decreasing the oscillatory exponent so that $Dc < n - 1$, the remaining bands satisfy

$$\sum_{\alpha > \alpha_0} A^{-c} \alpha^{n-1-Dc} \lesssim A^{-c}.$$

Thus the complete sum is $O(\min\{1, A^{-c'}\})$, where $c' = \min\{c, (n-1)/D\} > 0$, which is the right side of (194).

When $n = 1$, split the support into the finitely many signed intervals on which the two homogeneous kernel factors and radial profiles are of bounded variation. Their product has $L^\infty + \text{BV}$ norm at most $CB^C \|\Theta\|_\infty^2$. Lemma 7.3 gives the asserted unit-scale bound uniformly in h . $\square$

For a homogeneous polynomial $H(x, y)$, let $\mathcal{D}H$ denote the vector of all nonconstant-in- z coefficients of

$$H(y + z, y') - H(y + z, y). \quad (212)$$

Lemma 7.7 (Same-shell degree-block injectivity). *For every integer $m \geq 2$, on homogeneous degree- m polynomials modulo pure phases, the map $[H] \mapsto \mathcal{D}H$ is injective. Hence $\|\mathcal{D}H\|_{\text{coeff}} \geq c_{n,m} \|[H]\|_{\text{coeff}}$.*

Proof. If (212) is independent of z , then $\partial_x H(x, y') - \partial_x H(x, y) = 0$. Thus $\partial_x H$ is independent of its second variable, and integration makes H pure. Quantitative injectivity is finite-dimensional norm equivalence. $\square$

Proposition 7.8 (Flat-measure high block). *Fix a translate B_0 of a prescribed bounded unit-scale cube, and assume that every cube below is contained in B_0 ; the constants are uniform in the location of B_0 . Let $t \geq 0$, let $\mathcal{Q}$ be a finite family of pairwise disjoint q -cubes with $q = 2^{-t}$, and, for each $Q \in \mathcal{Q}$, let v_Q be a complex finite measure supported on Q . Assume $\Lambda \geq 0$ and*

$$|v_Q|(Q) \leq \Lambda|Q|, \quad M = \sum_Q |v_Q|(Q). \quad (213)$$

Suppose P is a real polynomial of degree at most d , its highest nonpure homogeneous block has coefficient norm $L \geq 1$, and $\Theta \in L^\infty(\mathbb{S}^{n-1})$. For every $|e_Q| \leq 1$ and every radial BV profile a , supported in a fixed compact subinterval of $(0, \infty)$ and of $L^\infty + \text{BV}$ norm at most B , where $B \geq 1$, write

$$A_a^{P,\Theta} v(x) = \int e^{iP(x,y)} k_a(x-y) dv(y).$$

Then

$$\left\| A_a^{P,\Theta} \sum_Q e_Q v_Q \right\|_2^2 \leq CB^C \|\Theta\|_\infty^2 \Lambda M (L^{-c} + 2^{-ct}). \quad (214)$$

Proof. After simultaneously translating the input and output variables by the center of B_0 , we may assume that B_0 lies in a fixed reference cube. This translation preserves the kernel geometry and the output L^2 norm; it also preserves the highest nonpure homogeneous block of P and its coefficient norm, since higher homogeneous blocks are pure and remain pure under translation. Put $w_Q = |v_Q|(Q)$. Since the cubes are disjoint and contained in B_0 , the data in (213) give

$$M = \sum_Q w_Q \leq \Lambda \sum_Q |Q| \leq \Lambda |B_0| \leq C\Lambda, \quad \sum_{Q,Q'} w_Q w_{Q'} = M^2 \leq C\Lambda M.$$

Cubes with $w_Q = 0$ may be omitted. Write $dv_Q = h_Q d|v_Q|$, where $|h_Q| = 1$ almost everywhere, and on the finite product probability space

$$\prod_Q \left(Q, \frac{|v_Q|}{w_Q} \right)$$

let Y_Q be the coordinate random variables. Testing against bounded Borel functions gives, in the weak sense of complex measures,

$$\mathbb{E}[w_Q h_Q(Y_Q) \delta_{Y_Q}] = v_Q.$$

The fixed radial support is separated from zero, so uniformly in y

$$\|A_a^{P,\Theta} \delta_y\|_2^2 \leq CB^2 \|\Theta\|_\infty^2.$$

Consequently the finite random sum

$$\mathcal{F}(\mathbf{Y}) = A_a^{P,\Theta} \sum_Q e_Q w_Q h_Q(Y_Q) \delta_{Y_Q}$$

is a strongly measurable, Bochner-integrable L^2 -valued random variable, and

$$A_a^{P,\Theta} \sum_Q e_Q v_Q = \mathbb{E} \mathcal{F} \quad \text{in } L^2.$$

Hilbert-space Jensen gives

$$\left\| A_a^{P,\Theta} \sum_Q e_Q v_Q \right\|_2^2 \leq \mathbb{E} \|\mathcal{F}\|_2^2.$$

It is therefore enough to prove the asserted bound for every fixed selection $y_Q \in Q$, with atomic coefficient $b_Q := e_Q h_Q(y_Q)$ of modulus at most one. Fix such a selection and colour the cubes by the bounded proximity coloring used in Lemma 7.1.

If $1 \leq L < 2$, the trivial correlation bound and $\sum_{Q,Q'} w_Q w_{Q'} \leq C\Lambda M$ on a fixed unit block prove (214). Hence assume $L \geq 2$.

After expanding the square, the correlation phase at a selected pair is $P(y+z, y') - P(y+z, y)$. To distinguish the two coefficient norms used here, write

$$P(y+z, y') - P(y+z, y) = C_0(y, y') + \sum_{1 \leq |\beta| \leq d} C_\beta(y, y') z^\beta, \quad \mathbf{C}(y, y') = (C_\beta(y, y'))_\beta.$$

For a fixed pair (y, y') , the nonconstant- z coefficient norm of the phase is the value norm $\|\mathbf{C}(y, y')\|_{\ell_\beta^2}$. By contrast,

$$N := \|\mathbf{C}\|_{\text{coeff}, (y, y'); \ell_\beta^2} = \left(\sum_{\beta, \gamma, \gamma'} |[y^\gamma(y')^\gamma] C_\beta|^2 \right)^{1/2} \quad (215)$$

is the coefficient norm after the vector-valued function $\mathbf{C}$ is again regarded as a polynomial in (y, y') . Lemma 7.7 and projection onto the top total-degree part give $N \geq cL$; lower-degree homogeneous blocks of P cannot enter that projection. Put $\mathbf{p} = \mathbf{C}/N$, so that this global coefficient norm of $\mathbf{p}$ is one, and define

$$\mathcal{B} = \{(Q, Q') : \|\mathbf{p}(y_Q, y_{Q'})\|_{\ell_\beta^2} \leq L^{-1/3}\}.$$

Choose a scalar component p_i with $\|p_i\|_{\text{coeff}} \geq c$ and apply (179) to $p_i/\|p_i\|_{\text{coeff}}$. The bad vector sublevel is contained in the resulting scalar $CL^{-1/3}$ -sublevel, so that equation bounds its weighted mass by $C\Lambda M(L^{-c} + q^c)$. On the complementary pairs the fixed-pair, nonconstant- z coefficient norm is at least $NL^{-1/3} \geq cL^{2/3}$, uniformly even when lower blocks make N arbitrarily large.

For clarity, this reduction enters the atomic square as follows. If $\mathcal{K}(y, y')$ denotes the correlation after the change $x = y + z$, then, with the convention $\langle F, G \rangle = \int F \overline{G}$ fixed above,

$$\left\| A_a^{P, \Theta} \sum_Q b_Q w_Q \delta_{y_Q} \right\|_2^2 = \sum_{Q, Q'} \overline{b_Q} b_{Q'} w_Q w_{Q'} \mathcal{K}(y_Q, y_{Q'}), \quad (216)$$

$$\mathcal{K}(y, y') = \int e^{i[P(y+z, y') - P(y+z, y)]} \overline{k_a(z)} k_a(z - (y' - y)) dz. \quad (217)$$

Thus absolute values are taken only after the exact finite square expansion, and the contribution of $\mathcal{B}$ is its weighted mass times the trivial correlation bound.

If $n = 1$, the one-dimensional conclusion of Lemma 7.5 has no separation factor. It therefore gives $CB^C \|\Theta\|_\infty^2 L^{-c}$ directly on these complementary pairs. Together with the preceding bad-pair mass bound and $\sum_{Q, Q'} w_Q w_{Q'} \leq C\Lambda M$, this proves (214) in dimension one. Hence assume $n \geq 2$.

For later use, the near-pair estimate is

$$\sum_{Q, Q'} w_Q w_{Q'} \mathbf{1}_{\{|y_Q - y_{Q'}| \leq r\}} \leq C\Lambda M(r + q)^n. \quad (218)$$

Indeed, use the bounded proximity coloring from Lemma 7.1. For one color $\mathfrak{c}$, spread w_Q uniformly on a ball B_Q of radius cq centered at y_Q and put

$$F_{\mathfrak{c}} = \sum_{Q \in \mathfrak{c}} \frac{w_Q}{|B_Q|} \mathbf{1}_{B_Q}.$$

The balls in one color are disjoint, $\|F_{\mathfrak{c}}\|_\infty \leq C\Lambda$, and $\int F_{\mathfrak{c}} \leq M$. A selected pair with $|y_Q - y_{Q'}| \leq r$ smears into $|x - x'| \leq r + Cq$. Thus, for every ordered pair of colors,

$$\iint_{\{|x - x'| \leq r + Cq\}} F_{\mathfrak{c}}(x) F_{\mathfrak{c}'}(x') dx dx' \leq C\Lambda(r + q)^n \int F_{\mathfrak{c}'} \leq C\Lambda M(r + q)^n.$$

Summing the boundedly many color pairs proves (218); when $r + q$ is bounded below, the same conclusion follows directly from $M^2 \leq C\Lambda M$. Take $r = L^{-1/(6d_*)}$, where $d_* = \max(d, 1)$. On a pair outside $\mathcal{B}$ with $|y_Q - y_{Q'}| > r$, if $D_0 \leq d$ is the degree of its nonconstant correlation phase, then

$$|\mathbf{C}(y_Q, y_{Q'})| |y_Q - y_{Q'}|^{D_0} \geq cL^{2/3} r^{D_0} \geq cL^{2/3 - d/(6d_*)} \geq cL^{1/2}.$$

Lemma 7.5 therefore bounds the remaining correlations by $CB^C \|\Theta\|_\infty^2 L^{-c}$. Splitting (216) into bad pairs, nonbad near pairs, and the remaining pairs, and then harmlessly enlarging the near sum to all near pairs, gives, after decreasing $c > 0$ once,

$$\begin{aligned} \left\| A_a^{P,\Theta} \sum_Q b_Q w_Q \delta_{y_Q} \right\|_2^2 &\leq CB^C \|\Theta\|_\infty^2 \left[\sum_{(Q,Q') \in \mathcal{B}} w_Q w_{Q'} + \sum_{|y_Q - y_{Q'}| \leq r} w_Q w_{Q'} + L^{-c} \sum_{Q,Q'} w_Q w_{Q'} \right] \\ &\leq CB^C \|\Theta\|_\infty^2 \Lambda M \{L^{-c} + q^c + (r+q)^n\} \\ &\leq CB^C \|\Theta\|_\infty^2 \Lambda M (L^{-c} + 2^{-ct}). \end{aligned}$$

This is the desired bound for every fixed atomic selection; Jensen returns to the original measures and proves (214). $\square$

7.3 A one-shell phase-adapted packet theorem

Lemma 7.9 (Fourier expansion of a low homogeneous block). *Let $\mathbf{H} = (H_1, \dots, H_N)$ be a fixed basis of one homogeneous degree block modulo pure phases. Fix smooth cutoffs $\chi, \tilde{\chi}$ supported on fixed bounded cubes and equal to one on the relevant smaller cubes. Then, for every $\mathbf{C} \in \mathbb{R}^N$ and every ξ in the specified fixed cube,*

$$\chi(X) \tilde{\chi}(Y) e^{i\mathbf{C} \cdot [\mathbf{H}(X,Y) - \mathbf{H}(\xi,Y)]} = \sum_{u,v \in \mathbb{Z}^n} a_{u,v}(\mathbf{C}, \xi) e^{iu \cdot X} e^{iv \cdot Y},$$

and there are numbers $A_{u,v}(\mathbf{C})$, independent of ξ , such that $|a_{u,v}(\mathbf{C}, \xi)| \leq A_{u,v}(\mathbf{C})$ and

$$\sum_{u,v} A_{u,v}(\mathbf{C}) (1 + |v|) \leq C(1 + \|\mathbf{C}\|)^{C_F}. \quad (219)$$

Proof. Extend the cutoff expression periodically to a fixed torus and integrate by parts more than $2n+2$ times. Every derivative is bounded by a fixed power of $1 + \|\mathbf{C}\|$, uniformly on the compact set of ξ . More precisely, for $N > 2n+2$ the integration-by-parts estimate is

$$|a_{u,v}(\mathbf{C}, \xi)| \leq C_N (1 + \|\mathbf{C}\|)^N (1 + |u| + |v|)^{-N}.$$

Taking the right side as $A_{u,v}(\mathbf{C})$ and summing proves the common-majorant assertion. $\square$

Proposition 7.10 (Phase-adapted one-shell packet principle). *There are $c_P, c_E > 0$ and t_0 , depending only on n, d , such that the following holds. Let P be a real polynomial of degree at most d , let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, and let $R > 0$, $t \in \mathbb{N}$, $q = R2^{-t}$, $t \geq t_0$, and let $\mathcal{Q}$ be a finite family of pairwise disjoint dyadic q -cubes (with R and q dyadic scales), with center c_Q for each $Q \in \mathcal{Q}$. Suppose*

$$\text{supp } d_Q \subset Q, \quad \int d_Q = 0, \quad \|d_Q\|_1 \leq \Lambda |Q|, \quad M = \sum_{Q \in \mathcal{Q}} \|d_Q\|_1. \quad (220)$$

For $R \leq \varepsilon \leq 2R$ put

$$\mathcal{P}_\varepsilon(x) = \sum_{Q \in \mathcal{Q}} \int_{\varepsilon < |x-y| \leq 2R} e^{i[P(x,y) - P(c_Q,y)]} K_\Theta(x-y) d_Q(y) dy.$$

Then $\mathcal{P} = G + E$ with

$$\|V^2(G_\varepsilon : R \leq \varepsilon \leq 2R)\|_2^2 \leq C \|\Theta\|_\infty^2 2^{-c_P t} \Lambda M, \quad (221)$$

$$\|V^1(E_\varepsilon : R \leq \varepsilon \leq 2R)\|_1 \leq C \|\Theta\|_1 2^{-c_E t} M. \quad (222)$$

No pointwise bound on d_Q is assumed. Both estimates remain uniform after replacing every d_Q by $e_Q d_Q$, where $|e_Q| \leq 1$.

Proof. The optional factors e_Q may be absorbed into the packets, so suppress them. Partition input into dyadic R -cubes J and output into dyadic R -cubes I ; only $O_n(1)$ relative positions interact. For one such pair put

$$M_{I,J} = \sum_{Q \subset J} \|d_Q\|_1.$$

Every input block has boundedly many interacting output blocks, and hence

$$\sum_{I,J} M_{I,J} \leq C_n M. \quad (223)$$

The same annular support gives

$$\sup_I \#\{J : I \sim J\} + \sup_J \#\{I : I \sim J\} \leq C_n,$$

where $I \sim J$ denotes an interacting block pair. Let a be the center of J , set $x = a + RX$, $y = a + RY$, and define

$$\Phi(X, Y) = P(a + RX, a + RY) - P(a, a + RY), \quad \xi_Q = \frac{c_Q - a}{R}. \quad (224)$$

Then $\Phi(0, Y) = 0$, and the packet phase is exactly $\Phi(X, Y) - \Phi(\xi_Q, Y)$. The rescaled packets $\tilde{d}_Q(Y) = d_Q(a + RY)$ are supported on disjoint dyadic 2^{-l} -cubes and satisfy

$$\|\tilde{d}_Q\|_1 = R^{-n} \|d_Q\|_1, \quad \|\tilde{d}_Q\|_1 \leq \Lambda |\tilde{Q}|, \quad \tilde{M}_{I,J} := \sum_{Q \subset J} \|\tilde{d}_Q\|_1 = R^{-n} M_{I,J}. \quad (225)$$

All normalized packet cubes lie in the fixed normalized input block $B_0 = R^{-1}(J - a)$, as required in Proposition 7.8. We prove the unit-scale estimate and suppress tildes on the packets, but retain $\tilde{M}_{I,J}$, until the last paragraph. Insert fixed smooth cutoffs which equal one on the normalized interacting input and output blocks; these are the cutoffs in Lemma 7.9.

Write the pure part of Φ as $p(X) + q(Y)$. Its mixed representative vanishes at $X = 0$, so $\Phi(0, Y) = 0$ forces q to be constant. In the anchored difference $p(X) - p(\xi_Q)$ is a common output modulation times a packet scalar and may be removed. Modulo these factors, write the common block phase in descending nonpure homogeneous degrees as

$$\Phi(X, Y) = H_1(X, Y) + \cdots + H_N(X, Y), \quad \deg H_1 > \cdots > \deg H_N \geq 2,$$

so the anchored phase is $\Phi(X, Y) - \Phi(\xi_Q, Y)$. Put $C_r = \|H_r\|_{\text{coeff}}$.

If $N = 0$, only the removed output modulation and packet scalars remain; Proposition 5.5 gives the required unit-block bound with its positive terminal decay exponent and $E = 0$. The rescaling and bounded-overlap summation in the last paragraph then give (221). Hence assume $N \geq 1$ below.

Let c_{flat}, C_B be respectively the decay exponent and the BV power in Proposition 7.8, let δ_T be from Proposition 5.5, and choose $C_{\text{Four}} \geq C_F$ large enough to cover all Fourier losses in Lemma 7.9. Decrease c_{flat} if necessary so that $0 < c_{\text{flat}} \leq 1$. Set

$$\theta = \frac{c_{\text{flat}}}{16(C_B + 1)}.$$

Choose $B_* > 1$ so large that

$$\frac{2C_{\text{Four}}}{B_* - 1} \leq \min\{\theta/2, c_{\text{flat}}/8\}, \quad (226)$$

and then choose $0 < \kappa_* \leq 1$ so small that, with

$$\kappa_r = \kappa_* B_*^{r-N}, \quad S_r = \sum_{i < r} \kappa_i, \quad \gamma_r = \theta \kappa_r, \quad (227)$$

one has

$$2C_{\text{Four}} \sum_{r=1}^N \kappa_r \leq \min\{\delta_T/2, 1/2\}. \quad (228)$$

Then $S_r \leq \kappa_r / (B_* - 1)$ and

$$\gamma_r - 2C_{\text{Four}}S_r \geq \gamma_r/2, \quad (229)$$

$$c_{\text{flat}}\kappa_r - C_B\gamma_r - 2C_{\text{Four}}S_r \geq c_{\text{flat}}\kappa_r/2, \quad (230)$$

$$c_{\text{flat}} - C_B\gamma_r - 2C_{\text{Four}}S_r \geq c_{\text{flat}}/2. \quad (231)$$

Call block r low if $C_r \leq 2^{\kappa_r t}$ and expand it by Lemma 7.9. The product of the preceding low expansions has weighted coefficient ℓ^1 norm at most

$$C2^{C_{\text{Four}}S_r t}. \quad (232)$$

For the nonlinear variation norms we use only finite Fourier sums and pass to the limit afterwards. Write $a_{u_i, v_i}^{(i)}$ and $A_{u_i, v_i}^{(i)}$ for the coefficient and its common majorant in the expansion of a low block i . For $K \in \mathbb{N}$ retain only

$$\mathfrak{F}_K = \{(u_i, v_i) \in \mathbb{Z}^n \times \mathbb{Z}^n : |u_i| + |v_i| \leq K\}$$

in each low expansion. If S is a set of expanded low blocks, define

$$\omega_K(S) = \sum_{(u_i, v_i)_{i \in S} : \max_{i \in S}(|u_i| + |v_i|) > K} \left(\prod_{i \in S} A_{u_i, v_i}^{(i)} \right) \left(1 + \left| \sum_{i \in S} v_i \right| \right), \quad \omega_K(\emptyset) = 0. \quad (233)$$

Since

$$1 + \left| \sum_{i \in S} v_i \right| \leq \prod_{i \in S} (1 + |v_i|),$$

Lemma 7.9 and dominated convergence for the resulting absolutely summable product series give

$$\omega_K(S) \longrightarrow 0 \quad (K \rightarrow \infty), \quad (234)$$

uniformly in all packet centers ξ_Q . Every estimate below is first applied with this finite-mode convention. Its constants are independent of K . Suppose r is the first high block. After a preceding Fourier tuple is fixed, denote its parent-dependent Fourier coefficient by $a_Q(\mathbf{u}, \mathbf{v})$ and the product of the corresponding common majorants in Lemma 7.9 by $A(\mathbf{u}, \mathbf{v})$. Then $|a_Q/A| \leq 1$, and summation of A gives exactly (232). Absorb this normalized scalar together with the factor

$$e^{-i\sum_{i \geq r} H_i(\xi_Q, Y)} e^{i\mathbf{v} \cdot Y}$$

into the complex measure on packet Q . This multiplier need not be constant; total variation and (220) are unchanged. The remaining common operator phase has H_r as its highest nonpure block, and all lower blocks are unrestricted.

For this fixed Fourier tuple denote the resulting measures by μ_Q , and write $\Phi_r = \sum_{i \geq r} H_i$ for the common phase. Partition $[1, 2]$ into $K_r \simeq 2^{\gamma_r t}$ radial cells I_ℓ and set

$$\mathcal{G}_\ell(X) = \sum_Q \int_{\{|X-Y| \in I_\ell\}} e^{i\Phi_r(X, Y)} K_\Theta(X-Y) d\mu_Q(Y).$$

Inserting all cell endpoints into an arbitrary variation partition gives the pointwise deterministic bound

$$\begin{aligned} V^2(\mathcal{H}_\varepsilon : 1 \leq \varepsilon \leq 2) &\leq CV^2 \left(\sum_{\ell \leq j} \mathcal{G}_\ell : 0 \leq j \leq K_r \right) \\ &\quad + C \left(\sum_\ell V^2(\mathcal{H}_\varepsilon : \varepsilon \in I_\ell)^2 \right)^{1/2}, \end{aligned} \quad (235)$$

where $\mathcal{H}$ is this component trajectory. Reversing the order of the grid partial sums, if necessary, does not change their variation. Inside a cell use $V^2 \leq V^1$. Lemma 5.1, whose proof uses only the average mass cap, gives

$$\sup_{\ell} \|W_{\ell}\|_{\infty} \leq C\Lambda \|\Theta\|_{\infty} (2^{-\gamma t} + 2^{-t}), \quad \sum_{\ell} \|W_{\ell}\|_1 \leq C\|\Theta\|_1 \tilde{M}_{I,J}.$$

Thus the squared L^2 norm of the within-cell square function is at most $C\|\Theta\|_{\infty}^2 \tilde{M}_{I,J} 2^{-\gamma t}$.

For the grid partial sums, fix $(\varepsilon_{\ell})_{\ell} \in \{-1, 0, 1\}^{K_r}$ before evaluation in X . The corresponding sharp radial profile $a_{\varepsilon} = \sum_{\ell} \varepsilon_{\ell} \mathbf{1}_{I_{\ell}}$ satisfies

$$\|a_{\varepsilon}\|_{\infty} + \text{Var}(a_{\varepsilon}) \leq CK_r.$$

Proposition 7.8 and Lemma 5.2 therefore give, uniformly over these fixed signs, using $\log^2(2 + K_r) \lesssim (1 + \gamma_r t)^2 \lesssim t^2$,

$$Ct^2 \|\Theta\|_{\infty}^2 2^{C_B \gamma_r t} \Lambda \tilde{M}_{I,J} \{2^{-c_{\text{flat}} K_r t} + 2^{-c_{\text{flat}} t}\}. \quad (236)$$

In particular, no X -dependent selector is inserted into the fixed-sign estimate; all such dependence was already confined to the positive within-cell term in (235). Denote by $G^{(r),I,J}$ the full component obtained after summing all Fourier tuples preceding the first high block r . The variation triangle inequality and the square of the common-majorant sum (232) give the complete estimate

$$\|V^2(G_{\varepsilon}^{(r),I,J})\|_2^2 \leq C\|\Theta\|_{\infty}^2 \Lambda \tilde{M}_{I,J} 2^{2C_{\text{Four}} S_r t} \left[2^{-\gamma_r t} + t^2 2^{C_B \gamma_r t} \{2^{-c_{\text{flat}} K_r t} + 2^{-c_{\text{flat}} t}\} \right]. \quad (237)$$

There are no suppressed exponential factors in this line: the first term is the within-cell square function and the other two are respectively the large-coefficient and mesh terms in (236). The three hierarchy inequalities say, term by term,

$$\begin{aligned} 2C_{\text{Four}} S_r - \gamma_r &\leq -\gamma_r/2, \\ 2C_{\text{Four}} S_r + C_B \gamma_r - c_{\text{flat}} K_r &\leq -c_{\text{flat}} K_r/2, \\ 2C_{\text{Four}} S_r + C_B \gamma_r - c_{\text{flat}} &\leq -c_{\text{flat}}/2. \end{aligned}$$

Since $t^2 2^{-at} \leq C_a 2^{-at/2}$ for $t \geq t_0$, the right side of (237) is bounded by

$$C\|\Theta\|_{\infty}^2 \Lambda \tilde{M}_{I,J} 2^{-c_r t}, \quad c_r = \frac{1}{4} \min\{\gamma_r, c_{\text{flat}} K_r, c_{\text{flat}}\} > 0.$$

This proves every high branch with $E = 0$ without changing any exponent used later.

If every block is low, expand them all. Under the normalized variables in (224), the center of the normalized packet cube is

$$\zeta_Q := \xi_Q = \frac{c_Q - a}{R}. \quad (238)$$

For a combined input frequency ν , write

$$e^{i\nu \cdot Y} d_Q(Y) = e^{i\nu \cdot \zeta_Q} d_Q(Y) + r_{Q,\nu}(Y), \quad r_{Q,\nu}(Y) = (e^{i\nu \cdot Y} - e^{i\nu \cdot \zeta_Q}) d_Q(Y),$$

and, since $|Y - \zeta_Q| \leq C2^{-t}$ on the normalized cube,

$$\|r_{Q,\nu}\|_1 \leq C|\nu| 2^{-t} \|d_Q\|_1.$$

The frozen term remains mean zero, so Proposition 5.5 applies with constant packet coefficients of modulus at most one: divide the parent-dependent Fourier coefficient by its product common majorant and absorb the normalized scalar into the packet. Summation of the remaining majorants is governed by (219), and the output Fourier factor is one fixed modulation. Equation (228) leaves at least $2^{-\delta_{rt}/2}$ after the squared Fourier loss. Young's inequality and the weighted Fourier sum give total L^1 variation at most $C\|\Theta\|_1 2^{-t/2} \tilde{M}_{I,J}$ for the remainders.

We now remove the finite-mode cutoff. For a normalized interacting block pair (I, J) , let

$$\mathcal{P}^{I,J,K} = G^{I,J,K} + E^{I,J,K}$$

be the finite-mode trajectory and decomposition constructed above, with the output restricted to the normalized cube corresponding to I . If r is the first high block, put $\mathcal{J}_r = \{i : i < r\}$ and

$$\mathfrak{B}_{r,t} = 2^{-\gamma_r t} + t^2 2^{C_B \gamma_r t} \{2^{-c_{\text{flat}} K_r t} + 2^{-c_{\text{flat}} t}\}.$$

For $K' > K$, applying the same within-cell and grid estimates only to the Fourier tuples omitted at level K , and then using the variation triangle inequality, gives

$$\|G^{I,J,K'} - G^{I,J,K}\|_{L^2(V^2)} \leq C \|\Theta\|_\infty (\Lambda \tilde{M}_{I,J})^{1/2} \mathfrak{B}_{r,t}^{1/2} \omega_K(\mathcal{J}_r), \quad E^{I,J,K'} - E^{I,J,K} = 0. \quad (239)$$

If every block is low, put $S = \{1, \dots, N\}$. The terminal-packet estimate for the frozen terms and Young's inequality for the remainders give

$$\|G^{I,J,K'} - G^{I,J,K}\|_{L^2(V^2)} \leq C \|\Theta\|_\infty 2^{-\delta_r t/2} (\Lambda \tilde{M}_{I,J})^{1/2} \omega_K(S), \quad (240)$$

$$\|E^{I,J,K'} - E^{I,J,K}\|_{L^1(V^1)} \leq C \|\Theta\|_1 2^{-t} \tilde{M}_{I,J} \omega_K(S). \quad (241)$$

Indeed, in the first line the unweighted coefficient tail is bounded by (233); in the second line the factor $1 + |\sum_i v_i|$ absorbs the freezing error. These are the same estimates used above, with the full coefficient sum replaced by its tail, so no decay exponent is changed.

There are only finitely many interacting block pairs. Hence (234) and the bounded-overlap inequalities show that (G^K) is Cauchy in $L^2(V^2)$ and (E^K) is Cauchy in $L^1(V^1)$. All finite-mode component paths vanish at the upper endpoint $\varepsilon = 2R$. Consequently the same is true of every difference path, and

$$\sup_{R \leq \varepsilon \leq 2R} |H_\varepsilon| \leq V^a(H_\varepsilon : R \leq \varepsilon \leq 2R), \quad a \in \{1, 2\},$$

so convergence in the anchored variation spaces also controls the supremum of the trajectories. Lemma 5.3 constructs anchored limits G, E for the full finite-mode sequences and gives the corresponding $L^2(V^2)$ and $L^1(V^1)$ convergence.

It remains to identify this limit. The common majorants in Lemma 7.9 imply that the truncated Fourier products converge absolutely and uniformly on the fixed normalized input, output, and packet-center boxes. Since the kernel is restricted to $1 < |X - Y| \leq 2$ and there are only finitely many packets, for each block pair

$$\sup_{1 \leq \varepsilon \leq 2} |\mathcal{P}_\varepsilon^{I,J,K}(X) - \mathcal{P}_\varepsilon^{I,J}(X)| \leq o_K(1) C \|\Theta\|_\infty \sum_{Q \subset J} \|d_Q\|_1, \quad o_K(1) \longrightarrow 0,$$

on its normalized output cube. The right side belongs to both L^1 and L^2 there, and the convergence is uniform in ε almost everywhere. The last assertion of Lemma 5.3 identifies $\mathcal{P}^{I,J} = G^{I,J} + E^{I,J}$ (after rescaling, at the physical scale). The Fatou conclusion of that lemma passes the uniform finite-mode bounds to these limits without changing an exponent.

Let $G_\varepsilon^{I,J} = \mathbf{1}_I G_{I,J,\varepsilon}$ and $E_\varepsilon^{I,J} = \mathbf{1}_I E_{I,J,\varepsilon}$ denote the resulting physical paths. Define

$$G_\varepsilon = \sum_I \sum_{J: J \sim I} G_\varepsilon^{I,J}, \quad E_\varepsilon = \sum_I \sum_{J: J \sim I} E_\varepsilon^{I,J}.$$

The block family is finite, and the limiting blockwise identities are exact on I . Therefore $\mathcal{P}_\varepsilon = G_\varepsilon + E_\varepsilon$ for the original trajectory.

Take $c_P > 0$ to be the minimum of $\delta_T/2$ and the finitely many high-branch exponents c_r , over all possible $N \leq d-1$, and take $c_E = 1/2$. At unit scale $\tilde{M}_{I,J} = R^{-n} M_{I,J}$; on returning to x , an L^2 -square and an L^1 norm both acquire the Jacobian R^n . Thus

$$\begin{aligned} \|V^2(G_\varepsilon^{I,J})\|_2^2 &\leq C \|\Theta\|_\infty^2 2^{-c_P t} \Lambda M_{I,J}, \\ \|V^1(E_\varepsilon^{I,J})\|_1 &\leq C \|\Theta\|_1 2^{-c_E t} M_{I,J}. \end{aligned}$$

Since the output cubes I are disjoint and each has only $O_n(1)$ input neighbors, the variation triangle inequality and Cauchy–Schwarz give

$$\|V^2(G_\varepsilon)\|_2^2 \leq C_n \sum_{I,J} \|V^2(G_\varepsilon^{I,J})\|_2^2, \quad \|V^1(E_\varepsilon)\|_1 \leq \sum_{I,J} \|V^1(E_\varepsilon^{I,J})\|_1.$$

Using (223) proves (221)–(222). $\square$

7.4 Exact oriented algebra and the collective high-shell estimate

Let

$$\mathcal{V}_m = \{\text{homogeneous degree-}m \text{ polynomials in } (x, y)\} / \{p(x) + q(y)\}.$$

Each class has a unique representative consisting only of monomials which have positive degree in both x and y . We use this representative whenever an element of $\mathcal{V}_m$ is evaluated. Let $\mathcal{W}_m$ be the space of homogeneous degree- m polynomials in (W, X) . For such a representative put

$$\mathcal{E}H(W, X) = \text{Mix}_{W,X} \{H(W + X, 0) - H(W + X, W)\}. \quad (242)$$

Lemma 7.11 (Oriented graph injectivity). *For every integer $m \geq 2$, the map $\mathcal{E} : \mathcal{V}_m \rightarrow \mathcal{W}_m$ is injective and*

$$\|\mathcal{E}H\|_{\text{coeff}} \geq c_{n,m} \|H\|_{\text{coeff}}. \quad (243)$$

Moreover

$$\mathcal{E}H = \sum_{\substack{A+B=m \\ A,B \geq 1}} E_{A,B}(W, X), \quad (244)$$

where $E_{A,B}$ is bihomogeneous of bidegree (A, B) .

Proof. If the mixed part in (242) vanishes, the expression in braces is $A(W) + B(X)$. At $W = 0$ it vanishes, and hence B is constant. Substituting $x = W + X$ and $y = W$ shows that $H(x, 0) - H(x, y)$ is a function of y alone. Thus H is pure. Quantitative injectivity gives (243) by norm equivalence on the fixed finite-dimensional spaces. Homogeneity and the identities at $W = 0$ and $X = 0$ give (244). $\square$

We next record the exact projective calculation used below. Its purpose is to identify ordinary coefficients of one common numerator; no formal asymptotic expansion in the scale ratio is used.

Lemma 7.12 (Exact projective numerator tags). *Fix $D_0 \geq 1$, a multi-index γ with $q = |\gamma|$, and a polynomial $Q(s) = \sum_{p=0}^P c_p s^p$, where $P \in \mathbb{N}_0$ and $P + q \leq D_0 - 1$. Let $\delta > 0$, $0 < \alpha \leq 1$, and let $\omega \in \mathbb{R}^k$, where k is the length of γ . On the set $D := V - \delta u \neq 0$, put*

$$t = \frac{\delta u}{D}, \quad r_\perp = \frac{\delta \alpha \omega}{D},$$

and put

$$F_{q,Q}(u, V, \omega) = r_\perp^\gamma Q(t).$$

Then

$$D^{D_0+2} \partial_u \partial_V F_{q,Q} = \alpha^q \omega^\gamma \sum_{j=0}^{D_0-q} \delta^{q+1+j} \tau_{q,j}(Q) u^j V^{D_0-q-j}, \quad (245)$$

where the functionals $\tau_{q,j}$ depend only on D_0, q, j . If $q = 0$ and $P \geq 1$, the map

$$(c_1, \dots, c_P) \longmapsto (\tau_{0,0}(Q), \dots, \tau_{0,P-1}(Q))$$

is invertible. If $q \geq 1$, the map

$$(c_0, \dots, c_P) \longmapsto (\tau_{q,0}(Q), \dots, \tau_{q,P}(Q))$$

is invertible. The inverse norms are bounded in terms of D_0 only. When $q = P = 0$, the first map is the empty map and the assertion is vacuous. In particular, every detecting tag in the case $q = 0$, $P \geq 1$ has order $\kappa = q + 1 + j \leq P$, and in the case $q \geq 1$ it has order $\kappa = q + 1 + j \leq q + P + 1$. After multiplication by D^{D_0+2} , the displayed equality is a polynomial identity. Although the derivation below uses the dense open set on which $DV \neq 0$, the cleared identity therefore extends across both $D = 0$ and $V = 0$.

Proof. Set $z = \delta u/V$ and

$$H_{q,Q}(z) = (1-z)^{-q} Q\left(\frac{z}{1-z}\right).$$

Apart from the factor $\alpha^q \omega^\gamma$, one has

$$F_{q,Q} = \delta^q V^{-q} H_{q,Q}(z).$$

Consequently

$$\partial_u \partial_V F_{q,Q} = -\alpha^q \omega^\gamma \delta^{q+1} V^{-q-2} [(q+1)H'_{q,Q}(z) + zH''_{q,Q}(z)]. \quad (246)$$

Multiplication by $D^{D_0+2} = V^{D_0+2}(1-z)^{D_0+2}$ gives a polynomial. The coefficient of z^j becomes the ordinary coefficient of $u^j V^{D_0-q-j}$, with the exact factor δ^{q+1+j} . This proves (245).

Write $H_{q,Q}(z) = \sum_{s \geq 0} h_s z^s$. The coefficient of z^{s-1} in the bracket in (246) is $s(q+s)h_s$. Multiplication by $(1-z)^{D_0+2}$ is triangular with diagonal one. It therefore suffices to inspect $(h_1, \dots, h_P)$ when $q = 0$, and $(h_1, \dots, h_{P+1})$ when $q \geq 1$. For $q = 0$, $Q(z/(1-z))$ is triangular with diagonal one on $(c_1, \dots, c_P)$. For $q \geq 1$, if the latter $P+1$ coefficients vanished, then, after $s = z/(1-z)$,

$$(1+s)^q Q(s) = c_0 + O(s^{P+2}).$$

The first P equations make Q equal to c_0 times the degree- P Taylor polynomial of $(1+s)^{-q}$, while the coefficient of s^{P+1} is a nonzero multiple of $c_0 \binom{q+P}{P+1}$. Hence $c_0 = 0$, and then all $c_p = 0$. Finite-dimensional norm equivalence gives the quantitative assertion. $\square$

We next define the unequal-shell phase appearing in the correlation. Throughout the remainder of this subsection assume that P has a highest nonpure homogeneous block P_m ; necessarily $m \geq 2$. Write, modulo pure phases,

$$P(a+X, a+Y) = p_a(X) + q_a(Y) + \sum_{\ell=2}^m H_{\ell,a}(X, Y), \quad (247)$$

where every $H_{\ell,a}$ is homogeneous of degree ℓ , and where $H_{m,a} = P_m$. For $W, X \in \mathbb{R}^n$, set

$$\Phi_{a,S}(W, X) = P(a+S(W+X), a) - P(a+S(W+X), a+SW). \quad (248)$$

Modulo a function independent of X ,

$$\Phi_{a,S}(W, X) \equiv \sum_{\ell=2}^m S^\ell \mathcal{E} H_{\ell,a}(W, X). \quad (249)$$

Indeed, the pure output phase cancels, the pure input phase is independent of X , and a mixed representative has no pure- X term in the difference defining $\mathcal{E}$.

Fix once and for all annular constants $0 < c_* < C_* < \infty$ containing all values of $|W|$ allowed by the two radial supports, and then choose C_0 sufficiently large, depending only on those fixed supports and annular constants. Thus $R \leq S/C_0$ makes $\delta = R/(Sp)$ uniformly small throughout the overlap region.

For completeness we now give a concrete rational atlas and frame. For $1 \leq j \leq n$ and $s \in \{-1, 1\}$ put $c_{j,s} = se_j$ and

$$U_{j,s} = \{\theta \in \mathbb{S}^{n-1} : c_{j,s} \cdot \theta > (2\sqrt{n})^{-1}\}, \quad U_{j,s}^* = \{\theta \in \mathbb{S}^{n-1} : c_{j,s} \cdot \theta > (4\sqrt{n})^{-1}\}.$$

The smaller sets cover the sphere and $U_{j,s} \in U_{j,s}^*$. Choose a fixed signed permutation $C_{j,s} \in O(n)$ with $C_{j,s}e_1 = c_{j,s}$. In stereographic coordinates $\vartheta \in \mathbb{R}^{n-1}$ from the pole $-c_{j,s}$, write

$$\theta_{j,s}(\vartheta) = C_{j,s} \left(\frac{1 - |\vartheta|^2}{1 + |\vartheta|^2}, \frac{2\vartheta}{1 + |\vartheta|^2} \right).$$

On $U_{j,s}^*$ the coordinate ϑ ranges over a fixed bounded set; the chart Jacobian and its inverse are therefore uniformly bounded. Let $a_{j,s} = -c_{j,s}$, choose a fixed signed permutation $B_{j,s} \in O(n)$ satisfying $B_{j,s}e_1 = a_{j,s}$, and define the Householder frame

$$O_{j,s}(\vartheta) = \left(I - 2 \frac{(a_{j,s} - \theta_{j,s}(\vartheta))(a_{j,s} - \theta_{j,s}(\vartheta))^T}{|a_{j,s} - \theta_{j,s}(\vartheta)|^2} \right) B_{j,s}. \quad (250)$$

This matrix is orthogonal and $O_{j,s}(\vartheta)e_1 = \theta_{j,s}(\vartheta)$. Moreover,

$$|a_{j,s} - \theta|^2 = 2(1 + c_{j,s} \cdot \theta) \geq 2(1 + (4\sqrt{n})^{-1}) \quad \text{on } U_{j,s}^*.$$

Thus every matrix entry is rational in ϑ ; after taking the fixed products needed as common denominators, those denominators and their reciprocals are uniformly bounded on the enlarged chart domains. This also remains true for the finitely many fixed powers that occur below.

Order these $2n$ charts as $U_1, \dots, U_{2n}$, relabeling the corresponding maps as θ_χ and O_χ , and make the smaller cover disjoint by putting

$$E_1 = U_1, \quad E_\chi = U_\chi \setminus \bigcup_{v < \chi} U_v.$$

These are measurable, disjoint, and cover the sphere. Write $\chi(W)$ for the unique active index satisfying $W/|W| \in E_{\chi(W)}$. Finally, the annular coordinate map $(\rho, \vartheta) \mapsto W = \rho\theta_\chi(\vartheta)$ is uniformly bi-Lipschitz for $c_* \leq \rho \leq C_*$ on each enlarged chart. We use the following precise convention. A *cleared chart-polynomial vector* on this annulus is a vector F_χ for which

$$\tilde{F}_\chi(\rho, \vartheta) := F_\chi(\rho\theta_\chi(\vartheta))$$

is an ordinary polynomial vector of degree at most a fixed structural bound, after the fixed common denominator has been incorporated into F_χ . Its chart coefficient norm is

$$\|F_\chi\|_{\text{ch}, \chi} := \|\tilde{F}_\chi\|_{\text{coeff}, (\rho, \vartheta)}. \quad (251)$$

On the fixed enlarged coordinate box, finite-dimensional norm equivalence gives

$$\|\tilde{F}_\chi\|_\infty + \|\nabla_{\rho, \vartheta} \tilde{F}_\chi\|_\infty \leq C\|F_\chi\|_{\text{ch}, \chi}. \quad (252)$$

Any two fixed clearing conventions give equivalent norms: after comparing supremum norms their bounded, nowhere-vanishing denominator ratio is absorbed, and coefficient/supremum norm equivalence is used on the fixed polynomial spaces. In particular, all later polynomial sublevel arguments are made in (ρ, ϑ) coordinates with (251); they do not regard $\rho = |W|$ as an ambient polynomial in W .

In frame coordinates define

$$G_{\ell, a, \chi}(\vartheta; T) = \rho^{-\ell} \mathcal{E}H_{\ell, a}(\rho O_\chi e_1, \rho O_\chi T). \quad (253)$$

This is independent of ρ , is a polynomial in T with no constant term, and has degree at most $\ell - 1$.

Proposition 7.13 (Oriented unequal-shell coefficient). *Assume $n \geq 2$. Let P_m be the highest nonpure homogeneous block of a real polynomial P of degree at most d , put*

$$L_R = R^m \|P_m\|_{\text{coeff}},$$

and suppose $0 < R \leq S/C_0$. For $y = a + SW$, assume $c_* \leq |W| \leq C_*$, where $a, W \in \mathbb{R}^n$; put $\rho = |W|$. On every chart χ there is a bounded-degree cleared chart-polynomial vector $\mathbf{p}_{a,R,S,\chi}(W)$, independent of the polar band parameter α , such that

$$\|\mathbf{p}_{a,R,S,\chi}\|_{\text{coeff}} \geq 1. \quad (254)$$

On the disjoint active-chart partition define

$$\mathbf{p}_{a,R,S}(W) := \mathbf{p}_{a,R,S,\chi(W)}(W). \quad (255)$$

Thus $\mathbf{p}$ is piecewise chart-polynomial, with a unique value at every W . Every later incidence argument is applied after splitting the atoms into the finitely many active sets E_χ . Here $\|\mathbf{p}\|_{\text{coeff}}$ means the chart norm (251), whereas $|\mathbf{p}(W)|$ denotes the Euclidean norm in its fixed finite-dimensional value space. The coefficient lower bound in (254) is not asserted to be a pointwise lower bound; small values of $|\mathbf{p}(W)|$ are handled by the incidence estimate.

More precisely, put $\delta = R/(Sp)$. For $0 < \alpha \leq 1$, $\omega \in \mathbb{S}^{n-2}$, and on each member of the fixed finite family of (u, V) -rectangles arising from the radial supports, use variables

$$D = V - \delta u, \quad T = T_{\delta, \alpha, \omega}(u, V) = \frac{(\delta u, \delta \alpha \omega)}{D}. \quad (256)$$

Every such rectangle has a fixed enlargement on which $D = V - \delta u$ is uniformly bounded above and away from zero. The actual unequal-shell phase is

$$\begin{aligned} \Psi_{a,R,S,W,\alpha,\omega}(u, V) &= \Phi_{a,S}(W, \rho O_\chi T_{\delta, \alpha, \omega}(u, V)) \\ &= P\left(a + Sp O_\chi \frac{(V, \delta \alpha \omega)}{D}, a\right) \\ &\quad - P\left(a + Sp O_\chi \frac{(V, \delta \alpha \omega)}{D}, a + Sp O_\chi e_1\right). \end{aligned} \quad (257)$$

Since $\deg P \leq d$, the phase in (257) has the canonical common denominator

$$Q(u, V) = D^d = (V - \delta u)^d.$$

Define its literal mixed numerator by

$$\mathcal{N}_\Psi(u, V) := D^{d+2} \partial_u \partial_V \Psi(u, V). \quad (258)$$

This is a polynomial of bounded degree. For the canonical presentation,

$$\mathfrak{M}_{D^d} \Psi = D^{3d} \partial_u \partial_V \Psi = D^{2d-2} \mathcal{N}_\Psi.$$

Multiplication by D^{2d-2} is uniformly bounded below in coefficient norm on these fixed polynomial spaces: this follows from the equivalence of coefficient and supremum norms and the lower bound for $|D|$ on the enlarged rectangle. Equation (189) therefore gives

$$\|[\Psi]\|_{\text{rat}, D^d} \geq c \|\mathcal{N}_\Psi\|_{\text{coeff}(u, V)}. \quad (259)$$

For every α there is a bounded-degree vector $\mathbf{q}_{\alpha, W}(\omega)$, made from selected ordinary coefficients of $\mathcal{N}_\Psi$, such that

$$\|[\Psi_{a,R,S,W,\alpha,\omega}]\|_{\text{rat}, D^d} \geq c |\mathbf{q}_{\alpha, W}(\omega)|, \quad (260)$$

$$\|\mathbf{q}_{\alpha, W}\|_{\text{coeff}, \omega} \geq c \alpha^d L_R |\mathbf{p}_{a,R,S,\chi}(W)|. \quad (261)$$

The constants are independent of a and of all coefficients of total degree less than m .

Proof. By (249) and (253), modulo $F(u) + G(V)$ the phase in (257) is

$$\sum_{\ell=2}^m S^\ell \rho^\ell G_{\ell,a,\chi} \left(\vartheta; \frac{(\delta u, \delta \alpha \omega)}{D} \right). \quad (262)$$

Expand

$$G_{\ell,a,\chi}(\vartheta; t, r_\perp) = \sum_{q, |\gamma|=q} r_\perp^\gamma Q_{\ell,a,\chi,\gamma}(\vartheta; t).$$

Here $\deg Q_{\ell,a,\chi,\gamma} + q \leq \ell - 1$, and the constant coefficient of $Q_{\ell,a,\chi,0}$ is zero because $G_{\ell,a,\chi}(\vartheta; 0) = 0$. Lemma 7.12, used with $D_0 = d$, shows that the coefficient of $\omega^\gamma u^j V^{d-q-j}$ in $\mathcal{N}_\Psi$ is exactly

$$C_{q,\gamma,j,\alpha}(\rho, \vartheta) = \alpha^q \sum_{\ell=2}^m R^\ell S^{\ell-\kappa} \rho^{\ell-\kappa} \tau_{q,j}(Q_{\ell,a,\chi,\gamma}), \quad \kappa = q + 1 + j. \quad (263)$$

Thus different tag orders are different ordinary numerator monomials; there is no appeal to a formal expansion in δ .

We make the finite-dimensional coefficient step explicit. Fix the mixed monomial basis of $\mathcal{V}_m$, and use ordinary monomial coefficients in the chart variable ϑ . Choose once and for all a nonvanishing polynomial $\Delta_\chi(\vartheta)$ which clears every denominator introduced by the frame O_χ in degrees at most d ; both Δ_χ and its reciprocal are bounded on U_χ^* . For $T = (t, r_\perp) \in \mathbb{R} \times \mathbb{R}^{n-1}$ define the cleared restriction map

$$(R_\chi H)(\vartheta; T) = \Delta_\chi(\vartheta) \mathcal{E}H(O_\chi(\vartheta)e_1, O_\chi(\vartheta)T). \quad (264)$$

By homogeneity this is precisely $\Delta_\chi G_{m,a,\chi}$ when $H = P_m$; in particular it is independent of a and ρ . Write

$$R_\chi H = \sum_\gamma r_\perp^\gamma \tilde{Q}_{\chi,\gamma}(\vartheta; t), \quad q = |\gamma|.$$

The $q = 0$ polynomial has zero constant coefficient, and $\deg_t \tilde{Q}_{\chi,\gamma} + q \leq m - 1$.

Let

$$\mathfrak{I}_m = \{(0, j) : 0 \leq j \leq m - 2\} \cup \{(\gamma, j) : 1 \leq |\gamma| \leq m - 1, 0 \leq j \leq m - 1 - |\gamma|\}, \quad (265)$$

where 0 denotes the zero multi-index, and put $\kappa(\gamma, j) = |\gamma| + 1 + j$. Let D_χ be a fixed degree bound for the cleared chart polynomials and set

$$\mathcal{L}_{m,\chi} = \bigoplus_{(\gamma,j) \in \mathfrak{I}_m} \mathcal{P}_{\leq D_\chi}(\vartheta),$$

with the Euclidean norm of all ordinary monomial coefficients. All tag functionals below are those of Lemma 7.12 with $D_0 = d$, extended coefficientwise in ϑ . Define

$$\mathsf{T}_\chi : \mathcal{V}_m \longrightarrow \mathcal{L}_{m,\chi}, \quad \mathsf{T}_\chi H = (\tau_{|\gamma|,j}(\tilde{Q}_{\chi,\gamma}(\vartheta; \cdot)))_{(\gamma,j) \in \mathfrak{I}_m}. \quad (266)$$

Multiplication by Δ_χ commutes with the tag functionals because it is independent of t and $r_\perp$. Lemma 7.12 therefore recovers, separately for each γ , all coefficients of $\tilde{Q}_{\chi,\gamma}$: for $q = 0$ it recovers the nonconstant t -coefficients, and for $q \geq 1$ it recovers all of them. Thus vanishing of $\mathsf{T}_\chi H$ implies $R_\chi H = 0$.

The restriction map R_χ is injective. Indeed, its vanishing implies $\mathcal{E}H(\theta, X) = 0$ for every X and every direction θ in the open chart. In the bidegree decomposition (244), distinct X -degrees separate the terms $E_{A,B}$; each is homogeneous in W , so vanishing on an open patch of the unit sphere extends to the corresponding open cone. Hence $\mathcal{E}H = 0$, and Lemma 7.11 gives $H = 0$ in $\mathcal{V}_m$. Consequently T_χ is injective. Its uniform injectivity verifies the application-specific hypothesis (182) for the oriented tag space; no blanket injectivity statement for ambient polynomials on the sphere is used. Its uniform constant is the genuine least singular value

$$\sigma_\chi = \min_{\|H\|_{\text{coeff}}=1} \|\mathsf{T}_\chi H\|_{\text{coeff}} > 0, \quad \sigma_* := \min_\chi \sigma_\chi > 0. \quad (267)$$

The minimum is taken on the unit sphere of the fixed finite-dimensional space $\mathcal{V}_m$; the maps depend only on n, m, d and the fixed atlas, not on a, R, S, α or any coefficient of a lower homogeneous block.

Now retain exactly the coordinates in (265) and multiply (263) by ρ^κ . This is a uniformly invertible diagonal change on $c_* \leq \rho \leq C_*$. For $\alpha = 1$, after the fixed denominator is cleared, the selected polynomial vector is

$$\widehat{C}_{q,\gamma,j}(\rho, \vartheta) = \Delta_\chi(\vartheta) \sum_{\ell=2}^m R^\kappa S^{\ell-\kappa} \rho^\ell \tau_{q,j}(\mathcal{Q}_{\ell,a,\chi,\gamma}). \quad (268)$$

Let π_m be ordinary coefficient projection onto radial degree ρ^m . Since the chart denominator and the tag selection are independent of ρ , both commute with π_m , and therefore

$$\pi_m \widehat{C}_{a,R,S,\chi} = D_{R,S} \top_\chi P_m, \quad (D_{R,S} v)_{\gamma,j} = R^{\kappa(\gamma,j)} S^{m-\kappa(\gamma,j)} v_{\gamma,j}. \quad (269)$$

This identity also shows directly why no translated lower-degree term can enter a protected coordinate: the block of total degree ℓ carries exactly the radial monomial ρ^ℓ , while $H_{m,a} = P_m$. For every retained coordinate, $1 \leq \kappa(\gamma, j) \leq m$; hence, since $R \leq S$,

$$\|D_{R,S} v\|_{\text{coeff}} \geq R^m \|v\|_{\text{coeff}}. \quad (270)$$

Combining (267)–(270) with the bounded coefficient projection gives

$$\|\widehat{C}_{a,R,S,\chi}\|_{\text{coeff}} \geq c_0 R^m \|P_m\|_{\text{coeff}} = c_0 L_R, \quad (271)$$

uniformly in all lower blocks. For orientation, the retained orders are $\{1, 2\}$ when $m = 2$ and $\{1, 2, 3\}$ when $m = 3$, giving respectively the scale factors RS, R^2 and $RS^2, R^2 S, R^3$. Define

$$\widetilde{\mathbf{p}}_{a,R,S,\chi}(\rho, \vartheta) = \frac{\widehat{C}_{a,R,S,\chi}}{c_0 L_R}.$$

This is an ordinary polynomial vector in (ρ, ϑ) because the fixed factor Δ_χ has already cleared all frame denominators. Define

$$\mathbf{p}_{a,R,S,\chi}(\rho \theta_\chi(\vartheta)) := \widetilde{\mathbf{p}}_{a,R,S,\chi}(\rho, \vartheta).$$

This proves (254).

For fixed W , group the selected coefficients in (263) according to (q, j) , leaving the homogeneous degree- q polynomial in ω unexpanded; after the harmless factors $\rho^\kappa \Delta_\chi$ are inserted, call the resulting vector $\mathbf{q}_{\alpha,W}(\omega)$. It consists of selected coefficients of the literal numerator $\mathcal{N}_\Psi$, up to a uniformly invertible diagonal change, and therefore (260) follows from (259). The dependence on α is exactly diagonal:

$$\widehat{C}_{q,\gamma,j,\alpha} = \alpha^q \widehat{C}_{q,\gamma,j,1}.$$

Since $0 < \alpha \leq 1$ and $0 \leq q \leq d$, the least singular value of this diagonal map is at least α^d . Consequently

$$\|\mathbf{q}_{\alpha,W}\|_{\text{coeff},\omega} \geq c \alpha^d |\widehat{C}_{a,R,S,\chi}(W)| \geq c \alpha^d L_R |\mathbf{p}_{a,R,S,\chi}(W)|.$$

This proves (261) and also proves that the same $\mathbf{p}$ works on every polar band. $\square$

Lemma 7.14 (Homogeneous angular restriction). *Let $k \geq 1$ and*

$$\mathbf{Q} = \bigoplus_{q=0}^d \mathbf{Q}^{[q]},$$

where the summands occupy disjoint value coordinates and every entry of $\mathbf{Q}^{[q]}$ is a homogeneous degree- q polynomial on $\mathbb{R}^k$. If $k \geq 2$, fix a finite rational atlas $\omega = \omega_v(\zeta)$ on $\mathbb{S}^{k-1}$ and fixed denominators Δ_v clearing all pullbacks through degree d . Then, on every nonempty chart,

$$\left\| \bigoplus_{q=0}^d \Delta_v(\zeta) \mathbf{Q}^{[q]}(\omega_v(\zeta)) \right\|_{\text{coeff},\zeta} \geq c_{k,d,v} \|\mathbf{Q}\|_{\text{coeff}}. \quad (272)$$

For $k = 1$, evaluation on $\mathbb{S}^0 = \{-1, 1\}$ has the same property.

Proof. The displayed pullback is a linear map between fixed finite-dimensional coefficient spaces. If it vanishes on a nonempty chart, the disjoint value coordinates show separately that every homogeneous block $\mathbf{Q}^{[q]}$ vanishes on an open patch of the sphere. Homogeneity extends this vanishing to the corresponding open cone in $\mathbb{R}^k$, so every block is the zero polynomial. Thus the pullback is injective, and its least singular value gives (272). When $k = 1$, an entry is $c\omega^q$, whose value at either point of $\mathbb{S}^0$ has modulus $|c|$. Notice that keeping distinct degrees in disjoint value coordinates is essential; no quotient by the sphere relation $|\omega|^2 = 1$ is being taken. $\square$

Corollary 7.15 (Rough unequal-shell alternative). *Let P be a real polynomial of degree at most d with highest nonpure homogeneous block P_m , let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, and let $0 < R \leq S/C_0$. Put $L_R = R^m \|P_m\|_{\text{coeff}}$, and let b_R, b_S be supported in a fixed compact subinterval of $(0, \infty)$, with $L^\infty + \text{BV}$ norm at most B , where $B \geq 1$. Put*

$$k_R(z) = |z|^{-n} \Theta(z/|z|) b_R(|z|/R), \quad k_S(z) = |z|^{-n} \Theta(z/|z|) b_S(|z|/S).$$

Fix $a, W \in \mathbb{R}^n$, put $y = a + SW$, and assume $c_ \leq |W| \leq C_*$. Use the active-chart vector $\mathbf{p}(W) = \mathbf{p}_{a,R,S,\chi(W)}(W)$ from (255). Then, for $n \geq 2$,*

$$\left| \int e^{i[P(y+z,a) - P(y+z,y)]} \overline{k_R(z)} k_S(z+y-a) dz \right| \leq CB^C \|\Theta\|_\infty^2 S^{-n} \min\{1, (L_R |\mathbf{p}(W)|)^{-c}\}. \quad (273)$$

If $\mathbf{p}(W) = 0$, the last factor is understood to be 1. For $n = 1$ the same conclusion holds, with S^{-1} , for a bounded-degree polynomial vector $\mathbf{p}(W)$ satisfying $\|\mathbf{p}\|_{\text{coeff}} \geq 1$.

Proof. If the two radial supports have no overlap after the translation by SW , the integral is zero in every dimension. Assume that the overlap is nonempty, and first let $n \geq 2$. Then $|z| \simeq R$, $|z + SW| \simeq S$, and the contributing set has measure $O(R^n)$; hence

$$\int |k_R(z) k_S(z + SW)| dz \leq CB^2 \|\Theta\|_\infty^2 S^{-n}. \quad (274)$$

This proves (273) when $\mathbf{p}(W) = 0$. Henceforth assume that $\mathbf{p}(W) \neq 0$. Write $W = \rho\theta$, choose the active chart so that $\theta = O_\chi(\vartheta)e_1$, and let $\omega \in \mathbb{S}^{n-2} \subset \mathbb{R}^{n-1}$ denote a transverse frame coordinate. Put

$$\omega_\chi = O_\chi(\vartheta)(0, \omega) \in \theta^\perp$$

and decompose $z = t\theta + r\omega_\chi$, where $r > 0$. Put

$$\xi = \frac{t}{r}, \quad \eta = \frac{t + S\rho}{r}.$$

Thus the two rays are $z = r(\xi\theta + \omega_\chi)$ and $z + SW = r(\eta\theta + \omega_\chi)$. Choose a zero-extended bounded-variation partition $\{\beta_k\}_{k \geq 0}$ in the dimensionless transverse radius such that, on the relevant range,

$$\sum_{k \geq 0} \beta_k(r/R) = 1, \quad \text{supp } \beta_k \subset [c2^{-k}, C2^{-k}], \quad \sup_k (\|\beta_k\|_\infty + \text{Var}(\beta_k)) \leq C,$$

and the supports have uniformly bounded overlap. Absorb the boundedly many largest transverse pieces into $k = 0$, put $\alpha = 2^{-k}$, and on the k -th band write $r \simeq \alpha R$. Set

$$\delta = \frac{R}{S\rho}, \quad u = \alpha\xi, \quad V = \alpha\delta\eta, \quad D = V - \delta u. \quad (275)$$

Then, exactly,

$$t = \frac{Ru}{D}, \quad r = \frac{\alpha R}{D}, \quad z = \frac{R(u\theta + \alpha\omega_\chi)}{D}, \quad z + SW = \frac{S\rho(V\theta + \alpha\delta\omega_\chi)}{D}. \quad (276)$$

Here $D = \alpha R/r > 0$. Moreover $z/S = \rho O_\chi T_{\delta, \alpha, \omega}(u, V)$, so the phase in these variables is exactly (257).

Indeed, cylindrical measure is $r^{n-2} dt dr d\omega$, and

$$\left| \frac{\partial(t, r)}{\partial(u, V)} \right| = \frac{\alpha R^2}{D^3}.$$

Since

$$|z| = \frac{R\sqrt{u^2 + \alpha^2}}{D}, \quad |z + SW| = \frac{S\rho\sqrt{V^2 + \alpha^2\delta^2}}{D},$$

the cylindrical Jacobian and the two homogeneous kernels give the exact factor

$$(S\rho)^{-n} \alpha^{n-1} D^{n-1} (\alpha^2 + u^2)^{-n/2} (V^2 + \alpha^2\delta^2)^{-n/2} du dV d\omega. \quad (277)$$

The radial profiles become

$$\overline{b_R \left(\frac{\sqrt{u^2 + \alpha^2}}{D} \right)}, \quad \overline{b_S \left(\frac{\rho\sqrt{V^2 + \alpha^2\delta^2}}{D} \right)}, \quad (278)$$

while the two rough factors are, respectively,

$$\overline{\Theta \left(\frac{u\theta + \alpha\omega_\chi}{\sqrt{u^2 + \alpha^2}} \right)}, \quad \overline{\Theta \left(\frac{V\theta + \alpha\delta\omega_\chi}{\sqrt{V^2 + \alpha^2\delta^2}} \right)}. \quad (279)$$

Thus neither rough angular factor is differentiated.

On the support of one transverse band and of the two radial profiles, u, V lie in fixed compact sets and D is bounded above and away from zero. The profile support also gives $\sqrt{u^2 + \alpha^2} \gtrsim 1$ and $\sqrt{V^2 + \alpha^2\delta^2} \gtrsim 1$; hence the two possibly singular weights in (277) and all of their sectional derivatives are uniformly bounded. After a bounded number of sign and monotonicity pieces, the remaining amplitude in (277)–(278), including the transverse band cutoff, has uniform row and column L^1 bounds and sectional $L^\infty + \text{BV}$ norm at most CB^C . This follows from the one-dimensional BV chain rule on each monotonicity piece. Explicitly, the two profile arguments are

$$\gamma_1(u, V) = \frac{\sqrt{u^2 + \alpha^2}}{V - \delta u}, \quad \gamma_2(u, V) = \frac{\rho\sqrt{V^2 + \alpha^2\delta^2}}{V - \delta u}.$$

Writing $s_1 = (u^2 + \alpha^2)^{1/2}$, $s_2 = (V^2 + \alpha^2\delta^2)^{1/2}$, and $D = V - \delta u$, direct differentiation gives

$$\partial_u \gamma_1 = \frac{uV + \delta\alpha^2}{s_1 D^2}, \quad \partial_V \gamma_1 = -\frac{s_1}{D^2}, \quad \partial_u \gamma_2 = \frac{\rho\delta s_2}{D^2}, \quad \partial_V \gamma_2 = -\frac{\rho\delta(uV + \delta\alpha^2)}{s_2 D^2}.$$

Thus the only nontrivial sectional critical-point equation is the linear equation $uV + \delta\alpha^2 = 0$ in the varying variable. Each section therefore has a uniformly bounded number of monotonicity pieces; on each piece $\text{Var}(b_R \circ \gamma_1) \leq \text{Var}(b_R)$ and $\text{Var}(b_S \circ \gamma_2) \leq \text{Var}(b_S)$. More explicitly, on an interval I on which γ is monotone,

$$\text{Var}_I(b \circ \gamma) \leq \text{Var}_{\gamma(I)}(b) \leq \text{Var}(b),$$

and the conclusion is trivial if γ is constant. The numerator $uV + \delta\alpha^2$ is affine in either varying coordinate, so each section requires at most one additional cut, independently of α, δ, ρ . Products are controlled by

$$\text{Var}(fg) \leq \|f\|_\infty \text{Var}(g) + \|g\|_\infty \text{Var}(f).$$

After extracting the explicit factor α^{n-1} in (277), the displayed lower bounds for D and the two square roots give uniformly bounded sectional $W^{1,\infty}$ norms for all remaining algebraic factors. Hence the sectional $L^\infty + \text{BV}$ bounds and the Schur row and column bounds remain uniform as $\alpha, \delta \downarrow 0$.

Fix a compact rectangle containing all the (u, V) -supports above, and partition it by a rectangular grid of sufficiently small fixed mesh. For each α, δ, ρ , retain only the grid rectangles meeting the corresponding support. Since on that support

$$D, \quad \sqrt{u^2 + \alpha^2}, \quad \sqrt{V^2 + \alpha^2 \delta^2}$$

are bounded below uniformly, and these functions are uniformly Lipschitz in (u, V) for $0 < \alpha \leq 1$ and $0 < \delta \leq \delta_0$, the mesh may be chosen so that the same quantities remain bounded below on a fixed enlargement of every retained rectangle. A fixed tensor-product partition subordinate to the grid has uniformly bounded sectional $L^\infty + \text{BV}$ norms. Refining each section at the at most one zero of $uV + \delta\alpha^2$ gives the monotonicity partitions required by Lemma 7.4. The number and interaction degree of all rectangles are uniform in α, δ, ρ , so their L^2 bounds synthesize by Cauchy–Schwarz with only a fixed loss.

Let $\mathcal{R}_{\alpha, \delta, \rho}^{\text{rect}}$ denote the uniformly finite family of retained rectangles, including the tensor-product partition just constructed. For a rectangle $\mathfrak{b} \in \mathcal{R}_{\alpha, \delta, \rho}^{\text{rect}}$ and fixed ω , let $T_{\alpha, W, \omega}^{\mathfrak{b}}$ be the corresponding operator from the V -interval to the u -interval, after the explicit factor $(S\rho)^{-n}\alpha^{n-1}$ in (277) has been removed. Put, with zero extension to the fixed ambient intervals,

$$\begin{aligned} f_{\alpha, W, \omega}(V) &= \Theta \left(\frac{V\theta + \alpha\delta\omega_\chi}{\sqrt{V^2 + \alpha^2\delta^2}} \right), \\ g_{\alpha, W, \omega}(u) &= \Theta \left(\frac{u\theta + \alpha\omega_\chi}{\sqrt{u^2 + \alpha^2}} \right). \end{aligned}$$

If the tensor cutoff of $\mathfrak{b}$ is $\eta_{\mathfrak{b}}(u)\tilde{\eta}_{\mathfrak{b}}(V)$, set $f_{\alpha, W, \omega}^{\mathfrak{b}} = \tilde{\eta}_{\mathfrak{b}} f_{\alpha, W, \omega}$ and $g_{\alpha, W, \omega}^{\mathfrak{b}} = \eta_{\mathfrak{b}} g_{\alpha, W, \omega}$; their L^2 norms on the fixed intervals are at most $C\|\Theta\|_\infty$. Lemma 7.4, (260), and the finite interaction degree give

$$\sup_{\mathfrak{b} \in \mathcal{R}_{\alpha, \delta, \rho}^{\text{rect}}} \|T_{\alpha, W, \omega}^{\mathfrak{b}}\|_{2 \rightarrow 2} \leq CB^C \min\{1, |\mathbf{q}_{\alpha, W}(\omega)|^{-c_0}\}. \quad (280)$$

Here and below the value at a zero of $\mathbf{q}_{\alpha, W}$ is defined to be one. If $\mathcal{J}_\alpha(W)$ is the contribution of the current transverse band, the changes of variables above and the tensor partition give the exact synthesis identity

$$\mathcal{J}_\alpha(W) = (S\rho)^{-n}\alpha^{n-1} \int_{\mathbb{S}^{n-2}} \sum_{\mathfrak{b} \in \mathcal{R}_{\alpha, \delta, \rho}^{\text{rect}}} \left\langle T_{\alpha, W, \omega}^{\mathfrak{b}} f_{\alpha, W, \omega}^{\mathfrak{b}}, g_{\alpha, W, \omega}^{\mathfrak{b}} \right\rangle d\omega. \quad (281)$$

All functions here are jointly measurable. The trivial Schur bound is a uniform integrable majorant, so ordinary Fubini justifies (281). Cauchy–Schwarz, (280), the uniformly finite rectangle sum, and $c_* \leq \rho \leq C_*$ now yield the explicit first half of the angular synthesis:

$$|\mathcal{J}_\alpha(W)| \leq CB^C \|\Theta\|_\infty^2 S^{-n} \alpha^{n-1} \int_{\mathbb{S}^{n-2}} \min\{1, |\mathbf{q}_{\alpha, W}(\omega)|^{-c_0}\} d\omega. \quad (282)$$

Write

$$\mathbf{q}_{\alpha, W} = \bigoplus_{q=0}^d \mathbf{q}_{\alpha, W}^{[q]},$$

where different homogeneous degrees q occupy disjoint value coordinates, and every entry of $\mathbf{q}_{\alpha, W}^{[q]}$ is homogeneous of degree q in ω . Set

$$A_\alpha = \|\mathbf{q}_{\alpha, W}\|_{\text{coeff}, \omega}.$$

Lemma 7.14, applied before any degrees are combined, shows that on every nonempty rational chart the cleared chart-polynomial vector has coefficient norm at least cA_α . Let $A_* \geq 1$ be fixed so large that the chart lower bound cA_α is at least one whenever $A_\alpha \geq A_*$. If $A_\alpha < A_*$, the integral in (282) is bounded by the surface measure. If $A_\alpha \geq A_*$, normalize the cleared vector on each chart by its own coefficient

norm and apply (178) with $\sigma = c_0$. The chart Jacobians and clearing denominators are bounded above and away from zero, so summing the finite atlas gives

$$\begin{aligned} \int_{\mathbb{S}^{n-2}} \min\{1, |\mathbf{q}_{\alpha, W}(\omega)|^{-c_0}\} d\omega &\leq C \min\{1, A_\alpha^{-c_1}\} \\ &\leq C \min\{1, (\alpha^d L_R |\mathbf{p}(W)|)^{-c_1}\}, \end{aligned} \quad (283)$$

where the last inequality is (261). When $n = 2$, the same calculation is the finite sum over $\mathbb{S}^0$: the $k = 1$ clause of Lemma 7.14 gives the required lower bound at either point. Combining (282) and (283) gives the band bound

$$CB^C \|\Theta\|_\infty^2 S^{-n} \alpha^{n-1} \min\{1, (L_R \alpha^d |\mathbf{p}(W)|)^{-c}\}. \quad (284)$$

Here ρ is confined to a fixed annulus and has been absorbed into the constant. If $A = L_R |\mathbf{p}(W)| \leq 1$, summability of α^{n-1} gives the trivial alternative. If $A > 1$, split the dyadic sum at $\alpha \simeq A^{-1/d}$; the two resulting geometric series are $O(A^{-c'})$ after decreasing the exponent. This proves (273).

For $n = 1$, work on one of the finitely many signed overlap intervals and write $z = Rt$. Let $\mathbf{C}(W)$ be the vector of all nonconstant-in- t coefficients of the complete translated correlation phase. Its degree- m part is

$$\sum_{\substack{A+B=m \\ A, B \geq 1}} c_{A,B} S^A R^B W^A t^B. \quad (285)$$

For a fixed t^B coefficient, every degree $\ell < m$ contribution has W -degree at most $\ell - B < m - B$, and cannot alter the coefficient of W^{m-B} . Since $S^A R^B \geq R^m$, Lemma 7.11 gives

$$\|\mathbf{C}\|_{\text{coeff}, W} \geq c R^m \|\mathbf{P}_m\|_{\text{coeff}} = c L_R.$$

Put $\mathbf{p} = \mathbf{C}/(c L_R)$. The product of the two kernels and dz is $S^{-1} b(t) dt$, on finitely many intervals, with $\|b\|_{L^\infty + \text{BV}} \leq CB^C \|\Theta\|_\infty^2$. Lemma 7.3 gives the asserted estimate when $\mathbf{p}(W) \neq 0$; when $\mathbf{p}(W) = 0$, the same conclusion is the trivial S^{-1} amplitude bound. $\square$

Lemma 7.16 (One-sided variable-mesh incidence). *Let $0 < \varepsilon \leq 1$, let $q_R = \varepsilon R$, let the scale- R packet cubes form a finite disjoint family $\mathcal{Q}_R$ of dyadic cubes in one aligned grid, where $R > 0$. For each $Q \in \mathcal{Q}_R$ choose a point $y_Q \in Q$, and attach a weight $0 \leq w_Q \leq \Lambda q_R^n$, where $\Lambda \geq 0$. Fix $a \in \mathbb{R}^n$ and a large shell scale $S \geq C_0 R$. If $\mathbf{p}$ is as in Proposition 7.13 for $n \geq 2$, or as constructed in Corollary 7.15 for $n = 1$, then, for the fixed annular-overlap constants $0 < c_* < C_* < \infty$ and every $0 < \tau \leq 1$,*

$$\sum_{\substack{Q: c_* S \leq |y_Q - a| \leq C_* S \\ |\mathbf{p}((y_Q - a)/S)| \leq \tau}} w_Q \leq C \Lambda S^n \{\tau^c + (\varepsilon R/S)^c\}. \quad (286)$$

Proof. For $n \geq 2$, first split the sum into the finitely many disjoint active sets E_χ from (255); on the χ -th piece use $\mathbf{p}_{a,R,S,\chi}$. (For $n = 1$ no chart split is needed.) There is no second denominator clearing at this stage. Indeed, for $n \geq 2$, equations (268)–(255) define

$$\tilde{\mathbf{p}}_{a,R,S,\chi}(\rho, \vartheta) = \mathbf{p}_{a,R,S,\chi}(\rho \theta_\chi(\vartheta))$$

as an ordinary coordinate-polynomial vector after the fixed denominator has already been cleared. For $n = 1$ the vector constructed in the proof of Corollary 7.15 is already an ordinary polynomial vector in W . In either case its vector coefficient norm is at least one. Hence one fixed scalar component on the current chart, called p_0 , satisfies $A_0 = \|p_0\|_{\text{coeff}} \geq c > 0$, where c depends only on the fixed vector length. Put $\tilde{p}_0 = p_0/A_0$. This last polynomial has coefficient norm one. At a bad point it satisfies $|\tilde{p}_0| \leq \tau/A_0 \leq C\tau$, regardless of how large the translated lower blocks make A_0 . After division by S , color the q_R grid and put disjoint balls of radius $c\varepsilon R/S$ around the selected points. Work in the fixed enlarged radial annulus $c_*/2 < |W| < 2C_*$. If this radius exceeds a fixed fraction of either the radial collar or, when $n \geq 2$, the angular chart collar $\text{dist}(U_\chi, \partial U_\chi^*)$, the mesh term on the right of (286) is already bounded below by a

fixed constant. Disjointness and the weight cap give total weight at most $C\Lambda S^n$ in the fixed annulus, so the assertion is then trivial. Otherwise the balls stay in the enlarged radial annulus and, for $n \geq 2$, in the enlarged chart U_χ^* . Uniform bi-Lipschitz equivalence between physical W -balls and chart-coordinate balls (ordinary W -intervals when $n = 1$) changes their radii and volumes by fixed factors only. The normalized, already-cleared polynomial $\tilde{p}_0$ has uniformly bounded gradient on the enlarged coordinate domain by (252), so a ball centered at a bad point lies in the $C(\tau + \varepsilon R/S)$ -sublevel. To record the density calculation, put

$$h = \frac{\varepsilon R}{S}, \quad w_Q = \frac{y_Q - a}{S}.$$

For $n \geq 2$ write $z_Q = (\rho_Q, \vartheta_Q)$ for the active chart coordinate of W_Q ; for $n = 1$ put $z_Q = W_Q$. For one color $\mathfrak{c}$ the coordinate balls $B_Q = B(z_Q, ch)$ are disjoint, after a fixed decrease of c , and

$$F_{\mathfrak{c}}(z) = \sum_{Q \in \mathfrak{c}} \frac{w_Q}{|B_Q|} \mathbf{1}_{B_Q}(z)$$

satisfies

$$\|F_{\mathfrak{c}}\|_\infty \leq C \frac{\Lambda(\varepsilon R)^n}{h^n} = C\Lambda S^n. \quad (287)$$

Every ball centered at a bad point is contained in the $C_1(\tau + h)$ -sublevel of $\tilde{p}_0$. If $C_1(\tau + h) \geq 1$, then $\tau^c + h^c \geq (\tau + h)^c \geq C_1^{-c}$. The disjoint packet cubes whose centers lie in the fixed S -annulus have total volume $O(S^n)$, and hence

$$\sum_{Q \text{ bad}} w_Q \leq \sum_Q w_Q \leq C\Lambda S^n \leq C\Lambda S^n(\tau^c + h^c).$$

This is the required trivial branch for the current color and chart, without applying a sublevel estimate outside its stated threshold range. For the remaining color-chart pieces we may therefore assume $C_1(\tau + h) < 1$. Lemma 7.1, now legitimately applied in the coordinate box, gives

$$\sum_{\substack{Q \in \mathfrak{c} \\ Q \text{ bad}}} w_Q \leq \int_{\{|\tilde{p}_0| \leq C_1(\tau + h)\}} F_{\mathfrak{c}}(z) dz \leq C\Lambda S^n(\tau + h)^{c_0}.$$

Sum the boundedly many colors and active charts, and decrease the exponent so that $(\tau + h)^{c_0} \leq C(\tau^c + h^c)$. This proves (286). The large atom was fixed before smearing, so $h = \varepsilon R/S$ and no mesh term of size $\varepsilon S/S$ occurs. $\square$

Proposition 7.17 (Collective fixed-sign high-shell estimate). *Let P be a real polynomial of degree at most d , let P_m be its highest nonpure homogeneous block, let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, fix $\kappa > 0$, let $s \in \mathbb{N}$, let $\varepsilon = 2^{-s}$, and suppose that $\mathcal{R}$ is a finite set of positive dyadic radii such that every $R \in \mathcal{R}$ satisfies*

$$L_R = R^m \|P_m\|_{\text{coeff}} \geq \varepsilon^{-\kappa}. \quad (288)$$

At radius $R \in \mathcal{R}$, let $\mathcal{Q}_R$ be a finite family of disjoint dyadic cubes of side εR in one aligned grid; assume the cubes over all radii are globally disjoint and each complex measure ν_Q is supported on Q and satisfies $|\nu_Q|(Q) \leq \Lambda|Q|$, where $\Lambda \geq 0$. Let

$$T_R \nu(x) = \int e^{iP(x,y)} K_\Theta(x-y) b_R(|x-y|/R) d\nu(y),$$

where $B \geq 1$ and all normalized radial profiles are supported in one fixed compact subinterval of $(0, \infty)$ with $L^\infty + \text{BV}$ norm at most B . Then, uniformly for arbitrary complex coefficients $|e_Q| \leq 1$,

$$\left\| \sum_{R \in \mathcal{R}} T_R \sum_{Q \in \mathcal{Q}_R} e_Q \nu_Q \right\|_2^2 \leq C B^C \|\Theta\|_\infty^2 \Lambda M 2^{-c_\kappa s}, \quad M = \sum_{R \in \mathcal{R}} \sum_{Q \in \mathcal{Q}_R} |\nu_Q|(Q). \quad (289)$$

Here $c_\kappa > 0$, and the implicit constant may depend on κ (in addition to the fixed structural data).

Proof. If $\mathcal{R} = \emptyset$, the assertion is immediate; hence assume $\mathcal{R} \neq \emptyset$. Let $c_{\text{flat}}, c_{\text{corr}}, c_{\text{inc}} > 0$ be fixed admissible exponents in Proposition 7.8, Corollary 7.15, and Lemma 7.16, respectively. Decrease them once and for all to the common exponent

$$c_0 = \min\{c_{\text{flat}}, c_{\text{corr}}/2, c_{\text{inc}}/2\} > 0, \quad c_{\kappa} = \frac{1}{2}c_0 \min\{1, \kappa\} > 0. \quad (290)$$

All occurrences of a decay exponent c in this proof are henceforth understood to have been lowered to c_0 ; the halves above account for the choice $\tau_R = L_R^{-1/2}$. Likewise, enlarge the harmless power of B to the maximum of the finitely many cited powers. Put $w_Q = |v_Q|(Q)$ and write the polar decomposition $dv_Q = h_Q d|v_Q|$, with $|h_Q| = 1$ $|v_Q|$ -almost everywhere; cubes with $w_Q = 0$ may be omitted. On the finite product probability space

$$(\mathfrak{X}, \mathbb{P}) = \prod_{R \in \mathcal{R}} \prod_{Q \in \mathcal{Q}_R} \left(Q, \frac{|v_Q|}{w_Q} \right),$$

let Y_Q be the coordinate random variables. Then, in the weak sense of complex measures,

$$\mathbb{E}[w_Q h_Q(Y_Q) \delta_{Y_Q}] = v_Q; \quad (291)$$

indeed, testing against a bounded Borel function gives the defining polar integral. Set

$$c_Q(\mathbf{Y}) = e_Q h_Q(Y_Q), \quad \mathcal{F}(\mathbf{Y}) = \sum_{R \in \mathcal{R}} T_R \sum_{Q \in \mathcal{Q}_R} c_Q(\mathbf{Y}) w_Q \delta_{Y_Q}.$$

The radial support is separated from zero, and hence, uniformly in y ,

$$\|T_R \delta_y\|_2^2 \leq C B^2 \|\Theta\|_\infty^2 R^{-n}. \quad (292)$$

Since all index sets are finite, this makes $\mathcal{F}$ a strongly measurable Bochner-integrable L^2 -valued random variable and justifies all exchanges of expectation, finite sums, and integration. By (291),

$$\mathbb{E}\mathcal{F} = \sum_R T_R \sum_Q e_Q v_Q \quad \text{in } L^2.$$

Hilbert-space Jensen therefore gives

$$\left\| \sum_R T_R \sum_Q e_Q v_Q \right\|_2^2 = \|\mathbb{E}\mathcal{F}\|_2^2 \leq \mathbb{E}\|\mathcal{F}\|_2^2. \quad (293)$$

It therefore suffices to prove the desired bound for every fixed selection $y_Q \in Q$, with weights $w_Q \leq \Lambda|Q|$ and arbitrary packet coefficients c_Q satisfying $|c_Q| \leq 1$. Fix such a selection and put

$$F_R = T_R \sum_{Q \in \mathcal{Q}_R} c_Q w_Q \delta_{y_Q}.$$

Partition physical input and output space into R -cubes; annular support leaves only boundedly many relative input–output positions. Write $J \sim I$ when the pair can interact. Then

$$\sup_I \#\{J : J \sim I\} + \sup_J \#\{I : J \sim I\} \leq C_n.$$

To make the scale normalization explicit, fix one interacting output–input pair (I, J) , let x_I be the chosen center of I , and let $\mathcal{Q}_{R,I,J}$ be the packet cubes in J which contribute to this pair. Put

$$M_{R,I,J} = \sum_{Q \in \mathcal{Q}_{R,I,J}} w_Q, \quad \sigma_{x_I,R}(y) = \frac{y - x_I}{R},$$

and, for each such Q , set

$$\tilde{v}_Q = R^{-n}(\sigma_{x_I,R})_\#(w_Q \delta_{y_Q}), \quad \tilde{Q} = \sigma_{x_I,R}(Q).$$

Then $\tilde{Q}$ has side length ε and, exactly,

$$|\tilde{v}_Q|(\tilde{Q}) = R^{-n} w_Q \leq R^{-n} \Lambda |Q| = \Lambda |\tilde{Q}|, \quad \sum_{Q \in \mathcal{Q}_{R,I,J}} |\tilde{v}_Q|(\tilde{Q}) = R^{-n} M_{R,I,J}.$$

All cubes $\tilde{Q}$ lie in the fixed normalized input block $B_0 = \sigma_{x_I, R}(J)$. Homogeneity of the kernel gives

$$T_R \sum_{Q \in \mathcal{Q}_{R,I,J}} c_Q w_Q \delta_{y_Q}(x_I + RX) = \tilde{T}_1 \sum_{Q \in \mathcal{Q}_{R,I,J}} c_Q \tilde{v}_Q(X),$$

where $\tilde{T}_1$ has phase $P(x_I + RX, x_I + RY)$ and normalized radial profile $b_R(|X - Y|)$. Modulo pure phases, the highest nonpure homogeneous block of the normalized phase is $R^m P_m$, whose coefficient norm is L_R . Proposition 7.8 therefore produces a normalized bound proportional to $\Lambda R^{-n} M_{R,I,J}$. Since $dx = R^n dX$, the output L^2 -square Jacobian cancels this R^{-n} factor exactly. For an interacting pair define

$$F_{R,I,J} = \mathbf{1}_I T_R \sum_{Q \in \mathcal{Q}_{R,I,J}} c_Q w_Q \delta_{y_Q}.$$

Then $F_R = \sum_I \sum_{J \sim I} F_{R,I,J}$, and disjointness of the output cubes together with the preceding adjacency bound gives

$$\|F_R\|_2^2 = \sum_I \left\| \sum_{J \sim I} F_{R,I,J} \right\|_2^2 \leq C_n \sum_I \sum_{J \sim I} \|F_{R,I,J}\|_2^2.$$

Moreover, every packet cube occurs for only $O_n(1)$ output blocks, so

$$\sum_I \sum_{J \sim I} M_{R,I,J} \leq C_n M_R,$$

where M_R is the total packet mass at radius R . Applying Proposition 7.8 to every block pair and using these two inequalities gives

$$\|F_R\|_2^2 \leq C B^C \|\Theta\|_\infty^2 \Lambda M_R (L_R^{-c} + \varepsilon^c).$$

The high threshold and (290) make the parenthesis at most $C(\varepsilon^{c_0 K} + \varepsilon^{c_0}) \leq C 2^{-c_K S}$. The finite squared-norm expansion is

$$\left\| \sum_{R \in \mathcal{R}} F_R \right\|_2^2 = \sum_R \|F_R\|_2^2 + 2 \operatorname{Re} \sum_{\substack{R, S \in \mathcal{R} \\ R < S}} \langle F_R, F_S \rangle. \quad (294)$$

For $R < S$, split the pairs into the comparable range $S < C_0 R$ and the separated range $R \leq S/C_0$; these ranges are disjoint and exhaustive. If R and S are comparable, then $2|\langle F_R, F_S \rangle| \leq \|F_R\|_2^2 + \|F_S\|_2^2$ reduces the cross term to those two diagonal bounds. Each radius has only boundedly many comparable dyadic neighbours, so their total is bounded by the right side of (289).

For $R \leq S/C_0$, fix a large-shell atom y_A of weight w_A and put

$$\tau_R = L_R^{-1/2}. \quad (295)$$

For this fixed atomic selection, the scale- R contribution to the inner product with the fixed scale- S atom has the form, up to packet coefficients of modulus at most one, as follows. Indeed, write $F_{S,A} = c_A w_A T_S \delta_{y_A}$, set $a = y_A$, $y = y_Q$, and $W = (y_Q - y_A)/S$, and change variables $x = y_Q + z$. Irrespective of the convention chosen for which argument of the inner product is complex-linear, taking the absolute value gives the exact identity

$$\begin{aligned} & |\langle T_R \delta_{y_Q}, T_S \delta_{y_A} \rangle| \\ &= \left| \int e^{i[P(y_Q+z, y_A) - P(y_Q+z, y_Q)]} \overline{k_R(z)} k_S(z + y_Q - y_A) dz \right|. \end{aligned}$$

On the support of this integral the annular overlap gives $c_* \leq |W| \leq C_*$, so this is exactly the left side of Corollary 7.15 with $y = a + SW$. Expanding the small-shell sum and using $|c_A|, |c_Q| \leq 1$ now gives

$$|\langle F_R, F_{S,A} \rangle| \leq w_A \sum_{Q \in \mathcal{Q}_R} w_Q |\mathcal{C}_{R,S}(y_Q, y_A)|,$$

where w_Q is the fixed weight attached to Q and $\mathcal{C}_{R,S}$ is precisely the correlation estimated in Corollary 7.15. For this fixed triple (S, A, R) , define the bad small-shell packets by

$$\text{Bad}_R(A) = \left\{ Q : c_* S \leq |y_Q - y_A| \leq C_* S, \quad \left| \mathbf{p} \left(\frac{y_Q - y_A}{S} \right) \right| \leq \tau_R \right\},$$

with the finite active-chart split from Lemma 7.16 understood when $n \geq 2$. On this bad set, that lemma and the trivial correlation bound give, before multiplication by w_A ,

$$CB^C \|\Theta\|_\infty^2 \Lambda \{ \tau_R^{c_{\text{inc}}} + (\varepsilon R/S)^{c_{\text{inc}}} \} = CB^C \|\Theta\|_\infty^2 \Lambda \{ L_R^{-c_{\text{inc}}/2} + (\varepsilon R/S)^{c_{\text{inc}}} \}.$$

On its complement, Corollary 7.15 gives

$$CB^C \|\Theta\|_\infty^2 S^{-n} (L_R \tau_R)^{-c_{\text{corr}}} = CB^C \|\Theta\|_\infty^2 S^{-n} L_R^{-c_{\text{corr}}/2}.$$

After lowering all exponents to the common c_0 from (290), these are the displayed L_R^{-c} and mesh terms used below. At one fixed scale the total small-atom weight in the interacting S -ball is at most $C\Lambda S^n$, by packing the disjoint εR -cubes. Thus the good row has the same bound $CB^C \|\Theta\|_\infty^2 \Lambda L_R^{-c}$.

Let R_0 be the smallest retained dyadic radius. Since $L_R = L_{R_0} (R/R_0)^m$ and $L_{R_0} \geq \varepsilon^{-\kappa}$,

$$\sum_{\substack{R \in \mathcal{R} \\ R \geq R_0}} L_R^{-c} \leq \varepsilon^{c\kappa} \sum_{\ell \geq 0} 2^{-m c \ell} \lesssim \varepsilon^{c\kappa}.$$

Likewise,

$$\sum_{R \leq S/C_0} (\varepsilon R/S)^c \lesssim \varepsilon^c \sum_{\ell \geq 0} 2^{-c \ell} \lesssim \varepsilon^c.$$

By (290) and $\varepsilon = 2^{-s}$,

$$\varepsilon^{c_0 \kappa} + \varepsilon^{c_0} \leq 2 2^{-c_0 \min\{1, \kappa\} s} \leq C 2^{-c \kappa s}.$$

Thus the complete small-shell row is at most $CB^C \|\Theta\|_\infty^2 \Lambda 2^{-c \kappa s}$. Multiply by w_A , sum the large atoms and then S . Equivalently, if $F_{S,A}$ denotes the contribution of the fixed large-shell atom, the preceding row estimate says

$$\sum_{R \leq S/C_0} |\langle F_R, F_{S,A} \rangle| \leq CB^C \|\Theta\|_\infty^2 \Lambda 2^{-c \kappa s} w_A.$$

Summing first in the atoms at scale S and then in S gives $\sum_{S,A} w_A = M$, so no additional large-shell counting factor occurs. The result is $CB^C \|\Theta\|_\infty^2 \Lambda M 2^{-c \kappa s}$ for every fixed atomic selection. Together with (294), this proves the deterministic atomic estimate; taking expectation in (293) proves (289) for the original complex measures. Every separated unordered pair has been oriented by its unique smaller and larger radius and counted once; the conjugate orientation is exactly the factor two in (294). All estimates used absolute values, so the coefficients may be fixed independently for every packet, as required by the later rank argument. $\square$

8 The unphased weak endpoint

We prove the unphased weak endpoint simultaneously in every dimension $n \geq 1$. The terminal one-shell estimate already has a direct one-dimensional branch in Proposition 5.5, and the long-variation fixed-sign estimate has the direct one-dimensional branch of Lemma 5.4. Thus the selector-free rank argument below requires no imported weak endpoint for Hilbert truncations. The good part of the Calderón–Zygmund decomposition is controlled by the $p = 2$ case of Proposition 2.9:

$$\|\mathbf{J}(T_{\Omega, \varepsilon}^0 h : \varepsilon > 0)\|_2 \leq C_n \mathcal{L}(\Omega) \|h\|_2, \quad \int \Omega = 0. \quad (296)$$

Proposition 8.1 (Unphased weak critical-jump inequality on core data). *For mean-zero $\Omega \in L \log L(\mathbb{S}^{n-1})$ and every bounded compactly supported f ,*

$$\|\mathbf{J}(T_{\Omega,\varepsilon}^0 f : \varepsilon > 0)\|_{L^{1,\infty}} \leq C_n \mathfrak{L}(\Omega) \|f\|_1. \quad (297)$$

Proof. Finite reduction and Calderón–Zygmund decomposition.

Throughout this proposition f is bounded and compactly supported. We impose the finite restrictions needed for every later ranking argument at the outset. After forming the full Calderón–Zygmund decomposition, fix a finite subfamily $\mathcal{Q}_0$ of its bad cubes, a finite interval of shell indices $j_- \leq j \leq j_+$, and a finite ordered set of positive rational truncation parameters, enlarged by the finitely many dyadic and cell endpoints used below. The bad sums, localized families, and trajectories are initially interpreted with these restrictions. In particular, every family $\mathcal{B}_{s,c}$ ranked below is finite when Lemma 5.2 is invoked. All bounds are uniform in the three cutoffs, which are suppressed from the notation.

If $\mathfrak{L}(\Omega) = 0$, then $\Omega = 0$ and the assertion is immediate. By homogeneity normalize $\mathfrak{L}(\Omega) = 1$. Fix a Calderón–Zygmund height $\alpha > 0$ and perform the ordinary dyadic Calderón–Zygmund decomposition at height α as follows. Take the maximal dyadic cubes Q for which $|Q|^{-1} \int_Q |f| > \alpha$. They are disjoint, $\sum_Q |Q| \leq \alpha^{-1} \|f\|_1$, and dyadic differentiation gives $|f| \leq \alpha$ off their union. Maximality makes the average over the dyadic parent at most α , hence $|Q|^{-1} \int_Q |f| \leq 2^n \alpha$. Put

$$g = f \mathbf{1}_{(\cup Q)^c} + \sum_Q f_Q \mathbf{1}_Q, \quad b_Q = (f - f_Q) \mathbf{1}_Q, \quad f_Q = |Q|^{-1} \int_Q f.$$

The preceding facts, followed by $\|g\|_2^2 \leq \|g\|_\infty \|g\|_1$, give

$$\|g\|_2^2 \leq C\alpha \|f\|_1, \quad \int b_Q = 0, \quad \|b_Q\|_1 \leq C\alpha |Q|, \quad \Sigma := \sum_Q \|b_Q\|_1 \leq C \|f\|_1. \quad (298)$$

By (296), (298), and Chebyshev,

$$|\{x : \mathbf{J}(T_{\Omega,\varepsilon}^0 g : \varepsilon > 0) > \alpha\}| \leq \alpha^{-2} \|\mathbf{J}(T_{\Omega,\varepsilon}^0 g)\|_2^2 \leq C\alpha^{-1} \|f\|_1. \quad (299)$$

Choose $s_0 \geq \max\{5, t_0^{\text{term}}\}$, where t_0^{term} denotes the constant t_0 in Proposition 5.5, and enlarge every bad cube by a fixed factor large enough that, outside the resulting set E_0 , an annulus of radius 2^j can meet a cube of side 2^m only if $s = j - m \geq s_0$. This factor depends only on n , and $|E_0| \leq C\alpha^{-1} \|f\|_1$. Put $b = \sum_Q b_Q$ and write $B_m = \sum_{\ell(Q)=2^m} b_Q$.

Angular decomposition and short variation.

For every $s \geq s_0$, make the mean-zero angular split (40) with a parameter $\delta_a > 0$. With the finite cutoffs still in force, define the bounded-angular and tail trajectories by

$$U_{s,\varepsilon} = \sum_{j=j_-}^{j_+} K\omega_s \mathbf{1}_{\{\max\{\varepsilon, 2^j\} < |z| \leq 2^{j+1}\}} * B_{j-s}, \quad (300)$$

$$\text{Tail}_{s,\varepsilon} = \sum_{j=j_-}^{j_+} K_{\Omega_s^\#} \mathbf{1}_{\{\max\{\varepsilon, 2^j\} < |z| \leq 2^{j+1}\}} * B_{j-s}. \quad (301)$$

Outside E_0 , the finite-cutoff bad trajectory is exactly $\sum_{s \geq s_0} (U_{s,\varepsilon} + \text{Tail}_{s,\varepsilon})$. The change of variables $(j, m) \mapsto (j, s = j - m)$ is a bijective reindexing of the shell–cube–scale pairs. For fixed s , each bad cube can therefore occur in at most one summand in j . We spell out the variation estimate, since neither positivity nor monotonicity of the complex kernel is being used. For fixed j, Q put

$$A_{j,Q}(\varepsilon, x) = \int_{\max\{\varepsilon, 2^j\} < |z| \leq 2^{j+1}} K_{\Omega_s^\#}(z) b_Q(x - z) dz.$$

If $0 < \varepsilon_0 < \dots < \varepsilon_N$, the radial regions occurring in the increments $A_{j,Q}(\varepsilon_k) - A_{j,Q}(\varepsilon_{k-1})$ are pairwise disjoint. Hence

$$\sum_{k=1}^N |A_{j,Q}(\varepsilon_k, x) - A_{j,Q}(\varepsilon_{k-1}, x)| \leq (|K_{\Omega_s^\sharp}| \mathbf{1}_{\{2^j < |z| \leq 2^{j+1}\}} * |b_Q|)(x).$$

Taking the supremum over the partition, summing in j, Q , and then using Tonelli gives the factor $\int_{2^j < |z| \leq 2^{j+1}} |K_{\Omega_s^\sharp}(z)| dz = \log 2 \|\Omega_s^\sharp\|_1$. Young's inequality consequently gives

$$\begin{aligned} \|V^1(\text{Tail}_{s,\varepsilon} : \varepsilon > 0)\|_1 &\leq C \|\Omega_s^\sharp\|_1 \sum_j \sum_{\ell(Q)=2^{j-s}} \|b_Q\|_1 \\ &\leq C \|\Omega_s^\sharp\|_1 \Sigma. \end{aligned} \quad (302)$$

There is no additional shell-counting factor. Summing in s and using Lemma 2.7 yields

$$\sum_{s \geq s_0} \|V^1(\text{Tail}_{s,\varepsilon} : \varepsilon > 0)\|_1 \leq C \mathfrak{L}(\Omega) \Sigma.$$

Under the present normalization this is at most $C\Sigma$.

For the bounded pieces ω_s , the short square function is controlled by Proposition 5.5:

$$\left\| \left(\sum_j V^2(K_{\omega_s} \mathbf{1}_{\{\varepsilon < |z| \leq 2^{j+1}\}} * B_{j-s} : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2} \right\|_2^2 \leq C 2^{2\delta_a s} 2^{-\delta_r s} \alpha \Sigma. \quad (303)$$

Choose $\delta_a < \delta_T/8$ and sum the square roots over s .

Fixed-rank control of the long variation.

It remains to control the dyadic long variation. Smooth each sharp annular kernel at relative radial scale $h_s = 2^{-\eta_0 s}$, where $N = 4n + 1$, $\eta_0 = 1/(192N)$, and choose

$$0 < \delta_a < \min(\delta_T/8, 1/192). \quad (304)$$

To make the smoothing precise, fix a nonnegative $\phi \in C_c^\infty((-1/10, 1/10))$ with $\int \phi = 1$, put $\phi_h(t) = h^{-1} \phi(t/h)$, and set

$$a_s = \mathbf{1}_{(1,2]} * \phi_{h_s}, \quad H_j^s(z) = K_{\omega_s}(z) a_s(2^{-j}|z|).$$

The difference between a_s and $\mathbf{1}_{(1,2]}$ is supported in two endpoint collars of total length $O(h_s)$ and is bounded by one. Hence, if E_j^s is the difference between the sharp and smoothed scale- j kernels, polar coordinates give

$$\|E_j^s\|_1 \leq C \|\omega_s\|_1 h_s.$$

The long paths are suffix sums of the shell contributions, so their V^1 distance is bounded by the sum of the shell errors. More precisely, at the finite-cutoff stage put $e_j = E_j^s * B_{j-s}$ and $D_k = \sum_{j \geq k} e_j$. Then $D_{k+1} - D_k = -e_k$, and insertion of the zero and total endpoints gives pointwise

$$V^1(D_k : k) = \sum_k |e_k|.$$

Young's inequality and the disjoint assignment of the bad cubes to the functions B_{j-s} therefore give

$$\|V^1(\text{sharp long path} - \text{smoothed long path})\|_1 \leq \sum_j \|E_j^s\|_1 \|B_{j-s}\|_1 \leq C \|\omega_s\|_1 h_s \Sigma.$$

This is summable in s , since $\|\omega_s\|_1 \leq 2\|\Omega\|_1$ and $h_s = 2^{-\eta_0 s}$. The kernels H_j^s are smooth in the radial variable and supported in a fixed annular enlargement. Leibniz' rule, $\|\partial_r^\ell a_s\|_\infty \leq C_\ell h_s^{-\ell}$, and $r \simeq 2^j$ on their support give

$$M_N(H^s) \leq C \|\omega_s\|_\infty h_s^{-N} \leq C 2^{(\delta_a + N\eta_0)s}. \quad (305)$$

We organize the long variation by a fixed global rank. This selector-free construction converts each point-dependent shell tail into a suffix of one finite ranked sequence. For a bad cube Q with $\ell(Q) = 2^{j-s}$, let $A_j(Q)$ be its ancestor of side 2^j in the original dyadic grid. Since H_j^s is supported in a fixed annular enlargement of radius $C2^j$, the function $H_j^s * b_Q$ is supported in a fixed enlargement of $A_j(Q)$. We use the following nested adjacent dyadic systems. For $a \in \{0, 1/3, 2/3\}^n$, put

$$\mathcal{D}^a = \left\{ 2^{-k}([0, 1]^n + m + (-1)^k a) : k \in \mathbb{Z}, m \in \mathbb{Z}^n \right\}. \quad (306)$$

These systems are nested. In one coordinate, if $s = (-1)^k$, the left endpoint of a child differs from a level- $(k-1)$ endpoint, in units of 2^{-k} , by $m - 2m' + 3sa$. Since $N = m + 3sa$ is integral, take $m' = \lfloor N/2 \rfloor$; then $N - 2m' \in \{0, 1\}$. Taking products proves nesting. They also have the adjacent covering property: choose a dyadic length L with $3\ell(I) < L \leq 6\ell(I)$ for a given interval I . At scale L , the three boundary classes are separated by $L/3$, so at least one class has no boundary in I and its length- L interval contains I . Choose this class in every coordinate. Thus every cube is contained in a member of some $\mathcal{D}^a$ whose side is at most six times larger.

Applying this property to a closed cube containing the fixed enlargement, and using the strict inequality $3\ell(I) < L$ in the covering argument, gives one of the 3^n systems and a cube $R(Q)$ in that system such that the closed enlargement is contained in the interior of $R(Q)$ and $\ell(R(Q)) \leq C_n 2^j$. Thus, after choosing the usual almost-everywhere representatives of the convolutions,

$$\text{ess supp}(H_j^s * b_Q) \subset R(Q). \quad (307)$$

All pointwise statements below are made off the countable union of the boundaries of the adjacent dyadic systems, a null set independent of the finite family. For the fixed s , assign each pair (j, Q) with $\ell(Q) = 2^{j-s}$ to exactly one of the finitely many classes

$$c = (a, \iota_{\text{gen}}) \in \mathfrak{C},$$

determined by the adjacent system and the generation offset, and make one fixed choice of $R(Q)$ in that class. The set $\mathfrak{C}$ depends only on the dimension. Write $\mathcal{Q}_{s,c}$ for the cubes assigned to c and

$$\Sigma_{s,c} = \sum_{Q \in \mathcal{Q}_{s,c}} \|b_Q\|_1, \quad \sum_{c \in \mathfrak{C}} \Sigma_{s,c} \leq \Sigma.$$

Fix a class c . All its output cubes belong to one nested dyadic system and satisfy $\ell(R(Q)) = 2^{j+\iota_c}$ for a fixed integer ι_c . For such an output cube define

$$F_{R,c} = H_j^s * \sum_{\substack{Q \in \mathcal{Q}_{s,c} : R(Q)=R \\ \ell(Q)=2^{j-s}}} b_Q, \quad \ell(R) = 2^{j+\iota_c}. \quad (308)$$

By (307), $\text{ess supp } F_{R,c} \subset R$. Let $\mathcal{R}_{s,c}$ be the finite family of nonzero R . Each assigned Q occurs once, and each nonempty R contains at least one assigned Q of side 2^{j-s} . Choosing one such cube for each R proves

$$\sum_{R \in \mathcal{R}_{s,c}} |R| \leq C 2^{ns} \sum_{Q \in \mathcal{Q}_{s,c}} |Q|. \quad (309)$$

Indeed $|R| \leq C 2^{ns} |Q|$, and the selected Q are distinct. Set

$$\mathcal{N}_{s,c}(x) = \sum_{R \in \mathcal{R}_{s,c}} \mathbf{1}_R(x), \quad D_s = \lceil 2^{(n+1)s} \rceil, \quad X_{s,c} = \{\mathcal{N}_{s,c} > D_s\},$$

and put $X_s = \bigcup_{c \in \mathfrak{C}} X_{s,c}$. Chebyshev, (309), the Calderón–Zygmund packing bound, and finiteness of $\mathfrak{C}$ give

$$|X_s| \leq C 2^{-s} \alpha^{-1} \|f\|_1. \quad (310)$$

Because $F_{R,c}$ is essentially supported in R , cubes $R \subset X_{s,c}$ contribute nothing almost everywhere on $X_{s,c}^c$ and may be discarded. Put

$$\mathcal{R}_{s,c}^{\text{ret}} = \{R \in \mathcal{R}_{s,c} : R \not\subset X_{s,c}\}.$$

Every chain in $\mathcal{R}_{s,c}^{\text{ret}}$ has length at most D_s : if $R_1 \supsetneq \cdots \supsetneq R_L$, choose $x \in R_L \setminus X_{s,c}$; then $L \leq \mathcal{N}_{s,c}(x) \leq D_s$.
Give a remaining cube the recursive classwise rank

$$\text{rk}_c(R) = 1 + \max\{\text{rk}_c(R') : R' \in \mathcal{R}_{s,c}^{\text{ret}}, R' \subsetneq R\}, \quad (311)$$

with the maximum of the empty set equal to zero. Thus $1 \leq \text{rk}_c(R) \leq D_s$, equal-rank cubes form an antichain, and strict inclusion strictly increases rank. Put

$$g_{r,c} = \sum_{\substack{R \in \mathcal{R}_{s,c}^{\text{ret}} \\ \text{rk}_c(R)=r}} F_{R,c}, \quad 1 \leq r \leq D_s.$$

Fix $x \notin X_{s,c}$ outside the null boundary set just specified. At each generation there is at most one cube of the current adjacent system containing x , and the active cubes over all generations form a chain. Write them in increasing shell order as

$$R_{j_1}(x) \subsetneq \cdots \subsetneq R_{j_L}(x), \quad j_1 < \cdots < j_L.$$

Their ranks strictly increase in this order. Every term arising from a cube not containing x vanishes by (307). Therefore, for every shell threshold k , there is an integer $r = r(k, x, c)$ such that

$$\sum_{j \geq k} \sum_{\ell(R)=2^{j+t_c}} F_{R,c}(x) = \sum_{q \geq r} g_{q,c}(x). \quad (312)$$

Conversely every distinct suffix of the active ranks is obtained by placing k between the corresponding consecutive generations. A rank not assumed by an active cube contributes zero at x and merely inserts a repeated suffix value. Repetitions do not affect variation, and inclusion of the zero and total endpoints shows that the classwise shell-tail 2-variation at x equals the 2-variation of the rank suffixes. Subtracting each suffix from the fixed total sum identifies suffix variation with prefix variation exactly. Notice also that the rank is a deterministic function of the finite cube family: once the Rademacher–Menshov signs ε_r are fixed, the induced packet coefficient $\varepsilon_{\text{rk}_c(R(Q))}$ is fixed before x is evaluated.

Apply Lemma 5.2 to $g_{1,c}, \dots, g_{D_s,c}$. For a fixed choice of rank signs, (308) becomes

$$\sum_r \varepsilon_r g_{r,c} = \sum_j H_j^s * \sum_{\ell(Q)=2^{j-s}} c_Q b_Q,$$

where

$$c_Q = \begin{cases} \varepsilon_{\text{rk}_c(R(Q))}, & Q \in \mathcal{Q}_{s,c}, R(Q) \in \mathcal{R}_{s,c}^{\text{ret}}, \\ 0, & \text{otherwise.} \end{cases}$$

Thus discarded cubes and cubes in other adjacent-grid classes have coefficient zero; in particular every displayed rank is defined. These coefficients are fixed before x is evaluated and satisfy the fixed-sign hypothesis of Lemma 5.4; the x -dependent threshold acts only through (312). Define

$$B_{j-s}^{s,c} = \sum_{\substack{Q \in \mathcal{Q}_{s,c} \\ \ell(Q)=2^{j-s}}} b_Q.$$

Lemma 5.4, (305), and $\log(2 + D_s) \lesssim s$ give

$$\|\mathbf{1}_{X_{s,c}^c} V^2(\sum_{j \geq k} H_j^s * B_{j-s}^{s,c} : k)\|_2^2 \leq C s^2 2^{-[1/24 - 2\delta_a - 2N\eta_0]s} \alpha \Sigma_{s,c}.$$

Finally, $B_{j-s} = \sum_{c \in \mathcal{C}} B_{j-s}^{s,c}$ and $X_s^c \subset X_{s,c}^c$ for every c . The V^2 triangle inequality, Cauchy–Schwarz over the finite class set, and $\sum_c \Sigma_{s,c} \leq \Sigma$ therefore yield

$$\|\mathbf{1}_{X_s^c} V^2(\sum_{j \geq k} H_j^s * B_{j-s} : k)\|_2^2 \leq C s^2 2^{-[1/24 - 2\delta_a - 2N\eta_0]s} \alpha \Sigma. \quad (313)$$

The exponent in brackets is positive by (304). Indeed, $2N\eta_0 = 1/96$, and $\delta_a < 1/192$ gives

$$\frac{1}{24} - 2\delta_a - 2N\eta_0 > \frac{1}{24} - \frac{1}{96} - \frac{1}{96} = \frac{1}{48}.$$

Equations (310)–(313) are summable in s .

Let $X = \bigcup_s X_s$. We now record explicitly the scalar majorants that make the passage to the pointwise supremum in the jump amplitude legitimate. Put

$$\begin{aligned} \mathcal{S}_s &= \left(\sum_j V^2(U_{s,\varepsilon} : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2}, \\ \tilde{U}_s(k) &= \sum_{j \geq k} H_j^s * B_{j-s}, \\ \mathcal{L}_s &= \mathbf{1}_{X_s^c} V^2(\tilde{U}_s(k) : k), \\ \mathcal{E}_s &= V^1(U_{s,2^k} - \tilde{U}_s(k) : k), \\ \mathcal{T}_s &= V^1(\text{Tail}_{s,\varepsilon} : \varepsilon > 0). \end{aligned}$$

The smoothing calculation preceding (305) gives $\|\mathcal{E}_s\|_1 \leq C \|\omega_s\|_1 h_s \Sigma$. Equations (303) and (313) give the corresponding L^2 estimates for $\mathcal{S}_s$ and $\mathcal{L}_s$, while (302) controls $\mathcal{T}_s$ in L^1 . Consequently, with

$$\mathcal{G} = \sum_{s \geq s_0} (\mathcal{S}_s + \mathcal{L}_s), \quad \mathcal{W} = \sum_{s \geq s_0} (\mathcal{E}_s + \mathcal{T}_s),$$

Minkowski’s inequality and the geometric decay in s yield

$$\|\mathcal{G}\|_2^2 \leq C \alpha \Sigma, \quad \|\mathcal{W}\|_1 \leq C \Sigma, \quad |X| \leq C \alpha^{-1} \|f\|_1. \quad (314)$$

Fix the finite truncation set I imposed at the start of the proof. The pointwise long–short inequality (23), followed by $\mathbf{J}^{\text{dyad}} \leq V^2$ and the V^2 triangle inequality, gives on X_s^c

$$\mathbf{J}_I(U_s) \leq C(\mathcal{S}_s + \mathcal{L}_s + \mathcal{E}_s), \quad \mathbf{J}_I(\text{Tail}_s) \leq \mathcal{T}_s.$$

Apply Lemma 2.3 to counting measure on the finite set of s -indices, and then use (27). Its constant is independent of the number of retained shells. Since the decomposition in (300)–(301) is exact, we obtain the pointwise, jump-amplitude-independent domination

$$\mathbf{1}_{(E_0 \cup X)^c} \mathbf{J}_I(T_{\Omega,\varepsilon}^0 b : \varepsilon \in I) \leq C(\mathcal{G} + \mathcal{W}). \quad (315)$$

This is the precise step that permits the supremum in λ to be placed before the spatial weak norm; no union over jump amplitudes is taken. Chebyshev and (314) now yield

$$|\{x \notin E_0 \cup X : \mathbf{J}_I(T_{\Omega,\varepsilon}^0 b : \varepsilon \in I) > C\alpha\}| \leq C\alpha^{-2} \|\mathcal{G}\|_2^2 + C\alpha^{-1} \|\mathcal{W}\|_1 \leq C\alpha^{-1} \Sigma.$$

Add (299), the measure bounds for E_0 and X , and use (18) for $f = g + b$. Thus there is an absolute C_0 such that, uniformly in all finite cutoffs,

$$|\{x : \mathbf{J}_I(T_{\Omega,\varepsilon}^0 f : \varepsilon \in I) > C_0 \alpha\}| \leq C\alpha^{-1} \|f\|_1.$$

This is an estimate for the pointwise supremum in the jump amplitude, not for the weaker mixed quasi-seminorm. After the cutoffs have been removed, putting $\alpha = \tau/C_0$ and taking the supremum over $\tau > 0$ gives the claimed $L^{1,\infty}$ quasi-norm bound.

Removal of the auxiliary cutoffs.

We remove the cutoffs in their declared order. With shells and parameters fixed, increase $\mathcal{Q}_0$ to the full bad family. Writing

$$f_{\mathcal{Q}_0} = g + \sum_{Q \in \mathcal{Q}_0} b_Q,$$

we have $f_{\mathcal{Q}_0} \rightarrow f$ in L^1 , since $\sum_Q \|b_Q\|_1 < \infty$. Every coordinate has a finite-annulus L^1 kernel, so Young's inequality gives the corresponding coordinatewise L^1 convergence. After a subsequence, convergence is pointwise simultaneously on the finite parameter set; write $F^{(v)}$ and F for the corresponding approximating and limiting finite paths. For finite I the functional $\mathbf{J}_I$ is a Borel function of finitely many measurable coordinates. The lower semicontinuity (37) therefore implies, for every $\tau > 0$,

$$\mathbf{1}_{\{\mathbf{J}_I(F) > \tau\}} \leq \liminf_{v \rightarrow \infty} \mathbf{1}_{\{\mathbf{J}_I(F^{(v)}) > \tau\}} \quad \text{almost everywhere.}$$

Indeed, if $\mathbf{J}_I(F) > \tau$, the definition of the supremum supplies one jump amplitude and one finite strict witness whose value exceeds τ by a positive amount. Coordinatewise convergence preserves all of its strict increments and its excess for every sufficiently large v . Fatou's lemma now preserves the distribution bound for the finite-parameter pointwise functional, with a constant independent of I .

Next send $j_- \rightarrow -\infty$ and $j_+ \rightarrow \infty$. For core data, each fixed spatial point x and positive truncation q sees only finitely many shells: if $2^{j+1} \leq q$, the shell lies below the truncation, while for all sufficiently large j the annulus centered at x no longer meets the compact support of the input. Thus every coordinate is eventually constant in the shell cutoff. The truncated coordinates converge pointwise on each fixed finite I , and the same strict-witness lower-semicontinuity and level-set Fatou argument applies again. Finally enumerate $\mathbb{Q}_+ = \{q_1, q_2, \dots\}$ and put $I_N = \{q_1, \dots, q_N\}$. These sets are nested, and finiteness of every strict witness gives, for each $\tau > 0$,

$$\{\mathbf{J}(T_{\Omega,q}^0 f : q \in \mathbb{Q}_+) > \tau\} = \bigcup_{N \geq 1} \{\mathbf{J}_{I_N}(T_{\Omega,q}^0 f : q \in I_N) > \tau\}.$$

Continuity from below transfers the uniform distribution bound. For the present bounded compactly supported data, Fubini on every finite annulus gives local absolute continuity of the truncation path (as recorded again in Section 12); hence Lemma 2.6 identifies the rational and full-parameter pointwise functionals: every finite full-parameter witness lies in one compact parameter interval, where its points can be perturbed to positive rationals while preserving all strict inequalities. Every Rademacher–Menshov ranking is therefore finite, and the cutoff limits recover the full trajectory from these finite fixed-rank sums. The resulting distribution bound is $C\|f\|_1/\alpha$ for $\mathbf{J}$, and hence gives (297) after restoring the homogeneous angular scale. $\square$

9 Phase-adapted Calderón–Zygmund decomposition

We modify the Calderón–Zygmund decomposition so that each bad block cancels against the input modulation generated by its parent cube. This is exactly the cancellation condition required by the packet estimates in the next section.

Fix a polynomial P and take the maximal dyadic family $\mathcal{Q}$ constructed in the proof of Proposition 8.1, at height α for f . For each $Q \in \mathcal{Q}$, let c_Q be its center and put

$$a_Q = |Q|^{-1} \int_Q e^{iP(c_Q, y)} f(y) dy, \quad g = f \mathbf{1}_{(\cup Q)^c} + \sum_Q a_Q e^{-iP(c_Q, \cdot)} \mathbf{1}_Q, \quad b_Q = (f - a_Q e^{-iP(c_Q, \cdot)}) \mathbf{1}_Q. \quad (316)$$

Then

$$\|g\|_\infty \leq C\alpha, \quad \|g\|_2^2 \leq C\alpha \|f\|_1, \quad \sum_Q \|b_Q\|_1 \leq C\|f\|_1, \quad \|b_Q\|_1 \leq C\alpha|Q|. \quad (317)$$

Indeed, maximality gives $|Q|^{-1} \int_Q |f| \leq 2^n \alpha$, and hence $|a_Q| \leq 2^n \alpha$ and $\|b_Q\|_1 \leq 2 \int_Q |f|$. The cubes are disjoint. Off their union $|f| \leq \alpha$ almost everywhere. Moreover,

$$|a_Q| |Q| = \left| \int_Q e^{iP(c_Q, y)} f(y) dy \right| \leq \int_Q |f(y)| dy$$

and, on Q , $|g|^2 = |a_Q|^2 \leq 2^n \alpha |a_Q|$. Consequently

$$\|g\|_2^2 \leq \alpha \int_{(\cup Q)^c} |f| + 2^n \alpha \sum_Q |a_Q| |Q| \leq C \alpha \|f\|_1.$$

These observations give all four estimates in (317). Most importantly, phase adaptation gives the exact cancellation

$$h_Q(y) := e^{iP(c_Q, y)} b_Q(y), \quad \int_Q h_Q(y) dy = 0, \quad |h_Q(y)| = |b_Q(y)|. \quad (318)$$

Indeed,

$$\int_Q h_Q(y) dy = \int_Q e^{iP(c_Q, y)} f(y) dy - a_Q |Q| = 0, \quad \|h_Q\|_1 = \|b_Q\|_1.$$

No pointwise bound for b_Q or h_Q is asserted. The packet estimates below are deliberately formulated with only the density $\|h_Q\|_1 / |Q|$ and are applied directly to the original blocks; thus no charge-neutralized stopping-tree decomposition is used. In the one-shell estimate below we take $d_Q = h_Q$ and use its cancellation. In the collective high-shell estimate we instead take $dv_Q = b_Q(y) dy$; that proposition requires the same mass-density bound but no cancellation.

10 The far-field weak endpoint

This section proves the weak critical-jump estimate for the normalized far-field trajectory. Assume that the highest nonpure homogeneous block P_m has been normalized by

$$\|P_m\|_{\text{coeff}} = 1. \quad (319)$$

Define the far family by

$$\mathcal{F}_\varepsilon^P f(x) = \int_{\substack{|x-y| > \varepsilon \\ |x-y| > 1}} e^{iP(x, y)} K_\Omega(x-y) f(y) dy,$$

extended constantly for $0 < \varepsilon \leq 1$. Equivalently,

$$\mathcal{F}_\varepsilon^P f = T_{\Omega, \max\{\varepsilon, 1\}}^P f. \quad (320)$$

Thus the far trajectory is a restriction of the full trajectory, with repeated values for $0 < \varepsilon \leq 1$, and hence

$$N_\lambda(\mathcal{F}_\varepsilon^P f : \varepsilon > 0) \leq N_\lambda(T_{\Omega, \varepsilon}^P f : \varepsilon > 0), \quad \mathbf{J}(\mathcal{F}_\varepsilon^P f : \varepsilon > 0) \leq \mathbf{J}(T_{\Omega, \varepsilon}^P f : \varepsilon > 0). \quad (321)$$

The strong $p = 2$ theorem gives

$$\|\mathbf{J}(T_{\Omega, \varepsilon}^P h : \varepsilon > 0)\|_2 \leq C_{n,d} \mathfrak{L}(\Omega) \|h\|_2. \quad (322)$$

Proposition 10.1 (Far weak endpoint on core data). *Let P be a real polynomial of degree at most d with highest nonpure homogeneous block P_m , and assume $\|P_m\|_{\text{coeff}} = 1$. Let $\Omega \in L \log L(\mathbb{S}^{n-1})$ have mean zero. Then, for every bounded compactly supported f ,*

$$\|\mathbf{J}(\mathcal{F}_\varepsilon^P f : \varepsilon > 0)\|_{L^{1,\infty}} \leq C_{n,d} \mathfrak{L}(\Omega) \|f\|_1. \quad (323)$$

Proof. We first take bounded compactly supported f and impose finite cube, shell, and rational-parameter cutoffs, exactly as in Proposition 8.1. Every estimate below is uniform in those cutoffs.

If $\mathfrak{L}(\Omega) = 0$, the assertion is immediate. By angular homogeneity normalize $\mathfrak{L}(\Omega) = 1$, fix a Calderón–Zygmund height $\alpha > 0$, and use (316). By (321), (322), (317), and Chebyshev,

$$|\{x : \mathbf{J}(\mathcal{F}_\varepsilon^P g : \varepsilon > 0) > \alpha\}| \leq \alpha^{-2} \|\mathbf{J}(\mathcal{F}_\varepsilon^P g)\|_2^2 \leq C\alpha^{-2} \|g\|_2^2 \leq C\alpha^{-1} \|f\|_1. \quad (324)$$

Thus the good part is controlled outside a set of the required measure. Enlarge the bad cubes by a fixed factor chosen as follows. Let $s_0 \geq \max\{5, t_0\}$, where t_0 is the threshold in Proposition 7.10, and use dilation factor $C_n 2^{s_0}$. This produces another exceptional set E_0 of measure $C\alpha^{-1} \|f\|_1$. Outside E_0 , a cube meeting a shell $R = 2^j \geq 1$ has

$$\ell(Q) = R2^{-s}, \quad s \geq s_0. \quad (325)$$

Put $b = \sum_Q b_Q$ and $\Sigma = \sum_Q \|b_Q\|_1$. For every fixed s , the cubes in (325), as j varies, form a subfamily of the original disjoint Calderón–Zygmund cubes and

$$\sum_j \sum_{\ell(Q)=2^{j-s}} \|h_Q\|_1 \leq \Sigma, \quad h_Q = e^{iP(c_Q, \cdot)} b_Q. \quad (326)$$

Split $\Omega = \omega_s + \Omega_s^\sharp$ by (40). With the finite shell cutoff understood, define the bounded-angular and angular-tail trajectories

$$U_{s,\varepsilon}^P(x) = \sum_{j \geq 0} \int_{\max\{\varepsilon, 2^j\} < |x-y| \leq 2^{j+1}} e^{iP(x,y)} K_{\omega_s}(x-y) \sum_{\ell(Q)=2^{j-s}} b_Q(y) dy, \quad (327)$$

$$\text{Tail}_{s,\varepsilon}^P(x) = \sum_{j \geq 0} \int_{\max\{\varepsilon, 2^j\} < |x-y| \leq 2^{j+1}} e^{iP(x,y)} K_{\Omega_s^\sharp}(x-y) \sum_{\ell(Q)=2^{j-s}} b_Q(y) dy. \quad (328)$$

Since $|e^{iP(x,y)}| = 1$, the same Fubini–Young calculation as in (302), together with (326), gives

$$\|V^1(\text{Tail}_{s,\varepsilon}^P : \varepsilon > 0)\|_1 \leq C\|\Omega_s^\sharp\|_1 \Sigma. \quad (329)$$

Here again the reindexing $(j, m) \mapsto (j, s = j - m)$ assigns each bad cube to at most one j for fixed s , so no shell-counting factor occurs. Summing (329) in s and using Lemma 2.7 proves that the angular tails have total $V^1 L^1$ norm at most $C\mathfrak{L}(\Omega)\Sigma$. For the bounded pieces,

$$\|\omega_s\|_\infty \leq C2^{\delta_a s} \|\Omega\|_1, \quad \|\omega_s\|_1 \leq 2\|\Omega\|_1. \quad (330)$$

Short variation.

We first control the short variation inside every dyadic annulus. Apply Proposition 7.10 directly to the h_Q at each (j, s) . The required phase matching is the exact identity

$$e^{iP(x,y)} b_Q(y) = e^{i[P(x,y) - P(c_Q, y)]} h_Q(y). \quad (331)$$

Thus the packet datum in that proposition is $d_Q = h_Q$; by (318) and (317) it is supported in Q , has integral zero, satisfies $\|d_Q\|_1 \leq C\alpha|Q|$, and the packet cubes are globally disjoint. We may therefore take $\Lambda = C\alpha$ and $t = s$. Square-summing in j and using (326) gives

$$\left\| \left(\sum_j V^2(G_{j,s})^2 \right)^{1/2} \right\|_2^2 \leq C\|\omega_s\|_\infty^2 2^{-c_P s} \alpha \Sigma, \quad (332)$$

and the packet errors satisfy

$$\sum_j \|V^1(E_{j,s})\|_1 \leq C\|\omega_s\|_1 2^{-c_E s} \Sigma. \quad (333)$$

Long variation: low shells.

It remains to control the dyadic long variation. Fix a small $\kappa > 0$ and split the shells by the single global top block:

$$\text{low: } 1 \leq R^m \leq 2^{\kappa s}, \quad \text{high: } R^m > 2^{\kappa s}. \quad (334)$$

Since $R = 2^j \geq 1$, the low condition implies $0 \leq j \leq \kappa s/m$; hence there are at most $1 + \kappa s/m \leq C_d(1+s)$ low shells. If $F_{j,s}$ is their full annular output, then

$$V^2 \left(\sum_{j \geq k, j \text{ low}} F_{j,s} : k \right) \leq 2 \sum_{j \text{ low}} |F_{j,s}|.$$

The endpoint difference of a one-shell trajectory is dominated by its full variation. More explicitly write $F_{j,s} = F_{j,s}^G + F_{j,s}^E$ using the endpoint differences of the decomposition in Proposition 7.10. Thus, with $R = 2^j$,

$$F_{j,s} = \mathcal{P}_R - \mathcal{P}_{2R} = (G_R - G_{2R}) + (E_R - E_{2R}),$$

and consequently

$$|F_{j,s}^G| \leq V^2(G_{j,s}), \quad |F_{j,s}^E| \leq V^1(E_{j,s}).$$

Cauchy–Schwarz in the $O(1+s)$ low shells and (221)–(222) therefore give

$$\left\| 2 \sum_{j \text{ low}} |F_{j,s}^G| \right\|_2^2 \leq C(1+s) \sum_{j \text{ low}} \|V^2(G_{j,s})\|_2^2 \leq C \|\omega_s\|_\infty^2 2^{-cs} \alpha \Sigma, \quad c = c_P/2, \quad (335)$$

where $(1+s)2^{-cps} \lesssim 2^{-cps/2}$ has been absorbed, together with a summable $V^1 L^1$ error. This is the only place where translated lower blocks are Fourier-expanded across low shells.

Long variation: high shells.

On the high shells no phase block is expanded. Return to the original form $e^{iP(x,y)} b_Q(y)$; every packet and shell now has the same global phase P , and (334) is precisely the threshold (288). Their short variation has already been included in the square sum (332); the collective estimate is needed only for the dyadic long variation.

For the high dyadic long variation use the output localization and fixed rank from the proof of Proposition 8.1. For every Q at a high shell 2^j , assign its natural annular output to a cube $R(Q)$ in one of the finitely many nested adjacent systems (306), with $\ell(R(Q)) \simeq 2^j$. As in the closed-enlargement construction preceding (307), choose $R(Q)$ so that the essential support of this sharp-annular output is contained in its interior. Let $\mathfrak{C}$ be the resulting finite set of adjacent-system and generation- offset classes; it depends only on the dimension. Make one fixed class assignment for each pair (j, Q) and write $\mathcal{Q}_{s,c}^{\text{high}}$ for the packets assigned to c . Within a fixed class, all $R(Q)$ lie in one nested system and $\ell(R(Q)) = 2^{j+\iota_c}$ for one fixed integer ι_c . Define the current bounded-angular annular operator by

$$T_j^{P, \omega_s} g(x) = \int_{2^j < |x-y| \leq 2^{j+1}} e^{iP(x,y)} K_{\omega_s}(x-y) g(y) dy,$$

and, for the classwise output cubes, put

$$F_{R,c} = T_j^{P, \omega_s} \sum_{\substack{Q \in \mathcal{Q}_{s,c}^{\text{high}}: R(Q)=R \\ \ell(Q)=2^{j-\iota_s}}} b_Q, \quad \ell(R) = 2^{j+\iota_c}.$$

Let $\mathcal{R}_{s,c}$ be the finite family of nonzero classwise output cubes. Then $\text{ess supp } F_{R,c} \subset R$ and

$$\sum_{R \in \mathcal{R}_{s,c}} |R| \leq C 2^{ns} \sum_{Q \in \mathcal{Q}_{s,c}^{\text{high}}} |Q|.$$

Let $D_s = \lceil 2^{(n+1)s} \rceil$ and define

$$X_{s,c} = \left\{ \sum_{R \in \mathcal{R}_{s,c}} \mathbf{1}_R > D_s \right\}, \quad X_s = \bigcup_{c \in \mathfrak{C}} X_{s,c}.$$

Since $\mathfrak{C}$ is finite, the packing estimate gives

$$|X_s| \leq C 2^{-s} \alpha^{-1} \|f\|_1. \quad (336)$$

For each class discard the cubes $R \subset X_{s,c}$ and rank the retained finite family by strict inclusion, as in (311). Put

$$\mathcal{Q}_{s,c}^{\text{high,ret}} = \{Q \in \mathcal{Q}_{s,c}^{\text{high}} : R(Q) \not\subset X_{s,c}\}.$$

At every $x \notin X_s$ outside the fixed null set of adjacent-grid boundaries, the high-shell tail in each class is a suffix of its rank sequence by the active-chain argument proving (312); gaps caused by low shells only repeat a suffix value. A fixed rank sign induces a fixed coefficient in $\{-1, 0, 1\}$ on each original packet, with coefficient zero on discarded cubes and on packets outside the current class. Thus no rank is ever invoked for a discarded or differently classified packet.

We verify explicitly the hypotheses of Proposition 7.17 class by class. For fixed s and c , the retained set of high radii is finite because the shell cutoff is finite. Across all of those radii, the packet cubes remain a subfamily of the original globally disjoint Calderón–Zygmund cubes. Take $d\nu_Q = b_Q(y) dy$; unlike the short estimate, this application uses no zero-mean hypothesis. Then

$$\text{supp } \nu_Q \subset Q, \quad |\nu_Q|(Q) = \|b_Q\|_1 = \|h_Q\|_1 \leq C\alpha|Q|.$$

The normalized profile is the sharp function $\mathbf{1}_{(1,2]}$, whose $L^\infty + \text{BV}$ norm is bounded by an absolute constant. Moreover, normalization of the top block and the high-shell condition give $R^m \|P_m\|_{\text{coeff}} = R^m > 2^{\kappa s}$, exactly (288). The Rademacher–Menshov signs are fixed packet by packet before the output point x is evaluated. If

$$\Sigma_{s,c}^{\text{high}} = \sum_{Q \in \mathcal{Q}_{s,c}^{\text{high,ret}}} \|b_Q\|_1,$$

then Lemma 5.2 and Proposition 7.17 give

$$\|\mathbf{1}_{X_{s,c}^c} V^2(\text{high dyadic tails in class } c)\|_2^2 \leq Cs^2 \|\omega_s\|_\infty^2 2^{-c\kappa s} \alpha \Sigma_{s,c}^{\text{high}}.$$

The V^2 triangle inequality and Cauchy–Schwarz over the finite class set, together with $\sum_c \Sigma_{s,c}^{\text{high}} \leq \Sigma$, yield the full high-shell estimate

$$\|\mathbf{1}_{X_s^c} V^2(\text{high dyadic tails})\|_2^2 \leq Cs^2 \|\omega_s\|_\infty^2 2^{-c\kappa s} \alpha \Sigma. \quad (337)$$

This is where the estimate must allow packetwise, rather than merely shellwise, signs.

Choose $0 < \delta_a < \frac{1}{4} \min(c_P, c_E, c_\kappa)$. Equations (332)–(337), polynomial factors in s , and (330) imply the following precise majorant bounds. Put

$$\begin{aligned} \mathcal{S}_s &= \left(\sum_j V^2(G_{j,s})^2 \right)^{1/2}, \\ \mathcal{L}_s &= 2 \sum_{j \text{ low}} |F_{j,s}^G|, \\ \mathcal{H}_s &= \mathbf{1}_{X_s^c} V^2(\text{high dyadic tails}). \end{aligned}$$

For later use retain explicitly the error produced by the one-shell decompositions and put

$$\mathcal{E}_s^* = \sum_j V^1(E_{j,s}). \quad (338)$$

At the present finite-cutoff stage this is a finite pointwise sum. In particular, no compatibility of the choices of $G_{j,s}$ and $E_{j,s}$ from one shell to another is being assumed. Let $\mathcal{W}$ be the total 1-variation majorant of all angular tail paths plus $C\sum_s \mathcal{E}_s^*$; the constant also covers the low-shell endpoint errors $2\sum_{j\text{ low}} |F_{j,s}^E|$. Set

$$\mathcal{G} = \sum_s (\mathcal{S}_s + \mathcal{L}_s + \mathcal{H}_s), \quad X = \bigcup_s X_s.$$

Minkowski's inequality, applied to the square roots of the fixed- s estimates, now gives

$$\|\mathcal{G}\|_2^2 \leq C\alpha\Sigma, \quad \|\mathcal{W}\|_1 \leq C\Sigma, \quad |X| \leq C\alpha^{-1}\|f\|_1. \quad (339)$$

The geometric decay makes all three sums finite. More precisely, at the finite-cutoff stage these are finite sums, while summability of their L^2 , respectively L^1 , norms makes the partial sums of $\mathcal{G}$, respectively $\mathcal{W}$, converge in those spaces with the displayed bounds; monotone subsequences may be used for the nonnegative majorants. We now spell out the passage from the shellwise decompositions to the full trajectory. Denote the moving bounded-angular one-shell path on $[2^j, 2^{j+1}]$ by $\mathcal{P}_{j,s} = G_{j,s} + E_{j,s}$, and denote its endpoint difference by $F_{j,s} = F_{j,s}^G + F_{j,s}^E$, as above. The outer-shell terms in (327) are constant while $\varepsilon \in [2^j, 2^{j+1}]$. Hence the variation triangle inequality, $V^2(E_{j,s}) \leq V^1(E_{j,s})$, and ℓ^2 -Minkowski give the pointwise estimate

$$\begin{aligned} & \left(\sum_j V^2(U_{s,\varepsilon}^P : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2} = \left(\sum_j V^2(\mathcal{P}_{j,s})^2 \right)^{1/2} \\ & \leq \left(\sum_j (V^2(G_{j,s}) + V^1(E_{j,s}))^2 \right)^{1/2} \\ & \leq \mathcal{S}_s + \left(\sum_j V^1(E_{j,s})^2 \right)^{1/2} \leq \mathcal{S}_s + \mathcal{E}_s^*. \end{aligned} \quad (340)$$

For the dyadic endpoint path, split the full suffix sum into its high- and low-shell parts. The exact endpoint identity is

$$U_{s,2^k}^P = \sum_{j \geq k} F_{j,s}.$$

Here and below $F_{j,s} = 0$ for $j < 0$, so the identity also records the constant part of the far trajectory for $2^k \leq 1$. On X_s^c , the high part is bounded by $\mathcal{H}_s$. The elementary suffix bound used in (335), together with $|F_{j,s}^E| \leq V^1(E_{j,s})$, gives

$$\begin{aligned} \mathbf{1}_{X_s^c} V^2 \left(\sum_{j \geq k} F_{j,s} : k \right) & \leq \mathcal{H}_s + V^2 \left(\sum_{\substack{j \geq k \\ j \text{ low}}} F_{j,s} : k \right) \\ & \leq \mathcal{H}_s + 2 \sum_{j \text{ low}} |F_{j,s}| \\ & \leq \mathcal{H}_s + \mathcal{L}_s + 2\mathcal{E}_s^*. \end{aligned} \quad (341)$$

Thus the deterministic full-variation long-short inequality (94), applied to the *original* bounded-angular path rather than to separately chosen global G/E paths, yields on X_s^c

$$V^2(U_{s,\varepsilon}^P : \varepsilon > 0) \leq C(\mathcal{S}_s + \mathcal{L}_s + \mathcal{H}_s + \mathcal{E}_s^*). \quad (342)$$

Summing (342) over s with (17), and adding the angular-tail V^1 paths, gives on $E_0^c \cap X^c$ a 2-variation majorant bounded by $C(\mathcal{G} + \mathcal{W})$. This construction uses only the displayed shellwise decompositions

and therefore makes no unstated cross-shell selection. Since the majorants do not depend on the jump amplitude, equations (18) and (19) give directly

$$\mathbf{1}_{E_0^c \cap X^c} \mathbf{J}(\mathcal{F}_\varepsilon^P b : \varepsilon > 0) \leq C(\mathcal{G} + \mathcal{W}). \quad (343)$$

Chebyshev applied to (343), together with (324), the measures of E_0 and X , (339), and the pointwise jump quasi-triangle inequality, gives

$$|\{x : \mathbf{J}(\mathcal{F}_\varepsilon^P f : \varepsilon > 0) > C_0 \alpha\}| \leq C \alpha^{-1} \|f\|_1.$$

Here C_0 is fixed and independent of every cutoff. Once the cutoffs are removed, setting $\alpha = \tau/C_0$ and taking the supremum over $\tau > 0$ gives precisely the $L^{1,\infty}$ quasi-norm of the pointwise critical-jump functional. The proof in dimension one is identical after using the one-dimensional parts of Corollary 7.15 and Proposition 7.17; mean zero on $\mathbb{S}^0$ makes the angular function a Hilbert-kernel multiple, so no angular-chart issue arises.

Removal of cutoffs.

Finally remove the cutoffs in the same order as in the proof of Proposition 8.1. With the shell interval and the finite rational parameter set I fixed, exhaust the phase-adapted bad-cube family. Since $\sum_Q \|b_Q\|_1 < \infty$, the corresponding inputs converge in L^1 ; Young's inequality for each finite-annulus coordinate, followed by a subsequence, gives pointwise convergence simultaneously on the finite parameter set; denote the approximating and limiting finite paths by $F^{(v)}$ and F . The finite functional is Borel in its coordinates. If $\mathbf{J}_I(F) > \tau$, one amplitude and one finite strict witness exceed τ by a positive amount, and coordinatewise convergence preserves that witness for all sufficiently large v . Hence the level-set form of (37),

$$\mathbf{1}_{\{\mathbf{J}_I(F) > \tau\}} \leq \liminf_{v \rightarrow \infty} \mathbf{1}_{\{\mathbf{J}_I(F^{(v)}) > \tau\}},$$

followed by Fatou preserves the distribution estimate for every $\tau > 0$.

Next exhaust the dyadic shell interval, still with I fixed. For a fixed output point x and truncation $q > 0$, every shell with $2^{j+1} \leq q$ lies below the truncation, while for all sufficiently large j the annulus centered at x misses the compact support of the input. Thus each coordinate is eventually a finite shell sum, and the shell-cutoff limit is pointwise on the finite set I . Apply the same strict-witness lower-semicontinuity and level-set Fatou argument once more. Finally enumerate $\mathbb{Q}_+ = \{q_1, q_2, \dots\}$, put $I_N = \{q_1, \dots, q_N\}$, and use

$$\{\mathbf{J}(\mathcal{F}_q^P f : q \in \mathbb{Q}_+) > \tau\} = \bigcup_{N \geq 1} \{\mathbf{J}_{I_N}(\mathcal{F}_q^P f : q \in I_N) > \tau\}.$$

The equality follows because every strict witness is finite; continuity from below therefore preserves the uniform distribution bound. For bounded compactly supported data, Fubini on finite annuli gives local continuity in the truncation parameter. Every full-parameter witness is contained in a compact interval, and Lemma 2.6 therefore identifies the rational and full-parameter functionals. The ranks, signs, and exceptional sets are auxiliary devices used only to prove estimates uniform in the cutoffs; none of them is passed to the limit. Only the original finite-cutoff trajectories are passed to the limit. In particular, no convergence of the ranks, signs, or exceptional sets is asserted or needed. This proves (323) for core data. Angular homogeneity restores $\mathcal{L}(\Omega)$. $\square$

11 Local weak estimates and homogeneous-degree induction

We close the weak endpoint with a coarser triangular order than in the strong argument: each step removes the entire highest nonpure homogeneous class rather than a single flag coordinate.

Proposition 11.1 (Local homogeneous-degree reduction). *Let P be a real polynomial of degree at most d whose highest nonpure homogeneous block P_m satisfies $\|P_m\|_{\text{coeff}} = 1$. Let $\Omega \in L^1(\mathbb{S}^{n-1})$ and let f be bounded and compactly supported. On each unit output cube, the local path (110) is, modulo pure input and output phases, the sum of*

- (i) a localized path of the form (112) whose phase has nonpure total degree at most $m - 1$;
- (ii) an error E satisfying

$$V^1(E)(x) \leq C_{n,d} \int_{|x-y|<1} \frac{|\Omega((x-y)/|x-y|)|}{|x-y|^{n-1}} |f(y)| dy. \quad (344)$$

The majorant has L^1 norm at most $C_{n,d} \|\Omega\|_1 \|f\|_1$.

Proof. Let Q_a be the output unit cube centered at a , and let Q_a^* be the fixed input enlargement relevant to Q_a . Write $x = a + X, y = a + Y$. Simultaneous translation preserves the degree- m class and creates only lower degrees. After separating pure terms, write

$$P(a + X, a + Y) = p_a(X) + q_a(Y) + H_m(X, Y) + R_{a, < m}(X, Y),$$

where H_m is the mixed representative of the normalized top class and $R_{a, < m}$ has nonpure degree at most $m - 1$. Replace H_m by the anchored representative

$$\widehat{H}_m(X, Y) = H_m(X, Y) - H_m(Y, Y),$$

and absorb the pure input polynomial $H_m(Y, Y)$ into q_a . Set

$$f_a(Y) = e^{iq_a(Y)} f(a + Y) \mathbf{1}_{Q_a^* - a}(Y).$$

The coordinate translation identifies the path at $a + X$ with the path at X and preserves Lebesgue measure. Multiplication of the output path by $e^{ip_a(X)}$ leaves pointwise jump counts unchanged, whereas the pure input factor is now part of f_a . In particular,

$$|f_a(Y)| = |f(a + Y)| \mathbf{1}_{Q_a^* - a}(Y), \quad \|f_a\|_1 = \|f \mathbf{1}_{Q_a^*}\|_1.$$

The anchored representative satisfies $\widehat{H}_m(Y, Y) = 0$. On the fixed bounded set relevant to $|X - Y| < 1$,

$$|\widehat{H}_m(X, Y)| \leq |X - Y| \int_0^1 |\nabla_X H_m(Y + t(X - Y), Y)| dt \leq C_{n,d} |X - Y|.$$

The last constant is uniform because the coefficient space $\mathcal{V}_m$ is finite dimensional, its normalized unit sphere is compact, and X, Y range over one fixed bounded set. Suppressing the harmless pure output factor, define, for $0 < \varepsilon < 1$,

$$E_{a,\varepsilon}(X) = \int_{\varepsilon < |X-Y| < 1} (e^{\widehat{H}_m(X,Y)} - 1) e^{iR_{a, < m}(X,Y)} K_\Omega(X - Y) f_a(Y) dY,$$

and put $E_{a,\varepsilon} = 0$ for $\varepsilon \geq 1$. For an arbitrary finite increasing truncation partition, the radial layers occurring in the successive increments are disjoint. Using $|e^{iu} - 1| \leq |u|$ and the preceding bound therefore gives

$$V^1(E_a)(X) \leq C_{n,d} \int_{|X-Y|<1} \frac{|\Omega((X-Y)/|X-Y|)|}{|X-Y|^{n-1}} |f_a(Y)| dY.$$

After translating back and discarding the input restriction this is (344). Since the output cubes are disjoint, the piecewise error satisfies the same pointwise majorant, and polar coordinates give

$$\int_{|z|<1} \frac{|\Omega(z/|z|)|}{|z|^{n-1}} dz = \|\Omega\|_1.$$

Tonelli thus proves the asserted global L^1 estimate. What remains after subtracting E_a is exactly the localized path with phase $R_{a, < m}$ applied to f_a , whose nonpure degree is at most $m - 1$. $\square$

Proposition 11.2 (Weak polynomial critical jumps on core data). *For every real polynomial P of degree at most d , every mean-zero $\Omega \in L \log L(\mathbb{S}^{n-1})$, and every bounded compactly supported f ,*

$$\|\mathbf{J}(T_{\Omega,\varepsilon}^P f : \varepsilon > 0)\|_{L^{1,\infty}} \leq C_{n,d} \mathfrak{L}(\Omega) \|f\|_1,$$

with a constant independent of the coefficients of P .

Proof. All weak estimates invoked in this induction are the core-data versions proved above. More precisely, the base case uses the bounded compactly supported conclusion of Proposition 8.1, and the far part uses the bounded compactly supported conclusion of Proposition 10.1. The modulated translated inputs f_a arising in Proposition 11.1 remain bounded and compactly supported, and satisfy $\|f_a\|_1 = \|f \mathbf{1}_{Q_a^*}\|_1$. Thus no operator value for a general L^1 input, and no result from Section 12, is used in the induction.

Induct on the largest total degree m carrying a nonpure homogeneous class. If there is no such class, the phase is pure and Proposition 8.1 applies after input and output conjugation. Assume the assertion for nonpure degree at most $m-1$ and let

$$C_m = \|P_m\|_{\text{coeff}} > 0, \quad \rho = C_m^{-1/m}.$$

Under $x = \rho X, y = \rho Y$, the entire degree- m class, rather than one scalar coordinate, has norm one. Proposition 10.1 controls the normalized far path.

For the local path partition the output into disjoint unit cubes Q_a , and let Q_a^* denote the fixed input enlargement relevant to Q_a . Proposition 11.1 gives on Q_a a residual localized path L_a with nonpure degree at most $m-1$, applied to the corresponding modulated translated input f_a , and an error E_a . The piecewise error $E = E_a$ on Q_a satisfies, by (344),

$$\|V^1(E)\|_1 \leq C_{n,d} \|\Omega\|_1 \|f\|_1 \leq C_{n,d} \mathfrak{L}(\Omega) \|f\|_1.$$

For the residual phase, subtracting the fixed truncation at one does not change increments below 1, and adjoining zero for $\varepsilon \geq 1$ creates no additional jump by (113). Taking the supremum in the jump amplitude in that pointwise domination, the induction hypothesis gives, for every $\alpha > 0$,

$$|\{x \in Q_a : \mathbf{J}(L_a : \varepsilon > 0) > \alpha\}| \leq C_{n,d} \alpha^{-1} \mathfrak{L}(\Omega) \|f_a\|_1 = C_{n,d} \alpha^{-1} \mathfrak{L}(\Omega) \|f \mathbf{1}_{Q_a^*}\|_1.$$

Let $L_{\text{res}} = L_a$ on Q_a . Since the output cubes are disjoint and the input enlargements have bounded overlap,

$$\begin{aligned} |\{x : \mathbf{J}(L_{\text{res}} : \varepsilon > 0) > \alpha\}| &= \sum_a |\{x \in Q_a : \mathbf{J}(L_a : \varepsilon > 0) > \alpha\}| \\ &\leq C_{n,d} \alpha^{-1} \mathfrak{L}(\Omega) \sum_a \|f \mathbf{1}_{Q_a^*}\|_1 \\ &\leq C_{n,d} \alpha^{-1} \mathfrak{L}(\Omega) \|f\|_1. \end{aligned}$$

For the error, (19) and $V^2 \leq V^1$ give $\mathbf{J}(E) \leq V^1(E)$; hence its L^1 majorant yields the required weak estimate. Thus the local residual and error paths have the desired pointwise critical-jump bound.

Combine local and far paths with (18). More explicitly, the two weak distribution estimates and the L^1 error majorant give a fixed C_0 such that, with $P_\rho(X, Y) = P(\rho X, \rho Y)$ and $f_\rho(X) = f(\rho X)$,

$$\left| \left\{ X : \mathbf{J}(T_{\Omega,\varepsilon}^{P_\rho} f_\rho : \varepsilon > 0)(X) > C_0 \alpha \right\} \right| \leq C \alpha^{-1} \mathfrak{L}(\Omega) \|f_\rho\|_1.$$

Putting $\alpha = \tau/C_0$ gives the normalized weak quasi-norm estimate. The degree- m block of P_ρ has coefficient norm $\rho^m C_m = 1$, and the exact conjugacy is

$$T_{\Omega,\varepsilon}^P f(\rho X) = T_{\Omega,\varepsilon/\rho}^{P_\rho} f_\rho(X), \quad f_\rho(X) = f(\rho X),$$

where the reparameterization $\varepsilon \mapsto \varepsilon/\rho$ preserves every jump count. The two scaling identities are, with H denoting any measurable scalar functional on $\mathbb{R}^n$ (in the application, the pointwise jump functional),

$$\|f_\rho\|_{L_X^1} = \rho^{-n} \|f\|_{L_X^1}, \quad \|H(\rho \cdot)\|_{L_X^{1,\infty}} = \rho^{-n} \|H\|_{L_X^{1,\infty}}.$$

Thus the factors ρ^{-n} on the normalized input and output sides cancel exactly, and the bound is independent of C_m . The finite degree induction proves the proposition for every polynomial phase. $\square$

12 Completion, consequences, and the direct radial BV transfer

12.1 Completion of the critical-jump trajectory

For core data, the weak endpoint propositions already remove the auxiliary cube, shell, and rational-parameter cutoffs. We pass to general L^1 data and construct the complete trajectory over all positive truncation parameters.

For bounded compactly supported f , put, for $N \geq 2$,

$$J_N(x) = \int_{N^{-1} < |z| < N} |K_\Omega(z)| |f(x-z)| dz.$$

Fubini gives

$$\int_{\mathbb{R}^n} J_N(x) dx = \|f\|_1 \|K_\Omega \mathbf{1}_{\{N^{-1} < |\cdot| < N\}}\|_1 < \infty.$$

After intersecting the countably many sets $\{J_N < \infty\}$, we obtain one full-measure set on which every finite-annulus integral is finite. Every compact interval $[a, b] \Subset (0, \infty)$ is contained in $[N^{-1}, N]$ for some N ; polar coordinates then show that, on this single set, the truncation path is locally absolutely continuous, hence continuous, simultaneously on all compact parameter intervals. Therefore the paired jump counts, and hence $\mathbf{J}$, computed on positive rationals agree there with those computed on all positive parameters by Lemma 2.6.

For such core data the path is zero for sufficiently large truncation at each fixed x . Equation (20) therefore gives pointwise

$$\sup_{\varepsilon > 0} |T_{\Omega, \varepsilon}^P h| \leq \mathbf{J}(T_{\Omega, \varepsilon}^P h : \varepsilon > 0).$$

The weak pointwise critical-jump estimate implies the maximal estimate

$$\left\| \sup_{q \in \mathbb{Q}_+} |T_{\Omega, q}^P h| \right\|_{L^{1, \infty}} \leq C_{n, d} \mathfrak{L}(\Omega) \|h\|_1. \quad (345)$$

Start with any bounded compactly supported approximating sequence and pass to a subsequence, still denoted by f_k , such that

$$f_k \rightarrow f \quad \text{in } L^1, \quad \|f_{k+1} - f_k\|_1 \leq 2^{-3k}.$$

In particular $\|f_k\|_1 \rightarrow \|f\|_1$. By (345),

$$\left| \left\{ \sup_{q \in \mathbb{Q}_+} |T_{\Omega, q}^P (f_{k+1} - f_k)| > 2^{-k} \right\} \right| \leq C \mathfrak{L}(\Omega) 2^k \|f_{k+1} - f_k\|_1 \leq C \mathfrak{L}(\Omega) 2^{-2k}.$$

The right side is summable in k . Borel–Cantelli gives almost-everywhere uniform convergence on $\mathbb{Q}_+$. More precisely, let

$$E_{\text{BC}} = \limsup_{k \rightarrow \infty} \left\{ x : \sup_{q \in \mathbb{Q}_+} |T_{\Omega, q}^P (f_{k+1} - f_k)(x)| > 2^{-k} \right\}.$$

This is a measurable null set, since the supremum is countable. For every rational $q > 0$, define on all of $\mathbb{R}^n$

$$T_{\Omega, q}^P f(x) = \begin{cases} \lim_{k \rightarrow \infty} T_{\Omega, q}^P f_k(x), & x \notin E_{\text{BC}}, \\ 0, & x \in E_{\text{BC}}. \end{cases} \quad (346)$$

Thus all rational coordinates are defined on the same set, every one is measurable, and every countable supremum formed from them is measurable. Outside E_{BC} there is $k_0(x)$ such that, for every $k \geq k_0(x)$,

$$\sup_{q \in \mathbb{Q}_+} |T_{\Omega, q}^P (f_{k+1} - f_k)(x)| \leq 2^{-k}.$$

Consequently the series

$$\mathcal{R}_m(q, x) := \sum_{k \geq m} T_{\Omega, q}^P(f_{k+1} - f_k)(x)$$

converges uniformly in $q \in \mathbb{Q}_+$ outside that exceptional set, and, for every $m \geq k_0(x)$,

$$\sup_{q \in \mathbb{Q}_+} |\mathcal{R}_m(q, x)| \leq \sum_{k \geq m} 2^{-k}. \quad (347)$$

By (346), outside E_{BC} , $T_{\Omega, q}^P f = T_{\Omega, q}^P f_m + \mathcal{R}_m(q, \cdot)$; thus no operator value on $f - f_m$ is being invoked before the completion is constructed. For the core function f_m , the trajectory is zero at x for all sufficiently large q . Combining this fact with (347) gives the stronger uniform rational anchor

$$\lim_{R \rightarrow \infty} \sup_{\substack{q \in \mathbb{Q}_+ \\ q \geq R}} |T_{\Omega, q}^P f(x)| = 0 \quad (348)$$

for almost every x : given m , take R beyond the support-dependent vanishing threshold for f_m , and then let $m \rightarrow \infty$. Equivalently, for every $m \geq k_0(x)$,

$$\limsup_{R \rightarrow \infty} \sup_{\substack{q \in \mathbb{Q}_+ \\ q \geq R}} |T_{\Omega, q}^P f(x)| \leq \sum_{k \geq m} 2^{-k};$$

letting $m \rightarrow \infty$ proves (348) with the quantifiers in the asserted order.

Select a further subsequence solely to obtain the pointwise finite-annulus identities below, and do not relabel it. This selection does not alter any of the rational limits already obtained, because the convergence on $\mathbb{Q}_+$ was uniform along the preceding sequence. For every rational pair $0 < a < b < \infty$, the absolute-value Young bound for the finite-annulus kernel,

$$\|K_{\Omega} \mathbf{1}_{\{a < |\cdot| \leq b\}}\|_1 = \|\Omega\|_1 \log(b/a),$$

gives L^1 convergence of

$$T_{\Omega, a}^P f_k - T_{\Omega, b}^P f_k \quad \text{to} \quad \int_{a < |z| \leq b} e^{iP(x, x-z)} K_{\Omega}(z) f(x-z) dz.$$

A diagonal choice over the countable set of rational pairs gives pointwise convergence of all these identities on one full-measure set. Separately, Fubini applied to the countable annuli $\{N^{-1} < |z| < N\}$, $N \geq 2$, gives a common full-measure set on which every finite-annulus integral is finite and depends locally absolutely continuously on its radial endpoints. Intersecting these sets with the complement of E_{BC} preserves full measure; denote the resulting measurable set by $\mathcal{X}$.

On $\mathcal{X}$ define the value at arbitrary ε from the base point 1 by

$$T_{\Omega, \varepsilon}^P f = T_{\Omega, 1}^P f + \begin{cases} \int_{\varepsilon < |z| \leq 1} e^{iP(x, x-z)} K_{\Omega}(z) f(x-z) dz, & \varepsilon < 1, \\ - \int_{1 < |z| \leq \varepsilon} e^{iP(x, x-z)} K_{\Omega}(z) f(x-z) dz, & \varepsilon > 1. \end{cases}$$

On $\mathcal{X}^c$ define $T_{\Omega, \varepsilon}^P f = 0$ simultaneously for every $\varepsilon > 0$. For each fixed ε , this is a measurable representative: the finite-annulus integral is measurable by the parameter integral theorem and $T_{\Omega, 1}^P f$ is measurable by (346). The modification on $\mathcal{X}^c$ merely replaces all rational representatives on one additional common null set. The resulting path is locally absolutely continuous, hence continuous, in ε and agrees with the already constructed rational values on $\mathcal{X}$ by the rational finite-annulus identities.

Fix an enumeration $\mathbb{Q}_+ = \{q_1, q_2, \dots\}$ and set $D_N = \{q_1, \dots, q_N\}$. For every N , (37) gives on $\mathcal{X}$, hence almost everywhere,

$$\mathbf{J}_{D_N}(T_{\Omega, \varepsilon}^P f : \varepsilon \in D_N) \leq \liminf_{k \rightarrow \infty} \mathbf{J}_{D_N}(T_{\Omega, \varepsilon}^P f_k : \varepsilon \in D_N).$$

The finite functional is Borel in its finitely many coordinates. The lower-semicontinuity statement is exactly the strict-witness statement: if the left side is $> \tau$, one may choose a jump amplitude and a finite strict witness with a positive excess over τ ; coordinatewise convergence preserves that same witness for all sufficiently large k . Consequently, for every $\tau > 0$,

$$\mathbf{1}_{\{\mathbf{J}_{D_N}(T_{\Omega,\varepsilon}^P f : \varepsilon \in D_N) > \tau\}} \leq \liminf_{k \rightarrow \infty} \mathbf{1}_{\{\mathbf{J}_{D_N}(T_{\Omega,\varepsilon}^P f_k : \varepsilon \in D_N) > \tau\}}.$$

Fatou's lemma and $\|f_k\|_1 \rightarrow \|f\|_1$ therefore transfer the core distribution estimate with the same right-hand side $C\tau^{-1}\mathfrak{L}(\Omega)\|f\|_1$, uniformly in N . Because the sets D_N are nested and every strict rational jump witness is finite,

$$\{\mathbf{J}(T_{\Omega,q}^P f : q \in \mathbb{Q}_+) > \tau\} = \bigcup_{N=1}^{\infty} \{\mathbf{J}_{D_N}(T_{\Omega,q}^P f : q \in D_N) > \tau\}. \quad (349)$$

Continuity from below therefore transfers the same distribution estimate to the measurable rational pointwise functional. Every finite full-parameter witness lies in a compact interval; local continuity and Lemma 2.6 then identify it almost everywhere with the full positive-parameter functional. Finally, for every $R > 0$, continuity and density of $\mathbb{Q}_+$ imply

$$\sup_{\varepsilon \geq R} |T_{\Omega,\varepsilon}^P f(x)| \leq \sup_{\substack{q \in \mathbb{Q}_+ \\ q \geq R}} |T_{\Omega,q}^P f(x)|.$$

Letting $R \rightarrow \infty$ in (348) proves the zero-at-infinity anchor for all real radii. In particular the construction is independent of the chosen core approximation: two resulting paths have the same finite-annulus increments, so on the intersection of their full-measure defining sets their difference is independent of ε ; their common zero anchor forces that difference to vanish. This also proves uniqueness among all paths with the finite-annulus identities and the zero anchor. This completes the unweighted weak critical-jump construction and, together with the strong theorem already proved, establishes Theorem 1.1.

12.2 Abstract consequences

The pointwise inequality (21) applied to the completed trajectory gives, for every $r > 2$,

$$V^r(T_{\Omega,\varepsilon}^P f : \varepsilon > 0) \leq \kappa_r \mathbf{J}(T_{\Omega,\varepsilon}^P f : \varepsilon > 0) \quad \text{almost everywhere.}$$

Taking the strong or weak spatial norm and using Theorem 1.1 proves (9) and

$$\|V^r(T_{\Omega,\varepsilon}^P f : \varepsilon > 0)\|_{L^{1,\infty}} \leq C_{n,d} \kappa_r \mathfrak{L}(\Omega) \|f\|_1. \quad (350)$$

No interchange of a spatial norm with the jump-amplitude supremum is used here. The zero anchor and (20) give the maximal estimate, while finite r -variation gives the limit at zero. This proves Corollary 1.2.

12.3 Radial BV transfer

Write

$$F_\varepsilon = T_{\Omega,\varepsilon}^P f, \quad G_\varepsilon = T_{\Omega,\varepsilon}^{P,b} f.$$

Fix throughout the right-continuous representative of b specified after (11). A right-continuous BV function determines a finite complex Borel measure by

$$db((s,t]) = b(t) - b(s), \quad |db|(0,\infty) \leq \text{Var}_{(0,\infty)}(b).$$

For test data the path F is locally absolutely continuous. The Lebesgue–Stieltjes product formula on $0 < \varepsilon < R < \infty$ reads

$$b(R)F_R - b(\varepsilon)F_\varepsilon = \int_{(\varepsilon,R]} F_s db(s) + \int_{(\varepsilon,R]} b(s-) dF_s.$$

Since $dF \ll ds$ has no atoms, the last integrand may equivalently be written $b(s)$. Moreover $G_\varepsilon - G_R = -\int_\varepsilon^R b(s) dF_s$. Rearranging gives

$$G_\varepsilon - G_R = b(\varepsilon)F_\varepsilon - b(R)F_R + \int_{(\varepsilon, R]} F_s db(s). \quad (351)$$

Altering b on a countable set of radii does not change the original Lebesgue shell integral, whereas the Stieltjes formula depends on the chosen representative. We next justify the passage to general data on one common full-measure set.

Use the representatives of F_ε constructed in the strong completion and in the preceding weak endpoint completion. In both cases the base coordinate F_1 is the almost-everywhere limit of core-data coordinates, and every other coordinate is fixed by its signed finite-annulus increment from 1. Choose the corresponding core approximation f_k as follows. In the strong L^p case, pass to a subsequence such that

$$\sum_k \|f_{k+1} - f_k\|_p < \infty.$$

Put

$$M_k = \sup_{q \in \mathbb{Q}_+} |T_{\Omega, q}^P(f_{k+1} - f_k)|.$$

The strong maximal estimate and Minkowski's inequality give

$$\sum_k \|M_k\|_p \leq C\mathfrak{L}(\Omega) \sum_k \|f_{k+1} - f_k\|_p < \infty.$$

Applying this bound to finite partial sums and then using monotone convergence shows that $\sum_k M_k \in L^p$, and hence is finite almost everywhere. In the weak L^1 case, choose instead $\|f_{k+1} - f_k\|_1 \leq 2^{-3k}$. By (345),

$$\sum_k \left| \left\{ \sup_{q \in \mathbb{Q}_+} |T_{\Omega, q}^P(f_{k+1} - f_k)| > 2^{-k} \right\} \right| < \infty,$$

so Borel–Cantelli gives the same uniform Cauchy conclusion outside one null set. Thus in either case $F_q^{(k)}$ converges uniformly in $q \in \mathbb{Q}_+$ on a common full-measure set. Since every core-data difference is locally continuous in the truncation parameter, the same sequence is uniformly Cauchy on each compact parameter interval: for a continuous difference, the supremum on such an interval is its supremum on the rational parameters there.

The locally uniform limit of the continuous core trajectories is locally continuous. Fix at this point Borel representatives of f and Ω . Also choose a Borel function F_1^B which agrees almost everywhere with the already completed base coordinate F_1 . All finite-annulus integrals below are formed with these fixed Borel representatives. For every rational pair $0 < a < c < \infty$, Young's inequality uses

$$\|K_\Omega \mathbf{1}_{\{a < |\cdot| \leq c\}}\|_1 \leq \|\Omega\|_1 \log(c/a) \quad (352)$$

to pass the core-data finite-annulus identity to the limit in L^p in the strong case and in L^1 in the weak case. Pass to one further diagonal subsequence. The locally uniform convergence is unaffected, and outside one null set the identity holds simultaneously for every rational pair.

For $j \geq 1$, put

$$J_j(x) = \int_{2^{-j} < |z| < 2^j} |K_\Omega(z)| |f(x-z)| dz.$$

The parameter integral theorem makes every J_j Borel measurable, and Young's inequality shows that it belongs to L^p in the strong case and to L^1 in the weak case. Form one exceptional set from the following null sets: the failure set for the locally uniform core limit; the set on which $F_1^B \neq F_1$; the simultaneous failure set for the rational finite-annulus identities just obtained; the sets where some J_j is infinite; and

the failure sets for the zero anchor and for finiteness of the pointwise critical-jump functional. This is a countable union of Lebesgue null sets. Contain it in a Borel null set $\mathcal{N}_{\text{BV}}$, and put

$$\mathcal{X}_{\text{BV}} = \mathbb{R}^n \setminus \mathcal{N}_{\text{BV}}.$$

Replace F_1^{B} by zero on $\mathcal{N}_{\text{BV}}$, and henceforth denote this Borel base coordinate simply by F_1 . Thus, at every point of $\mathcal{X}_{\text{BV}}$, it is the previously completed base coordinate, all rational finite-annulus identities hold for the fixed Borel versions of f and Ω , the core trajectories converge locally uniformly, and the zero anchor and finite critical-jump bound hold.

The function

$$\Phi(x, r, \theta) = e^{iP(x, x-r\theta)} \Omega(\theta) f(x-r\theta), \quad (x, r, \theta) \in \mathbb{R}^n \times (0, \infty) \times \mathbb{S}^{n-1},$$

is jointly Borel measurable. By the parameter integral theorem, both $(x, r) \mapsto \int_{\mathbb{S}^{n-1}} |\Phi(x, r, \theta)| d\sigma(\theta)$ and the integrals of the real and imaginary parts of Φ , wherever absolutely convergent, are Borel measurable. Define on the entire product space

$$\mathcal{A}(x, r) = \begin{cases} \int_{\mathbb{S}^{n-1}} \Phi(x, r, \theta) d\sigma(\theta), & x \in \mathcal{X}_{\text{BV}} \text{ and } \int_{\mathbb{S}^{n-1}} |\Phi(x, r, \theta)| d\sigma(\theta) < \infty, \\ 0, & \text{otherwise.} \end{cases} \quad (353)$$

Thus $(x, r) \mapsto \mathcal{A}(x, r)$ is a jointly Borel version, not merely a separately defined family of radial sections. Polar coordinates and the finiteness of the J_j give, for $x \in \mathcal{X}_{\text{BV}}$ and on every compact $0 < a < c < \infty$,

$$\int_a^c |\mathcal{A}(x, r)| \frac{dr}{r} \leq \int_{a < |z| < c} |K_{\Omega}(z)| |f(x-z)| dz < \infty.$$

Thus the sphere integral is absolutely convergent for dr/r -almost every r , which is exactly the representative needed below; no assertion is made about every individual spherical restriction of an L^p function. For rational $a < c$, the finite-annulus identity is

$$F_a(x) - F_c(x) = \int_a^c \mathcal{A}(x, r) \frac{dr}{r}.$$

The left-hand side is continuous in the endpoints, while the indefinite integral on the right is locally absolutely continuous. Density of the rational endpoints therefore extends the identity to all real $0 < a < c < \infty$. Thus, on this same full-measure set, F is locally absolutely continuous and

$$\int_a^c |\mathcal{A}(x, r)| \frac{dr}{r} < \infty, \quad F_a(x) - F_c(x) = \int_a^c \mathcal{A}(x, r) \frac{dr}{r}. \quad (354)$$

Using the Borel base coordinate F_1 fixed above, define on the entire product space

$$F(x, s) = F_1(x) + \mathbf{1}_{\{s < 1\}} \int_s^1 \mathcal{A}(x, r) \frac{dr}{r} - \mathbf{1}_{\{s > 1\}} \int_1^s \mathcal{A}(x, r) \frac{dr}{r}. \quad (355)$$

The parameter integral theorem makes $(x, s) \mapsto F(x, s)$ jointly Borel measurable. On $\mathcal{X}_{\text{BV}}$ it agrees with every previously constructed coordinate by (354); on its complement it is identically zero. We henceforth use this common representative. Finally, the zero anchor and (20) give

$$F_R \longrightarrow 0 \quad (R \rightarrow \infty), \quad M_F := \sup_{s > 0} |F_s| \leq \mathbf{J}(F) < \infty.$$

On the entire product space define

$$G(x, \varepsilon) = b(\varepsilon)F(x, \varepsilon) + \int_{(\varepsilon, \infty)} F(x, s) db(s). \quad (356)$$

The Stieltjes integral is absolutely convergent because $|db|(0, \infty) < \infty$ and $M_F < \infty$ on $\mathcal{X}_{BV}$, and both F and G are zero on its complement. Moreover, the integrand $(x, \varepsilon, s) \mapsto \mathbf{1}_{\{s > \varepsilon\}} F(x, s)$ is jointly Borel. Applying the parameter integral theorem to the total-variation measure $|db|$, and then to the real and imaginary parts of the complex measure db , proves that $(x, \varepsilon) \mapsto G(x, \varepsilon)$ is jointly Borel measurable. For $0 < \varepsilon < R < \infty$, integration by parts for the locally absolutely continuous path F and the fixed right-continuous representative of b uses the product formula

$$b(R)F_R - b(\varepsilon)F_\varepsilon = \int_{(\varepsilon, R]} F_s db(s) + \int_\varepsilon^R b(s) dF_s. \quad (357)$$

There is no ambiguity between $b(s)$ and $b(s-)$ in the second integral, because dF has no atoms. By (354),

$$dF_s = -\mathcal{A}(x, s) \frac{ds}{s}.$$

This equality is an equality of locally finite absolutely continuous measures and uses the preceding $L^1_{\text{loc}}(ds/s)$ representative of $\mathcal{A}$; in particular it is only an almost-everywhere density assertion. Consequently, (357) gives

$$\begin{aligned} G_\varepsilon - G_R &= b(\varepsilon)F_\varepsilon - b(R)F_R + \int_{(\varepsilon, R]} F_s db(s) \\ &= - \int_{(\varepsilon, R]} b(s) dF_s \\ &= \int_{\varepsilon < |z| \leq R} e^{iP(x, x-z)} K_\Omega(z) b(|z|) f(x-z) dz. \end{aligned} \quad (358)$$

Indeed, $-\int_\varepsilon^R b(s) dF_s = \int_\varepsilon^R b(s) \mathcal{A}(x, s) ds/s$, and polar coordinates give exactly the last line, including its sign and half-open endpoint convention. The last integral is absolutely meaningful on the same set; its kernel has L^1 norm at most $\|b\|_\infty \|\Omega\|_1 \log(R/\varepsilon)$. Hence (356) has exactly the required finite-annulus increments for general data. For clarity, it also agrees with the original operator on core data, rather than merely defining an abstract extension. Indeed, if f is bounded and compactly supported, then

$$\tilde{G}_\varepsilon(x) = \int_{|z| > \varepsilon} e^{iP(x, x-z)} K_\Omega(z) b(|z|) f(x-z) dz$$

is absolutely convergent for each $\varepsilon > 0$, has the same finite-annulus increments as G by (358), and satisfies $\tilde{G}_R(x) = 0$ for all sufficiently large R . Since $G_R(x) \rightarrow 0$, letting $R \rightarrow \infty$ in the common increment identity gives $G_\varepsilon(x) = \tilde{G}_\varepsilon(x)$ on the common full-measure set. For general data, the finite-annulus increments and zero anchor uniquely determine the extension. Right continuity fixes the Stieltjes representative, while altering b at one radius does not change the original Lebesgue shell integral. Finally,

$$|G_\varepsilon| \leq \|b\|_\infty |F_\varepsilon| + M_F |db|((\varepsilon, \infty)) \longrightarrow 0 \quad (\varepsilon \rightarrow \infty), \quad (359)$$

so the zero anchor uniquely determines the extension.

Put $B = \|b\|_\infty$, $V = \text{Var}(b)$, and $M_F = \sup_{s>0} |F_s|$. The zero anchor gives $M_F \leq \mathbf{J}(F)$ by (20). For every $r \geq 1$, the same anchor also gives

$$M_F \leq V^r(F). \quad (360)$$

Indeed, for fixed $s > 0$ and $R > s$, the two-point partition gives $|F_s - F_R| \leq V^r(F)$; letting $R \rightarrow \infty$ and then taking the supremum in s proves (360). For a variation partition $t_0 < \dots < t_N$, use the exact product increment

$$b(t_i)F_{t_i} - b(t_{i-1})F_{t_{i-1}} = b(t_i)(F_{t_i} - F_{t_{i-1}}) + (b(t_i) - b(t_{i-1}))F_{t_{i-1}}.$$

Minkowski's inequality and $(\sum_i |\Delta_i b|^r)^{1/r} \leq \sum_i |\Delta_i b| \leq V$ now give the first line below. Together with (356) this yields, pointwise,

$$\begin{aligned} V^r(bF) &\leq BV^r(F) + VM_F, \\ V^r\left(\int_{(\varepsilon, \infty)} F_s db(s) : \varepsilon > 0\right) &\leq VM_F, \\ V^r(G) &\leq (B + 2V)V^r(F). \end{aligned} \quad (361)$$

Indeed, an increment of the Stieltjes-tail path over $(t_{i-1}, t_i]$ is the integral of F over that half-open interval. These intervals are pairwise disjoint, so its ℓ^r variation is bounded by the sum of their $|db|$ -masses times M_F , hence by VM_F .

We record separately the critical-jump transfer. If $M_F = 0$, then $F_s = 0$ for every $s > 0$, and (356) gives $G_\varepsilon = 0$; all assertions are then immediate. Hence assume $M_F > 0$. Suppose first that $B > 0$. For an ordered family of λ -jumps of bF , write

$$b(t)F_t - b(s)F_s = b(t)(F_t - F_s) + (b(t) - b(s))F_s.$$

Assign the pair to the first term if its modulus is $> \lambda/2$, and otherwise to the second. The first subfamily contributes at most $N_{\lambda/(2B)}(F)$. The parameter intervals in the second subfamily are disjoint, and hence

$$\frac{\lambda}{2M_F} N_{\text{second}} < \sum_{\text{second}} |b(t) - b(s)| \leq V.$$

This proves

$$N_\lambda(bF) \leq N_{\lambda/(2B)}(F) + \frac{2M_F V}{\lambda}. \quad (362)$$

Use $\sqrt{u+v} \leq \sqrt{u} + \sqrt{v}$. On the set where the left side is nonzero, $\lambda < 2BM_F$. With $\mu = \lambda/(2B)$, the contribution of the second term after taking square roots is bounded by

$$\sqrt{2\lambda M_F V} \leq 2M_F \sqrt{BV} \leq (B + V)M_F.$$

It follows that

$$\lambda N_\lambda(bF)^{1/2} \leq 2B\mu N_\mu(F)^{1/2} + C(B + V)M_F. \quad (363)$$

For $B = 0$ the path is zero. Define the Stieltjes-tail path by

$$H_\varepsilon = \int_{(\varepsilon, \infty)} F_s db(s).$$

It satisfies $V^1(H) \leq VM_F$. The disjoint jump intervals give $\lambda N_\lambda(H) < V^1(H) \leq VM_F$, whereas a nonzero jump also forces $\lambda < 2 \sup_\varepsilon |H_\varepsilon| \leq 2VM_F$. Hence

$$\lambda N_\lambda(H)^{1/2} \leq CVM_F. \quad (364)$$

Taking the supremum in λ in (363)–(364) gives

$$\mathbf{J}(bF) \leq 2B\mathbf{J}(F) + C(B + V)M_F, \quad \mathbf{J}(H) \leq CVM_F.$$

Since $G = bF + H$, equations (18) and (20) imply the pointwise transfer

$$\mathbf{J}(G) \leq C(B + V)\mathbf{J}(F). \quad (365)$$

The strong and weak pointwise critical-jump estimates for G now follow directly from Theorem 1.1. Formula (361) gives the asserted strong and weak r -variation bounds. This proves the estimates in Corollary 1.4.

The almost-everywhere zero anchor was proved in (359). For $f \in L^p$, the strong zero anchor for F , the strong maximal estimate, and continuity from above for the finite measure $|db|$ give the explicit bound

$$\|G_\varepsilon\|_p \leq B\|F_\varepsilon\|_p + |db|((\varepsilon, \infty))\|M_F\|_p \longrightarrow 0.$$

Thus both terms also tend to zero in L^p . Since a scalar path of finite r -variation has one-sided limits at both endpoints, G_ε has an almost-everywhere limit as $\varepsilon \downarrow 0$. The zero anchor gives $\sup_{\varepsilon > 0} |G_\varepsilon| \leq \mathbf{J}(G)$ by (20), and hence the maximal truncation assertion. This completes the proof of Corollary 1.4.

13 Finite radial variation beyond bounded variation

For $1 \leq \rho < \infty$ recall the radial norm

$$\|b\|_{\mathcal{V}^\rho} = \|b\|_\infty + V^\rho(b; (0, \infty)).$$

The purpose of this section is twofold. First, we record a direct pathwise Young–Stieltjes transfer when $\rho < 2$. Second, we prove Theorem 1.3 for every fixed finite ρ by showing that each oscillatory, packet, endpoint, and completion component of the unweighted proof is stable under a radial V^ρ amplitude. The latter argument is genuinely operator-specific at and above $\rho = 2$.

13.1 A pathwise Young–Stieltjes transfer below the radial exponent two

For radial variation below two, the full operator-specific proof is not needed. The following abstract statement isolates a direct consequence of the pointwise critical-jump theorem. It is also useful for identifying the precise obstruction at the exponent two.

Throughout this subsection, for a right-continuous integrator b we use the Young increment notation

$$db((s, t]) = b(t) - b(s).$$

This is only a half-open interval-increment convention; when $\rho > 1$, it does not assert that db is a countably additive Stieltjes measure. Such a measure is available in the BV case, whereas the argument below uses Young sums and the Young–Loeve estimate. Young sums on a finite annulus use the corresponding half-open interval $(a, c]$: a tagged sum over $a = t_0 < \dots < t_N = c$ is

$$\sum_{j=1}^N F(\xi_j) \{b(t_j) - b(t_{j-1})\}, \quad \xi_j \in [t_{j-1}, t_j].$$

Thus a jump of the right-continuous integrator at the right endpoint is included and one at the left endpoint is not. For $F = F_1 + iF_2$ and $b = b_1 + ib_2$, the complex Young integral is defined by

$$\int F db := \int F_1 db_1 - \int F_2 db_2 + i \left(\int F_1 db_2 + \int F_2 db_1 \right).$$

Thus it is the complex-bilinear extension of the four real integrals; the real Young–Loeve estimate therefore gives the same bound up to an absolute constant. Since the path F below is locally absolutely continuous and hence continuous, there is no common-jump correction in integration by parts. The integral against dF is the ordinary Lebesgue–Stieltjes integral; since local absolute continuity makes dF atomless, its value is unchanged if $b(s)$ is replaced by $b(s-)$.

Proposition 13.1 (Young–Stieltjes critical-jump transfer). *Let $F = (F_\varepsilon)_{\varepsilon>0}$ be a locally absolutely continuous scalar path such that*

$$F_\varepsilon \longrightarrow 0 \quad (\varepsilon \rightarrow \infty), \quad \mathbf{J}(F) < \infty.$$

Fix $1 \leq \rho < 2$, and let $b : (0, \infty) \rightarrow \mathbb{C}$ be bounded and of finite ρ -variation. Choose its right-continuous representative. Then the improper Young integral

$$H_\varepsilon := \int_{(\varepsilon, \infty)} F_s db(s) \tag{366}$$

exists, and the anchored path

$$G_\varepsilon := b(\varepsilon)F_\varepsilon + H_\varepsilon \tag{367}$$

satisfies

$$\mathbf{J}(G) \leq C_\rho (\|b\|_\infty + V^\rho(b)) \mathbf{J}(F). \tag{368}$$

Moreover $G_\varepsilon \rightarrow 0$ as $\varepsilon \rightarrow \infty$, and for every $0 < a < c < \infty$,

$$G_a - G_c = - \int_a^c b(s) dF_s. \tag{369}$$

Proof. Use the norm-nonincreasing right-continuous representative constructed after (11). To keep every constant quantitative, fix once and for all

$$u = u_\rho := \begin{cases} 3, & \rho = 1, \\ \frac{1}{2}(2 + \rho'), & 1 < \rho < 2, \end{cases} \quad \rho' = \frac{\rho}{\rho - 1}.$$

Then

$$2 < u < \rho', \quad \frac{1}{u} + \frac{1}{\rho} > 1, \quad (370)$$

where $\rho' = \infty$ when $\rho = 1$. Put

$$J = \mathbf{J}(F), \quad R = V^u(F), \quad Q = V^\rho(b), \quad B = \|b\|_\infty, \quad M = \sup_{s>0} |F_s|.$$

Equations (21) and (20) give

$$R \leq \kappa_u J, \quad M \leq J. \quad (371)$$

Because F is continuous and (370) holds, the real-valued Young theorem [31], applied componentwise as fixed above, applies on every compact interval. Its Young–Loeve estimate gives

$$\left| \int_{(a,c]} (F_s - F_a) db(s) \right| \leq C_{u,\rho} V^u(F; [a, c]) V^\rho(b; [a, c]). \quad (372)$$

Consequently,

$$\left| \int_{(a,c]} F_s db(s) \right| \leq \sup_{a \leq s \leq c} |F_s| |b(c) - b(a)| + C_{u,\rho} V^u(F; [a, c]) V^\rho(b; [a, c]). \quad (373)$$

Finite ρ -variation gives a limit of b at infinity, while finite u -variation and the zero anchor imply that both $\sup_{s \geq R} |F_s|$ and $V^u(F; [R, \infty))$ tend to zero as $R \rightarrow \infty$. For completeness, the latter assertion follows directly from finite variation. If it failed, there would be an $\eta > 0$ and, recursively, disjoint finite intervals $[R_k, S_k]$, tending to infinity, with

$$V^u(F; [R_k, S_k]) > \eta.$$

Choose on each such interval a finite partition whose u -power sum is larger than η^u . Concatenating the first N partitions gives $V^u(F)^u \geq N\eta^u$, a contradiction. Thus the two terms on the right of (373), with $a, c \geq R$, tend to zero uniformly as $R \rightarrow \infty$. Hence that estimate makes the compact Young integrals Cauchy and proves (366). It also yields

$$\sup_{\varepsilon > 0} |H_\varepsilon| \leq C_{u,\rho} Q(M + R) \leq C_\rho QJ. \quad (374)$$

With the half-open convention fixed above, Young integration by parts (valid because F is continuous) gives for $0 < \varepsilon < R_0 < \infty$,

$$b(R_0)F_{R_0} - b(\varepsilon)F_\varepsilon = \int_{(\varepsilon, R_0]} F_s db(s) + \int_\varepsilon^{R_0} b(s) dF_s.$$

As $R_0 \rightarrow \infty$, the first integral on the right converges to H_ε by its Cauchy construction, while $|b(R_0)F_{R_0}| \leq B|F_{R_0}| \rightarrow 0$. The identity therefore also proves the existence of the improper integral against dF and gives

$$G_\varepsilon = b(\varepsilon)F_\varepsilon + H_\varepsilon = - \int_\varepsilon^\infty b(s) dF_s. \quad (375)$$

Separately, as $\varepsilon \rightarrow \infty$,

$$|b(\varepsilon)F_\varepsilon| \leq B \sup_{s \geq \varepsilon} |F_s| \longrightarrow 0, \quad |H_\varepsilon| \leq Q \sup_{s \geq \varepsilon} |F_s| + C_{u,\rho} V^u(F; [\varepsilon, \infty))Q \longrightarrow 0.$$

Thus $G_\varepsilon \rightarrow 0$ at infinity. For two finite endpoints, the tail definition and finite-interval integration by parts give explicitly

$$\begin{aligned} G_a - G_c &= b(a)F_a - b(c)F_c + \int_{(a,c]} F_s db(s) \\ &= - \int_a^c b(s) dF_s. \end{aligned}$$

This proves (369), including its sign and half-open endpoint convention.

It remains to estimate jumps. Consider an ordered family of strict λ -jumps of the product path bF . On each jump interval (s, t) ,

$$b(t)F_t - b(s)F_s = b(t)(F_t - F_s) + (b(t) - b(s))F_s.$$

Assign the interval to the first term when its modulus exceeds $\lambda/2$, and otherwise to the second. For $B > 0$ this gives

$$N_\lambda(bF) \leq N_{\lambda/(2B)}(F) + \left(\frac{2MQ}{\lambda} \right)^\rho. \quad (376)$$

A nonzero jump also forces $\lambda \leq 2BM$. Since $\rho < 2$,

$$\lambda \left(\frac{2MQ}{\lambda} \right)^{\rho/2} \leq C_\rho M B^{1-\rho/2} Q^{\rho/2} \leq C_\rho M(B+Q).$$

Moreover,

$$\lambda N_{\lambda/(2B)}(F)^{1/2} = 2B \frac{\lambda}{2B} N_{\lambda/(2B)}(F)^{1/2} \leq 2BJ.$$

Using $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ in (376), and then $M \leq J$, therefore gives

$$\mathbf{J}(bF) \leq C_\rho(B+Q)J. \quad (377)$$

The case $B = 0$ is immediate.

Now let (s_i, t_i) be ordered λ -jump intervals for H , and put

$$A_i = V^u(F; [s_i, t_i]), \quad C_i = V^\rho(b; [s_i, t_i]).$$

By (373),

$$|H_{s_i} - H_{t_i}| \leq M|b(t_i) - b(s_i)| + C_{u,\rho} A_i C_i.$$

If $Q = 0$, the fixed right-continuous representative of b is constant, the Young tail H vanishes, and there is nothing to prove. Hence assume $Q > 0$ below. Assign each interval to a family $\mathcal{J}_1$ or $\mathcal{J}_2$ according as the first or the second term on the right exceeds a fixed positive multiple of λ , breaking ties arbitrarily. If $N_v = \#\mathcal{J}_v$, then

$$N_1 \leq C \left(\frac{MQ}{\lambda} \right)^\rho.$$

Whenever a jump exists, (374) gives $\lambda \leq C_\rho QJ$. Since $\rho < 2$ and $M \leq J$, the precise interpolation is

$$\lambda N_1^{1/2} \leq C(MQ)^{\rho/2} \lambda^{1-\rho/2} \leq C_\rho (MQ)^{\rho/2} (QJ)^{1-\rho/2} \leq C_\rho QJ.$$

For the second family define $0 < \theta < 1$ by

$$\frac{1}{\theta} = \frac{1}{u} + \frac{1}{\rho}.$$

Disjointness of the jump intervals and Hölder's inequality give

$$N_2(c\lambda)^\theta \leq \sum_{i \in \mathcal{J}_2} (A_i C_i)^\theta \leq \left(\sum_{i \in \mathcal{J}_2} A_i^u \right)^{\theta/u} \left(\sum_{i \in \mathcal{J}_2} C_i^\rho \right)^{\theta/\rho} \leq R^\theta Q^\theta, \quad N_2 \leq C_{u,\rho} \left(\frac{RQ}{\lambda} \right)^\theta.$$

Using again $\lambda \leq C_\rho QJ$ and $R \leq \kappa_\mu J$ gives

$$\lambda N_2^{1/2} \leq C_{u,\rho} (RQ)^{\theta/2} \lambda^{1-\theta/2} \leq C_\rho (RQ)^{\theta/2} (QJ)^{1-\theta/2} \leq C_\rho QJ.$$

For the original witness $N = N_1 + N_2$, and hence $\lambda \sqrt{N} \leq \lambda \sqrt{N_1} + \lambda \sqrt{N_2} \leq C_\rho QJ$. Taking the supremum over all finite strict witnesses and then over $\lambda > 0$ gives

$$\mathbf{J}(H) \leq C_\rho QJ. \quad (378)$$

Finally apply the pointwise quasi-triangle inequality (18) to $G = bF + H$ and combine (377) with (378). $\square$

13.2 The operator-specific theorem for every finite radial exponent

We now prove Theorem 1.3. The argument retains the notation of the preceding sections so that every dependency can be checked directly. By homogeneity in b , it is enough to impose

$$\|b\|_{\gamma\rho} \leq 1. \quad (379)$$

The factor $\|b\|_{\gamma\rho}$ is restored only at the end. Notice also the exact dilation invariance

$$\|b(R\cdot)\|_\infty = \|b\|_\infty, \quad V^\rho(b(R\cdot)) = V^\rho(b), \quad R > 0. \quad (380)$$

Quantitative convention. Fix n, d , and $1 \leq \rho < \infty$ for the remainder of this section. Every polynomial space, rational chart, compact interval, cutoff, and finite-dimensional norm below belongs to one of the finitely many structural families already fixed in Sections 3–7. We therefore take the minimum of the finitely many positive sublevel exponents in Lemma 7.1 and denote it by $c_{\text{sub}} = c_{\text{sub}}(n, d) > 0$. Constants denoted by C_ρ may increase from line to line and depend only on n, d, ρ and the fixed structural data. Polynomial powers of an amplitude norm are structural and finite; whenever such a power enters a later parameter choice, it is fixed once and for all at that point. Positive exponents are always named before they are used. Replacing an exponent by a smaller positive number is allowed without further comment. This convention makes the parameter choices below finite and noncircular: first the oscillatory exponents are fixed, then the packet hierarchy, and only afterwards the angular splitting parameters.

13.3 Finite ρ -variation calculus

We first collect the elementary facts used everywhere below. Throughout this subsection a compactly supported radial profile is first replaced by its fixed right-continuous representative and then extended by zero to the real line. Variations without an indicated interval refer to this zero extension. The endpoint convention for consecutive cells is specified in each displayed partition, and no endpoint value is counted twice. Thus every ordinary supremum and every endpoint jump refers to one fixed representative.

Lemma 13.2 (Disjoint intervals, pullbacks, products, and step profiles). *Let $1 \leq \rho < \infty$.*

- (i) *If a is a scalar function with $V^\rho(a) < \infty$ and $I_1, \dots, I_M$ are pairwise disjoint intervals arranged from left to right, then*

$$\sum_{m=1}^M V^\rho(a; I_m)^\rho \leq V^\rho(a)^\rho. \quad (381)$$

- (ii) *If $a : I \rightarrow \mathbb{C}$ has finite ρ -variation and $\gamma : J \rightarrow I$ is monotone, then*

$$V^\rho(a \circ \gamma; J) \leq V^\rho(a; I). \quad (382)$$

- (iii) *For bounded f, g with $V^\rho(f) + V^\rho(g) < \infty$,*

$$V^\rho(fg) \leq \|f\|_\infty V^\rho(g) + \|g\|_\infty V^\rho(f). \quad (383)$$

- (iv) Let $x_0 < \dots < x_K$ and $I_k = [x_{k-1}, x_k)$ for $1 \leq k \leq K$. Extend σ by zero off $[x_0, x_K)$, and suppose that σ is constant on each I_k and $|\sigma| \leq 1$. Then

$$V^p(\sigma) \leq 2K^{1/p}. \quad (384)$$

Consequently, if additionally $0 < x_0 < \dots < x_K < \infty$ and (379) holds, then, with $b\sigma$ extended by zero off $[x_0, x_K)$,

$$\|b\sigma\|_\infty + V^p(b\sigma) \leq C_p(1 + K^{1/p}). \quad (385)$$

Proof. For (i), choose on every I_m a partition which is within an arbitrary ε of its variation and concatenate these partitions in increasing order. The within- I_m increments are a subcollection of the increments in the resulting global partition. Letting $\varepsilon \downarrow 0$ proves (381). Assertion (ii) follows by pulling an ordered partition of J through γ and deleting repetitions; if γ is decreasing, reverse the resulting list, which does not change the moduli of its successive increments. For (iii), on every increment

$$f(t)g(t) - f(s)g(s) = f(t)(g(t) - g(s)) + g(s)(f(t) - f(s)),$$

and Minkowski in the finite ℓ^p norm gives (383). Finally, fix an arbitrary variation partition for the zero-extended step profile. Delete zero increments and all but one point from each constant cell that the partition visits. The remaining ordered list has at most $K - 1$ increments between two cells, each of modulus at most two, and at most two increments between the exterior zero value and an interior cell, each of modulus at most one. Thus every such partition satisfies

$$\sum_j |\sigma(t_j) - \sigma(t_{j-1})|^p \leq 2 + (K - 1)2^p \leq K2^p.$$

Taking the supremum proves (384). More explicitly, the zero extension creates at most two endpoint increments, and hence

$$V^p(b\sigma; \mathbb{R}) \leq \|b\|_{L^\infty(\text{supp } \sigma)} V^p(\sigma; \mathbb{R}) + \|\sigma\|_\infty \{V^p(b; \text{supp } \sigma) + 2\|b\|_\infty\}.$$

This proves (385) without extending b itself outside $(0, \infty)$. $\square$

Lemma 13.3 (Translation modulus and mollification). *Let $I \Subset \mathbb{R}$ be fixed and let a be bounded, supported in the interior of I , and extended by zero to $\mathbb{R}$. All variation seminorms in this lemma are taken for that zero extension. Then for $|h| \leq 1$,*

$$\|a(\cdot + h) - a\|_{L^1(\mathbb{R})} \leq C_{I,p} |h|^{1/p} (\|a\|_\infty + V^p(a; I)). \quad (386)$$

If $0 < h \leq 1$, ϕ is a standard compactly supported mollifier, $\phi_h(t) := h^{-1}\phi(t/h)$, and $a_h := a * \phi_h$, then, for every integer $N \geq 0$,

$$\|a - a_h\|_1 \leq C_{I,p} h^{1/p} (\|a\|_\infty + V^p(a; I)), \quad (387)$$

$$V^p(a_h) \leq V^p(a), \quad (388)$$

$$\|\partial^N a_h\|_\infty \leq C_N h^{-N} \|a\|_\infty. \quad (389)$$

Proof. It is enough to treat $h > 0$. For $x \in [0, h)$ consider the ordered chain $x + kh$. Only $O_I(1 + h^{-1})$ increments meet the support of the zero extension. Hölder's inequality and the definition of V^p give

$$\sum_k |a(x + (k+1)h) - a(x + kh)| \leq C_I h^{-(1-1/p)} (\|a\|_\infty + V^p(a; I)).$$

Integrating over $x \in [0, h)$ proves (386). Averaging this inequality against the mollifier proves (387). For any finite partition (t_i) , Minkowski in ℓ^p gives

$$\left(\sum_i |a_h(t_i) - a_h(t_{i-1})|^p \right)^{1/p} \leq \int \phi_h(u) \left(\sum_i |a(t_i - u) - a(t_{i-1} - u)|^p \right)^{1/p} du,$$

which yields (388); the derivative estimate is standard. $\square$

The next elementary upper-sum estimate is the missing ingredient in the terminal packet argument. Small L^1 error alone is not enough there, because packet inputs are controlled by average mass rather than pointwise density.

Lemma 13.4 (Upper Riemann sums for finite variation). *Let $I \subseteq \mathbb{R}$ be a fixed bounded interval, and let $a : \mathbb{R} \rightarrow \mathbb{C}$ be bounded, supported in I , and satisfy $V^p(a; \mathbb{R}) < \infty$. Partition I into consecutive intervals J_m whose lengths are comparable to $\delta \in (0, 1]$. Then*

$$\delta \sum_m \sup_{J_m} |a| \leq C_I \left[\|a\|_{L^1(\mathbb{R})} + \delta^{1/p} (\|a\|_\infty + V^p(a; \mathbb{R})) \right]. \quad (390)$$

If J_m^ is obtained from J_m by an enlargement about its centre by a fixed factor, then the same estimate holds with $\sup_{J_m^*} |a|$ on the left; the constant may additionally depend on that fixed factor.*

Proof. For an un-enlarged interval,

$$\sup_{J_m} |a| \leq |J_m|^{-1} \int_{J_m} |a| + \text{osc}_{J_m}(a).$$

For each m , choose two ordered points whose increment approaches $\text{osc}_{J_m}(a)$. Concatenating these pairs in increasing order and then taking limits shows

$$\sum_m \text{osc}_{J_m}(a)^p \leq V^p(a; \mathbb{R})^p.$$

Hölder's inequality and the fact that there are $O_I(\delta^{-1})$ intervals give

$$\delta \sum_m \text{osc}_{J_m}(a) \leq C_I \delta^{1/p} V^p(a; \mathbb{R}),$$

which proves (390). For fixed enlargements, the interval graph of $(J_m^*)_m$ has uniformly bounded degree and hence a bounded proper coloring. Intervals in one color have disjoint interiors. Apply the preceding oscillation argument to each color; after using the same average-plus-oscillation inequality on J_m^* , the average terms are controlled by bounded overlap. This proves the enlarged version without identifying values at adjacent endpoints. $\square$

Lemma 13.5 (Radial discretization for packet inputs). *Fix $0 < c_- < c_+ < \infty$. Let $\Theta \in L^\infty(\mathbb{S}^{n-1})$, $\Lambda \geq 0$, and let w be a bounded measurable function supported in $[c_-, c_+]$ with $V^p(w) < \infty$. Let $0 < \delta \leq 1$, let $\{U\}$ be pairwise disjoint dyadic δ -cubes in $\mathbb{R}^n$, and suppose functions $d_U \in L^1(\mathbb{R}^n)$ satisfy*

$$\text{supp } d_U \subset U, \quad \|d_U\|_1 \leq \Lambda |U|.$$

Then

$$\left\| K_\Theta(\cdot) w(|\cdot|) * \sum_U d_U \right\|_\infty \leq C \Lambda \|\Theta\|_\infty \left[\|w\|_1 + \delta^{1/p} (\|w\|_\infty + V^p(w)) \right], \quad (391)$$

$$\left\| K_\Theta(\cdot) w(|\cdot|) * \sum_U d_U \right\|_1 \leq C \|\Theta\|_1 \|w\|_1 \sum_U \|d_U\|_1. \quad (392)$$

Here the L^1 norm of w is one-dimensional; on the fixed annulus it is equivalent to the natural dr/r norm.

Proof. Only (391) is new. Fix x . Since $|z|$ stays in a fixed annulus,

$$\left| K_\Theta w * \sum_U d_U(x) \right| \leq C \Lambda \|\Theta\|_\infty \delta^n \sum_U \sup_{y \in U} |w(|x - y|)|.$$

Group the cubes according to intervals of length $C\delta$ containing the distance from x to their centers. If their centers lie in one such layer, the union of the corresponding disjoint cubes is contained in a fixed enlargement of a radial layer of volume $O(\delta)$. Since every cube has volume δ^n , the layer contains at most $C\delta^{1-n}$ cubes. Hence the last display is at most

$$C\Lambda\|\Theta\|_\infty\delta\sum_m\sup_{J_m^*}|w|,$$

where the J_m^* are fixed enlargements of consecutive radial intervals of length comparable to δ . Lemma 13.4 proves (391). Equation (392) is Young's inequality and polar coordinates. $\square$

13.4 Oscillatory integrals with a finite-variation amplitude

Lemma 13.6 (First derivative estimate with a V^p amplitude). *Let I be a fixed bounded interval, let $\phi \in C^1(I; \mathbb{R})$, and let $a : I \rightarrow \mathbb{C}$ be bounded with $V^p(a; I) < \infty$. Suppose ϕ' is monotone and*

$$|\phi'(t)| \geq L \geq 1 \quad (t \in I).$$

Then

$$\left| \int_I e^{i\phi(t)} a(t) dt \right| \leq C_{L,p} L^{-1/p} (\|a\|_\infty + V^p(a; I)). \quad (393)$$

The same constant is valid when I is replaced by any subinterval of the fixed ambient interval.

Proof. Partition I into consecutive intervals $I_1, \dots, I_M$ of length at most L^{-1} and, except possibly for the last interval, at least $(2L)^{-1}$. Thus $M \leq C_L L$ because $L \geq 1$. Choose $t_m \in I_m$ in increasing order. The error after replacing a by $a(t_m)$ on I_m is bounded by

$$\sum_m |I_m| V^p(a; I_m) \leq CL^{-1} M^{1-1/p} \left(\sum_m V^p(a; I_m)^p \right)^{1/p} \leq CL^{-1/p} V^p(a; I),$$

using Lemma 13.2(i). Put $w_m = \int_{I_m} e^{i\phi(t)} dt$. The ordinary first derivative van der Corput estimate implies

$$\left| \sum_{m=r}^s w_m \right| = \left| \int_{\cup_{m=r}^s I_m} e^{i\phi(t)} dt \right| \leq CL^{-1}$$

for every consecutive block. Put $a_m = a(t_m)$ and $W_m = \sum_{j=1}^m w_j$. Discrete Abel summation gives

$$\sum_{m=1}^M a_m w_m = a_M W_M + \sum_{m=1}^{M-1} (a_m - a_{m+1}) W_m, \quad |W_m| \leq CL^{-1}.$$

Consequently,

$$\left| \sum_m a(t_m) w_m \right| \leq CL^{-1} \left(\|a\|_\infty + \sum_{m=1}^{M-1} |a(t_{m+1}) - a(t_m)| \right).$$

Since the samples are ordered,

$$\left(\sum_{m=1}^{M-1} |a(t_{m+1}) - a(t_m)|^p \right)^{1/p} \leq V^p(a; I).$$

Hölder therefore bounds the last sum by $M^{1-1/p} V^p(a; I)$, proving (393). $\square$

Lemma 13.7 (Rational van der Corput with a V^p amplitude). *Let $\psi = N/Q$ on a fixed compact interval, where N, Q are real polynomials of degree at most D , no bound is imposed on the coefficients of N , and*

$$\|Q\|_{\text{coeff}} \leq M, \quad 0 < \eta \leq |Q| \leq M$$

on a fixed enlargement of the interval. Put

$$G := \|Q^2\psi'\|_{\text{coeff}} = \|N'Q - NQ'\|_{\text{coeff}}.$$

Let $c_0 = c_0(D) > 0$ be a valid normalized one-dimensional polynomial sublevel exponent for the derived degree bound of $N'Q - NQ'$ (at most $2D - 1$ when $D \geq 1$). Then, with

$$c_{\text{vdc},\rho,D} := \frac{c_0}{1 + c_0\rho} > 0, \quad (394)$$

every bounded amplitude a supported in the interval with $V^\rho(a) < \infty$ satisfies

$$\left| \int e^{i\psi(v)} a(v) dv \right| \leq C_{\rho,D,\eta,M}(\|a\|_\infty + V^\rho(a)) \min\{1, G^{-c_{\text{vdc},\rho,D}}\}. \quad (395)$$

When $G = 0$, the minimum on the right is understood to be 1. The same assertion holds after a uniformly bounded sectional partition. In particular, the constants are independent of the coefficients of the numerator N .

Proof. Put $p = Q^2\psi' = N'Q - NQ'$ and choose

$$\vartheta = \frac{c_0\rho}{1 + c_0\rho} \in (0, 1).$$

The trivial L^1 bound proves the assertion for $G \leq 1$. If $1 < G < (M^2)^{1/\vartheta}$, the same bound is still dominated by the right side of (395) after increasing its structural constant. We may therefore assume $M^{-2}G^\vartheta \geq 1$. The normalized polynomial p/G has coefficient norm one. Hence the fixed-degree sublevel estimate gives

$$E = \{|p| \leq G^\vartheta\}, \quad |E| \leq CG^{-c_0(1-\vartheta)}.$$

The contribution of E is at most $C\|a\|_\infty G^{-c_0(1-\vartheta)}$. On E^c the denominator bound implies

$$|\psi'| = |p|/|Q|^2 \geq M^{-2}G^\vartheta.$$

The boundary of E is contained in the zero sets of the two bounded-degree polynomials $p \pm G^\vartheta$, and the numerator of ψ'' also has uniformly bounded degree. If that numerator vanishes identically, then ψ' is constant and no additional cut is needed; otherwise it has only $O_D(1)$ zeros. Splitting at all these points produces $O_D(1)$ intervals on each of which ψ' is monotone and bounded away from zero. Lemma 13.6 on every such interval yields

$$C_{\rho,D,\eta,M}(\|a\|_\infty + V^\rho(a)) G^{-\vartheta/\rho}.$$

The chosen value of ϑ gives the exact equality

$$c_0(1 - \vartheta) = \vartheta/\rho = \frac{c_0}{1 + c_0\rho} = c_{\text{vdc},\rho,D}.$$

Summing the boundedly many intervals proves (395). Indeed, if these intervals are J_k , then

$$\sum_k V^\rho(a; J_k) \leq C_D^{1-1/\rho} \left(\sum_k V^\rho(a; J_k)^\rho \right)^{1/\rho} \leq C_{D,\rho} V^\rho(a),$$

and the finitely many supremum terms cost at most $C_D\|a\|_\infty$. Equivalently, if the restrictions are zero extended, their endpoint jumps cost at most two additional supremum terms per interval. A uniformly bounded initial partition is handled by the same refinement. $\square$

Corollary 13.8 (Coefficient-norm polynomial van der Corput). *For every fixed degree D , every real polynomial p of degree at most D , and every bounded measurable amplitude a supported on a fixed compact interval with $V^P(a) < \infty$,*

$$\left| \int e^{ip(t)} a(t) dt \right| \leq C_{D,p} (\|a\|_\infty + V^P(a)) \min\{1, \|p - p(0)\|_{\text{coeff}}^{-c_{D,p}}\}. \quad (396)$$

When p is constant, the minimum on the right is understood to be 1.

Proof. Apply Lemma 13.7 with denominator identically one. That lemma imposes no bound on the numerator coefficients, so the resulting estimate is uniform over all coefficients of p . Finally use finite-dimensional norm equivalence between the coefficient norm of p' and that of p modulo constants. $\square$

Lemma 13.9 (Uniform rational oscillatory operator with sectional V^P). *In Lemma 7.4, retain $B \geq 1$ and replace the sectional hypothesis $\|A(u, \cdot)\|_\infty + \text{Var}(A(u, \cdot)) \leq B$ by*

$$\|A(u, \cdot)\|_\infty + V^P(A(u, \cdot)) \leq B$$

on at most M_0 intervals, and retain the same row and column L^1 bounds. Let $c_{\text{vdc},p} > 0$ be the minimum of the finitely many exponents in (394) that occur in the structural rational families, and put

$$\delta_{\text{rat},p} := \frac{2c_{\text{sub}}c_{\text{vdc},p}}{c_{\text{sub}} + 2c_{\text{vdc},p}}, \quad c_{\text{rat},p} := \frac{\delta_{\text{rat},p}}{4} > 0. \quad (397)$$

Then

$$\|S\|_{2 \rightarrow 2} \leq C_p B \min\{1, L^{-c_{\text{rat},p}}\}, \quad (398)$$

where $L = \|\Psi\|_{\text{rat},Q}$. The constants are independent of all phase coefficients.

Proof. The Schur bound handles $L \leq 1$, so assume $L > 1$. The kernel of SS^* is

$$K(u, u') = \int e^{i[\Psi(u,v) - \Psi(u',v)]} A(u, v) \overline{A(u', v)} dv.$$

Refine the two sectional partitions belonging to u and u' . On each refined interval Lemma 13.2(iii) gives

$$\|A(u, \cdot) \overline{A(u', \cdot)}\|_\infty + V^P(A(u, \cdot) \overline{A(u', \cdot)}) \leq CB^2.$$

With $G(u, u') = \|\mathfrak{C}_Q \Psi(u, u', \cdot)\|_{\text{coeff},v}$, Lemma 13.7 therefore yields

$$|K(u, u')| \leq C_p B^2 \min\{1, G(u, u')^{-c_{\text{vdc},p}}\}. \quad (399)$$

Let $\mathbf{p}(u, u')$ be the vector of coefficients in v whose Euclidean length is $G(u, u')$. Regrouping coefficients is an isomorphism between two fixed finite-dimensional coefficient spaces, so

$$A_0 := \|\mathbf{p}\|_{\text{coeff},(u,u')} \asymp \|\mathfrak{C}_Q \Psi\|_{\text{coeff},(u,u',v)} = L.$$

Apply (178) on $I \times I$ to the square of (399), with $\sigma = 2c_{\text{vdc},p}$ and scale parameter $L \geq 1$. Since $A_0 \asymp L$, its normalized factor $L|\mathbf{p}(u, u')|/A_0$ is comparable to $G(u, u')$, so its integrand is comparable, with structural constants, to $\min\{1, G(u, u')^{-2c_{\text{vdc},p}}\}$. The definition (397) gives

$$\iint |K(u, u')|^2 du du' \leq C_p B^4 L^{-\delta_{\text{rat},p}}.$$

Consequently

$$\|S\|_{2 \rightarrow 2}^2 = \|SS^*\|_{2 \rightarrow 2} \leq \|K\|_{L^2(I \times I)} \leq C_p B^2 L^{-\delta_{\text{rat},p}/2}.$$

Thus the negative-moment estimate has first been square-rooted to pass from $\iint |K|^2$ to $\|K\|_2$, and the SS^* identity supplies the second square root. Taking that final square root proves (398). Algebraically degenerate points and the diagonal are already included in the negative-moment integral, so no coefficient-dependent deletion is made. $\square$

Lemma 13.10 (Sectional pullback calculus). *Let I, J be fixed compact intervals and let*

$$A(u, v) = w(u, v) \prod_{\ell=1}^m a_\ell(\gamma_\ell(u, v)).$$

Assume that w and every γ_ℓ are jointly Borel measurable on $I \times J$; in particular, after choosing the fixed regulated representatives of the a_ℓ , the function A is jointly measurable. Assume that every a_ℓ is extended by zero from one fixed compact radial interval and

$$\|a_\ell\|_\infty + V^P(a_\ell; \mathbb{R}) \leq B, \quad B \geq 1.$$

Suppose that, in either variable with the other fixed, each γ_ℓ has at most J_0 monotonicity intervals, uniformly in the fixed variable. Suppose also that w has uniformly bounded L^∞ norm and sectional $W^{1,\infty}$ norms on a fixed partition into at most J_0 rectangles. Fixed half-open cutoffs with a uniformly bounded number of sectional jumps may be included in w . Then, for each fixed $u \in I$, the interval J has a partition into at most C_{m,J_0} pieces $J_\alpha(u)$ on each of which

$$\|A(u, \cdot)\|_{L^\infty(J_\alpha(u))} + V^P(A(u, \cdot); J_\alpha(u)) \leq C_{m,J_0,\rho} B^m. \quad (400)$$

Separately, for each fixed $v \in J$, the interval I has a partition into at most C_{m,J_0} pieces $I_\beta(v)$ on each of which

$$\|A(\cdot, v)\|_{L^\infty(I_\beta(v))} + V^P(A(\cdot, v); I_\beta(v)) \leq C_{m,J_0,\rho} B^m.$$

Moreover

$$\sup_u \int_J |A(u, v)| dv + \sup_v \int_I |A(u, v)| du \leq C_{m,J_0} B^m. \quad (401)$$

Proof. For a fixed row take the common refinement of the structural partition and all monotonicity partitions of the functions $v \mapsto \gamma_\ell(u, v)$. Its cardinality is bounded solely in terms of m, J_0 . On every resulting interval, monotone pullback gives

$$V^P(a_\ell \circ \gamma_\ell) \leq V^P(a_\ell; \mathbb{R}) \leq B.$$

The zero extension is essential here: crossings of the radial support are already among the increments counted by the variation of a_ℓ . The sectional $W^{1,\infty}$ bound gives $V^P(w) \leq V^1(w) \leq C$. Repeated application of Lemma 13.2(iii), including the boundedly many cutoff jumps, proves the row part of (400). The column proof is identical. Finally the rectangles have fixed length and $|A| \leq CB^m$, which proves the two independent Schur bounds (401). No two-variable or mixed variation norm is used. $\square$

13.5 Stability of the correlation and high-shell estimates

Proposition 13.11 (Rough correlation with finite radial variation). *Fix $1 \leq \rho < \infty$ and put $d_* = \max(d, 1)$. Uniformly for every polynomial degree $1 \leq D \leq d_*$, there are positive exponents $c_{\text{eq},\rho}$ and $c_{\text{uneq},\rho}$ and a finite power $C_{\text{corr},\rho}$, depending only on n, d, ρ , such that the following two statements hold.*

First, the conclusion of Lemma 7.5 remains valid for phases of degree at most D and profiles satisfying $\|a\|_\infty + V^P(a) \leq B$ and $\|b\|_\infty + V^P(b) \leq B$, where the profiles are zero extended and $B \geq 1$, with

$$\left| \int e^{iS(z)} \overline{k_a(z)} k_b(z-h) dz \right| \leq C_\rho B^{C_{\text{corr},\rho}} \|\Theta\|_\infty^2 \min\{1, (\Lambda|h|^D)^{-c_{\text{eq},\rho}}\} \quad (402)$$

when $n \geq 2$, and with $\min\{1, \Lambda^{-c_{\text{eq},\rho}}\}$ in dimension one.

Second, Corollary 7.15 remains valid for V^P profiles, with

$$\left| \int e^{i[P(y+z,a)-P(y+z,y)]} \overline{k_R(z)} k_S(z+y-a) dz \right| \leq C_\rho B^{C_{\text{corr},\rho}} \|\Theta\|_\infty^2 S^{-n} \min\{1, (L_R |\mathbf{p}(W)|)^{-c_{\text{uneq},\rho}}\} \quad (403)$$

for $n \geq 2$, with the stated S^{-1} analogue in dimension one.

Proof. We follow the coordinate calculations in the two cited proofs and record all places where radial regularity enters.

For the equal-shell estimate, on each normalized (X, Y) block the two profile arguments are, with the notation of the proof of Lemma 7.5,

$$\gamma_1(X, Y) = \frac{\sqrt{U^2 + \alpha^2}}{|q(X, Y)|}, \quad \gamma_2(X, Y) = \frac{\sqrt{V^2 + \alpha^2}}{|q(X, Y)|},$$

where $U = U_0 + HX$, $V = V_0 + HY$, and $q = \Delta + X - Y$ has fixed sign and is bounded away from zero on a fixed enlargement. More precisely, if $K = [r_-, r_+] \Subset (0, \infty)$ is the common fixed support from Lemma 7.5, then the zero-extended amplitude is supported where both pullbacks lie in K . Choose $\chi \in C_c^\infty((0, \infty))$ equal to one on K and supported in a slightly larger fixed interval $K^+ \Subset (0, \infty)$. Because the profiles are zero outside K , inserting $\chi(\gamma_1)\chi(\gamma_2)$ does not change the amplitude; it does, however, give a smooth support factor on whose transition region the same lower bounds remain valid. The fixed block cutoffs and this enlarged support condition give, uniformly on a fixed enlargement,

$$|U| + |V| \leq C_K, \quad c_K \leq |q| \leq C_K, \quad c_K \leq s_U, s_V \leq C_K, \quad s_U = (U^2 + \alpha^2)^{1/2}, \quad s_V = (V^2 + \alpha^2)^{1/2}.$$

Thus every denominator occurring in the normalized Jacobian and in the fixed rational cutoffs is separated from zero on the actual support. Equations displayed immediately before (207) give

$$\partial_X \frac{s_U}{q} = \frac{HUq - s_U^2}{s_U q^2}, \quad \partial_Y \frac{s_U}{q} = \frac{s_U}{q^2},$$

and the analogous two formulas for s_V/q . Thus the only possible critical-point equations are polynomials of degree at most two. Each row and column therefore splits into at most $C_{n,d}$ intervals on which both arguments are monotone. On every such interval Lemma 13.2(ii) gives

$$V^\rho(a \circ \gamma) \leq V^\rho(a), \quad V^\rho(b \circ \tilde{\gamma}) \leq V^\rho(b).$$

Consequently the quotient rule, the preceding lower bounds, and the fixed size of the normalized rectangles give a uniform sectional $W^{1,\infty}$ bound for every smooth algebraic Jacobian and fixed cutoff (including cutoffs of the form $\beta(1/q)$). These smooth factors are differentiated only inside their fixed support; a sharp half-open cutoff contributes instead only a bounded number of sectional jumps. Lemma 13.10 therefore gives both the required sectional $L^\infty + V^\rho$ bound and the separate row and column L^1 bounds, with size $C_\rho B^{C_{\text{corr},\rho}}$. The two rough angular factors remain the input and output L^2 functions in (206); they are not part of this amplitude and are never differentiated. Thus (207) is replaced by

$$\|T_{\alpha,\omega}\|_{2 \rightarrow 2} \leq C_\rho B^{C_{\text{corr},\rho}} \min\{1, [\Gamma(\omega)(\alpha H)^D]^{-C_{\text{rat},\rho}}\}.$$

When $n = 2$, the transverse sphere is $\mathbb{S}^0$ and the two restrictions have coefficient norm comparable to the full coefficient norm, exactly as in the proof of Lemma 7.5. When $n \geq 3$, the chart coefficient lower bound (210) and (178), with $\sigma = c_{\text{rat},\rho}$, give an angular exponent

$$c_{\text{ang},\rho} := \frac{c_{\text{sub}} c_{\text{rat},\rho}}{c_{\text{sub}} + c_{\text{rat},\rho}} > 0. \quad (404)$$

The $n = 2$ estimate is stronger, so after decreasing its exponent the same $c_{\text{ang},\rho}$ is valid in every dimension $n \geq 2$. Hence one transverse band is bounded by

$$C_\rho B^{C_{\text{corr},\rho}} \|\Theta\|_\infty^2 \alpha^{n-1} \min\{1, [\Lambda(\alpha H)^D]^{-c_{\text{ang},\rho}}\}.$$

For $n \geq 2$ choose

$$c_{\text{eq},\rho}^{(n \geq 2)} := \frac{1}{2} \min\left\{c_{\text{ang},\rho}, \frac{n-1}{d_*}\right\} > 0. \quad (405)$$

The two geometric sums in the proof of Lemma 7.5 then give (402). In dimension one, Corollary 13.8 applies on the finitely many signed overlap intervals and gives an exponent $c_{\text{eq},\rho}^{(n=1)} > 0$. Define, for the fixed ambient dimension,

$$c_{\text{eq},\rho} := \begin{cases} c_{\text{eq},\rho}^{(n=1)}, & n = 1, \\ c_{\text{eq},\rho}^{(n \geq 2)}, & n \geq 2. \end{cases}$$

This number is strictly positive.

For unequal shells, the two zero-extended profile arguments are exactly (278). In the formulas cited in this paragraph, write the geometric parameter $|W|$ as r_W (so the earlier ρ , $S\rho$, and $\delta = R/(S\rho)$ are read as r_W , Sr_W , and $R/(Sr_W)$); this keeps it distinct from the fixed radial-variation exponent ρ of the present section. On the support of a transverse band and both profiles, the calculation following (277) gives

$$\begin{aligned} |u| + |V| &\leq C, & 0 < c \leq D = V - \delta u &\leq C, & 0 < \delta &\leq C, \\ (u^2 + \alpha^2)^{1/2} &\geq c, & (V^2 + \alpha^2 \delta^2)^{1/2} &\geq c. \end{aligned}$$

As in the equal-shell case, insert fixed smooth cutoffs of the two pullback variables which equal one on the radial profile support and are supported in a slightly larger compact subinterval of $(0, \infty)$. Zero extension makes this insertion exact, and all the displayed lower bounds persist on the cutoff transition regions. All constants here are independent of the shell radii and of α, δ . It follows directly from the product and quotient rules that the factor

$$D^{n-1} (u^2 + \alpha^2)^{-n/2} (V^2 + \alpha^2 \delta^2)^{-n/2}$$

and every smooth rational/chart cutoff have uniformly bounded sectional $W^{1,\infty}$ norm on each fixed normalized rectangle. A half-open rectangle or band boundary produces only one of a bounded number of sectional jumps. The two pullbacks have only the nontrivial sectional critical equation $uV + \delta\alpha^2 = 0$, so each section requires at most one additional cut. Take the common refinement for the two arguments, the fixed rational rectangles, and the transverse cutoff. Lemma 13.10 then supplies both sectional V^ρ and the independent Schur row/column bounds. The zero-extended transverse cutoffs satisfy $V^\rho(\beta_k) \leq V^1(\beta_k) \leq C$. Thus the normalized rational operator contributes the same exponent $c_{\text{rat},\rho}$. For $n = 2$ the transverse sphere again consists of two points; for $n \geq 3$ use the finite rational atlas and (178). The oriented coefficient lower bound and this angular argument give, in either case after decreasing the exponent, a band estimate

$$C_\rho B^{C_{\text{corr},\rho}} \|\Theta\|_\infty^2 S^{-n} \alpha^{n-1} \min\{1, (L_R \alpha^d |\mathbf{p}(W)|)^{-c_{\text{ang},\rho}}\}.$$

For $n \geq 2$ set

$$c_{\text{uneq},\rho}^{(n \geq 2)} := \frac{1}{2} \min\left\{c_{\text{ang},\rho}, \frac{n-1}{d_*}\right\} > 0. \quad (406)$$

Splitting the dyadic α -sum at $\alpha \simeq (L_R |\mathbf{p}(W)|)^{-1/d_*}$ proves (403). The one-dimensional proof uses Corollary 13.8 and gives an exponent $c_{\text{uneq},\rho}^{(n=1)} > 0$. Define, for the fixed ambient dimension,

$$c_{\text{uneq},\rho} := \begin{cases} c_{\text{uneq},\rho}^{(n=1)}, & n = 1, \\ c_{\text{uneq},\rho}^{(n \geq 2)}, & n \geq 2. \end{cases}$$

This number is strictly positive and completes the proof. No rough angular factor has been differentiated. $\square$

Corollary 13.12 (Quantitative flat high block and collective high shells). *There are $c_{\text{flat},\rho} > 0$ and a finite power $C_{B,\rho}$ such that Proposition 7.8, with radial profiles controlled in $L^\infty + V^\rho$ by $B \geq 1$, has the quantitative form*

$$\left\| A_a^{P,\Theta} \sum_Q e_Q v_Q \right\|_2^2 \leq C_\rho B^{C_{B,\rho}} \|\Theta\|_\infty^2 \Lambda M(L^{-c_{\text{flat},\rho}} + 2^{-c_{\text{flat},\rho} t}). \quad (407)$$

For every fixed $\kappa > 0$ there is $c_{\kappa,\rho} > 0$ such that Proposition 7.17 has the form

$$\left\| \sum_{R \in \mathcal{R}} T_R \sum_{Q \in \mathcal{Q}_R} e_Q v_Q \right\|_2^2 \leq C_{\rho,\kappa} B^{C_{B,\rho}} \|\Theta\|_\infty^2 \Lambda M 2^{-c_{\kappa,\rho} S}. \quad (408)$$

Both estimates are uniform under packetwise coefficients $|e_Q| \leq 1$.

Proof. Let $c_{\text{pair}} > 0$ be the exponent in (179), reduced if necessary so that it also works in the finitely many rational charts used in the unequal-shell incidence estimate. In the proof of Proposition 7.8, the bad coefficient sublevel has weighted mass

$$C\Lambda M(L^{-c_{\text{pair}}/3} + 2^{-c_{\text{pair}} t}).$$

On its complement the actual oscillatory coefficient is at least $cL^{2/3}$. For $n \geq 2$, the near-pair choice $r = L^{-1/(6d_*)}$ contributes $C\Lambda M(L^{-n/(6d_*)} + 2^{-nt})$, while on the remaining pairs (402) gives $C_\rho B^{C_{\text{corr},\rho}} \|\Theta\|_\infty^2 L^{-c_{\text{eq},\rho}/2}$. The one-dimensional branch omits the near-pair split. Therefore one may take

$$c_{\text{flat},\rho} := \begin{cases} \frac{1}{2} \min \left\{ \frac{c_{\text{pair}}}{3}, c_{\text{pair}}, \frac{c_{\text{eq},\rho}}{2}, \frac{n}{6d_*} \right\}, & n \geq 2, \\ \frac{1}{2} \min \left\{ \frac{c_{\text{pair}}}{3}, c_{\text{pair}}, \frac{c_{\text{eq},\rho}}{2} \right\}, & n = 1. \end{cases} \quad (409)$$

Both values are strictly positive. Taking $C_{B,\rho}$ larger than the finite radial powers in the correlation estimate proves (407).

For the collective estimate, the diagonal and comparable-scale terms are controlled by (407). For $R \leq S/C_0$, retain the original choice $\tau_R = L_R^{-1/2}$. The one-sided incidence lemma bounds the bad row by

$$C\Lambda \{L_R^{-c_{\text{pair}}/2} + (2^{-s} R/S)^{c_{\text{pair}}}\},$$

whereas (403) bounds the good row by $C_\rho B^{C_{B,\rho}} \Lambda L_R^{-c_{\text{uneq},\rho}/2}$. Put

$$c_{\text{row},\rho} := \frac{1}{2} \min \left\{ c_{\text{flat},\rho}, \frac{c_{\text{pair}}}{2}, \frac{c_{\text{uneq},\rho}}{2} \right\} > 0.$$

Since $L_R \geq 2^{\kappa s}$ on every high shell, the geometric row sums in Proposition 7.17 are bounded by a constant times

$$2^{-\kappa c_{\text{row},\rho} S} + 2^{-c_{\text{pair}} S} + 2^{-c_{\text{flat},\rho} S}.$$

Thus the explicit choice

$$c_{\kappa,\rho} := \frac{1}{2} \min \{ \kappa c_{\text{row},\rho}, c_{\text{pair}}, c_{\text{flat},\rho} \} > 0 \quad (410)$$

absorbs all diagonal, comparable, incidence, and separated-scale terms. The original Jensen reduction and row orientation use absolute values and are unchanged, so packetwise signs remain admissible. This proves (408). $\square$

13.6 Terminal packets with a finite-variation radial profile

Lemma 13.13 (Smooth fixed-sign packet kernel). *Put $N_0 = 4n + 1$ and $\eta = (16N_0)^{-1}$. In the unit-scale packet setting of Proposition 5.5, let c_t be supported in one fixed annular enlargement and satisfy*

$$|c_t| \leq 1, \quad \|\partial_r^\ell c_t\|_\infty \leq C_\ell 2^{\eta \ell t} \quad (0 \leq \ell \leq N_0).$$

For arbitrary fixed packet coefficients of modulus at most one, let S_t be the convolution output with kernel $K_\Theta(z)c_t(|z|)$. Then

$$\|S_t\|_2^2 \leq C_n \|\Theta\|_\infty^2 2^{-3t/32} \Lambda M. \quad (411)$$

The assertion is unchanged by the input and output modulations allowed in Proposition 5.5.

Proof. This is the smooth part of the proof of Proposition 5.5, but we record precisely which hypotheses are used. Put $H_t(z) = K_\Theta(z)c_t(|z|)$. If $n = 1$, choose the center c_U of each packet interval. Packet cancellation gives

$$H_t * d_U(x) = \int_U [H_t(x-y) - H_t(x-c_U)] d_U(y) dy.$$

The annular support stays a fixed distance from the origin, so the stated radial derivative bounds imply

$$\|H'_t\|_1 \leq C\|\Theta\|_\infty 2^{\eta t}.$$

Since $\text{diam } U \lesssim 2^{-t}$, translation continuity in $W^{1,1}$ and summation over the disjoint packets yield

$$\left\| H_t * \sum_U a_U d_U \right\|_1 \leq C\|\Theta\|_\infty 2^{-(1-\eta)t} M.$$

The positive thin-layer majorant gives simultaneously $\|S_t\|_\infty \leq C\Lambda\|\Theta\|_\infty$. Hence

$$\|S_t\|_2^2 \leq \|S_t\|_1 \|S_t\|_\infty \leq C\|\Theta\|_\infty^2 \Lambda M 2^{-(1-\eta)t}.$$

Here $\eta = 1/80$, so this is stronger than (411). We may therefore assume $n \geq 2$. The one-shell kernel has normalized radial C^{N_0} size at most $C\|\Theta\|_\infty 2^{\eta N_0 t}$. Seeger's decomposition with $\kappa = 1/2$ and $\varepsilon = 1/4$ therefore gives a microlocal part with squared L^2 norm

$$C\|\Theta\|_\infty^2 2^{-t/2} \Lambda M$$

and a remainder with L^1 norm

$$C\|\Theta\|_\infty 2^{-(1/4-\eta N_0)t} M = C\|\Theta\|_\infty 2^{-3t/16} M.$$

The support and the bound $|c_t| \leq 1$ give, by the positive thin-layer majorant, the simultaneous L^∞ bound $C\Lambda\|\Theta\|_\infty$. Splitting the remainder at height $\Lambda\|\Theta\|_\infty 2^{-3t/32}$ and interpolating exactly as in (147) proves (411). No property of a step function, or of the set on which an unsmoothing error is supported, enters these three estimates. The proof uses absolute packet coefficients and packet cancellation, so the modulation statements are preserved. $\square$

Proposition 13.14 (Radially weighted selector-free terminal packet). *Fix $1 \leq \rho < \infty$ and put*

$$N_0 = 4n + 1, \quad \eta = \frac{1}{16N_0}, \quad \gamma = \frac{\eta}{2}, \quad c_{\text{sm}} = \frac{3}{32}. \quad (412)$$

There is $t_{0,\rho} \in \mathbb{N}$ such that Proposition 5.5 remains valid for $t \geq t_{0,\rho}$ after inserting a radial factor $a(|x-y|/R)$ satisfying $\|a\|_\infty + V^\rho(a; [1, 2]) \leq 1$. More precisely, one may take

$$\delta_{T,\rho} := \frac{1}{4} \min \left\{ \gamma, c_{\text{sm}}, \frac{\eta}{\rho} \right\} > 0 \quad (413)$$

so that, with the same packet hypotheses and $R \leq \varepsilon \leq 2R$,

$$\|V^2(F_\varepsilon : R \leq \varepsilon \leq 2R)\|_2^2 \leq C_{n,\rho} \|\Theta\|_\infty^2 2^{-\delta_{T,\rho} t} \Lambda M. \quad (414)$$

The modulation invariances in Proposition 5.5 remain valid.

Proof. Scale to $R = 1$ and partition $[1, 2]$ into $K_t \simeq 2^{\eta t}$ consecutive cells I_ℓ . If G_ℓ denotes the full shell output over I_ℓ , inserting all cell endpoints into an arbitrary variation partition gives the deterministic decomposition

$$V^2(F_\varepsilon : 1 \leq \varepsilon \leq 2) \leq C V^2 \left(\sum_{\ell \leq m} G_\ell : 0 \leq m \leq K_t \right) + C \left(\sum_{\ell} V^2(F_\varepsilon : \varepsilon \in I_\ell)^2 \right)^{1/2}.$$

The within-cell argument uses $V^2 \leq V^1$, the positive thin-layer majorant, and only $\|a\|_\infty \leq 1$. It therefore gives

$$\left\| \left(\sum_{\ell} V^2(F_{\varepsilon} : \varepsilon \in I_{\ell})^2 \right)^{1/2} \right\|_2^2 \leq C \|\Theta\|_\infty^2 \Lambda M 2^{-\gamma t}. \quad (415)$$

For the grid partial sums fix signs $\varepsilon_{\ell} \in \{-1, 0, 1\}$ and put

$$c(r) = a(r) \sum_{\ell=1}^{K_t} \varepsilon_{\ell} \mathbf{1}_{I_{\ell}}(r).$$

Lemma 13.2 gives

$$\|c\|_\infty + V^\rho(c) \leq C_\rho (1 + K_t^{1/\rho}). \quad (416)$$

Extend c by zero and mollify at scale $h = 2^{-\eta t}$. Lemma 13.3 yields

$$\|c - c_h\|_1 \leq C_\rho h^{1/\rho} (1 + K_t^{1/\rho}) \leq C_\rho 2^{-(\eta-\gamma)t/\rho} = C_\rho 2^{-\eta t/(2\rho)}. \quad (417)$$

Moreover $\|c - c_h\|_\infty \leq 2$ and $V^\rho(c - c_h) \leq C_\rho (1 + 2^{\gamma/\rho})$. With packet side 2^{-t} , Lemma 13.5 gives for the unsmoothing output D

$$\|D\|_\infty \leq C_\rho \Lambda \|\Theta\|_\infty 2^{-\eta t/(2\rho)}, \quad (418)$$

$$\|D\|_1 \leq C_\rho \|\Theta\|_1 M 2^{-\eta t/(2\rho)}. \quad (419)$$

Indeed the additional discretization term is $2^{-t/\rho} (1 + 2^{\gamma/\rho})$, whose exponent is no smaller than $\eta/(2\rho)$ because $\gamma = \eta/2 < 1$. Since $\|\Theta\|_1 \leq C_n \|\Theta\|_\infty$, it follows that

$$\|D\|_2^2 \leq C_{n,\rho} \|\Theta\|_\infty^2 \Lambda M 2^{-\eta t/\rho}. \quad (420)$$

It remains to estimate the smoothed signed sum. Since $|c| \leq 1$, $|c_h| \leq 1$, and $\|\partial_t^\ell c_h\|_\infty \leq C_\ell h^{-\ell}$, the smoothed kernel satisfies Lemma 13.13. Hence

$$\|S\|_2^2 \leq C \|\Theta\|_\infty^2 \Lambda M 2^{-c_{\text{sm}} t}. \quad (421)$$

For $n = 1$, the elementary cancellation argument in the proof of Proposition 5.5 gives the stronger exponent $1 - \eta$; thus the same common choice $c_{\text{sm}} = 3/32$ is valid in every dimension.

Lemma 5.2 costs at most $C(1+t)^2$ after squaring. The elementary bound $(1+t)^2 2^{-at} \leq C_a 2^{-at/2}$ for $a > 0$, together with (415), (420), and (421), proves (414) with the smaller exponent in (413). Increase $t_{0,\rho}$ once so that the support enlargements caused by mollification remain in the fixed radial annulus and so that the finitely many polynomial factors in t are absorbed by the displayed geometric decays. The proof uses only absolute packet coefficients and packetwise cancellation, so the stated input and output modulation invariances are preserved. $\square$

13.7 The unphased strong theorem for every finite radial exponent

We next prove the terminal strong input in a form with the exact angular conjugate-exponent loss needed for the $L \log L$ endpoint.

For $j \in \mathbb{Z}$ define the annular measure

$$dv_j(x) = \frac{\Omega(x/|x|)}{|x|^n} b(|x|) \mathbf{1}_{\{2^j < |x| \leq 2^{j+1}\}} dx. \quad (422)$$

The mean-zero condition on Ω implies $v_j(\mathbb{R}^n) = 0$.

Lemma 13.15 (Shell Fourier envelope). *Assume (379) and $\int_{\mathbb{S}^{n-1}} \Omega d\sigma = 0$. There exists $\beta_\rho > 0$ such that $\widehat{v}_j(0) = 0$ and, for $t = 2^j|\xi| > 0$,*

$$|\widehat{v}_j(\xi)| \leq Ct \|\Omega\|_1, \quad 0 < t \leq 1, \quad (423)$$

$$|\widehat{v}_j(\xi)| \leq C_\rho t^{-\beta_\rho} \|\Omega\|_2, \quad t \geq 1. \quad (424)$$

One may take

$$\beta_\rho = \min \left\{ \frac{1}{4}, \frac{1}{2\rho} \right\}.$$

Consequently, if $1 < s \leq 2$, $s' = s/(s-1)$, and Ω has mean zero, then, for $t > 0$,

$$|\widehat{v}_j(\xi)| \leq C_\rho \|\Omega\|_s a_{s,\rho}(\log_2 t), \quad (425)$$

where $a_{s,\rho} : \mathbb{R} \rightarrow (0, \infty)$ is defined by

$$a_{s,\rho}(r) = \begin{cases} C2^r, & r \leq 0, \\ C_\rho 2^{-2\beta_\rho r/s'}, & r > 0. \end{cases} \quad (426)$$

For every $\delta > 0$,

$$\sum_{r \in \mathbb{Z}} a_{s,\rho}(r) \delta \leq C_{\rho,\delta} s'. \quad (427)$$

Proof. For $\xi \neq 0$, set $e = \xi/|\xi|$ and make the change of variables $r = 2^j u$. Then the multiplier is

$$\widehat{v}_j(\xi) = \int_{\mathbb{S}^{n-1}} \Omega(\theta) \int_1^2 b(2^j u) e^{-itue \cdot \theta} \frac{du}{u} d\sigma(\theta).$$

At frequency zero the same formula before introducing e has a radial integral independent of θ , so mean zero of Ω gives $\widehat{v}_j(0) = 0$. For $\xi \neq 0$, subtracting that zero-frequency value and using $|e^{-iz} - 1| \leq |z|$ proves (423). Thus no value of $\log_2 t$ is invoked when $t = 0$.

The amplitude $u \mapsto b(2^j u)/u$ has uniformly bounded $L^\infty + V^\rho$ norm by Lemma 13.2. Applying Lemma 13.6 to the linear radial phase gives

$$\left| \int_1^2 b(2^j u) e^{-itue \cdot \theta} \frac{du}{u} \right| \leq C_\rho \min\{1, (t|\tau|)^{-1/\rho}\}.$$

For $n \geq 2$, Cauchy–Schwarz in θ gives the square root of

$$\int_{\mathbb{S}^{n-1}} \min\{1, (t|e \cdot \theta|)^{-2/\rho}\} d\sigma(\theta).$$

The distribution of $|e \cdot \theta|$ near zero has bounded density. Thus this integral is $O(t^{-1})$ for $\rho < 2$, is $O(t^{-1} \log(2+t))$ for $\rho = 2$, and is $O_\rho(t^{-2/\rho})$ for $\rho > 2$. The weaker uniform bound $C_\rho t^{-2\beta_\rho}$ follows. When $n = 1$, there is no equatorial integration and the direct bound is even stronger. This proves (424).

For high frequencies interpolate the linear functional of Ω between the trivial L^1 bound and (424). Since the L^2 endpoint has interpolation weight $2/s'$, one obtains the second line of (426); the low-frequency line follows from (423) and $L^s \hookrightarrow L^1$ on the sphere. Finally (427) is the sum of two geometric series; the positive-frequency series has ratio $2^{-2\beta_\rho \delta/s'}$ and hence costs $O_{\rho,\delta}(s')$. $\square$

We shall also use the following standard vector-valued shell estimate. A proof is included because it is uniform down to angular L^1 and therefore is exactly what allows spatial interpolation below.

Lemma 13.16 (Uniform vector-valued annular estimate). *For $1 < p < \infty$ and arbitrary functions f_j ,*

$$\left\| \left(\sum_j |v_j * f_j|^2 \right)^{1/2} \right\|_p \leq C_p \|b\|_\infty \|\Omega\|_1 \left\| \left(\sum_j |f_j|^2 \right)^{1/2} \right\|_p. \quad (428)$$

Proof. For a direction θ , let M_θ denote the one-dimensional Hardy–Littlewood maximal operator along lines parallel to θ . Polar coordinates give pointwise

$$|v_j * f_j(x)| \leq C \|b\|_\infty \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| M_\theta f_j(x) d\sigma(\theta).$$

Minkowski first in ℓ^2 and then in θ , followed by the vector-valued one-dimensional maximal inequality on almost every line, gives

$$\left\| \left(\sum_j |v_j * f_j|^2 \right)^{1/2} \right\|_p \leq C_p \|b\|_\infty \|\Omega\|_1 \left\| \left(\sum_j |f_j|^2 \right)^{1/2} \right\|_p.$$

The constant is uniform in θ by rotation. $\square$

Fix a smooth Littlewood–Paley function ψ supported in $\{1/2 \leq |\xi| \leq 2\}$ and satisfying $\sum_{\ell \in \mathbb{Z}} \psi(2^{-\ell} \xi)^2 = 1$ for $\xi \neq 0$. Let Δ_ℓ be the corresponding multiplier. For a relative frequency $r \in \mathbb{Z}$ put

$$g_{j,r} := v_j * \Delta_{r-j}^2 f.$$

On the Fourier support of $g_{j,r}$ one has $2^j |\xi| \simeq 2^r$.

Lemma 13.17 (Frequency-separated tail representation). *For every $1 < p < \infty$ and every $f \in L^p(\mathbb{R}^n)$, there is a fixed integer $M \geq 5$ such that, after splitting the shell indices j into their M residue classes, the tail*

$$\sum_{\substack{j \geq k \\ j \equiv c \pmod{M}}} g_{j,r}$$

*is, up to repetitions as k varies, exactly a subsequence of the smooth low-pass family $\{\phi_\ell * h_{r,c}\}_{\ell \in \mathbb{Z}}$ used in Proposition 2.1 of Chen–Ding–Hong–Liu, applied to*

$$h_{r,c} := \sum_{j \equiv c \pmod{M}} g_{j,r}.$$

Here $\widehat{\phi} = 1$ on $|\xi| \leq 2$ and $\widehat{\phi} = 0$ on $|\xi| > 4$, and $\widehat{\phi}_\ell(\xi) = \widehat{\phi}(2^\ell \xi)$. Consequently, Proposition 2.1 of [9] gives

$$\left\| \mathbf{J} \left(\sum_{\substack{j \geq k \\ j \equiv c \pmod{M}}} g_{j,r} : k \in \mathbb{Z} \right) \right\|_p \leq C_p \|h_{r,c}\|_p. \quad (429)$$

Proof. The Fourier support of $g_{j,r}$ lies in

$$2^{r-j-1} \leq |\xi| \leq 2^{r-j+1}.$$

Fix $M \geq 5$. If $j_0 \equiv c \pmod{M}$, then ϕ_{j_0-r} equals one on every support with $j \geq j_0$: indeed the largest such frequency is 2^{r-j_0+1} , precisely the inner plateau of $\widehat{\phi}(2^{j_0-r} \xi)$. On the other hand, the preceding index in the same residue class is $j_0 - M$, whose support begins at $2^{r-j_0+M-1} > 2^{r-j_0+2}$; hence ϕ_{j_0-r} vanishes on it and on all earlier supports. Therefore

$$\phi_{j_0-r} * h_{r,c} = \sum_{\substack{j \geq j_0 \\ j \equiv c \pmod{M}}} g_{j,r}$$

exactly. When k moves between two consecutive indices in the residue class, the tail is merely repeated. Thus the residue-class tail is a subsequence, with repetitions, of the very family covered by Proposition 2.1 of [9]; repetitions cannot create jumps. The jump count in that proposition is the paired count used in the definition of $\mathbf{J}$, so it proves (429) directly. The first inequality in (27) also gives the corresponding chain bound with only an absolute loss; this latter form is used below solely to enter the normed quotient of Lemma 2.3. $\square$

Lemma 13.18 (s' -tracked dyadic terminal jump). *Let $1 < s \leq 2$, assume $\Omega \in L^s(\mathbb{S}^{n-1})$ has mean zero, and fix $1 < p < \infty$. Then, for every $f \in L^p(\mathbb{R}^n)$, under (379),*

$$\left\| \mathbf{J}(T_{\Omega, 2^k}^{0,b} f : k \in \mathbb{Z}) \right\|_p \leq C_{n,p,\rho} s' \|\Omega\|_s \|f\|_p. \quad (430)$$

Proof. First impose finite shell and frequency cutoffs. For one residue class, Littlewood–Paley theory, Lemma 13.16, and bounded frequency overlap give, for every $1 < p_0 < \infty$,

$$\|h_{r,c}\|_{p_0} \leq C_{p_0} \|\Omega\|_1 \|f\|_{p_0} \leq C_{p_0} \|\Omega\|_s \|f\|_{p_0}. \quad (431)$$

On L^2 , the separated Fourier supports and Lemma 13.15 give the following bound. Indeed, on $\widehat{\text{supp } \Delta_{r-j} f}$ one has $|\log_2(2^j |\xi|) - r| \leq 1$. The two exponential slopes in (426) are uniformly bounded for $1 < s \leq 2$; including the possible crossing of $r = 0$, this gives $a_{s,p}(\log_2(2^j |\xi|)) \leq C_p a_{s,p}(r)$ with a constant independent of s . Hence

$$\|h_{r,c}\|_2 \leq C_p a_{s,p}(r) \|\Omega\|_s \|f\|_2. \quad (432)$$

If $p < 2$, interpolate (432) with (431) at one fixed $p_0 < p$; if $p > 2$, use one fixed $p_0 > p$. For $p = 2$ use (432). Thus for some $\theta_p > 0$ independent of s, r ,

$$\|h_{r,c}\|_p \leq C_{p,\rho} a_{s,p}(r)^{\theta_p} \|\Omega\|_s \|f\|_p. \quad (433)$$

Lemma 13.17 gives the same estimate for the paired jump of the corresponding residue-class tail. By $\mathbf{J}^{\text{ch}} \leq \sqrt{2} \mathbf{J}$ from (27), and then the uniform equivalence in Lemma 2.3, the same bound holds in the norm $\|\cdot\|_{\mathfrak{J}_D}$ for every finite ordered parameter set D . Summing the finitely many residue classes changes only the constant.

We now remove the cutoffs rather than merely invoking lower semicontinuity. Fix a finite ordered set $D \subset \mathbb{Z}$ of dyadic truncation indices and use the norm $\|\cdot\|_{\mathfrak{J}_D}$ supplied by Lemma 2.3. The preceding paired-to-chain conversion is what permits this normed summation. If $R_2 > R_1$, its triangle inequality and (433) give, uniformly in the finite shell cutoff,

$$\left\| \sum_{R_1 < |r| \leq R_2} \left[\sum_{j \geq k} g_{j,r} \right]_{k \in D} \right\|_{L^p(\mathfrak{J}_D)} \leq C_{p,\rho} \|\Omega\|_s \|f\|_p \sum_{R_1 < |r| \leq R_2} a_{s,p}(r)^{\theta_p}.$$

By (427) the right side tends to zero as $R_1 \rightarrow \infty$, so the frequency sums are Cauchy in $L^p(\mathfrak{J}_D)$. For a fixed shell j , the Littlewood–Paley identity $\sum_r \Delta_{r-j}^2 f = f$ holds in L^p ; convolution with the finite-annulus measure v_j identifies every finite-shell coordinate of the limit.

Next remove the shell cutoff with the order of the two tails made explicit. Let $\mathcal{J}_J \uparrow \mathbb{Z}$ be the growing finite shell sets and put $E_J = \mathbb{Z} \setminus \mathcal{J}_J$. Given $\varepsilon > 0$, first choose R so large that

$$C_{p,\rho} \|\Omega\|_s \|f\|_p \sum_{|r| > R} a_{s,p}(r)^{\theta_p} < \frac{\varepsilon}{2};$$

this estimate is uniform in the shell cutoff. For the finitely many $|r| \leq R$ and residue classes c , put

$$h_{r,c}^{E_J} = \sum_{\substack{j \in E_J \\ j \equiv c \pmod{M}}} v_j * \Delta_{r-j}^2 f.$$

The separated-tail estimate applies to an arbitrary subset $E \subset \{j : j \equiv c \pmod{M}\}$, not only to the full residue class: the low-pass multiplier used in its proof still gives, term by term,

$$\phi_{j_0-r} * h_{r,c}^E = \sum_{\substack{j \in E, j \geq j_0 \\ j \equiv c \pmod{M}}} g_{j,r}.$$

Thus the same CDHL bound, with the same constant, holds for every such E . Interpret this sum first over finite subsets of E_J . The following uniform estimate makes those partial sums Cauchy in L^p and hence defines $h_{r,c}^{E_J}$; the finite-parameter tail estimate then passes to that limit through the norm $\mathfrak{J}_D$. The proof of (431), restricted to E_J , gives

$$\|h_{r,c}^{E_J}\|_p \leq C_p \|\Omega\|_1 \left\| \left(\sum_{j \in E_J} |\Delta_{r-j}^2 f|^2 \right)^{1/2} \right\|_p \longrightarrow 0.$$

Indeed, for fixed r the map $j \mapsto r - j$ is a bijection of $\mathbb{Z}$, and

$$\left(\sum_{j \in E_J} |\Delta_{r-j}^2 f|^2 \right)^{1/2} \longrightarrow 0 \quad \text{pointwise,} \quad \left(\sum_{j \in E_J} |\Delta_{r-j}^2 f|^2 \right)^{1/2} \leq \left(\sum_{\ell \in \mathbb{Z}} |\Delta_\ell^2 f|^2 \right)^{1/2}.$$

The dominating full Littlewood–Paley square function belongs to L^p ; dominated convergence proves the asserted norm limit. This also shows that the limit is independent of the particular exhaustion $\mathcal{J}_J$. Choose J so that the sum of these finitely many norms is less than $\varepsilon/2$. Lemma 13.17, the triangle inequality in $L^p(\mathfrak{J}_D)$, and the preceding frequency-tail choice then show that the full omitted contribution has $L^p(\mathfrak{J}_D)$ norm below ε . This proves joint Cauchy convergence, independently of the order in which the two cutoffs are removed.

To identify the quotient limit without losing its constant coordinate, fix $k_0 \in D$ and choose the representative whose k_0 -coordinate is the ordinary L^p limit supplied by the finite-annulus shell series and the Littlewood–Paley identity. Finite-dimensional quotient convergence controls every difference between a k -coordinate and the k_0 -coordinate; the same coordinatewise Littlewood–Paley identity therefore identifies all coordinates with $T_{\Omega, 2^k}^{0,b} f$. Convergence in $L^p(\mathfrak{J}_D)$ gives, after a subsequence, convergence in the finite-dimensional quotient almost everywhere. The equivalence in Lemma 2.3 first returns the resulting bound to $\mathbf{J}_D^{\text{ch}}$; the inequality $\mathbf{J}_D \leq 2\mathbf{J}_D^{\text{ch}}$ in (27), or equivalently finite strict-witness lower semicontinuity, then gives the paired estimate

$$\|\mathbf{J}(T_{\Omega, 2^k}^{0,b} f : k \in D)\|_p \leq C_{p,p} \sum_r a_{s,p}(r)^{\theta_p} \|\Omega\|_s \|f\|_p \leq C_{p,p} s' \|\Omega\|_s \|f\|_p.$$

Finally exhaust D . To reiterate the interface: Proposition 2.1 of [9] is stated for the paired jump count and therefore enters without conversion. The paired-to-chain comparison is used only to enter the MSZK normed quotient for summation, and the chain-to-paired comparison is used on exit. Both losses are absolute and independent of D . $\square$

Lemma 13.19 (s' -tracked short terminal variation). *Under the hypotheses of Lemma 13.18, and for every $f \in L^p(\mathbb{R}^n)$, let $S_2(T_{\Omega}^{0,b} f)$ denote the square sum of the within-dyadic-shell 2-variations. Then*

$$\|S_2(T_{\Omega}^{0,b} f)\|_p \leq C_{n,p,p} s' \|\Omega\|_s \|f\|_p. \quad (434)$$

Proof. For $1 \leq u \leq 2$, let

$$A_{j,u} f(x) = \int_{u2^j < |z| \leq 2^{j+1}} b(|z|) K_{\Omega}(z) f(x-z) dz$$

and put

$$S_r f = \left(\sum_j V^2(A_{j,u} \Delta_{r-j}^2 f : 1 \leq u \leq 2)^2 \right)^{1/2}.$$

The path $u \mapsto A_{j,u}f(x)$ is locally absolutely continuous for almost every x . Since $V^2 \leq V^1$, differentiation of the moving boundary and integration in u give

$$\begin{aligned} V^2(A_{j,u}\Delta_{r-j}^2 f : u) &\leq \int_1^2 \left| \frac{d}{du} A_{j,u}\Delta_{r-j}^2 f \right| du \\ &\leq C\|b\|_\infty \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| \left[2^{-j} \int_{2^j}^{2^{j+1}} |\Delta_{r-j}^2 f(x-t\theta)| dt \right] d\sigma(\theta) \\ &\leq C\|b\|_\infty \int_{\mathbb{S}^{n-1}} |\Omega(\theta)| M_\theta(\Delta_{r-j}^2 f)(x) d\sigma(\theta). \end{aligned}$$

The vector-valued directional maximal inequality used in Lemma 13.16 therefore gives, uniformly in r and for every $1 < p_0 < \infty$,

$$\|S_r f\|_{p_0} \leq C_{p_0} \|\Omega\|_1 \|f\|_{p_0} \leq C_{p_0} \|\Omega\|_s \|f\|_{p_0}. \quad (435)$$

We now justify the L^2 frequency envelope uniformly in the moving endpoint u . On the relative frequency $t = 2^j|\xi| \simeq 2^r$, the partial-shell multiplier is

$$m_{j,u}(\xi) = \int_{\mathbb{S}^{n-1}} \Omega(\theta) \int_u^2 b(2^j v) e^{-itue \cdot \theta} \frac{dv}{v} d\sigma(\theta), \quad e = \xi/|\xi|.$$

The zero-frequency value is independent of θ , so mean zero of Ω gives the low-frequency bound $Ct\|\Omega\|_1$, uniformly in u . For high frequency, the zero-extended amplitude

$$v \mapsto b(2^j v) v^{-1} \mathbf{1}_{[u,2]}(v)$$

has uniformly bounded $L^\infty + V^\rho$ norm by Lemma 13.2(iii)–(iv). Lemma 13.6, followed by the same equatorial integration and L^1 – L^2 interpolation as in Lemma 13.15, therefore yields

$$\sup_{1 \leq u \leq 2} |m_{j,u}(\xi)| \leq C_\rho \|\Omega\|_{s,a_{s,\rho}}(r). \quad (436)$$

Moreover,

$$\frac{d}{du} m_{j,u}(\xi) = -u^{-1} \int_{\mathbb{S}^{n-1}} \Omega(\theta) b(2^j u) e^{-itue \cdot \theta} d\sigma(\theta),$$

so the derivative operator has L^2 norm at most $C\|b\|_\infty \|\Omega\|_1$. Apply the anchored Sobolev–variation argument of Lemma 3.7, with the operator norm in (436) playing the role of a and the derivative bound playing the role of b . This gives

$$\|V^2(A_{j,u}\Delta_{r-j}^2 f : u)\|_2 \leq C_\rho a_{s,\rho}(r)^{1/4} \|\Omega\|_s \|\Delta_{r-j}^2 f\|_2.$$

After squaring and summing in j ,

$$\|S_r f\|_2 \leq C_\rho a_{s,\rho}(r)^{1/4} \|\Omega\|_s \|f\|_2. \quad (437)$$

Interpolate the sublinear operator S_r between (435) and (437). For the target p this gives an exponent $\vartheta_p > 0$, independent of s, r :

$$\|S_r f\|_p \leq C_{p,\rho} a_{s,\rho}(r)^{\vartheta_p} \|\Omega\|_s \|f\|_p.$$

For finite j - and r -sets, the variation triangle inequality and Minkowski in the outer ℓ^2 sum give

$$\left(\sum_j V^2 \left(\sum_{r \in E} A_{j,u} \Delta_{r-j}^2 f : u \right)^2 \right)^{1/2} \leq \sum_{r \in E} S_r f.$$

Consequently (427), with $\delta = \vartheta_p$, makes the r -sum Cauchy in L^p , uniformly over finite shell sets and finite u -partitions. We again record the two-tail argument. Given $\varepsilon > 0$, choose R so that

$$\sum_{|r| > R} \|S_r f\|_p < \frac{\varepsilon}{2}.$$

For the finitely many $|r| \leq R$, let $\mathcal{J}_J \uparrow \mathbb{Z}$ be the retained shell sets, put $E_J = \mathbb{Z} \setminus \mathcal{J}_J$, and restrict the outer square sum defining S_r to E_J . The vector-valued estimates above, with the same restriction, bound this quantity by the corresponding Littlewood–Paley square-function tail; hence

$$\left\| \left(\sum_{j \in E_J} V^2(A_{j,u} \Delta_{r-j}^2 f : u \in \mathcal{U})^2 \right)^{1/2} \right\|_p \rightarrow 0$$

for every finite $\mathcal{U} \subset [1, 2]$, uniformly in the choice of its ordered partition. Choose J so that the sum of these finitely many norms is below $\varepsilon/2$. Minkowski in the outer ℓ^2 sum now shows that the combined frequency–shell approximation is Cauchy in the finite-path variation norm. Here the asserted tail convergence is not formal: for each fixed r , $j \mapsto r - j$ is a bijection and the restricted square function is pointwise dominated by $(\sum_\ell |\Delta_\ell^2 f|^2)^{1/2} \in L^p$, while it tends pointwise to zero. Dominated convergence therefore removes the shell cutoff uniformly over the already fixed finite set of r 's and finite u -partitions.

For each fixed j, u , the Littlewood–Paley identity and the fixed-annulus L^1 kernel identify its coordinate limit with $A_{j,u} f$. First exhaust the shell sets and then the finite rational u -partitions. Finite-partition lower semicontinuity and Fatou, followed by local absolute continuity of $u \mapsto A_{j,u} f(x)$, give $S_2(T_\Omega^{0,b} f) \leq \sum_r S_r f$ in the limiting sense and complete the proof. $\square$

Proposition 13.20 (Terminal s' -tracked critical jump). *For $1 < s \leq 2$, mean-zero $\Omega \in L^s(\mathbb{S}^{n-1})$, and $1 < p < \infty$, the following holds for every $f \in L^p(\mathbb{R}^n)$:*

$$\|\mathbf{J}(T_{\Omega,\varepsilon}^{0,b} f : \varepsilon > 0)\|_p \leq C_{n,p,\rho} s' \|\Omega\|_s \|f\|_p \quad (438)$$

under normalization (379).

Proof. The deterministic long–short inequality, equivalently (1.13) in [9], gives pointwise

$$\mathbf{J}(T_{\Omega,\varepsilon}^{0,b} f) \leq C \left[S_2(T_\Omega^{0,b} f) + \mathbf{J}(T_{\Omega,2^k}^{0,b} f : k \in \mathbb{Z}) \right].$$

Apply Lemmas 13.18 and 13.19. $\square$

Proposition 13.21 (Unphased terminal $L \log L$ theorem). *For mean-zero $\Omega \in L \log L(\mathbb{S}^{n-1})$, $1 < p < \infty$, and every $f \in L^p(\mathbb{R}^n)$,*

$$\|\mathbf{J}(T_{\Omega,\varepsilon}^{0,b} f : \varepsilon > 0)\|_p \leq C_{n,p,\rho} \mathfrak{L}(\Omega) \|f\|_p \quad (439)$$

under (379).

Proof. We use exactly the finite-trajectory Orlicz extrapolation used in the proof of Proposition 4.3, but record the cancellation issue. Let

$$\Pi \Theta = \Theta - \sigma(\mathbb{S}^{n-1})^{-1} \int_{\mathbb{S}^{n-1}} \Theta d\sigma.$$

For a finite ordered truncation set D , let E_D be the scalar finite-path quotient endowed with a norm uniformly equivalent to the pointwise critical jump functional. Define the linear map

$$\mathcal{A}_{f,D} \Theta = [T_{\Pi\Theta,\varepsilon}^{0,b} f]_{\varepsilon \in D} \in L^p(\mathbb{R}^n; E_D).$$

Since Π is bounded on every L^s on the finite-measure sphere, Proposition 13.20 gives uniformly in D

$$\|\mathcal{A}_{f,D}\|_{L^s(\mathbb{S}^{n-1}) \rightarrow L^p(E_D)} \leq C_{p,\rho} s' \|f\|_p, \quad 1 < s \leq 2.$$

Pair with a norm-one functional in $[L^p(E_D)]^*$ and take $s = k/(k-1)$ for integers $k \geq 2$. For each k the resulting scalar functional on the sphere is represented by $h_k \in L^k$ satisfying $\|h_k\|_k \leq C_{p,\rho} k \|f\|_p$. These representatives agree almost everywhere: two of them represent the same functional on bounded angular functions, their difference lies in L^1 on the finite-measure sphere, and testing against its bounded complex sign makes that difference zero. Denote the common representative by h . If $\|f\|_p > 0$, Tonelli and the exponential series give, for a sufficiently large structural L ,

$$\int_{\mathbb{S}^{n-1}} \Psi_0 \left(\frac{|h|}{L\|f\|_p} \right) d\sigma \leq \sum_{k=2}^{\infty} \frac{(C_{p,\rho} k/L)^k}{k!} \leq 1.$$

The case $f = 0$ is immediate. Hence

$$\|h\|_{\exp L} \leq C_{p,\rho} \|f\|_p.$$

Orlicz–Hölder therefore yields

$$\|\mathcal{A}_{f,D}\Omega\|_{L^p(E_D)} \leq C_{p,\rho} \mathfrak{L}(\Omega) \|f\|_p$$

for mean-zero Ω , uniformly in D . Increase finite rational sets and use finite-annulus local absolute continuity to recover all positive parameters. This proves (439). $\square$

13.8 Strong polynomial phases

Proposition 13.22 (Radially weighted annular decay). *Let P be a real polynomial on $\mathbb{R}^n \times \mathbb{R}^n$ of total degree at most d , and let $\Theta \in L^\infty(\mathbb{S}^{n-1})$. For $R > 0$ and $1/2 \leq u \leq 1$ put*

$$A_{R,u}^{P,\Theta,b} f(x) = \int_{uR < |x-y| < R} e^{iP(x,y)} b(|x-y|) K_\Theta(x-y) f(y) dy.$$

Under (379), for every nonterminal flag position μ there are $a_{\mu,\rho} > 0$ and $C_{n,d,\rho}$ such that, if the first nonzero flag coordinate of P is ± 1 at position μ , then

$$\sup_{1/2 \leq u \leq 1} \|A_{R,u}^{P,\Theta,b}\|_{2 \rightarrow 2} \leq C_{n,d,\rho} R^{-a_{\mu,\rho}} \|\Theta\|_\infty, \quad R \geq 1. \quad (440)$$

Every later phase coefficient remains unrestricted.

Proof. We revisit the three branches in Proposition 3.6.

For a transverse pivot, the kernel entering $A_Q^* A_Q$ is obtained from the original $g_h(v)$ by multiplication by $\overline{b(|v|)} b(|v-h|)$. The compact polynomial operator is applied to this function only through its L^2 norm. Since $\|b\|_\infty \leq 1$, the bound (72), the polynomial negative moment in h , and both Hilbert–Schmidt square roots are unchanged. Thus the original transverse exponent remains valid.

For a skew pivot, the twisted-convolution argument uses only the L^2 size of the annular kernel. Inserting b decreases that size by at most $\|b\|_\infty$; the original R^{-1} decay is unchanged.

For a convolution pivot, write the first nonzero homogeneous convolution term as Q_s , with $\|Q_s\|_{\text{coeff}} \geq c_{n,d}$. After the change $r = Rv$, the one-dimensional fibre amplitude is

$$v \mapsto b(Rv) v^{-1} \mathbf{1}_{[u,1]}(v).$$

Its zero extension has uniformly bounded $L^\infty + V^p$ norm by Lemma 13.2. Let $c_{\text{poly},\rho} > 0$ be the exponent in Corollary 13.8 for degrees at most d . Plancherel on each line therefore gives the fibre bound

$$C_\rho \min\{1, (R^s |Q_s(\theta)|)^{-c_{\text{poly},\rho}}\}.$$

Choose

$$\beta_{s,\rho} := \frac{1}{2} \min \left\{ c_{\text{poly},\rho}, \frac{1}{s} \right\} > 0. \quad (441)$$

The last display is bounded by $C_\rho R^{-s\beta_{s,\rho}} |Q_s(\theta)|^{-\beta_{s,\rho}}$, and (63) makes the angular integral finite uniformly in all lower coefficients. Thus the convolution branch has exponent $s\beta_{s,\rho} > 0$.

Take $a_{\mu,\rho}$ to be the corresponding branch exponent. There are only finitely many flag positions, so all constants are structural and coefficient-uniform. $\square$

For test functions, polar coordinates show that $u \mapsto A_{R,u}^{P,\Theta,b} f$ is an absolutely continuous L^2 -valued path even when b has jumps. For almost every u ,

$$\partial_u A_{R,u}^{P,\Theta,b} f(x) = -u^{-1} b(uR) \int_{\mathbb{S}^{n-1}} \Theta(\theta) e^{iP(x,x-uR\theta)} f(x-uR\theta) d\sigma(\theta).$$

Consequently

$$\sup_{1/2 \leq u \leq 1} \|\partial_u A_{R,u}^{P,\Theta,b} f\|_q \leq C \|b\|_\infty \|\Theta\|_1 \|f\|_q, \quad 1 \leq q \leq \infty.$$

The anchored Sobolev–variation argument of Lemma 3.7 therefore gives

$$\|V^2(A_{R,u}^{P,\Theta,b} f : 1/2 \leq u \leq 1)\|_2 \leq C_{n,d,\rho} R^{-a_{\mu,\rho}/4} \|\Theta\|_\infty \|f\|_2. \quad (442)$$

Put

$$a_{0,\rho} := \min \left(\left\{ \frac{a_{\mu,\rho}}{4} : \mu \text{ nonterminal} \right\} \cup \{1\} \right) > 0. \quad (443)$$

The added value 1 is used only when there is no nonterminal position.

Lemma 13.23 (Weighted q' -tracked partial-annulus estimates). *Let P be normalized and nonterminal, $R \geq 1$, $1 < q \leq 2$, and $q' = q/(q-1)$. Let $\Theta \in L^q(\mathbb{S}^{n-1})$. Under (379), for every $1 < p < \infty$ set*

$$\gamma_{p,\rho} := \frac{a_{0,\rho}}{2 \max(p, p')} > 0.$$

Then, for every $f \in L^p(\mathbb{R}^n)$,

$$\sup_{1/2 \leq u \leq 1} \|A_{R,u}^{P,\Theta,b} f\|_p \leq C_{n,d,p,\rho} R^{-\gamma_{p,\rho}/q'} \|\Theta\|_q \|f\|_p, \quad (444)$$

$$\|V^2(A_{R,u}^{P,\Theta,b} f : 1/2 \leq u \leq 1)\|_p \leq C_{n,d,p,\rho} R^{-\gamma_{p,\rho}/q'} \|\Theta\|_q \|f\|_p. \quad (445)$$

All constants are uniform in the later phase coefficients and in the location of the variation of b .

Proof. For fixed P, R, u, b , the map $\Theta \mapsto A_{R,u}^{P,\Theta,b}$ is linear into $\mathcal{B}(L^2)$. Its angular L^1 bound follows from the annular L^1 kernel and $\|b\|_\infty \leq 1$; its angular L^∞ bound is Proposition 13.22. Complex interpolation, whose L^∞ weight is $1/q'$, gives, after weakening the finite minimum of the nonterminal exponents to $a_{0,\rho}$,

$$\sup_u \|A_{R,u}^{P,\Theta,b}\|_{2 \rightarrow 2} \leq C_{n,d,\rho} R^{-a_{0,\rho}/q'} \|\Theta\|_q.$$

The preceding a.e. derivative estimate and the anchor $A_{R,1}^{P,\Theta,b} = 0$ give, exactly as in (89),

$$\|V^2(A_{R,u}^{P,\Theta,b} f : u)\|_2 \leq C_{n,d,\rho} R^{-a_{0,\rho}/(4q')} \|\Theta\|_q \|f\|_2.$$

On the other hand $V^2 \leq V^1$ and the same derivative formula give nondecaying strong $(1, 1)$ and (∞, ∞) bounds with norm $C \|\Theta\|_1$. Marcinkiewicz interpolation puts weight $2/p'$ on the L^2 endpoint when $p < 2$ and weight $2/p$ when $p > 2$. This proves (445) with the displayed, slightly weakened, $\gamma_{p,\rho}$. The same spatial interpolation starting from the fixed- u L^2 estimate proves the stronger fixed- u bound and hence (444). Approximation in f and Θ is justified by the nondecaying V^1 estimate. $\square$

Apply this lemma to the dyadic far path. The annular pieces and their short variations have L^p norm $C2^{-\gamma_{p,\rho}j/q'}\|\Theta\|_q\|f\|_p$; hence their geometric sum is at most $Cq'\|\Theta\|_q\|f\|_p$. The deterministic dyadic long-short decomposition therefore proves the weighted analogue of Proposition 4.2. On every finite truncation set, pair the resulting linear angular map with a norm-one dual functional. Taking $q = k/(k-1)$ gives the compatible bounds $\|h\|_k \leq Ck\|f\|_p$; the common-representative and exponential-series argument used in Proposition 13.21 yields the weighted analogue of Proposition 4.3.

Finally, the local triangular reduction of Proposition 4.5 acquires the factor $b(|x-y|)$ in its error kernel; its absolute value is bounded by one, and the residual later-phase path carries the same b . Backward induction over the finite flag, with the unphased terminal estimate Proposition 13.21, therefore proves

$$\|\mathbf{J}(T_{\Omega,\varepsilon}^{P,b}f)\|_p \leq C_{n,d,p,\rho}\mathfrak{L}(\Omega)\|f\|_p \quad (446)$$

under (379). The normalizing dilation sends b to $b(R\cdot)$, and (380) preserves its radial norm; hence the estimate is uniform in every coefficient of P .

13.9 Phase-adapted packets with finite radial variation

Proposition 13.24 (Radially weighted phase-adapted one-shell packet). *For every fixed $1 \leq \rho < \infty$, Proposition 7.10 remains valid with the radial factor $b(|x-y|)$ under (379). Namely, there are $c_{P,\rho}, c_{E,\rho} > 0$ and $t_{0,\rho}$ such that, whenever the packet separation parameter satisfies $t \geq t_{0,\rho}$,*

$$\|V^2(G_\varepsilon : R \leq \varepsilon \leq 2R)\|_2^2 \leq C_{n,d,p}\|\Theta\|_\infty^2 2^{-c_{P,\rho}t} \Lambda M, \quad (447)$$

$$\|V^1(E_\varepsilon : R \leq \varepsilon \leq 2R)\|_1 \leq C_{n,d,p}\|\Theta\|_1 2^{-c_{E,\rho}t} M. \quad (448)$$

The estimates are uniform under packetwise scalar coefficients of modulus at most one.

Proof. We repeat the finite-mode low/high phase-block induction in Proposition 7.10, but choose all parameters after the radial exponents have been fixed. Let $c_{\text{flat},\rho}$ and $C_{B,\rho}$ be the constants in Corollary 13.12, let $\delta_{T,\rho}$ be given by (413), and let C_{Four} dominate all powers in Lemma 7.9. Decrease $c_{\text{flat},\rho}$ if necessary so that it is at most one. Here and below $C_{B,\rho}$ has first been enlarged once to dominate every radial-amplitude power in the equal-shell, unequal-shell, flat-block, and collective-shell estimates; it is not enlarged after the next parameter is chosen. Put

$$\theta_\rho := \frac{c_{\text{flat},\rho}}{16(C_{B,\rho} + 1)}. \quad (449)$$

Choose $B_{*,\rho} > 1$ so large that

$$\frac{2C_{\text{Four}}}{B_{*,\rho} - 1} \leq \min\{\theta_\rho/2, c_{\text{flat},\rho}/8\}. \quad (450)$$

Put $N_d = \max(d-1, 0)$ and choose $0 < \kappa_{*,\rho} \leq 1$ so small that, for every possible number $0 \leq N \leq N_d$ of nonpure homogeneous blocks,

$$2C_{\text{Four}} \sum_{r=1}^N \kappa_{*,\rho} B_{*,\rho}^{r-N} \leq \min\{\delta_{T,\rho}/2, 1/2\}. \quad (451)$$

Set

$$\kappa_r = \kappa_{*,\rho} B_{*,\rho}^{r-N}, \quad S_r = \sum_{i < r} \kappa_i, \quad \gamma_r = \theta_\rho \kappa_r.$$

Then $S_r \leq \kappa_r/(B_{*,\rho} - 1)$ and

$$\gamma_r - 2C_{\text{Four}}S_r \geq \gamma_r/2, \quad (452)$$

$$c_{\text{flat},\rho}\kappa_r - C_{B,\rho}\gamma_r - 2C_{\text{Four}}S_r \geq c_{\text{flat},\rho}\kappa_r/2, \quad (453)$$

$$c_{\text{flat},\rho} - C_{B,\rho}\gamma_r - 2C_{\text{Four}}S_r \geq c_{\text{flat},\rho}/2. \quad (454)$$

These are the complete compatibility conditions; the choice is noncircular because all quantities on their right were fixed before $B_{*,\rho}$ and $\kappa_{*,\rho}$.

Suppose block r is the first high block. The preceding low blocks have squared Fourier loss at most $2^{2C_{\text{Four}}S_r t}$. Partition the normalized shell into $K_r \simeq 2^{\gamma_r t}$ cells. The within-cell square function uses only $\|b\|_\infty \leq 1$ and contributes $C2^{-\gamma_r t} \|\Theta\|_\infty^2 \Lambda M$. A fixed Rademacher–Menshov choice of cell signs produces the radial profile

$$b(Ru) \sum_{\ell=1}^{K_r} \varepsilon_\ell \mathbf{1}_{I_\ell}(u).$$

By (385), its $L^\infty + V^\rho$ norm is at most $C_\rho(1 + K_r^{1/\rho})$. Applying (407) and Lemma 5.2 therefore gives, before the preceding Fourier loss is included,

$$C_\rho t^2 \|\Theta\|_\infty^2 2^{(C_{B,\rho}/\rho)\gamma_r t} \Lambda M \{2^{-c_{\text{flat},\rho} K_r t} + 2^{-c_{\text{flat},\rho} t}\}. \quad (455)$$

Since $\rho \geq 1$, replace $C_{B,\rho}/\rho$ by $C_{B,\rho}$. Equations (452)–(454), after multiplication by the squared Fourier loss, leave the three exponents $\gamma_r/2$, $c_{\text{flat},\rho} K_r/2$, and $c_{\text{flat},\rho}/2$. Thus, after absorbing the polynomial factor t^2 , one may take

$$c_{r,\rho} := \frac{1}{4} \min\{\gamma_r, c_{\text{flat},\rho} K_r, c_{\text{flat},\rho}\} > 0. \quad (456)$$

Every high branch therefore has $E = 0$ and geometric $L^2(V^2)$ decay.

If all blocks are low, expand them all. Freezing the combined input frequency at each packet centre leaves a mean-zero packet, so Proposition 13.14 applies. The squared Fourier loss is at most $2^{2C_{\text{Four}} \sum_r K_r t}$, and (451) gives, at the level of the full squared estimate,

$$2^{-\delta_{T,\rho} t} (2^{C_{\text{Four}} \sum_r K_r t})^2 \leq 2^{-\delta_{T,\rho} t/2}.$$

The input-modulation remainder is estimated in total truncation 1-variation exactly as in Proposition 7.10; multiplication by b costs only $\|b\|_\infty \leq 1$. The weighted Fourier sum and (451) leave the bound $C\|\Theta\|_1 2^{-t/2} M$.

All Fourier expansions are first truncated. If $K' > K$ and block r is the first high block, define

$$\mathfrak{B}_{r,t,\rho} := 2^{-\gamma_r t} + t^2 2^{(C_{B,\rho}/\rho)\gamma_r t} \{2^{-c_{\text{flat},\rho} K_r t} + 2^{-c_{\text{flat},\rho} t}\}.$$

Applying the within-cell and signed-grid estimates only to Fourier tuples omitted at level K gives, on every normalized interacting block pair,

$$\|G^{K'} - G^K\|_{L^2(V^2)} \leq C_\rho \|\Theta\|_\infty (\Lambda \tilde{M})^{1/2} \mathfrak{B}_{r,t,\rho}^{1/2} \omega_K(\{i : i < r\}), \quad E^{K'} - E^K = 0. \quad (457)$$

If all blocks are low, the radial terminal-packet estimate and the freezing remainder give the following sharper tail estimates. Here one does *not* replace the omitted Fourier coefficients by the global majorant $2^{C_{\text{Four}} \sum_r K_r t}$: the exact common tail ω_K is retained. Consequently the terminal estimate contributes its raw square-root factor $2^{-\delta_{T,\rho} t/2}$ before the omitted tuples are summed:

$$\|G^{K'} - G^K\|_{L^2(V^2)} \leq C_\rho \|\Theta\|_\infty 2^{-\delta_{T,\rho} t/2} (\Lambda \tilde{M})^{1/2} \omega_K(\{1, \dots, N\}), \quad (458)$$

$$\|E^{K'} - E^K\|_{L^1(V^1)} \leq C \|\Theta\|_1 2^{-t} \tilde{M} \omega_K(\{1, \dots, N\}). \quad (459)$$

Here $\tilde{M}$ is the normalized block mass. Since $\omega_K(S) \rightarrow 0$ by (234), the finite-mode decompositions are Cauchy in the anchored $L^2(V^2)$ and $L^1(V^1)$ spaces. The absolute common Fourier majorants give uniform convergence of the phase factors on the compact normalized blocks; dominated convergence therefore identifies the variation-space limit with the original trajectory. Fatou on finite parameter partitions preserves all displayed bounds.

For the branch $N = 0$ there is no high block and we assign its squared decay exponent to be $\delta_{T,\rho}/2$. Thus the final minimum is never over an empty set: take $c_{P,\rho}$ to be the minimum of $\delta_{T,\rho}/2$ and all

high-branch exponents $c_{r,\rho}$ arising for the finitely many cases $1 \leq N \leq N_d$, and take $c_{E,\rho} = 1/2$. Increase $t_{0,\rho}$ so that it dominates the terminal-packet threshold and absorbs the finitely many polynomial factors used above. Rescaling and the bounded-overlap block summation are exactly those in the final paragraph of Proposition 7.10; the L^2 -square and L^1 Jacobians cancel the normalized mass factor. This proves (447)–(448). $\square$

13.10 The weak endpoint

We first treat the unphased long path, where one additional modification is essential.

Lemma 13.25 (Whole-profile smoothing in the unphased weak long path). *Fix a shell $2^j < |z| \leq 2^{j+1}$ and put*

$$a_j(u) = b(2^j u) \mathbf{1}_{(1,2]}(u), \quad 1/2 < u < 3.$$

*Under (379), $\|a_j\|_\infty + V^\rho(a_j) \leq C$ uniformly in j . If $a_{j,h} = a_j * \phi_h$, then*

$$\|a_j - a_{j,h}\|_1 \leq C_\rho h^{1/\rho}, \quad (460)$$

$$\|\partial^N a_{j,h}\|_\infty \leq C_N h^{-N}. \quad (461)$$

Consequently, in the long-variation part of the proof of Proposition 8.1 one may replace the smoothing of the sharp shell indicator by smoothing the entire profile a_j . The sharp–smooth difference has total $V^1 L^1$ norm $O_\rho(h^{1/\rho} \Sigma)$, while the smoothed full-kernel estimate has exactly the derivative hypothesis used in the proof of Lemma 5.4.

Proof. The uniform variation bound follows from Lemma 13.2; the two estimates are Lemma 13.3. Polar coordinates give the shell-kernel L^1 difference $C\|\omega_s\|_1 h^{1/\rho}$. If E_j denotes this kernel difference, the sharp and smooth long paths are suffix sums. Inserting the zero and total endpoints in an arbitrary variation partition gives the deterministic bound

$$V^1 \left(\sum_{j \geq k} E_j * B_{j-s} : k \right) \leq 2 \sum_j |E_j * B_{j-s}|.$$

Since, for fixed s , every bad cube is assigned to at most one shell, Young’s inequality gives

$$\|V^1(\text{sharp long path} - \text{smooth long path})\|_1 \leq C\|\omega_s\|_1 h^{1/\rho} \sum_j \|B_{j-s}\|_1 \leq C\|\omega_s\|_1 h^{1/\rho} \Sigma.$$

At physical scale the smoothed kernel is

$$H_j^s(z) = K_{\omega_s}(z) a_{j,h}(2^{-j}|z|).$$

The chain rule and (461) give

$$\|\partial_r^N [a_{j,h}(2^{-j}r)]\|_\infty \leq C_N 2^{-jN} h^{-N},$$

while polar coordinates give $\|H_j^s\|_1 \leq C\|\omega_s\|_1$. Thus its normalized full-kernel constant satisfies $M_N \leq C\|\omega_s\|_\infty h^{-N}$, exactly as required. $\square$

Lemma 13.26 (One-dimensional fixed-sign full-kernel estimate). *Let H_j be odd C^2 kernels on $\mathbb{R}$, supported where $2^{j-2} \leq |x| \leq 2^{j+2}$, and assume*

$$\|H_j\|_1 \leq M, \quad \|H_j'\|_\infty \leq M 2^{-2j}, \quad \|H_j''\|_\infty \leq M 2^{-3j}.$$

Let $\mathcal{Q}$ be a finite family of pairwise disjoint intervals and suppose

$$\text{supp } b_Q \subset Q, \quad \int b_Q = 0, \quad \|b_Q\|_1 \leq \Lambda |Q|.$$

For $s \geq 5$, pair H_j with the intervals of length 2^{j-s} . Then for arbitrary fixed coefficients $|\varepsilon_Q| \leq 1$,

$$\left\| \sum_j H_j * \sum_{|Q|=2^{j-s}} \varepsilon_Q b_Q \right\|_2^2 \leq C M^2 2^{-s} \Lambda \sum_Q \|b_Q\|_1. \quad (462)$$

Proof. Write $B_j = \sum_{|Q|=2^{j-s}} \varepsilon_Q b_Q$ and $L_{jk} = \tilde{H}_j * H_k$. Since H_j is odd, $\int H_j = 0$. The family $\mathcal{Q}$ is finite, so only finitely many B_j are nonzero and all expansions below are finite. With $\tilde{H}_j(x) = \overline{H_j(-x)}$, oddness and integrability also give $\int \tilde{H}_j = 0$. For $j \leq k$,

$$L'_{jk}(x) = \int \tilde{H}_j(y) [H'_k(x-y) - H'_k(x)] dy,$$

and therefore

$$\|L'_{jk}\|_\infty \leq CM^2 2^j 2^{-3k}. \quad (463)$$

Moreover L_{jk} is supported in an interval of length $C2^k$. Because the intervals supporting B_k are disjoint and have length 2^{k-s} , the total $|B_k|$ mass in any interval of length $C2^k$ is at most $C2^s \Lambda 2^{k-s} = C\Lambda 2^k$. Hence

$$\|(L_{jk} * B_k)'\|_\infty \leq CM^2 \Lambda 2^{j-2k}.$$

For each interval Q contributing to B_j , cancellation at its center x_Q gives

$$\left| \int_Q \varepsilon_Q b_Q(x) (L_{jk} * B_k)(x) dx \right| \leq \|b_Q\|_1 |Q| \|(L_{jk} * B_k)'\|_\infty,$$

where $|\varepsilon_Q| \leq 1$. Since $|Q| = 2^{j-s}$,

$$|\langle H_j * B_j, H_k * B_k \rangle| \leq CM^2 \Lambda 2^{-s} 2^{-2(k-j)} \sum_{|Q|=2^{j-s}} \|b_Q\|_1.$$

This estimate includes $j = k$. Sum first in $k \geq j$ and then in j ; the geometric factor $2^{-2(k-j)}$ and the identity $\sum_j \sum_{|Q|=2^{j-s}} \|b_Q\|_1 = \sum_Q \|b_Q\|_1$ prove (462) after expanding the squared norm and using conjugate symmetry for $j > k$. $\square$

Proposition 13.27 (Unphased weak endpoint with radial V^p). *For every mean-zero $\Omega \in L \log L(\mathbb{S}^{n-1})$ and every bounded compactly supported f , under (379),*

$$\|\mathbf{J}(T_{\Omega, \varepsilon}^{0,b} f)\|_{L^{1,\infty}} \leq C_{n,p} \mathfrak{L}(\Omega) \|f\|_1. \quad (464)$$

Proof. Impose the finite bad-cube, shell, and rational-parameter cutoffs from Proposition 8.1; all estimates below are uniform in those cutoffs. If $\mathfrak{L}(\Omega) = 0$, then $\Omega = 0$ and there is nothing to prove. By homogeneity normalize $\mathfrak{L}(\Omega) = 1$ and perform the ordinary Calderón–Zygmund decomposition at height α . Write

$$b_{CZ} = \sum_Q b_Q, \quad \Sigma = \sum_Q \|b_Q\|_1 \leq C \|f\|_1.$$

The good function is controlled by the $p = 2$ case of Proposition 13.21. Thus, for the current finite parameter set I , the set

$$E_{\text{good}} = \{x : \mathbf{J}_I(T_{\Omega, \varepsilon}^{0,b} g) > \alpha\}$$

has measure at most $C\alpha^{-1} \|f\|_1$.

Choose

$$N = 4n + 1, \quad \eta_0 = \frac{1}{192N}, \quad 0 < \delta_a < \min \left\{ \frac{\delta_{T,p}}{8}, \frac{1}{192} \right\}. \quad (465)$$

For the bad function use the mean-zero angular split $\Omega = \omega_s + \Omega_s^\sharp$ of (40) with this value of δ_a . The angular tails satisfy the original $V^1 L^1$ estimate because multiplication by b costs only $\|b\|_\infty \leq 1$. For the bounded pieces,

$$\|\omega_s\|_\infty \leq C 2^{\delta_a s}, \quad \|\omega_s\|_1 \leq C.$$

Proposition 13.14 gives the bounded-angular short square function with squared L^2 bound

$$C 2^{-[\delta_{T,p} - 2\delta_a]s} \alpha \Sigma, \quad (466)$$

which is geometrically summable by (465).

For the dyadic long path take $h_s = 2^{-\eta_0 s}$ and apply Lemma 13.25. The sharp–smooth difference has total $V^1 L^1$ norm

$$C_\rho 2^{-\eta_0 s / \rho} \Sigma. \quad (467)$$

The smoothed kernels satisfy

$$M_N \leq C \|\omega_s\|_\infty h_s^{-N} \leq C 2^{(\delta_a + N\eta_0)s}.$$

Use exactly the nested adjacent grids, exceptional sets X_s , fixed ranks, and suffix representation from (306)–(312). A fixed rank sign is still chosen before the output point is evaluated, so Lemma 5.4 applies. After the Rademacher–Menshov factor is squared, the classwise long bound is

$$C s^2 2^{-[1/24 - 2\delta_a - 2N\eta_0]s} \alpha \Sigma. \quad (468)$$

Here

$$\frac{1}{24} - 2\delta_a - 2N\eta_0 > \frac{1}{24} - \frac{1}{96} - \frac{1}{96} = \frac{1}{48}.$$

Thus (468) is summable. In dimension one, (40) gives $\omega_s(1) + \omega_s(-1) = 0$. Since the smoothed radial profile is even as a function of the spatial variable, the kernel H_j^s is odd. Lemma 13.26 therefore applies; the C^2 smoothing norm is at most $C 2^{(\delta_a + 2\eta_0)s}$, so its exponent $1 - 2\delta_a - 4\eta_0$ is larger than the common positive exponent $1/48$. Hence the same summable majorant works in every dimension.

For clarity, here is the exact weighted counterpart of the majorants in Proposition 8.1. Denote by $U_{s,\varepsilon}^b$ the bounded-angular level- s path, by $H_j^{s,b}$ its whole-profile smoothed annular kernel, by B_{j-s} the same bad-cube input as there, and by $\text{Tail}_{s,\varepsilon}^b$ the angular-tail path. Put

$$\begin{aligned} \mathcal{S}_s^b &= \left(\sum_j V^2(U_{s,\varepsilon}^b : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2}, \\ \tilde{U}_s^b(k) &= \sum_{j \geq k} H_j^{s,b} * B_{j-s}, \\ \mathcal{L}_s^b &= \mathbf{1}_{X_s^c} V^2(\tilde{U}_s^b(k) : k), \\ \mathcal{E}_s^b &= V^1(U_{s,2^k}^b - \tilde{U}_s^b(k) : k), \\ \mathcal{T}_s^b &= V^1(\text{Tail}_{s,\varepsilon}^b : \varepsilon > 0). \end{aligned}$$

The terminal packet estimate (466) controls $\mathcal{S}_s^b$ in L^2 , the fixed-rank whole-profile estimate (468) controls $\mathcal{L}_s^b$ in L^2 , (467) controls $\mathcal{E}_s^b$ in L^1 , and the unchanged angular-tail estimate controls $\mathcal{T}_s^b$ in L^1 . Thus, exactly as in (314), set

$$\mathcal{G} = \sum_{s \geq s_0} (\mathcal{S}_s^b + \mathcal{L}_s^b), \quad \mathcal{W} = \sum_{s \geq s_0} (\mathcal{E}_s^b + \mathcal{T}_s^b).$$

Their sums satisfy

$$\|\mathcal{G}\|_2^2 \leq C_\rho \alpha \Sigma, \quad \|\mathcal{W}\|_1 \leq C_\rho \Sigma, \quad |X| \leq C \alpha^{-1} \|f\|_1.$$

All majorants and exceptional sets are independent of the jump amplitude. The finite-path long–short inequality and finite-path jump normability therefore give, off the fixed exceptional set,

$$\mathbf{J}_I(T_{\Omega,\varepsilon}^{0,b} b_{CZ} : \varepsilon \in I) \leq C_\rho (\mathcal{G} + \mathcal{W}).$$

Let E_{exc} be the union of E_{good} , the fixed enlarged Calderón–Zygmund set, and $X = \bigcup_s X_s$ (an empty component is simply omitted). Its measure is at most $C \alpha^{-1} \|f\|_1$. Choose a fixed $A_\rho > 2$ larger than the constants in the pointwise jump quasi-triangle inequality and in the last display. Outside E_{exc} , the inclusion

$$\{\mathbf{J}_I(T_{\Omega,\varepsilon}^{0,b} f) > A_\rho \alpha\} \subset \{\mathcal{G} > c_\rho \alpha\} \cup \{\mathcal{W} > c_\rho \alpha\}$$

holds for a fixed $c_\rho > 0$. Consequently Chebyshev, with its quadratic form on $\mathcal{G}$ and linear form on $\mathcal{W}$, gives

$$\begin{aligned} |\{\mathbf{J}_I(T_{\Omega,\varepsilon}^{0,b}f) > A_\rho \alpha\}| &\leq |E_{\text{exc}}| + C_\rho \alpha^{-2} \|\mathcal{G}\|_2^2 + C_\rho \alpha^{-1} \|\mathcal{W}\|_1 \\ &\leq C_\rho \alpha^{-1} \|f\|_1. \end{aligned}$$

This estimate is uniform in every cutoff. For an arbitrary requested threshold $\tau > 0$, run the decomposition at $\alpha = \tau/A_\rho$; no majorant or exceptional set depends on a jump amplitude chosen afterward.

Finally remove the bad-cube and shell cutoffs, then exhaust the positive rational parameters, in the order used in Proposition 8.1. At each cutoff-removal step, fix $\tau > 0$ and a finite parameter set I . The finite-annulus L^1 kernel gives, after a subsequence, coordinatewise almost-everywhere convergence $F_N \rightarrow F$ on I . If $\mathbf{J}_I(F) > \tau$, a finite strict witness has positive excess over τ and persists for all sufficiently large N ; hence

$$\mathbf{1}_{\{\mathbf{J}_I(F) > \tau\}} \leq \liminf_{N \rightarrow \infty} \mathbf{1}_{\{\mathbf{J}_I(F_N) > \tau\}}.$$

Fatou preserves the distribution estimate at the same threshold. Increasing finite sets exhaust $\mathbb{Q}_+$, and local absolute continuity in ε identifies the rational and full-parameter functionals. Thus the cutoff estimates pass to the full pointwise functional. Restoring angular homogeneity proves (464). $\square$

Proposition 13.28 (Polynomial weak endpoint with radial V^ρ). *Under (379), for every polynomial P of degree at most d , every mean-zero $\Omega \in L \log L(\mathbb{S}^{n-1})$, and every bounded compactly supported f ,*

$$\|\mathbf{J}(T_{\Omega,\varepsilon}^{P,b}f)\|_{L^{1,\infty}} \leq C_{n,d,\rho} \mathfrak{L}(\Omega) \|f\|_1. \quad (469)$$

Proof. If $\mathfrak{L}(\Omega) = 0$, then $\Omega = 0$ and the assertion is immediate. By angular homogeneity assume throughout the induction that $\mathfrak{L}(\Omega) = 1$. At each inductive stage impose finite cube and shell cutoffs and fix a finite rational truncation set I ; every bound below is uniform in these choices. We induct on the largest total degree m carrying a nonpure homogeneous class. If no such class exists, write $P(x,y) = p(x) + q(y)$ and use Proposition 13.27 after input and output modulation.

Suppose the result is known through nonpure degree $m-1$. Let P_m be the highest nonpure homogeneous block and put

$$C_m = \|P_m\|_{\text{coeff}}, \quad r_0 = C_m^{-1/m}.$$

The exact conjugacy

$$T_{\Omega,r_0\varepsilon}^{P,b}f(r_0X) = T_{\Omega,\varepsilon}^{\tilde{P},\tilde{b}}f_{r_0}(X), \quad \tilde{b}(u) = b(r_0u), \quad (470)$$

and (380) reduce the proof to $\|P_m\|_{\text{coeff}} = 1$ without changing the normalized radial bound.

We now make the far-field parameter choice explicit. Put

$$d_* = \max(d, 1), \quad \kappa_0 = \frac{1}{100d_*}, \quad c_{H,\rho} = c_{\kappa_0,\rho}, \quad (471)$$

where $c_{\kappa_0,\rho} > 0$ is given by (410), and choose

$$0 < \delta_a < \frac{1}{8} \min\{c_{P,\rho}, c_{E,\rho}, c_{H,\rho}\}. \quad (472)$$

Perform the phase-adapted Calderón–Zygmund decomposition and write

$$b_{\text{CZ}} = \sum_Q b_Q, \quad \Sigma = \sum_Q \|b_Q\|_1 \leq C \|f\|_1.$$

The good part is controlled by (446): since the Calderón–Zygmund decomposition gives $\|g\|_2^2 \leq C\alpha \|f\|_1$, the finite-path set

$$E_{\text{good}}^P = \{x : \mathbf{J}_I(T_{\Omega,\varepsilon}^{P,b}g) > \alpha\}$$

has measure at most $C_{n,d,\rho} \alpha^{-1} \|f\|_1$. Use the mean-zero angular split (40) with the value δ_a in (472). The angular tails have the same summable $V^1 L^1$ bound as in the unweighted proof.

For the bounded angular part, Proposition 13.24 gives the short square function with squared L^2 decay $2^{-[c_{P,\rho}-2\delta_a]s}$ and the packet error with L^1 decay $2^{-c_{E,\rho}s}$. Split the dyadic shells by

$$1 \leq R^m \leq 2^{K_0 s} \quad \text{and} \quad R^m > 2^{K_0 s}.$$

There are $O(1+s)$ low shells. Cauchy–Schwarz and the endpoint differences supplied by Proposition 13.24 give a low-shell squared L^2 bound

$$C(1+s)2^{-[c_{P,\rho}-2\delta_a]s}\alpha\Sigma,$$

plus a summable $V^1 L^1$ error.

On the high shells use the same output localization, fixed adjacent-grid classes, exceptional sets, and rank suffixes as in Proposition 10.1. The normalized sharp-annulus profiles are products of $b(R\cdot)$ with fixed step cutoffs and hence have uniformly bounded $L^\infty + V^\rho$ norm. Corollary 13.12, followed by Lemma 5.2, gives

$$Cs^2 2^{-[c_{H,\rho}-2\delta_a]s}\alpha\Sigma$$

for the high dyadic tails. By (472), every displayed exponent is strictly positive and all polynomial factors in s are summable.

Here is the precise comparison with the majorants in Proposition 10.1. Write $G_{j,s}^b + E_{j,s}^b$ for the weighted one-shell decomposition supplied by Proposition 13.24, and $F_{j,s}^{G,b} + F_{j,s}^{E,b}$ for its endpoint difference. Define

$$\begin{aligned} \mathcal{S}_s^b &= \left(\sum_j V^2 (G_{j,s}^b)^2 \right)^{1/2}, & \mathcal{L}_s^b &= 2 \sum_{j \text{ low}} |F_{j,s}^{G,b}|, \\ \mathcal{H}_s^b &= \mathbf{1}_{X_s^c} V^2 (\text{weighted high dyadic tails}), & \mathcal{E}_s^{*,b} &= \sum_j V^1 (E_{j,s}^b). \end{aligned}$$

Let $\mathcal{W}$ be the total V^1 majorant of the weighted angular-tail paths plus $C \sum_s \mathcal{E}_s^{*,b}$, including the low-shell endpoint errors, and set

$$\mathcal{G} = \sum_s (\mathcal{S}_s^b + \mathcal{L}_s^b + \mathcal{H}_s^b), \quad X = \bigcup_s X_s.$$

These are exactly the four objects in (338)–(339), with the new direct-packet and collective-high estimates substituted; multiplication by $|b| \leq 1$ leaves the angular-tail comparison unchanged. Hence the resulting nonnegative majorants satisfy

$$\|\mathcal{G}\|_2^2 \leq C_{n,d,\rho} \alpha \Sigma, \quad \|\mathcal{W}\|_1 \leq C_{n,d,\rho} \Sigma, \quad |X| \leq C \alpha^{-1} \|f\|_1,$$

and, off the fixed complement,

$$\mathbf{J}(\mathcal{F}^{P,b} b_{\text{CZ}}) \leq C_{n,d,\rho} (\mathcal{G} + \mathcal{W}).$$

The signs are fixed packetwise before the output point is evaluated, and none of these objects depends on the jump threshold.

For the local path apply Proposition 11.1. Its error kernel is simply multiplied by $|b(|x-y|)| \leq 1$, while the residual phase has nonpure degree at most $m-1$ and carries the same radial function. On each unit output cube the induction hypothesis applies to the translated, purely modulated input; the fixed input enlargements have bounded overlap. The piecewise error satisfies

$$\|V^1(E)\|_1 \leq C_{n,d} \|\Omega\|_1 \|f\|_1.$$

Combining the far path, local residual, and error with the pointwise jump quasi-triangle inequality now gives a literal threshold argument. Namely, choose a fixed $A_{n,d,\rho}$ larger than all quasi-triangle constants. Outside E_{good}^P , the enlarged-CZ set, and X , a strict $A_{n,d,\rho} \alpha$ witness for the full path forces either $\mathcal{G} > c_{n,d,\rho} \alpha$, $\mathcal{W} > c_{n,d,\rho} \alpha$, a strict $c_{n,d,\rho} \alpha$ witness for the lower-degree residual, or $V^1(E) > c_{n,d,\rho} \alpha$.

Chebyshev, the two displayed majorant bounds, the induction hypothesis, and the $V^1(E)$ estimate therefore give

$$|\{\mathbf{J}_I(T_{\Omega,\varepsilon}^{P,b}f : \varepsilon \in I) > A_{n,d,\rho}\alpha\}| \leq C_{n,d,\rho}\alpha^{-1}\|f\|_1.$$

Putting $\alpha = \tau/A_{n,d,\rho}$ proves the required distribution estimate for every strict threshold $\tau > 0$, uniformly at the finite-cutoff stage and for the normalized top block.

All estimates above were first proved with finite cube, shell, and rational-parameter cutoffs. To remove them, fix $\tau > 0$ and a finite rational parameter set I . Finite-annulus coordinate convergence gives, after a subsequence, $F_N \rightarrow F$ pointwise on I . A strict witness for $\mathbf{J}_I(F) > \tau$ has positive excess and survives for all sufficiently large N , so

$$\mathbf{1}_{\{\mathbf{J}_I(F) > \tau\}} \leq \liminf_N \mathbf{1}_{\{\mathbf{J}_I(F_N) > \tau\}}.$$

Fatou preserves the finite-parameter distribution estimate without changing its threshold. Increasing finite sets exhaust $\mathbb{Q}_+$, and local absolute continuity recovers all positive parameters, exactly as in the final part of Proposition 10.1. Finally the input L^1 norm and weak output quasi-norm acquire the same factor under the dilation r_0 , so the constant is independent of C_m . This closes the finite degree induction under the normalization $\mathfrak{L}(\Omega) = 1$; angular homogeneity restores the factor $\mathfrak{L}(\Omega)$. $\square$

Remark 13.29 (Quantitative closure of the finite-radial proof). For clarity, the order of every small-parameter choice in this section is as follows. Once n, d, ρ are fixed, the polynomial sublevel constants are structural. Equation (394) first fixes $c_{\text{vdc},\rho} > 0$, then (397) fixes $c_{\text{rat},\rho} > 0$. These determine $c_{\text{eq},\rho}$ and $c_{\text{uneq},\rho}$ in (405) and (406), hence $c_{\text{flat},\rho}$ and $c_{\kappa,\rho}$ in (409) and (410). Independently, (413) fixes $\delta_{T,\rho} > 0$. Only after these exponents are known are $B_{*,\rho}$ and $\kappa_{*,\rho}$ selected in (450)–(451); the three inequalities (452)–(454) then fix $c_{P,\rho}, c_{E,\rho} > 0$. Finally the angular splitting parameters are chosen in (465) and (472). Thus no decay exponent is defined using a parameter that is chosen later, and every geometric series in the strong and weak arguments has a strictly positive exponent for each fixed finite ρ . The resulting constants may deteriorate as $\rho \rightarrow \infty$, as they must in view of Proposition 14.3.

13.11 Completion of the proof

We give the completion because it also explains why point jumps of b do not create jumps in the truncation trajectory. Assume first (379). For bounded compactly supported f set

$$\mathcal{A}_x(r) := \int_{\mathbb{S}^{n-1}} e^{iP(x, x-r\theta)} \Omega(\theta) f(x-r\theta) d\sigma(\theta).$$

For every $0 < a < c < \infty$,

$$\|b(|\cdot|)K_\Omega(\cdot)\mathbf{1}_{\{a < |\cdot| \leq c\}}\|_1 \leq \|b\|_\infty \|\Omega\|_1 \log(c/a). \quad (473)$$

Fubini, first on the countable annuli $\{N^{-1} < |z| < N\}$, gives one full-measure set on which

$$\int_a^c |b(r)\mathcal{A}_x(r)| \frac{dr}{r} < \infty, \quad T_{\Omega,a}^{P,b}f(x) - T_{\Omega,c}^{P,b}f(x) = \int_a^c b(r)\mathcal{A}_x(r) \frac{dr}{r} \quad (474)$$

for every $0 < a < c < \infty$. Thus the core path is locally absolutely continuous. In particular, a jump of the regulated function b at one radius changes neither the Lebesgue integral nor the truncation path: no derivative of b occurs in (474). For core data the path is also zero for all sufficiently large truncations at each fixed x .

We next construct all coordinates for general data. In the strong case choose bounded compactly supported $f_m \rightarrow f$ in L^p so that

$$\sum_m \|f_{m+1} - f_m\|_p < \infty.$$

For $g_m = f_{m+1} - f_m$, the zero anchor for core data and (20), followed by (446), give

$$\left\| \sup_{q \in \mathbb{Q}_+} |T_{\Omega,q}^{P,b} g_m| \right\|_p \leq \| \mathbf{J}(T_{\Omega,\varepsilon}^{P,b} g_m) \|_p \leq C_{n,d,p,\rho} \mathfrak{L}(\Omega) \|g_m\|_p. \quad (475)$$

Hence the sum of the rational suprema is finite almost everywhere, and $T_{\Omega,q}^{P,b} f_m$ is uniformly Cauchy in $q \in \mathbb{Q}_+$ outside one null set.

In the weak case choose instead bounded compactly supported $f_m \rightarrow f$ in L^1 with $\|g_m\|_1 \leq 2^{-3m}$. The core weak estimate and the same anchor imply

$$\sum_m \left| \left\{ x : \sup_{q \in \mathbb{Q}_+} |T_{\Omega,q}^{P,b} g_m(x)| > 2^{-m} \right\} \right| \leq C_{n,d,\rho} \mathfrak{L}(\Omega) \sum_m 2^m \|g_m\|_1 < \infty. \quad (476)$$

Borel–Cantelli again gives almost-everywhere uniform Cauchy convergence on $\mathbb{Q}_+$. In either case we define every rational coordinate by this uniform limit. Moreover, if

$$\mathcal{R}_m(q, x) := \sum_{k \geq m} T_{\Omega,q}^{P,b} g_k(x),$$

then $\sup_{q \in \mathbb{Q}_+} |\mathcal{R}_m(q, x)| \rightarrow 0$ almost everywhere. For fixed m the core path of f_m vanishes for all sufficiently large q ; consequently

$$\lim_{R \rightarrow \infty} \sup_{\substack{q \in \mathbb{Q}_+ \\ q \geq R}} |T_{\Omega,q}^{P,b} f(x)| = 0 \quad (477)$$

almost everywhere in both completions.

It remains to identify these coordinates with the desired finite-shell increments. By (473), for every rational $0 < a < c < \infty$ the core finite-annulus expressions converge to the corresponding expression for f in L^p in the strong case and in L^1 in the weak case. A diagonal subsequence over the rational pairs therefore gives, without changing the preceding uniform rational limit, the identity

$$T_{\Omega,a}^{P,b} f(x) - T_{\Omega,c}^{P,b} f(x) = \int_a^c b(r) \mathcal{A}_x(r) \frac{dr}{r}$$

on one full-measure set. Fubini on the same countable annuli shows that the right-hand density is locally integrable for general f as well (Young’s inequality applies in the strong case). Define the real coordinates from the base point 1 by these finite-annulus integrals. The resulting path is locally absolutely continuous, agrees with the rational coordinates, and (477), by continuity and density of $\mathbb{Q}_+$, yields its zero-at-infinity anchor for all real radii.

For every finite rational parameter set, coordinatewise convergence and (37) transfer the strong estimate by Fatou. For the weak estimate, fix its threshold $\tau > 0$ before taking the limit: if the limiting finite path has $\mathbf{J} > \tau$, one of its finite strict witnesses has positive excess and therefore persists along the coordinatewise convergent subsequence. The corresponding level-set indicator is bounded by the liminf of the approximating indicators, so Fatou transfers the distribution estimate at the same τ . Increasing through finite sets exhausting $\mathbb{Q}_+$ and then using local continuity and Lemma 2.6 recovers the full positive-parameter functional. This also settles measurability: all coordinates are measurable, and the completed jump functional is a countable supremum over finite rational witnesses. Two locally continuous paths with the same finite-annulus increments differ by a constant in the parameter, and the zero anchor makes the completion unique.

We have proved (12) and (13) under (379). If $\|b\|_{\mathcal{V}\rho} = 0$, then $b = 0$ almost everywhere and the theorem is immediate. Otherwise apply the normalized result to $b/\|b\|_{\mathcal{V}\rho}$; linearity of every coordinate and homogeneity of $\mathbf{J}$ restore exactly one factor $\|b\|_{\mathcal{V}\rho}$. Finally the pointwise weak-2 jump-to- r -variation inequality from Lemma 2.2 gives every $r > 2$ consequence, while the zero anchor gives the maximal assertion. This completes the proof of Theorem 1.3.

Remark 13.30 (Dependence on the radial exponent). The proof uses the positive exponents $1/\rho$ in both the translation modulus and the first-derivative oscillatory estimate. These exponents tend to zero as $\rho \rightarrow \infty$, and no constant uniform in ρ is asserted. Proposition 14.3 shows that a merely bounded radial endpoint is false even for the one-dimensional Hilbert kernel.

13.12 A shellwise quadratic radial-variation refinement for the Hilbert transform

The abstract transfer in Proposition 13.1 stops at $\rho = 2$. In the one-dimensional Hilbert model, Fourier orthogonality allows a sharper shellwise statement.

For bounded $b : (0, \infty) \rightarrow \mathbb{C}$, put

$$H_\varepsilon^b f(x) = \int_{|y| > \varepsilon} b(|y|) f(x-y) \frac{dy}{y},$$

where on Schwartz data the integral is understood as the symmetric improper integral at infinity, and on L^2 the compatible strong- L^2 representative is constructed in the proof below, and define

$$q_j = V^2(b; (2^j, 2^{j+1})), \quad R_2(b) = \left(\sum_{j \in \mathbb{Z}} q_j^2 \right)^{1/2}. \quad (478)$$

Concatenating disjoint shell partitions gives $R_2(b) \leq V^2(b)$ whenever the latter is finite.

Proposition 13.31 (Shellwise radial V^2 Hilbert estimate). *For every bounded b with $R_2(b) < \infty$ and every $f \in L^2(\mathbb{R})$,*

$$\|\mathbf{J}(H_\varepsilon^b f : \varepsilon > 0)\|_2 \leq C(\|b\|_\infty + R_2(b)) \|f\|_2. \quad (479)$$

In particular, the right-hand side may be replaced by $C(\|b\|_\infty + V^2(b)) \|f\|_2$.

Proof. Put $B = \|b\|_\infty$. On $I_j = (2^j, 2^{j+1})$, finite 2-variation makes b regulated. Introduce the closed-shell representative

$$\bar{b}_j(2^j) = b(2^j+), \quad \bar{b}_j(r) = b(r) \quad (r \in I_j), \quad \bar{b}_j(2^{j+1}) = b(2^{j+1}-).$$

Approximating the two endpoints from inside I_j in every finite partition gives

$$V^2(\bar{b}_j; [2^j, 2^{j+1}]) \leq q_j. \quad (480)$$

Changing the values at the two endpoints does not change any of the Lebesgue kernels below. Set $c_j = \bar{b}_j(2^j)$ and $e_j = \bar{b}_j - c_j$ on the closed shell. Then

$$|c_j| \leq B, \quad e_j(2^j) = 0, \quad \|e_j\|_{L^\infty([2^j, 2^{j+1}])} \leq q_j, \quad V^2(e_j; [2^j, 2^{j+1}]) \leq q_j. \quad (481)$$

For $2^j \leq \varepsilon \leq 2^{j+1}$, the within-shell part of the path is

$$A_{j,\varepsilon}^b f(x) = \int_{\varepsilon < |y| \leq 2^{j+1}} b(|y|) f(x-y) \frac{dy}{y}.$$

For the constant component, write

$$\mathcal{H}_\varepsilon f(x) = \int_{|y| > \varepsilon} f(x-y) \frac{dy}{y}.$$

Then $A_{j,\varepsilon}^1 f = \mathcal{H}_\varepsilon f - \mathcal{H}_{2^{j+1}} f$ for $\varepsilon \in I_j$, and hence the classical Hilbert short-2-variation square-function estimate [7, Theorem 1.4] gives

$$\left\| \left(\sum_j V^2(A_{j,\varepsilon}^1 f : \varepsilon \in I_j)^2 \right)^{1/2} \right\|_2 \leq C \|f\|_2.$$

The finite-annulus integral is continuous in ε for almost every x on locally integrable data, so the same estimate holds after adjoining both dyadic endpoints. Together with $|c_j| \leq B$, this controls the constant components. For the error component $E_{j,\varepsilon}$, every finite partition of I_j gives pointwise

$$V^1(E_{j,\varepsilon}f : \varepsilon \in I_j) \leq q_j \int_{2^j < |y| \leq 2^{j+1}} |f(x-y)| \frac{dy}{|y|}.$$

Put

$$K_j(y) = |y|^{-1} \mathbf{1}_{\{2^j < |y| \leq 2^{j+1}\}}, \quad \|K_j\|_1 = 2 \log 2.$$

Since $V^2 \leq V^1$, Tonelli in $L^2(\ell^2)$ and Young's inequality give

$$\begin{aligned} \left\| \left(\sum_j V^2(E_{j,\varepsilon}f : \varepsilon \in I_j)^2 \right)^{1/2} \right\|_2^2 &\leq \sum_j q_j^2 \|K_j * |f|\|_2^2 \\ &\leq (2 \log 2)^2 R_2(b)^2 \|f\|_2^2. \end{aligned}$$

The endpoint continuity noted above and the triangle inequality in $L^2(\ell^2)$ therefore give

$$\left\| \left(\sum_j V^2(A_{j,\varepsilon}^b f : 2^j \leq \varepsilon \leq 2^{j+1})^2 \right)^{1/2} \right\|_2 \leq C(B + R_2(b)) \|f\|_2. \quad (482)$$

For the dyadic path define the odd shell kernels

$$v_j^b(y) = \frac{b(|y|)}{y} \mathbf{1}_{\{2^j < |y| \leq 2^{j+1}\}}. \quad (483)$$

We claim that, with $t = 2^j |\xi|$,

$$|\widehat{v_j^b}(\xi)| \leq C(B + q_j) \min\{t, t^{-1/2}\}. \quad (484)$$

Oddness gives the bound CBt for $t \leq 1$. For $t \geq 1$, scale the radial integral to $[1, 2]$ and write $e(u) = e_j(2^j u)$. By (480), $e(1) = 0$ and $V^2(e; [1, 2]) \leq q_j$. The constant part in (481) is $O(Bt^{-1})$ by integration by parts. It remains to prove

$$\left| \int_1^2 e(u) \frac{\sin(tu)}{u} du \right| \leq CV^2(e; [1, 2]) t^{-1/2} \quad (485)$$

when $e(1) = 0$. Partition $[1, 2]$ at the zeros of $\sin(tu)$. There are $M \leq C(1+t)$ consecutive intervals J_m . Choose $u_m \in J_m$ and split $e = e(u_m) + (e - e(u_m))$ on each interval. The error is at most

$$Ct^{-1} \sum_m V^2(e; J_m) \leq Ct^{-1} M^{1/2} \left(\sum_m V^2(e; J_m)^2 \right)^{1/2} \leq Ct^{-1/2} V^2(e).$$

For the last inequality, choose near-extremizing partitions on the consecutive J_m and concatenate them; this gives

$$\sum_m V^2(e; J_m)^2 \leq V^2(e; [1, 2])^2.$$

For the frozen part put $\omega_m = \int_{J_m} \sin(tu) u^{-1} du$. Integration by parts shows that every consecutive partial sum of the ω_m is $O(t^{-1})$. Discrete Abel summation, followed by Cauchy-Schwarz, gives

$$\left| \sum_m e(u_m) \omega_m \right| \leq Ct^{-1} \left(|e(u_1)| + \sum_m |e(u_{m+1}) - e(u_m)| \right) \leq Ct^{-1/2} V^2(e).$$

Indeed, take the u_m in increasing order. Since $e(1) = 0$, the partition $1 < u_1 < \dots < u_M$ gives

$$|e(u_1)|^2 + \sum_{m=1}^{M-1} |e(u_{m+1}) - e(u_m)|^2 \leq V^2(e; [1, 2])^2;$$

Cauchy–Schwarz over at most M terms proves the last bound and also accounts for the endpoint term in Abel summation. This proves (485) and hence (484).

For $\xi \neq 0$ define

$$m_b(\xi) = \sum_{j \in \mathbb{Z}} \widehat{v_j^b}(\xi), \quad m_b(0) = 0.$$

This series is absolutely convergent at each nonzero frequency. More precisely, for every finite $J \subset \mathbb{Z}$, the geometric summation of the B part of (484) and Cauchy–Schwarz for the q_j part give

$$\begin{aligned} \sum_{j \in J} |\widehat{v_j^b}(\xi)| &\leq CB \sum_j \min\{2^j |\xi|, (2^j |\xi|)^{-1/2}\} \\ &\quad + CR_2(b) \left(\sum_j \min\{2^j |\xi|, (2^j |\xi|)^{-1/2}\}^2 \right)^{1/2} \\ &\leq C(B + R_2(b)), \end{aligned}$$

uniformly in ξ and J . If $J_N \uparrow \mathbb{Z}$, the corresponding multipliers converge pointwise to m_b and remain uniformly bounded by the displayed estimate. Plancherel and dominated convergence therefore show, for every $f \in L^2$, that

$$\sum_{j \in J_N} v_j^b * f \longrightarrow T^b f := \mathcal{F}^{-1}(m_b \widehat{f}) \quad \text{strongly in } L^2.$$

In particular,

$$\|T^b f\|_2 \leq C(B + R_2(b)) \|f\|_2. \quad (486)$$

Choose a smooth radial low-pass multiplier $\widehat{\phi}$ equal to one on $[-2, 2]$ and supported in $[-4, 4]$, and write $\widehat{\phi}_k(\xi) = \widehat{\phi}(2^k \xi)$. The exact identity

$$H_{2^k}^b f = \phi_k * T^b f + \sum_{s \geq 0} (\delta_0 - \phi_k) * v_{k+s}^b * f - \sum_{s < 0} \phi_k * v_{k+s}^b * f \quad (487)$$

holds first for finite shell sums. The T^b term passes to the limit by the strong convergence just proved, while the two relative-frequency series pass to the limit by the summable square estimates below. Thus the identity holds in L^2 and, in particular, defines the dyadic truncations $H_{2^k}^b f$ compatibly with the strong shell limit. For $2^k < \varepsilon < 2^{k+1}$ we use the compatible representative

$$H_\varepsilon^b f = H_{2^{k+1}}^b f + A_{k,\varepsilon}^b f.$$

On Schwartz data this agrees with the symmetric improper integral. The estimates below are first proved with finite shell, relative-frequency, and parameter cutoffs; the strong L^2 limits just constructed, followed by the finite-parameter lower-semicontinuity argument below, remove those cutoffs. Proposition 2.1 of Chen–Ding–Hong–Liu [9] and (486) control the critical jump of the first term. Here Proposition 2.1 uses the paired jump count, so this estimate already follows the convention of **J**. The chain functional below is introduced only to sum the two relative-frequency series in the MSZK normed quotient, and (27) returns that sum to the paired convention after the cutoffs are removed.

For a scalar sequence (h_k) ,

$$\mathbf{J}(h_k : k \in \mathbb{Z}) \leq V^2(h_k : k \in \mathbb{Z}) \leq 2 \left(\sum_k |h_k|^2 \right)^{1/2}. \quad (488)$$

Using (484), the support and first-order vanishing of $1 - \widehat{\phi}$, and dyadic summation, one obtains uniformly in ξ

$$\sum_k |1 - \widehat{\phi}(2^k \xi)|^2 |\widehat{v_{k+s}^b}(\xi)|^2 \leq C 2^{-s} (B^2 + R_2(b)^2), \quad s \geq 0, \quad (489)$$

$$\sum_k |\widehat{\phi}(2^k \xi)|^2 |\widehat{v_{k+s}^b}(\xi)|^2 \leq C 2^{2s} (B^2 + R_2(b)^2), \quad s < 0. \quad (490)$$

For completeness, the B^2 terms follow by summing the dyadic envelope in (484); for the q_{k+s}^2 terms one uses the corresponding pointwise envelope bound, respectively $C2^{-s}$ and $C2^{2s}$, and then sums q_{k+s}^2 . Plancherel and (488) now give geometric bounds $C2^{-s/2}(B + R_2(b))\|f\|_2$ for $s \geq 0$ and $C2^s(B + R_2(b))\|f\|_2$ for $s < 0$.

To sum the relative-frequency pieces in (487), fix $K_R = \{-R, \dots, R\}$ and first restrict s to a finite set. Use the chain functional and the normed finite-path quotient from Lemma 2.3; its triangle inequality has a constant independent of K_R , and (27) converts between the chain and paired functionals. Write

$$U_k^{(s)} = (\delta_0 - \phi_k) * v_{k+s}^b * f \quad (s \geq 0), \quad L_k^{(s)} = \phi_k * v_{k+s}^b * f \quad (s < 0).$$

The preceding estimates and the two geometric series give, uniformly in both cutoffs,

$$\left\| \mathbf{J}_{K_R}^{\text{ch}} \left(\left(\sum_{s \geq 0} U_k^{(s)} \right)_{k \in K_R} \right) \right\|_2 + \left\| \mathbf{J}_{K_R}^{\text{ch}} \left(\left(\sum_{s < 0} L_k^{(s)} \right)_{k \in K_R} \right) \right\|_2 \leq C(B + R_2(b))\|f\|_2,$$

where each sum is finite at this stage. Moreover, the tails tend to zero in $L_x^2(\ell_k^2)$: by Minkowski and the geometric estimates their norms are bounded by the corresponding tails of $\sum_{s \geq 0} 2^{-s/2}$ and $\sum_{s < 0} 2^s$. Thus the relative-frequency sums converge in $L_x^2(\ell_k^2)$ and, after passing to a subsequence, at every k almost everywhere. Lemma 2.6, (38), and Fatou remove the s cutoff on K_R . Now let $R \rightarrow \infty$. The finite-parameter chain functionals increase to the full dyadic functional, so monotone convergence, followed by (27), proves

$$\|\mathbf{J}(H_{2^k}^b f : k \in \mathbb{Z})\|_2 \leq C(B + R_2(b))\|f\|_2. \quad (491)$$

Finally combine (482) and (491) through the pointwise long–short inequality (23). There is a common full-measure set on which all dyadic coordinates are defined and every finite-shell path is continuous in ε ; the compatible representative above is used on this set. This justifies the pointwise long–short step for the full continuum path. $\square$

Remark 13.32. The condition $R_2(b) < \infty$ is strictly weaker than global finite $V^2(b)$. A bounded function may be constant on each half-open dyadic shell, so that $R_2(b) = 0$, while its jumps from shell to shell have infinite global 2-variation. Thus even this basic model identifies a radial class strictly larger than the global V^2 class of Theorem 1.3.

14 Obstructions and sharpness

This section records four complementary negative results. They distinguish the pointwise critical-jump functional from the weaker mixed quasi-seminorm, show why the soft radial transfer cannot reach exponent two, prove that the finite-radial theorem has no $\rho = \infty$ endpoint, and show that the output critical jump cannot be strengthened to true 2-variation.

14.1 The mixed-norm exchange is false

Proposition 14.1 (Finite genuine-jump mixed-norm counterexamples). *For every $1 \leq p < \infty$ and every integer $M \geq 2$ there exist a finite measure space X_M and a two-point scalar path F such that*

$$\sup_{\lambda > 0} \|\lambda N_\lambda(F)^{1/2}\|_{L^p(X_M)} \leq C_p, \quad (492)$$

whereas

$$\left\| \sup_{\lambda > 0} \lambda N_\lambda(F)^{1/2} \right\|_{L^p(X_M)} \simeq (\log M)^{1/p}. \quad (493)$$

In addition, for every $M \geq 2$ there is a scalar path G on one common finite ordered parameter set such that

$$\sup_{\lambda > 0} \|\lambda N_\lambda(G)^{1/2}\|_{L^{1,\infty}(X_M)} \leq \sqrt{2}, \quad (494)$$

but

$$\left\| \sup_{\lambda > 0} \lambda N_\lambda(G)^{1/2} \right\|_{L^{1,\infty}(X_M)} = \sqrt{2}M. \quad (495)$$

Thus neither the strong nor the weak spatial norm can in general be interchanged with the supremum in the jump amplitude.

Proof. For the strong example, take $X_M = \{1, \dots, M\}$ with counting measure and set $a_k = k^{-1/p}$. On the ordered parameter set $\{0, 1\}$ define $F_0(k) = 0$ and $F_1(k) = a_k$. Then $N_\lambda(F(k)) = 1$ for $0 < \lambda < a_k$ and vanishes for $\lambda \geq a_k$. Thus the pointwise supremum equals a_k , and its p th power sums to $\sum_{k \leq M} k^{-1} \simeq \log M$. For fixed λ ,

$$\|\lambda N_\lambda(F)^{1/2}\|_p = \lambda \#\{k : a_k > \lambda\}^{1/p} \leq C_p,$$

which proves (492)–(493).

For the weak endpoint use the same counting space and put

$$I_M = \{0, 1, \dots, 2 \cdot 4^M\}, \quad a_k = 2^{-k}, \quad L_k = 4^k.$$

For $j \in I_M$ define

$$G_j(k) = a_k \mathbf{1}_{\{j < 2L_k\}} \mathbf{1}_{\{j \text{ odd}\}}.$$

Thus, at the point k , the path alternates $0, a_k, 0, a_k, \dots, 0$ through the first $2L_k + 1$ coordinates and is zero thereafter. For $0 < \lambda < a_k$, its $2L_k$ consecutive changes are admissible paired strict jumps (successive pairs may share an endpoint). Conversely, every strict jump interval must contain one of these changes, and ordered jump intervals cannot use the same change twice. Hence, exactly,

$$N_\lambda(G(k)) = \begin{cases} 2L_k, & 0 < \lambda < a_k, \\ 0, & \lambda \geq a_k. \end{cases}$$

It follows that

$$\mathbf{J}(G)(k) = a_k \sqrt{2L_k} = \sqrt{2}$$

for every $1 \leq k \leq M$, proving (495).

Fix $\lambda > 0$. If the path is nonzero at no spatial point there is nothing to prove. Otherwise let K be the largest integer in $\{1, \dots, M\}$ for which $\lambda < 2^{-K}$. The nonzero values of $\lambda N_\lambda(G)^{1/2}$ have decreasing rearrangement

$$A, \frac{A}{2}, \dots, \frac{A}{2^{K-1}}, \quad A = \sqrt{2} \lambda 2^K < \sqrt{2}.$$

Consequently

$$\|\lambda N_\lambda(G)^{1/2}\|_{\ell^{1,\infty}} = \sup_{1 \leq m \leq K} m \frac{A}{2^{m-1}} \leq A < \sqrt{2},$$

uniformly in λ and M . This proves (494). $\square$

The first construction shows that an outer supremum in the jump amplitude cannot be upgraded to the pointwise functional by an abstract exchange of strong spatial norms. The second construction proves the same strict failure in $L^{1,\infty}$ with a genuine jump trajectory on a common finite parameter set. In particular, the pointwise-inside weak endpoint in Theorems 1.1 and 1.3 is genuinely stronger than the conventional mixed weak quasi-seminorm.

14.2 No black-box endpoint radial-variation path transfer

Proposition 14.2 (Abstract endpoint transfer obstruction). *For every $N \geq 2$ there exist a finite scalar path F , a scalar coefficient path β , and the weighted-tail path*

$$G_i = \sum_{k=i}^{2N-1} \beta_k (F_k - F_{k+1}), \quad G_{2N} = 0, \quad (496)$$

such that

$$\mathbf{J}(F) \leq C, \quad \|\beta\|_\infty + V^2(\beta) \leq C, \quad (497)$$

but

$$\mathbf{J}(G) \geq c\sqrt{\log N}. \quad (498)$$

Hence no universal scalar-path estimate of the form

$$\mathbf{J}(G) \leq C(\|\beta\|_\infty + V^2(\beta))\mathbf{J}(F)$$

can hold for the operation (496). Moreover, with absolute changes in the constants, this discrete example embeds into the same weighted-tail structural class that appears in Proposition 13.1, now at the formal endpoint $\rho = 2$: there are a locally absolutely continuous path $\tilde{F} : (0, \infty) \rightarrow \mathbb{C}$ with $\tilde{F}_s \rightarrow 0$ as $s \rightarrow \infty$ and a bounded right-continuous $\tilde{\beta}$ of finite 2-variation such that

$$\tilde{G}_s = - \int_s^\infty \tilde{\beta}(u) d\tilde{F}_u$$

has the analogues of (497) and (498). Thus local absolute continuity of the input path does not restore an abstract endpoint transfer at $\rho = 2$.

Proof. Let

$$H_N = \sum_{j=1}^N \frac{1}{j}, \quad a_j = j^{-1/2}, \quad \sigma_j = (-1)^j,$$

and define

$$F_{2j} = 0 \quad (0 \leq j \leq N), \quad F_{2j-1} = \sigma_j a_j \quad (1 \leq j \leq N). \quad (499)$$

If both endpoints of an increment have modulus at most $\lambda/2$, the increment cannot be a strict λ -jump. Each nonzero endpoint can occur in at most two ordered paired jumps, and therefore

$$N_\lambda(F) \leq 2\#\{j : a_j > \lambda/2\} \leq C\lambda^{-2}.$$

Thus $\mathbf{J}(F) \leq C$.

Put $\delta_j = (jH_N)^{-1/2}$, take $\beta_0 = 0$, and prescribe

$$\beta_{2j-1} - \beta_{2j-2} = \sigma_j \delta_j, \quad \beta_{2j} - \beta_{2j-1} = 0. \quad (500)$$

The nonzero increments alternate in sign and decrease in modulus. Every consecutive block sum is therefore bounded by its first nonzero term. Every increment selected by a variation partition is such a consecutive block, and these blocks are disjoint and occur in order. Their first nonzero indices are consequently distinct, so

$$V^2(\beta)^2 \leq \sum_{j=1}^N \delta_j^2 = 1, \quad \|\beta\|_\infty \leq 1.$$

Discrete summation by parts in (496), using $\beta_0 = F_{2N} = 0$, gives

$$G_0 = \sum_{k=1}^{2N-1} (\beta_k - \beta_{k-1}) F_k = \sum_{j=1}^N \delta_j a_j = \sqrt{H_N}.$$

Since $G_{2N} = 0$, one ordered jump yields $\mathbf{J}(G) \geq \sqrt{H_N} \simeq \sqrt{\log N}$.

It remains to realize the same obstruction inside the locally absolutely continuous path class. Define $\tilde{F}$ by $\tilde{F}(s) = 0$ for $0 < s \leq 1$, by affine interpolation from F_k to F_{k+1} on every interval $[k+1, k+2]$, $0 \leq k < 2N$, and by $\tilde{F}(s) = 0$ for $s \geq 2N+1$. Thus $\tilde{F}$ is locally absolutely continuous and has the zero anchor. Define the right-continuous step function

$$\tilde{\beta}(s) = \beta_k \quad (k+1 \leq s < k+2, \ 0 \leq k < 2N),$$

take $\tilde{\beta}(s) = \beta_0 = 0$ for $0 < s < 1$, and take $\tilde{\beta}(s) = \beta_{2N}$ for $s \geq 2N+1$. Since $\beta_{2N} = \beta_{2N-1}$, this extension adds no terminal jump. Hence

$$\|\tilde{\beta}\|_\infty + V^2(\tilde{\beta}) \leq \|\beta\|_\infty + V^2(\beta) \leq C.$$

For every $0 \leq i \leq 2N$, absolute continuity on the finitely many affine pieces gives the exact identity

$$-\int_{i+1}^\infty \tilde{\beta}(s) d\tilde{F}_s = \sum_{k=i}^{2N-1} \beta_k (F_k - F_{k+1}) = G_i. \quad (501)$$

For completeness, the interpolation does not enlarge the input critical jump by a factor depending on N . Given $\lambda > 0$, let $\tilde{F}^{(\lambda)}$ be obtained by flattening to zero every triangular spike whose height satisfies $a_j \leq \lambda/4$. Then

$$\|\tilde{F} - \tilde{F}^{(\lambda)}\|_\infty \leq \lambda/4, \quad V^1(\tilde{F}^{(\lambda)}) \leq 2 \sum_{a_j > \lambda/4} a_j \leq C\lambda^{-1}.$$

Every strict λ -jump of $\tilde{F}$ is therefore a strict $\lambda/2$ -jump of $\tilde{F}^{(\lambda)}$. Disjoint jump intervals consume disjoint portions of total variation, and consequently

$$N_\lambda(\tilde{F}) \leq N_{\lambda/2}(\tilde{F}^{(\lambda)}) \leq \frac{2}{\lambda} V^1(\tilde{F}^{(\lambda)}) \leq C\lambda^{-2}.$$

Thus $\mathbf{J}(\tilde{F}) \leq C$. On the integer subsequence $s = i+1$, (501) retains $\tilde{G}_1 = G_0 = \sqrt{H_N}$ and $\tilde{G}_{2N+1} = G_{2N} = 0$, so $\mathbf{J}(G) \geq c\sqrt{\log N}$. This proves the asserted locally absolutely continuous embedding. $\square$

This obstruction is the sequence-space failure $\ell^{2,\infty} \cdot \ell^2 \not\subset \ell^1$. Proposition 13.31 and the general argument of Section 13 escape it by using Fourier and packet orthogonality absent from an arbitrary scalar path.

14.3 Failure at the bounded-radial endpoint

For the limiting seminorm put $V^\infty(b) = \sup_{s < t} |b(t) - b(s)|$.

Proposition 14.3 (No uniform merely bounded radial theory). *There is no uniform L^2 bound for the one-dimensional radial Hilbert operator in terms of $\|b\|_\infty + V^\infty(b)$ alone. Consequently, Theorem 1.3 cannot be extended to $\rho = \infty$.*

Proof. Let

$$\text{sgn}_+(\sin r) = \begin{cases} \text{sgn}(\sin r), & \sin r \neq 0, \\ \lim_{h \downarrow 0} \text{sgn}(\sin(r+h)), & \sin r = 0. \end{cases}$$

For $N > 2$ take the right-continuous representative

$$b_N(r) = \mathbf{1}_{[1,N)}(r) \text{sgn}_+(\sin r).$$

The half-open cutoff makes the value at N agree with the right limit, and the definition at each zero of $\sin r$ makes all interior values right continuous; the value at 1 also agrees with its right limit. Changing

these countably many endpoint values does not change any Lebesgue integral below. Then $\|b_N\|_\infty \leq 1$ and $V^\infty(b_N) \leq 2$. The convolution operator

$$T_N f(x) = \int_{\mathbb{R}} \frac{b_N(|y|)}{y} f(x-y) dy$$

has an integrable kernel. With $\widehat{f}(\xi) = \int f(x) e^{-ix\xi} dx$, its continuous multiplier at $\xi = 1$ is

$$m_N(1) = -2i \int_1^N \frac{|\sin r|}{r} dr.$$

Splitting the integral into intervals of length π and comparing r^{-1} with its endpoint values gives $\int_1^N |\sin r| r^{-1} dr = (2/\pi) \log N + O(1)$. Hence $\|T_N\|_{2 \rightarrow 2} \geq |m_N(1)| \geq c \log N$.

For the associated truncation path, $T_\varepsilon^{b_N} = T_N$ when $0 < \varepsilon < 1$ and $T_\varepsilon^{b_N} = 0$ when $\varepsilon > N$. Therefore $|T_N f| \leq \mathbf{J}(T_\varepsilon^{b_N} f : \varepsilon > 0)$. A jump estimate with a constant depending only on $\|b_N\|_\infty + V^\infty(b_N)$ would contradict the preceding multiplier lower bound. $\square$

14.4 Failure of quadratic output variation for sharp Hilbert truncations

Let

$$H_\varepsilon f(x) = \int_{|y| > \varepsilon} f(x-y) \frac{dy}{y}, \quad \widehat{P_s f}(\xi) = e^{-s|\xi|} \widehat{f}(\xi).$$

For $u, t > 0$ define

$$\Phi(u) = \int_u^\infty \frac{\sin v}{v} dv, \quad K(t) = \frac{1}{\pi} \operatorname{sech}^2(\log t) = \frac{4t^2}{\pi(1+t^2)^2}. \quad (502)$$

Lemma 14.4 (Mellin identity). *For every $s > 0$,*

$$e^{-s} = \int_0^\infty K(t) \Phi(s/t) \frac{dt}{t}, \quad \int_0^\infty K(t) \frac{dt}{t} = \frac{2}{\pi}. \quad (503)$$

Proof. Let $I(s)$ denote the first integral. Since $|\Phi|$ is bounded and $K(t)dt/t$ is integrable, the integral is absolutely convergent. Moreover $\Phi'(u) = -\sin(u)/u$. To justify differentiation, fix $0 < a < b < \infty$. Uniformly for $s \in [a, b]$,

$$|\partial_s \{K(t) \Phi(s/t)\}| \frac{1}{t} = \frac{K(t)}{s} |\sin(s/t)| \frac{1}{t} \leq C_{a,b} \begin{cases} t, & 0 < t \leq 1, \\ t^{-4}, & t \geq 1, \end{cases}$$

where for $t \geq 1$ we used $|\sin(s/t)| \leq s/t$. The right-hand side is integrable, so local dominated convergence gives

$$I'(s) = -\frac{4}{\pi s} \int_0^\infty \frac{t}{(1+t^2)^2} \sin(s/t) dt = -\frac{4}{\pi s} \int_0^\infty \frac{y}{(1+y^2)^2} \sin(sy) dy.$$

The standard contour integral

$$\int_0^\infty \frac{\cos(sy)}{(1+y^2)^2} dy = \frac{\pi}{4} (1+s) e^{-s}$$

and differentiation in s show that the last sine integral equals $\frac{\pi}{4} s e^{-s}$. Hence $I'(s) = -e^{-s}$. Dominated convergence gives $I(s) \rightarrow 0$ as $s \rightarrow \infty$, so $I(s) = e^{-s}$. Finally,

$$\int_0^\infty K(t) \frac{dt}{t} = \frac{4}{\pi} \int_0^\infty \frac{t}{(1+t^2)^2} dt = \frac{2}{\pi}.$$

$\square$

For $\xi > 0$, the multiplier of H_ε is $-2i\Phi(\varepsilon\xi)$. Lemma 14.4 therefore gives, first for Schwartz functions with Fourier support compactly contained in $(0, \infty)$,

$$P_s f = \frac{i}{2} \int_0^\infty K(t) H_{s/t} f \frac{dt}{t}. \quad (504)$$

Here the integral is initially a Bochner integral in $L^2(\mathbb{R})$. For fixed s , the map $t \mapsto H_{s/t} f$ is strongly measurable (on the Schwartz core this follows from joint measurability, or from continuity of the multiplier in L^2), the Hilbert truncation multipliers are uniformly bounded, and $K(t)dt/t$ has finite mass. The multiplier identity identifies this Bochner integral with $P_s f$ in L^2 .

Proposition 14.5 (Poisson variation dominated by Hilbert variation on the core). *If $f \in \mathcal{S}(\mathbb{R})$ and $\text{supp } \widehat{f} \Subset (0, \infty)$, then*

$$V^2(P_s f : s > 0) \leq \frac{1}{\pi} V^2(H_\varepsilon f : \varepsilon > 0) \quad (505)$$

pointwise.

Proof. For the present Schwartz data, $(t, x) \mapsto H_{s/t} f(x)$ is jointly measurable. Define

$$G_s(x) = \int_0^\infty K(t) |H_{s/t} f(x)| \frac{dt}{t}.$$

The uniform L^2 multiplier bound and Minkowski give

$$\|G_s\|_2 \leq \int_0^\infty K(t) \|H_{s/t} f\|_2 \frac{dt}{t} \leq C \|f\|_2 \int_0^\infty K(t) \frac{dt}{t} < \infty.$$

Thus $G_s(x) < \infty$ almost everywhere. The Bochner–Fubini theorem (equivalently, approximation of $t \mapsto K(t)H_{s/t} f/t$ in $L^1((0, \infty); L_x^2)$ by simple functions) says that its Bochner integral is represented by the pointwise absolutely convergent integral off this null set. Since the multiplier computation identifies the same Bochner integral with $P_s f$ in L^2 , (504) holds there after choosing representatives. Taking the union of these null sets over any fixed finite set of s -parameters gives one common full-measure set. For a finite partition $0 < s_0 < \dots < s_J$, apply (504) and Minkowski's inequality in ℓ_J^2 :

$$\begin{aligned} \left(\sum_{j=1}^J |P_{s_j} f - P_{s_{j-1}} f|^2 \right)^{1/2} &\leq \frac{1}{2} \int_0^\infty K(t) \left(\sum_{j=1}^J |H_{s_j/t} f - H_{s_{j-1}/t} f|^2 \right)^{1/2} \frac{dt}{t} \\ &\leq \frac{1}{\pi} V^2(H_\varepsilon f : \varepsilon > 0). \end{aligned}$$

Both the Poisson and sharp Hilbert trajectories of this Schwartz input are continuous in their positive parameters. It is therefore enough to take the supremum over rational finite partitions, for which one common full-measure set suffices. This proves the pointwise inequality. No infinite-parameter L^2 closure is used here; the finite-parameter L^2 closure needed below is carried out explicitly in the proof of Theorem 14.7. $\square$

We record explicitly the finite-parameter transference used below.

Lemma 14.6 (Finite-dimensional de Leeuw restriction). *Let E be a finite-dimensional Banach space and let $M : \mathbb{R} \rightarrow E$ be continuous. If*

$$\left\| \mathcal{F}^{-1}(M\widehat{f}) \right\|_{L^2(\mathbb{R}; E)} \leq C \|f\|_{L^2(\mathbb{R})}$$

for Schwartz f , then for every trigonometric polynomial g ,

$$\left\| \sum_{n \in \mathbb{Z}} M(n) \widehat{g}(n) e^{inx} \right\|_{L^2(\mathbb{T}; E)} \leq C \|g\|_{L^2(\mathbb{T})}. \quad (506)$$

Proof. This is the restriction argument of de Leeuw [13], applied with a finite-dimensional range. Let $\Lambda \subset \mathbb{Z}$ be the finite Fourier support of g . Choose a nonzero $\eta \in \mathcal{S}(\mathbb{R})$ with $\text{supp } \hat{\eta} \subset (-1/4, 1/4)$, periodically extend g , and put $f_\delta(x) = \eta(\delta x)g(x)$, $0 < \delta < 1$. Then

$$\hat{f}_\delta(\xi) = \sum_{n \in \Lambda} \hat{g}(n) \delta^{-1} \hat{\eta}((\xi - n)/\delta),$$

and these packets are disjoint. On the packet centered at n , continuity gives

$$\sup_{|\zeta| \leq 1/4} \|M(n + \delta \zeta) - M(n)\|_E \longrightarrow 0.$$

There are only finitely many packets. To record the normalization of the error, choose functionals $\ell_1, \dots, \ell_m \in E^*$ and a constant C_E such that

$$\|v\|_E \leq C_E \left(\sum_{a=1}^m |\ell_a(v)|^2 \right)^{1/2}.$$

Set

$$\omega_\Lambda(\delta) = \max_{n \in \Lambda} \sup_{|\zeta| \leq 1/4} \|M(n + \delta \zeta) - M(n)\|_E,$$

so $\omega_\Lambda(\delta) \rightarrow 0$. Packet disjointness and scalar Plancherel applied to the ℓ_a give, up to the fixed Fourier normalization,

$$\delta \left\| \mathcal{F}^{-1} \left(\sum_{n \in \Lambda} [M(\xi) - M(n)] \hat{g}(n) \delta^{-1} \hat{\eta}((\xi - n)/\delta) \right) \right\|_{L^2(\mathbb{R}; E)}^2 \leq C_{E, \eta} \omega_\Lambda(\delta)^2 \sum_{n \in \Lambda} |\hat{g}(n)|^2.$$

Consequently

$$\delta^{1/2} \left\| \mathcal{F}^{-1}(M \hat{f}_\delta) - \eta(\delta \cdot) \sum_{n \in \Lambda} M(n) \hat{g}(n) e^{inx} \right\|_{L^2(\mathbb{R}; E)} \longrightarrow 0.$$

We also use the elementary periodic averaging formula

$$\delta \int_{\mathbb{R}} |\eta(\delta x)|^2 A(x) dx \longrightarrow \left(\int_{\mathbb{R}} |\eta(u)|^2 du \right) \frac{1}{2\pi} \int_0^{2\pi} A(x) dx$$

for every continuous 2π -periodic scalar function A . Apply it first to $A = |g|^2$ and then to the squared E -norm of the finite E -valued trigonometric polynomial above. With the harmless Fourier normalization incorporated into $c_\eta > 0$, this gives

$$\begin{aligned} \delta \|f_\delta\|_2^2 &\longrightarrow c_\eta \|g\|_{L^2(\mathbb{T})}^2, \\ \delta \|\mathcal{F}^{-1}(M \hat{f}_\delta)\|_{L^2(E)}^2 &\longrightarrow c_\eta \left\| \sum_n M(n) \hat{g}(n) e^{inx} \right\|_{L^2(\mathbb{T}; E)}^2. \end{aligned}$$

The packet error tends to zero at precisely this normalization, and the same constant $c_\eta > 0$ occurs in both limits. Applying the assumed Euclidean inequality before taking the limit proves (506) without loss in the operator norm. $\square$

Theorem 14.7 (Failure of true V^2 for sharp Hilbert truncations). *There is no constant C such that*

$$\|V^\varepsilon(H_\varepsilon f : \varepsilon > 0)\|_2 \leq C \|f\|_2 \quad (507)$$

for every $f \in L^2(\mathbb{R})$.

Proof. Assume (507), and denote its constant by C_H . For every

$$f \in \mathcal{S}_+ := \{g \in \mathcal{S}(\mathbb{R}) : \text{supp } \widehat{g} \subseteq (0, \infty)\},$$

Proposition 14.5 and the assumed estimate give

$$\|V^2(P_s f : s > 0)\|_2 \leq \frac{C_H}{\pi} \|f\|_2. \quad (508)$$

We now pass only on a fixed finite parameter set. Let $D = \{s_0 < \dots < s_J\} \subset (0, \infty)$ and write V_D^2 for the corresponding finite variation. For every scalar D -path z ,

$$V_D^2(z) \leq 2 \left(\sum_{s \in D} |z_s|^2 \right)^{1/2}; \quad (509)$$

indeed, in any increasing variation chain each selected coordinate occurs in at most two consecutive differences. Since every P_s is an L^2 contraction, (509) implies

$$\|V_D^2(P_s h : s \in D)\|_2 \leq 2|D|^{1/2} \|h\|_2. \quad (510)$$

The class $\mathcal{S}_+$ is dense in the positive-frequency subspace $L_+^2(\mathbb{R})$. Given $f \in L_+^2$, choose $f_m \in \mathcal{S}_+$ with $f_m \rightarrow f$ in L^2 . Equation (510) gives convergence in $L^2(V_D^2)$, while (508) has a constant independent of D and m . Hence

$$\|V_D^2(P_s f : s \in D)\|_2 \leq \frac{C_H}{\pi} \|f\|_2, \quad f \in L_+^2(\mathbb{R}). \quad (511)$$

Reflection gives the same estimate on $L_-^2(\mathbb{R})$. Decomposing $f = f_+ + f_-$, using subadditivity of V_D^2 , and using orthogonality of the frequency projections, we obtain

$$\|V_D^2(P_s f : s \in D)\|_2 \leq \frac{\sqrt{2}C_H}{\pi} \|f\|_2 \quad (512)$$

for every $f \in L^2(\mathbb{R})$, uniformly in the finite set D .

Let $E_D = \mathbb{C}^{J+1}/\mathbb{C}\mathbf{1}$ with norm

$$\|[z]\|_{E_D} := \sup_{0 \leq j_0 < \dots < j_m \leq J} \left(\sum_{\ell=1}^m |z_{j_\ell} - z_{j_{\ell-1}}|^2 \right)^{1/2}.$$

This is the finite V^2 norm and is a genuine norm after quotienting by constant vectors. Equation (512) says that the continuous E_D -valued symbol

$$M_D(\xi) = [(e^{-s|\xi|})_{s \in D}]$$

defines an $L^2(\mathbb{R})$ to $L^2(\mathbb{R}; E_D)$ multiplier with norm independent of D . Notice that it is continuous also at $\xi = 0$ because the vector $(1)_{s \in D}$ is zero in the quotient.

Lemma 14.6 restricts M_D to the integers with the same uniform bound. The restricted coordinates are the periodic Poisson multipliers $e^{-s|n|}$. If trigonometric polynomials g_m tend to g in $L^2(\mathbb{T})$, then (509) and the L^2 contractivity of each periodic P_s give

$$\|V_D^2(P_s(g_m - g) : s \in D)\|_{L^2(\mathbb{T})} \leq 2|D|^{1/2} \|g_m - g\|_{L^2(\mathbb{T})}.$$

Thus scalar density extends the restricted bound to every $g \in L^2(\mathbb{T})$ and every finite D , with the original bound uniform in D .

Enumerate $\mathbb{Q}_+$ and let D_m be the increasing finite sets formed by the first m rationals. Pointwise,

$$V_{D_m}^2(P_s g : s \in D_m) \uparrow V_{\mathbb{Q}_+}^2(P_s g : s \in \mathbb{Q}_+).$$

For s in a compact subinterval of $(0, \infty)$, Cauchy–Schwarz in the Fourier series shows that $s \mapsto P_s g(x)$ is continuous (indeed the series is uniformly absolutely convergent). Consequently

$$V_{\mathbb{Q}_+}^2(P_s g : s \in \mathbb{Q}_+) = V^2(P_s g : s > 0).$$

Monotone convergence applied to the squares and the uniform finite- D_m bounds now yields

$$\|V^2(P_s g : s > 0)\|_{L^2(\mathbb{T})} \leq \frac{\sqrt{2}C_H}{\pi} \|g\|_{L^2(\mathbb{T})}.$$

Jones and Wang [20, Theorem 1.13, statement on p. 4495 and proof on p. 4517] give a bounded function on $\mathbb{T}$ whose periodic Poisson trajectory has infinite 2-variation almost everywhere. This function lies in $L^2(\mathbb{T})$, so it is in the domain of the estimate just obtained. Their V_2 is the supremum of the same quadratic increment sum over finite ordered parameter chains; adding or omitting a single base-point term does not affect the assertion that it is infinite. Their periodic Poisson operator has multiplier $r^{|n|}$ for $0 < r < 1$. Under $r = e^{-s}$ this is exactly $e^{-s|n|}$. Although this change of variables reverses order, reversing each finite chain leaves the set of increments, and hence its quadratic variation, unchanged. Thus their counterexample contradicts the preceding periodic estimate. $\square$

Remark 14.8 (Sharpness of the output fluctuation exponent). The failure already occurs for $n = 1$, $P = 0$, $b \equiv 1$, and the odd kernel on $\mathbb{S}^0$. Thus the pointwise critical jump $\mathbf{J}(F)$ —the paired/chain weak- ℓ^2 increment functional, up to the absolute comparison constants in (27)—cannot be replaced in the main theorems by the true strong variation $V^2(F)$.

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